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Ketohexokinase (KHK) iRNA compositions and methods of use thereof

Abstract

The present invention relates to RNAi agents, e.g., dsRNA agents, targeting the ketohexokinase (KHK) gene. The invention also relates to methods of using such RNAi agents to inhibit expression of a KHK gene and to methods of treating or preventing a KHK-associated disorder in a subject.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/060,990, filed on Dec. 2, 2022, which is a 35 § U.S.C. 111(a) continuation application which claims the benefit of priority to PCT/US2022/016890, filed on Feb. 18, 2022, which, in turn, claims the benefit of priority to U.S. Provisional Application No. 63/154,005, filed on Feb. 26, 2021, U.S. Provisional Application No. 63/223,581, filed on Jul. 20, 2021, and U.S. Provisional Application No. 63/280,668, filed on Nov. 18, 2021. The entire contents of each of the foregoing applications are incorporated herein by reference.

SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted electronically in XML file format and is hereby incorporated by reference in its entirety. Said XML copy, created on Oct. 11, 2023, is named 121301_14705_SL.xml and is 14,669,763 bytes in size.

BACKGROUND OF THE INVENTION

(2) Epidemiological studies have shown that a western diet is one of the leading causes of the modern obesity pandemic. Increase in fructose uptake, associated with the use of enriched soft drinks and processed food, is proposed to be a major contributing factor to the epidemic. High fructose corn sweeteners started gaining widespread use in the food industry by 1967. Although glucose and fructose have the same caloric value per molecule, the two sugars are metabolized differently and utilize different GLUT transporters. Fructose is almost exclusively metabolized in the liver, and unlike the glucose metabolism pathway, the fructose metabolism pathway is not regulated by feedback inhibition by the product (Khaitan Z et al., (2013) *J. Nutr. Metab.* 2013, Article ID 682673, 1-12). While hexokinase and phosphofructokinase (PFK) regulate the production of glyceraldehyde-3-P from glucose, fructokinase or ketohexokinase (KHK), which is responsible for phosphorylation of fructose to fructose-1-phosphate in the liver, it is not down regulated by increasing concentrations of fructose-1-phosphate. As a result, all fructose entering the cell is rapidly phosphorylated. (Cirillo P. et al., (2009) *J. Am. Soc. Nephrol.* 20: 545-553). Continued utilization of ATP to phosphorylate the fructose to fructose-1-phosphate results in intracellular phosphate depletion, ATP depletion, activation of AMP deaminase and formation of uric acid (Khaitan Z. et al., (2013) *J. Nutr. Metab.* Article ID 682673, 1-12). Increased uric acid further stimulates the up-regulation of KHK (Lanaspa M. A. et al., (2012) *PLOS ONE* 7(10): 1-11) and causes endothelial cell and adipocyte dysfunction. Fructose-1-phosphate is subsequently converted to glyceraldehyde by the action of aldolase B and is phosphorylated to glyceraldehyde-3-phosphate. The latter proceeds downstream to the glycolysis pathway to form pyruvate, which enters the citric acid cycle, wherefrom, under well-fed conditions, citrate is exported to the cytosol from the mitochondria, providing Acetyl Coenzyme A for lipogenesis (FIG. 1).

(3) The phosphorylation of fructose by KHK, and subsequent activation of lipogenesis leads to, for example, fatty liver, hypertriglyceridemia, dyslipidemia, and insulin resistance. Proinflammatory changes in renal proximal tubular cells have also been shown to be induced by KHK activity (Cirillo P. et al., (2009) *J. Am. Soc. Nephrol.* 20: 545-553). The phosphorylation of fructose by KHK is associated with diseases, disorders or conditions such as liver disease (e.g., fatty liver, steatohepatitis), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, hyperuricemia, gout, eating disorders, and excessive sugar craving. Accordingly, there is a need in the art for compositions and methods for treating diseases, disorders, and conditions associated with KHK activity.

SUMMARY OF THE INVENTION

(4) The present invention provides iRNA compositions which affect the RNA-induced silencing

complex (RISC)-mediated cleavage of RNA transcripts of a gene encoding ketohexokinase (KHK). The KHK gene may be within a cell, e.g., a cell within a subject, such as a human subject.

(5) In an aspect, the invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of ketohexokinase (KHK) in a cell, wherein the dsRNA agent comprises a sense strand and an antisense strand forming a double stranded region, wherein the sense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, or 21, contiguous nucleotides differing by no more than 0, 1, 2, or 3 nucleotides from the nucleotide sequence of SEQ ID NO:1 and the antisense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, 21, 22 or 23, contiguous nucleotides differing by no more than 1, 2, or 3 nucleotides from the nucleotide sequence of SEQ ID NO:2. In one embodiment, the dsRNA agent comprises at least one thermally destabilizing nucleotide modification, e.g., an abasic modification; a mismatch with the opposing nucleotide in the duplex; and destabilizing sugar modification, a 2'-deoxy modification, an acyclic nucleotide, an unlocked nucleic acids (UNA), or a glycerol nucleic acid (GNA), e.g., the antisense strand comprises at least one thermally destabilizing nucleotide modification.

(6) In another aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) for inhibiting expression of ketohexokinase (KHK) in a cell, wherein said dsRNA comprises a sense strand and an antisense strand forming a double stranded region, wherein the antisense strand comprises a region of complementarity to an mRNA encoding KHK, and wherein the region of complementarity comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, 21, 22 or 23, contiguous nucleotides differing by no more than 0, 1, 2, or 3 nucleotides from any one of the antisense nucleotide sequences in any one of Tables 2, 3, 5, 6, and 8-13.

(7) In one embodiment, the dsRNA agent comprises a sense strand comprising a contiguous nucleotide sequence which has at least 85%, e.g., 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100%, nucleotide sequence identity over its entire length to any one of the nucleotide sequences of the sense strands in any one of Tables 2, 3, 5, 6 and 8-13 and an antisense strand comprising a contiguous nucleotide sequence which has at least 85%, e.g., 85%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, 99%, or 100%, nucleotide sequence identity over its entire length to any one of the nucleotide sequences of the antisense strands in any one of Tables 2, 3, 5, 6 and 8-13.

(8) In one embodiment, the dsRNA agent comprises a sense strand comprising at least 15, e.g., 15, 16, 17, 18, 19, 20, or 21, contiguous nucleotides differing by no more than three nucleotides from any one of the nucleotide sequences of the sense strands in any one of Tables 2, 3, 5, 6 and 8-13 and an antisense strand comprising at least 15, e.g., 15, 16, 17, 18, 19, 20, 21, 22 or 23, contiguous nucleotides differing by no more than three nucleotides from any one of the nucleotide sequences of the antisense strands in any one of Tables 2, 3, 5, 6 and 8-13.

(9) In one embodiment, the dsRNA agent comprises a sense strand comprising at least 15, e.g., 15, 16, 17, 18, 19, 20, or 21, contiguous nucleotides differing by no more than two nucleotides from any one of the nucleotide sequences of the sense strands in any one of Tables 2, 3, 5, 6 and 8-13 and an antisense strand comprising at least 15, e.g., 15, 16, 17, 18, 19, 20, 21, 22 or 23, contiguous nucleotides differing by no more than two nucleotides from any one of the nucleotide sequences of the antisense strands in any one of Tables 2, 3, 5, 6 and 8-13.

(10) In one embodiment, the dsRNA agent comprises a sense strand comprising at least 15, e.g., 15, 16, 17, 18, 19, 20, or 21, contiguous nucleotides differing by no more than one nucleotide from any one of the nucleotide sequences of the sense strands in any one of Tables 2, 3, 5, 6 and 8-13 and an antisense strand comprising at least 15, e.g., 15, 16, 17, 18, 19, 20, 21, 22 or 23, contiguous nucleotides differing by no more than one nucleotide from any one of the nucleotide sequences of the antisense strands in any one of Tables 2, 3, 5, 6 and 8-13.

(11) In one embodiment, the dsRNA agent comprises a sense strand comprising or consisting of a nucleotide sequence selected from the group consisting of any one of the nucleotide sequences of the sense strands in any one of Tables 2, 3, 5, 6 and 8-13 and an antisense strand comprising or

consisting of a nucleotide sequence selected from the group consisting of any one of the nucleotide sequences of the antisense strands in any one of Tables 2, 3, 5, 6 and 8-13.

(12) In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) for inhibiting expression of ketohexokinase (KHK) in a cell, wherein said dsRNA comprises a sense strand and an antisense strand forming a double stranded region, wherein the sense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, or 21, contiguous nucleotides differing by no more than 0, 1, 2, or 3 nucleotides from any one of the nucleotide sequence of nucleotides 120-162; 164-188; 181-207; 193-217; 209-231; 283-306; 508-546; 568-603; 596-632; 640-674; 746-806; 806-835; 917-942; 936-084; 1016-1041; 1100-1123; 1149-1175; 1160-1193; 1205-1229; 1252-1283; 1334-1356; 1407-1429; 1472-1497; 1506-1533; 1539-1561; 1704-1727; 1747-1787; 1850-1873; 1936-1964; 1960-1990; 2015-2048; 2060-2095; 2090-2118; 2124-2160; 2181-2200; 2221-2262 of SEQ ID NO: 1, and the antisense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, 21, 22 or 23, contiguous nucleotides differing by no more than three, e.g., 3, 2, 1, or 0, from the corresponding nucleotide sequence of SEQ ID NO:2.

(13) In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) for inhibiting expression of ketohexokinase (KHK) in a cell, wherein said dsRNA comprises a sense strand and an antisense strand forming a double stranded region, wherein the sense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, or 21, contiguous nucleotides differing by no more than 0, 1, 2, or 3 nucleotides from any one of the nucleotide sequence of nucleotides 517-539; 521-543; 524-546; 517-546; 581-603; 610-632; 747-769; 749-771; 752-774; 755-777; 757-779; 758-780; 764-786; 776-798; 781-803; 747-803; 920-942; 941-963; 944-966; 950-972; 962-984; 920-984; 1149-1171; 1161-1183; 1165-1187; 1171-1193; 1149-1193; 1205-1227; 1206-1228; 1205-1228; 1334-1356; 1472-1494; 1475-1497; 1472-1497 of SEQ ID NO: 1, and the antisense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, 21, 22 or 23, contiguous nucleotides differing by no more than three, e.g., 3, 2, 1, or 0, from the corresponding nucleotide sequence of SEQ ID NO:2.

(14) In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) for inhibiting expression of ketohexokinase (KHK) in a cell, wherein said dsRNA comprises a sense strand and an antisense strand forming a double stranded region, wherein the sense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, or 21, contiguous nucleotides differing by no more than 0, 1, 2, or 3 nucleotides from any one of the nucleotide sequence of nucleotides 517-539; 524-546; 517-546; 753-775; 757-779; 753-779; 764-786; 767-789; 768-790; 769-791; 764-791; 773-795; 781-803; 773-803; 753-803; 808-830; 937-959; 941-963; 944-966; 941-966; 948-970; 950-972; 948-972; 1160-1182; 1161-1183; 1160-1183; and 1207-1229 of SEQ ID NO: 1, and the antisense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, 21, 22 or 23, contiguous nucleotides differing by no more than three, e.g., 3, 2, 1, or 0, from the corresponding nucleotide sequence of SEQ ID NO:2.

(15) In one embodiment, the antisense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, 21, 22 or 23, contiguous nucleotides differing by no more than 0, 1, 2, or 3 nucleotides from any one of the antisense strand nucleotide sequences of a duplex selected from the group consisting of AD-1613400; AD-1613243; AD-1290757.3; AD-1290878.3; AD-1290969.3; AD-1423317.2; AD-1423327.2; AD-1423336.2; AD-1290599.3; AD-1523172.1; AD-1290837.3; AD-1523173.1; AD-1290884.3; AD-1523174.1; AD-1290959.3; AD-1523175.1; AD-1423311.2; AD-1423324.2; AD-1523176.1; AD-1423329.2; AD-1423333.2; AD-1423330.2; AD-1523177.1; AD-1290885.3; AD-1523178.1; AD-1423334.2; AD-1523179.1; AD-1523180.1; AD-1290539.3; and AD-1523181.1.

(16) In another aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of ketohexokinase (KHK) in a cell, wherein said dsRNA comprises a sense strand and an antisense strand forming a double stranded region, wherein the sense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, or 21, contiguous nucleotides differing by no more than three, e.g., 3, 2, 1, or 0, nucleotides from any one of the nucleotide sequence of

nucleotides 495-517, 492-517, 500-529, 514-551, 517-539, 524-546, 517-548, 614-643, 625-647, 625-660, 642-664, 642-672, 753-811, 754-780, 762-791, 764-786, 772-800, 781-803, 805-827, 808-830, 809-831, 792-838, 931-982, 944-966, 947-969, 948-970, 948-982, 1011-1035, 1021-1043, 1019-1050, 1063-1091, 1150-1192, 1152-1176, 1160-1192, 1160-1182, 1162-1184, 1198-1230, 1198-1221, and 1202-1230 of SEQ ID NO: 1, and the antisense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, 21, 22 or 23, contiguous nucleotides differing by no more than 3, e.g., 3, 2, 1, or 0, nucleotides from the corresponding nucleotide sequence of SEQ ID NO:2.

(17) In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of ketohexokinase (KHK) in a cell, wherein said dsRNA comprises a sense strand and an antisense strand forming a double stranded region, wherein the sense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, or 21, contiguous nucleotides differing by no more than three, e.g., 3, 2, 1, or 0, nucleotides from any one of the nucleotide sequence of nucleotides 513-556, 753-813, 936-981, 1155-1193, 1200-1229, 1704-1727, 1747-1787, 1850-1873, 1936-1964, 1960-1990, 1936-1990, 2015-2048, 2060-2095, 2090-2118, 2060-2118, 2124-2160, 2181-2220, 2221-2249, 2181-2249, 2240-2262, 2221-2262, and 2181-2262 of SEQ ID NO: 1, and the antisense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, 21, 22 or 23, contiguous nucleotides differing by no more than 3, e.g., 3, 2, 1, or 0, nucleotides from the corresponding nucleotide sequence of SEQ ID NO:2.

(18) In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of ketohexokinase (KHK) in a cell, wherein said dsRNA comprises a sense strand and an antisense strand forming a double stranded region, wherein the sense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, or 21, contiguous nucleotides differing by no more than 3, e.g., 3, 2, 1, or 0, nucleotides from the nucleotide sequence of nucleotides 1207-1229 or 937-959 of SEQ ID NO: 1, and the antisense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, 21, 22, or 23, contiguous nucleotides differing by no more than 3, e.g., 3, 2, 1, or 0, from the corresponding nucleotide sequence of SEQ ID NO:2.

(19) In one embodiment, the sense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, or 21, contiguous nucleotides differing by no more than three, e.g., 3, 2, 1, or 0, nucleotides from the nucleotide sequence of nucleotides 1207-1229 of SEQ ID NO: 1, and the antisense strand comprises at least 15, e.g., 15, 16, 17, 18, 19, 20, 21, 22, or 23, contiguous nucleotides differing by no more than 3, e.g., 3, 2, 1, or 0, from the corresponding nucleotide sequence of SEQ ID NO:2.

(20) In one embodiment, the dsRNA agent comprises at least one modified nucleotide.

(21) In one embodiment, substantially all of the nucleotides of the sense strand; substantially all of the nucleotides of the antisense strand comprise a modification; or substantially all of the nucleotides of the sense strand and substantially all of the nucleotides of the antisense strand comprise a modification.

(22) In one embodiment, all of the nucleotides of the sense strand comprise a modification; all of the nucleotides of the antisense strand comprise a modification; or all of the nucleotides of the sense strand and all of the nucleotides of the antisense strand comprise a modification.

(23) In one embodiment, at least one of the modified nucleotides is selected from the group consisting of a deoxy-nucleotide, a 3'-terminal deoxythymidine (dT) nucleotide, a 2'-O-methyl modified nucleotide, a 2'-fluoro modified nucleotide, a 2'-deoxy-modified nucleotide, a locked nucleotide, an unlocked nucleotide, a conformationally restricted nucleotide, a constrained ethyl nucleotide, an abasic nucleotide, a 2'-amino-modified nucleotide, a 2'-O-allyl-modified nucleotide, 2'-C-alkyl-modified nucleotide, 2'-hydroxyl-modified nucleotide, a 2'-methoxyethyl modified nucleotide, a 2'-O-alkyl-modified nucleotide, a morpholino nucleotide, a phosphoramidate, a non-natural base comprising nucleotide, a tetrahydropyran modified nucleotide, a 1,5-anhydrohexitol modified nucleotide, a cyclohexenyl modified nucleotide, a nucleotide comprising a phosphorothioate group, a nucleotide comprising a methylphosphonate group, a nucleotide comprising a 5'-phosphate, a nucleotide comprising a 5'-phosphate mimic, a thermally destabilizing

nucleotide, a glycol modified nucleotide (GNA), a nucleotide comprising a 2' phosphate, and a 2-O—(N-methylacetamide) modified nucleotide; and combinations thereof.

(24) In one embodiment, the modifications on the nucleotides are selected from the group consisting of LNA, HNA, CeNA, 2'-methoxyethyl, 2'-O-alkyl, 2'-O-allyl, 2'-C-allyl, 2'-fluoro, 2'-deoxy, 2'-hydroxyl, and glycol; and combinations thereof.

(25) In one embodiment, at least one of the modified nucleotides is selected from the group consisting of a deoxy-nucleotide, a 2'-O-methyl modified nucleotide, a 2'-fluoro modified nucleotide, a 2'-deoxy-modified nucleotide, a glycol modified nucleotide (GNA), e.g., Ggn, Cgn, Tgn, or Agn, a nucleotide comprising a 2' phosphate, e.g., G2p, C2p, A2p, U2p, a vinyl-phosphonate nucleotide; and combinations thereof.

(26) In another embodiment, at least one of the modifications on the nucleotides is a thermally destabilizing nucleotide modification.

(27) In one embodiment, the thermally destabilizing nucleotide modification is selected from the group consisting of an abasic modification; a mismatch with the opposing nucleotide in the duplex; and destabilizing sugar modification, a 2'-deoxy modification, an acyclic nucleotide, an unlocked nucleic acids (UNA), and a glycerol nucleic acid (GNA).

(28) In some embodiments, the modified nucleotide comprises a short sequence of 3'-terminal deoxythymidine nucleotides (dT).

(29) In some embodiments, the modifications on the nucleotides are 2'-O-methyl, GNA and 2'-fluoro modifications.

(30) In some embodiments, the dsRNA agent further comprises at least one phosphorothioate internucleotide linkage. In some embodiments, the dsRNA agent comprises 6-8 phosphorothioate internucleotide linkages. In one embodiment, the phosphorothioate or methylphosphonate internucleotide linkage is at the 3'-terminus of one strand. Optionally, the strand is the antisense strand. In another embodiment, the strand is the sense strand. In a related embodiment, the phosphorothioate or methylphosphonate internucleotide linkage is at the 5'-terminus of one strand. Optionally, the strand is the antisense strand. In another embodiment, the strand is the sense strand. In another embodiment, the phosphorothioate or methylphosphonate internucleotide linkage is at the both the 5'- and 3'-terminus of one strand. Optionally, the strand is the antisense strand. In another embodiment, the strand is the sense strand.

(31) The double stranded region may be 19-30 nucleotide pairs in length; 19-25 nucleotide pairs in length; 19-23 nucleotide pairs in length; 23-27 nucleotide pairs in length; or 21-23 nucleotide pairs in length.

(32) In one embodiment, each strand is independently no more than 30 nucleotides in length.

(33) In one embodiment, the sense strand is 21 nucleotides in length and the antisense strand is 23 nucleotides in length.

(34) The region of complementarity may be at least 17 nucleotides in length; between 19 and 23 nucleotides in length; or 19 nucleotides in length.

(35) In one embodiment, at least one strand comprises a 3' overhang of at least 1 nucleotide. In another embodiment, at least one strand comprises a 3' overhang of at least 2 nucleotides.

(36) In one embodiment, the dsRNA agent further comprises a ligand.

(37) In one embodiment, the ligand is conjugated to the 3' end of the sense strand of the dsRNA agent.

(38) In one embodiment, the ligand is an N-acetylgalactosamine (GalNAc) derivative.

(39) In one embodiment, the ligand is one or more GalNAc derivatives attached through a monovalent, bivalent, or trivalent branched linker.

(40) In one embodiment the ligand is

(41) ##STR00001##

(42) In one embodiment, the dsRNA agent is conjugated to the ligand as shown in the following schematic

(43) ##STR00002##

and, wherein X is O or S.

(44) In one embodiment, the X is O.

(45) In one embodiment, the dsRNA agent further comprises at least one phosphorothioate or methylphosphonate internucleotide linkage.

(46) In one embodiment, the phosphorothioate or methylphosphonate internucleotide linkage is at the 3'-terminus of one strand, e.g., the antisense strand or the sense strand.

(47) In another embodiment, the phosphorothioate or methylphosphonate internucleotide linkage is at the 5'-terminus of one strand, e.g., the antisense strand or the sense strand.

(48) In one embodiment, the phosphorothioate or methylphosphonate internucleotide linkage is at the both the 5'- and 3'-terminus of one strand. In one embodiment, the strand is the antisense strand.

(49) In one embodiment, the base pair at the 1 position of the 5'-end of the antisense strand of the duplex is an AU base pair.

(50) In one aspect, the present invention provides a double stranded ribonucleic acid (dsRNA) agent for inhibiting expression of ketohexokinase (KHK) in a cell, or a pharmaceutically acceptable salt thereof, comprising a sense strand and an antisense strand forming a double stranded region, wherein a) the nucleotide sequence of the sense strand differs by no more than 4 bases from the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' (SEQ ID NO: 2042) and the nucleotide sequence of the antisense strand differs by no more than 4 bases from the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' (SEQ ID NO: 1532), wherein a, g, c and u are 2'-O-methyl (2'-OMe) A, G, C, and U; Cf and Uf are 2'-deoxy-2'-fluoro (2'-F) C and U; dC, dA, and dT are 2'-deoxy C, A, and T; and s is a phosphorothioate linkage; or b) the nucleotide sequence of the sense strand differs by no more than 4 bases from the nucleotide sequence 5'-csusacggagAfCfGfugguguuugu-3' (SEQ ID NO: 2043) and the nucleotide sequence of the antisense strand differs by no more than 4 bases from the nucleotide sequence 5'-asdCsaadAcdAccacdGuCfuccguagscsc-3' (SEQ ID NO: 1511), wherein a, g, c and u are 2'-O-methyl (2'-OMe) A, G, C, and U; Cf and Uf are 2'-deoxy-2'-fluoro (2'-F) C and U; dC, dA, and dG are 2'-deoxy C, A, and G; and s is a phosphorothioate linkage.

(51) In one embodiment, a) the nucleotide sequence of the sense strand differs by no more than 3 bases from the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' (SEQ ID NO: 2042) and the nucleotide sequence of the antisense strand differs by no more than 3 bases from the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' (SEQ ID NO: 1532), wherein a, g, c and u are 2'-O-methyl (2'-OMe) A, G, C, and U; Cf and Uf are 2'-deoxy-2'-fluoro (2'-F) C and U; dC, dA, and dT are 2'-deoxy C, A, and T; and s is a phosphorothioate linkage; or b) the nucleotide sequence of the sense strand differs by no more than 3 bases from the nucleotide sequence 5'-csusacggagAfCfGfugguguuugu-3' (SEQ ID NO: 2043) and the nucleotide sequence of the antisense strand differs by no more than 3 bases from the nucleotide sequence 5'-asdCsaadAcdAccacdGuCfuccguagscsc-3' (SEQ ID NO: 1511), wherein a, g, c and u are 2'-O-methyl (2'-OMe) A, G, C, and U; Cf and Uf are 2'-deoxy-2'-fluoro (2'-F) C and U; dC, dA, and dG are 2'-deoxy C, A, and G; and s is a phosphorothioate linkage.

(52) In one embodiment, a) the nucleotide sequence of the sense strand differs by no more than 2 bases from the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' (SEQ ID NO: 2042) and the nucleotide sequence of the antisense strand differs by no more than 2 bases from the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' (SEQ ID NO: 1532), wherein a, g, c and u are 2'-O-methyl (2'-OMe) A, G, C, and U; Cf and Uf are 2'-deoxy-2'-fluoro (2'-F) C and U; dC, dA, and dT are 2'-deoxy C, A, and T; and s is a phosphorothioate linkage; or b) the nucleotide sequence of the sense strand differs by no more than 2 bases from the nucleotide sequence 5'-csusacggagAfCfGfugguguuugu-3' (SEQ ID NO: 2043) and the nucleotide sequence of the antisense strand differs by no more than 2 bases from the nucleotide sequence 5'-asdCsaadAcdAccacdGuCfuccguagscsc-3' (SEQ ID NO: 1511), wherein a, g, c and u are 2'-O-

methyl (2'-OMe) A, G, C, and U; Cf and Uf are 2'-deoxy-2'-fluoro (2'-F) C and U; dC, dA, and dG are 2'-deoxy C, A, and G; and s is a phosphorothioate linkage.

(53) In one embodiment, a) the nucleotide sequence of the sense strand differs by no more than 1 base from the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' (SEQ ID NO: 2042) and the nucleotide sequence of the antisense strand differs by no more than 1 base from the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' (SEQ ID NO: 1532), wherein a, g, c and u are 2'-O-methyl (2'-OMe) A, G, C, and U; Cf and Uf are 2'-deoxy-2'-fluoro (2'-F) C and U; dC, dA, and dT are 2'-deoxy C, A, and T; and s is a phosphorothioate linkage; or b) the nucleotide sequence of the sense strand differs by no more than 1 base from the nucleotide sequence 5'-csusacggagAfCfGfugguguuugu-3' (SEQ ID NO: 2043) and the nucleotide sequence of the antisense strand differs by no more than 1 base from the nucleotide sequence 5'-asdCsaadAcdAccacdGuCfuccguagscsc-3' (SEQ ID NO: 1511), wherein a, g, c and u are 2'-O-methyl (2'-OMe) A, G, C, and U; Cf and Uf are 2'-deoxy-2'-fluoro (2'-F) C and U; dC, dA, and dG are 2'-deoxy C, A, and G; and s is a phosphorothioate linkage.

(54) In one embodiment, a) the nucleotide sequence of the sense strand comprises the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' (SEQ ID NO: 2042) and the nucleotide sequence of the antisense strand comprises the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' (SEQ ID NO: 1532), wherein a, g, c and u are 2'-O-methyl (2'-OMe) A, G, C, and U; Cf and Uf are 2'-deoxy-2'-fluoro (2'-F) C and U; dC, dA, and dT are 2'-deoxy C, A, and T; and s is a phosphorothioate linkage; or b) the nucleotide sequence of the sense strand comprises the nucleotide sequence 5'-csusacggagAfCfGfugguguuugu-3' (SEQ ID NO: 2043) and the nucleotide sequence of the antisense strand comprises the nucleotide sequence 5'-asdCsaadAcdAccacdGuCfuccguagscsc-3' (SEQ ID NO: 1511), wherein a, g, c and u are 2'-O-methyl (2'-OMe) A, G, C, and U; Cf and Uf are 2'-deoxy-2'-fluoro (2'-F) C and U; dC, dA, and dG are 2'-deoxy C, A, and G; and s is a phosphorothioate linkage.

(55) In one embodiment, the dsRNA agent further comprises a ligand.

(56) In one embodiment, the ligand is conjugated to the 3' end of the sense strand of the dsRNA agent.

(57) In one embodiment, the ligand is an N-acetylgalactosamine (GalNAc) derivative.

(58) In one embodiment, the ligand is one or more GalNAc derivatives attached through a monovalent, bivalent, or trivalent branched linker.

(59) In one embodiment, the ligand is

(60) ##STR00003##

(61) In one embodiment, the dsRNA agent is conjugated to the ligand as shown in the following schematic

(62) ##STR00004##

and, wherein X is O or S.

(63) In one embodiment, X is O.

(64) The present invention also provides cells containing any of the dsRNA agents of the invention and pharmaceutical compositions comprising any of the dsRNA agents of the invention.

(65) The pharmaceutical composition of the invention may include dsRNA agent in an unbuffered solution, e.g., saline or water, or the pharmaceutical composition of the invention may include the dsRNA agent in a buffer solution, e.g., a buffer solution comprising acetate, citrate, prolamine, carbonate, or phosphate or any combination thereof; or phosphate buffered saline (PBS).

(66) In one aspect, the present invention provides a method of inhibiting expression of a ketohexokinase (KHK) gene in a cell. The method includes contacting the cell with any of the dsRNAs of the invention or any of the pharmaceutical compositions of the invention, thereby inhibiting expression of the KHK gene in the cell.

(67) In one embodiment, the cell is within a subject, e.g., a human subject, e.g., a subject having a ketohexokinase (KHK)-associated disorder, such as a KHK-associated disorder selected from the

group consisting of liver disease (e.g., fatty liver, steatohepatitis, non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, hyperuricemia, gout, eating disorders, and excessive sugar craving.

(68) In one embodiment, contacting the cell with the dsRNA agent inhibits the expression of KHK by at least 50%, 60%, 70%, 80%, 90%, or 95%.

(69) In one embodiment, inhibiting expression of KHK decreases KHK protein level in serum of the subject by at least 50%, 60%, 70%, 80%, 90%, or 95%.

(70) In one aspect, the present invention provides a method of treating a subject having a disorder that would benefit from reduction in ketohexokinase (KHK) expression. The method includes administering to the subject a therapeutically effective amount of any of the dsRNAs of the invention or any of the pharmaceutical compositions of the invention, thereby treating the subject having the disorder that would benefit from reduction in KHK expression.

(71) In another aspect, the present invention provides a method of preventing at least one symptom in a subject having a disorder that would benefit from reduction in ketohexokinase (KHK) expression. The method includes administering to the subject a prophylactically effective amount of any of the dsRNAs of the invention or any of the pharmaceutical compositions of the invention, thereby preventing at least one symptom in the subject having the disorder that would benefit from reduction in KHK expression.

(72) In certain embodiments, the KHK-associated disorder is a liver disease, e.g., fatty liver disease such as NAFLD or NASH. In certain embodiments, the KHK-associated disorder is dyslipidemia, e.g., elevated serum triglycerides, elevated serum LDL, elevated serum cholesterol, lowered serum HDL, postprandial hypertriglyceridemia. In another embodiment, the KHK-associated disorder is a disorder of glycemic control, e.g., insulin resistance not resulting from an immune response against insulin, glucose resistance, type 2 diabetes. In certain embodiments, the KHK-associated disorder is a cardiovascular disease, e.g., hypertension, endothelial cell dysfunction. In certain embodiments, the KHK-associated disorder is a kidney disease, e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease. In certain embodiments, the disease is metabolic syndrome. In certain embodiments, the KHK-associated disorder is a disease of lipid deposition or dysfunction, e.g., visceral adipose deposition, fatty liver, obesity. In certain embodiments, the KHK-associated disorder is a disease of elevated uric acid, e.g., gout, hyperuricemia. In certain embodiments the KHK-associated disorder is an eating disorder such as excessive sugar craving.

(73) In certain embodiments, the administration of the dsRNA to the subject causes a decrease in fructose metabolism. In certain embodiments, the administration of the dsRNA causes a decrease in the level of KHK in the subject, especially hepatic KHK, especially KHK-C in a subject with elevated KHK. In certain embodiments, the administration of the dsRNA causes a decrease in fructose metabolism in the subject. In certain embodiments, the administration of the dsRNA causes a decrease in the level of uric acid, e.g., serum uric acid, in a subject with elevated serum uric acid, e.g., elevated serum uric acid associated with gout. In certain embodiments, the administration of the dsRNA causes a normalization of serum lipids, e.g., triglycerides including postprandial triglycerides, LDL, HDL, or cholesterol, in a subject with at least one abnormal serum lipid level. In certain embodiments, the administration of the dsRNA causes a normalization of lipid deposition, e.g., a decrease of lipid deposition in the liver (e.g., decrease of NAFLD or NASH), a decrease of visceral fat deposition, a decrease in body weight. In certain embodiments, the administration of the dsRNA causes a normalization of insulin or glucose response in a subject

with abnormal insulin response not related to an immune response to insulin, or abnormal glucose response. In certain embodiments, the administration of the dsRNA results in an improvement of kidney function, or a stoppage or reduction of the rate of loss of kidney function. In certain embodiments, the dsRNA causes a reduction of hypertension, i.e., elevated blood pressure.

(74) In certain embodiments, the invention further comprises administering an additional agent to a subject with a KHK-associated disease. In certain embodiments, treatments known in the art for the various KHK-associated diseases are used in combination with the RNAi agents of the invention. Such treatments are discussed below.

(75) In one embodiment, the subject is human.

(76) In one embodiment, the dsRNA agent is administered to the subject at a dose of about 0.01 mg/kg to about 50 mg/kg.

(77) In one embodiment, the dsRNA agent is administered to the subject subcutaneously.

(78) In one embodiment, the methods of the invention include further determining the level of KHK in a sample(s) from the subject.

(79) In one embodiment, the level of KHK in the subject sample(s) is a KHK protein level in a blood or serum sample(s).

(80) In one embodiment, the methods of the invention further comprise measuring the uric acid level, especially serum uric acid level, in the subject. In one embodiment, the methods of the invention further comprise measuring the urine fructose level in the subject. In one embodiment, the methods of the invention further comprise measuring a serum lipid level in a subject. In certain embodiments, the methods of the invention further include measuring insulin or glucose sensitivity in a subject. In certain embodiments, a decrease in the levels of expression or activity of fructose metabolism indicates that the KHK-associated disease is being treated or prevented.

(81) The present invention also provides kits comprising any of the dsRNAs of the invention or any of the pharmaceutical compositions of the invention, and optionally, instructions for use.

(82) The present invention further provides an RNA-induced silencing complex (RISC) comprising an antisense strand of any of the dsRNA agents of the invention.

(83) In another embodiment, the RNAi agent is a pharmaceutically acceptable salt thereof. "Pharmaceutically acceptable salts" of each of RNAi agents herein include, but are not limited to, a sodium salt, a calcium salt, a lithium salt, a potassium salt, an ammonium salt, a magnesium salt, an mixtures thereof. One skilled in the art will appreciate that the RNAi agent, when provided as a polycationic salt having one cation per free acid group of the optionally modified phosphodiester backbone and/or any other acidic modifications (e.g., 5'-terminal phosphonate groups). For example, an oligonucleotide of "n" nucleotides in length contains n-1 optionally modified phosphodiesters, so that an oligonucleotide of 21 nt in length may be provided as a salt having up to 20 cations (e.g, 20 sodium cations). Similarly, an RNAi agents having a sense strand of 21 nt in length and an antisense strand of 23 nt in length may be provided as a salt having up to 42 cations (e.g, 42 sodium cations). In the preceding example, where the RNAi agent also includes a 5'-terminal phosphate or a 5'-terminal vinylphosphonate group, the RNAi agent may be provided as a salt having up to 44 cations (e.g, 44 sodium cations).

(84) The present invention is further illustrated by the following detailed description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 depicts the classic and alternative lipogenic pathways of fructose. In the classical pathway, triglycerides (TG) are a direct product of fructose metabolism by the action of multiple enzymes including aldolase B (Aldo B) and fatty acid synthase (FAS). In an alternative pathway, uric acid produced from the nucleotide turnover that occurs during the phosphorylation of fructose

to fructose-1-phosphate (F-1-P) results in the generation of mitochondrial oxidative stress (mtROS), which causes a decrease in the activity of aconitase (ACO2) in the Krebs cycle. As a consequence, the ACO2 substrate, citrate, accumulates and is released to the cytosol where it acts as substrate for TG synthesis through the activation of ATP citrate lyase (ACL) and fatty acid synthase. AMPD2, AMP deaminase 2; IMP, inosine monophosphate; PO.sub.4, phosphate (from Johnson et al. (2013) *Diabetes*. 62:3307-3315).

(2) FIG. 2 is a graph depicting the effect of administration of a single 3 mg/kg dose of the indicated duplexes on the level of KHK mRNA and KHK protein at Day 29 post-dose. The level of mRNA and protein are depicted as % remaining normalized to the pre-dose level of mRNA and protein.

(3) FIG. 3 is a graph depicting the effect of administration of a single 3 mg/kg dose of the indicated duplexes on the level of KHK mRNA at Day 50 post-dose. The level of mRNA is depicted as % remaining normalized to the pre-dose level of mRNA.

(4) FIG. 4 is a graph depicting the effect of administration of a single 3 mg/kg dose of the indicated duplexes on the level of KHK mRNA at Day 78 post-dose. The level of mRNA is depicted as % remaining normalized to the pre-dose level of mRNA.

DETAILED DESCRIPTION OF THE INVENTION

(5) The present invention provides iRNA compositions which effect the RNA-induced silencing complex (RISC)-mediated cleavage of RNA transcripts of a ketohexokinase (KHK) gene. The gene may be within a cell, e.g., a cell within a subject, such as a human. The use of these iRNAs enables the targeted degradation of mRNAs of the corresponding gene (KHK gene) in mammals.

(6) The iRNAs of the invention have been designed to target the human ketohexokinase (KHK) gene, including portions of the gene that are conserved in the ketohexokinase (KHK) orthologs of other mammalian species. Without intending to be limited by theory, it is believed that a combination or sub-combination of the foregoing properties and the specific target sites or the specific modifications in these iRNAs confer to the iRNAs of the invention improved efficacy, stability, potency, durability, and safety.

(7) Accordingly, the present invention provides methods for treating and preventing a ketohexokinase (KHK)-associated disorder, e.g., liver disease (e.g., fatty liver, steatohepatitis, non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, hyperuricemia, gout, eating disorders, and excessive sugar craving, using iRNA compositions which effect the RNA-induced silencing complex (RISC)-mediated cleavage of RNA transcripts of a KHK gene.

(8) The iRNAs of the invention include an RNA strand (the antisense strand) having a region which is up to about 30 nucleotides or less in length, e.g., 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length, which region is substantially complementary to at least part of an mRNA transcript of a KHK gene.

(9) In certain embodiments, one or both of the strands of the double stranded RNAi agents of the invention is up to 66 nucleotides in length, e.g., 36-66, 26-36, 25-36, 31-60, 22-43, 27-53 nucleotides in length, with a region of at least 19 contiguous nucleotides that is substantially complementary to at least a part of an mRNA transcript of a KHK gene. In some embodiments, such iRNA agents having longer length antisense strands can, for example, include a second RNA strand (the sense strand) of 20-60 nucleotides in length wherein the sense and antisense strands form a duplex of 18-30 contiguous nucleotides.

(10) The use of iRNAs of the invention enables the targeted degradation of mRNAs of the corresponding gene (KHK gene) in mammals. Using in vitro assays, the present inventors have

demonstrated that iRNAs targeting a KHK gene can potentially mediate RNAi, resulting in significant inhibition of expression of a KHK gene. Thus, methods and compositions including these iRNAs are useful for treating a subject having a KHK-associated disorder, e.g., liver disease (e.g., fatty liver, steatohepatitis, non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, hyperuricemia, gout, eating disorders, and excessive sugar craving.

(11) Accordingly, the present invention provides methods and combination therapies for treating a subject having a disorder that would benefit from inhibiting or reducing the expression of a KHK gene, e.g., a KHK-associated disorder, such as liver disease (e.g., fatty liver, steatohepatitis, non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, hyperuricemia, gout, eating disorders, and excessive sugar craving, using iRNA compositions which effect the RNA-induced silencing complex (RISC)-mediated cleavage of RNA transcripts of a KHK gene.

(12) The present invention also provides methods for preventing at least one symptom in a subject having a disorder that would benefit from inhibiting or reducing the expression of a KHK gene, e.g., liver disease (e.g., fatty liver, steatohepatitis, non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, hyperuricemia, gout, eating disorders, and excessive sugar craving.

(13) The following detailed description discloses how to make and use compositions containing iRNAs to inhibit the expression of a KHK gene as well as compositions, uses, and methods for treating subjects that would benefit from inhibition and/or reduction of the expression of a KHK gene, e.g., subjects susceptible to or diagnosed with a KHK-associated disorder.

I. Definitions

(14) In order that the present invention may be more readily understood, certain terms are first defined. In addition, it should be noted that whenever a value or range of values of a parameter are recited, it is intended that values and ranges intermediate to the recited values are also intended to be part of this invention.

(15) The articles “a” and “an” are used herein to refer to one or to more than one (i.e., to at least one) of the grammatical object of the article. By way of example, “an element” means one element or more than one element, e.g., a plurality of elements.

(16) The term “including” is used herein to mean, and is used interchangeably with, the phrase “including but not limited to”.

(17) The term “or” is used herein to mean, and is used interchangeably with, the term “and/or,” unless context clearly indicates otherwise. For example, “sense strand or antisense strand” is understood as “sense strand or antisense strand or sense strand and antisense strand.”

(18) The term “about” is used herein to mean within the typical ranges of tolerances in the art. For example, “about” can be understood as about 2 standard deviations from the mean. In certain

embodiments, about means $\pm 10\%$. In certain embodiments, about means $\pm 5\%$. When about is present before a series of numbers or a range, it is understood that “about” can modify each of the numbers in the series or range.

(19) The term “at least”, “no less than”, or “or more” prior to a number or series of numbers is understood to include the number adjacent to the term “at least”, and all subsequent numbers or integers that could logically be included, as clear from context. For example, the number of nucleotides in a nucleic acid molecule must be an integer. For example, “at least 19 nucleotides of a 21 nucleotide nucleic acid molecule” means that 19, 20, or 21 nucleotides have the indicated property. When at least is present before a series of numbers or a range, it is understood that “at least” can modify each of the numbers in the series or range.

(20) As used herein, “no more than” or “or less” is understood as the value adjacent to the phrase and logical lower values or integers, as logical from context, to zero. For example, a duplex with an overhang of “no more than 2 nucleotides” has a 2, 1, or 0 nucleotide overhang. When “no more than” is present before a series of numbers or a range, it is understood that “no more than” can modify each of the numbers in the series or range. As used herein, ranges include both the upper and lower limit.

(21) As used herein, methods of detection can include determination that the amount of analyte present is below the level of detection of the method.

(22) In the event of a conflict between an indicated target site and the nucleotide sequence for a sense or antisense strand, the indicated sequence takes precedence.

(23) In the event of a conflict between a sequence and its indicated site on a transcript or other sequence, the nucleotide sequence recited in the specification takes precedence.

(24) As used herein, “ketohexokinase,” used interchangeably with the term “KHK,” refers to the naturally occurring gene that encodes an enzyme that catalyzes conversion of fructose to fructose-1-phosphate. The product of this gene is the first enzyme in the pathway that catabolizes dietary fructose. Alternatively spliced transcript variants encoding different isoforms have been identified. The gene is also known as fructokinase.

(25) Exemplary nucleotide and amino acid sequences of KHK can be found, for example, at GenBank Accession No. XM_017004061.1 (*Homo sapiens* KHK; SEQ ID NO:1; reverse complement, SEQ ID NO:2); NM_006488.3 (*Homo sapiens* KHK; SEQ ID NO:3; reverse complement, SEQ ID NO:4); NM_000221.3 (*Homo sapiens* KHK; SEQ ID NO:5; reverse complement, SEQ ID NO:6); GenBank Accession No. NM_001310524.1 (*Mus musculus* KHK; SEQ ID NO:7; reverse complement, SEQ ID NO:8); GenBank Accession No. NM_031855.3 (*Rattus norvegicus* KHK; SEQ ID NO:9; reverse complement, SEQ ID NO:10); GenBank Accession No. XM_005576322.2 (*Macaca fascicularis* KHK, SEQ ID NO:11; reverse complement, SEQ ID NO:12).

(26) The KHK (Ketohexokinase) gene is located on chromosome 2p23 and encodes ketohexokinase, also known as fructokinase. KHK is a phosphotransferase enzyme with an alcohol as the phosphate acceptor. KHK belongs to the ribokinase family of carbohydrate kinases (Trinh et al., *ACTA Cryst.*, D65: 201-211). Two isoforms of ketohexokinase have been identified, KHK-A and KHK-C, that result from alternative splicing of the full length mRNA. These isoforms differ by inclusion of either exon 3a or 3c, and differ by 32 amino acids between positions 72 and 115. KHK-C mRNA is expressed at high levels, predominantly in the liver, kidney and small intestine. KHK-C has a much lower $K_{sub.m}$ for fructose binding than KHK-A, and as a result, is highly effective in phosphorylating dietary fructose. The sequence of a human KHK-C mRNA transcript may be found at, for example, GenBank Accession No. NM_006488.3 (*Homo sapiens* KHK; SEQ ID NO:3). The sequence of a human KHK-A mRNA transcript may be found at, for example GenBank Accession No. NM_000221.3; SEQ ID NO:5). The sequence of full-length human KHK mRNA is provided in GenBank Accession No. GI: XM_017004061.1 (SEQ ID NO:1).

(27) The iRNA agents provided herein can be capable of silencing one or both KHK isoforms.

(28) Additional examples of KHK mRNA sequences are readily available through publicly available databases, e.g., GenBank, UniProt, OMIM, and the Macaca genome project web site.

(29) Further information on KHK can be found, for example, at www.ncbi.nlm.nih.gov/gene/?term=khk.

(30) The entire contents of each of the foregoing GenBank Accession numbers and the Gene database numbers are incorporated herein by reference as of the date of filing this application.

(31) The term KHK, as used herein, also refers to variations of the KHK gene including variants provided in the SNP database. Numerous sequence variations within the KHK gene have been identified and may be found at, for example, NCBI dbSNP and UniProt (see, e.g., www.ncbi.nlm.nih.gov/snp/?term=KHK, the entire contents of which is incorporated herein by reference as of the date of filing this application.

(32) As used herein, “target sequence” refers to a contiguous portion of the nucleotide sequence of an mRNA molecule formed during the transcription of a KHK gene, including mRNA that is a product of RNA processing of a primary transcription product. The target portion of the sequence will be at least long enough to serve as a substrate for iRNA-directed cleavage at or near that portion of the nucleotide sequence of an mRNA molecule formed during the transcription of a KHK gene. In one embodiment, the target sequence is within the protein coding region of KHK.

(33) The target sequence may be from about 19-36 nucleotides in length, e.g., about 19-30 nucleotides in length. For example, the target sequence can be about 19-30 nucleotides, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length. In certain embodiments, the target sequence is 19-23 nucleotides in length, optionally 21-23 nucleotides in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the disclosure.

(34) As used herein, the term “strand comprising a sequence” refers to an oligonucleotide comprising a chain of nucleotides that is described by the sequence referred to using the standard nucleotide nomenclature.

(35) “G,” “C,” “A,” “T,” and “U” each generally stand for a nucleotide that contains guanine, cytosine, adenine, thymidine, and uracil as a base, respectively. However, it will be understood that the term “ribonucleotide” or “nucleotide” can also refer to a modified nucleotide, as further detailed below, or a surrogate replacement moiety (see, e.g., Table 1). The skilled person is well aware that guanine, cytosine, adenine, and uracil can be replaced by other moieties without substantially altering the base pairing properties of an oligonucleotide comprising a nucleotide bearing such replacement moiety. For example, without limitation, a nucleotide comprising inosine as its base can base pair with nucleotides containing adenine, cytosine, or uracil. Hence, nucleotides containing uracil, guanine, or adenine can be replaced in the nucleotide sequences of dsRNA featured in the invention by a nucleotide containing, for example, inosine. In another example, adenine and cytosine anywhere in the oligonucleotide can be replaced with guanine and uracil, respectively to form G-U Wobble base pairing with the target mRNA. Sequences containing such replacement moieties are suitable for the compositions and methods featured in the invention.

(36) The terms “iRNA”, “RNAi agent,” “iRNA agent,” “RNA interference agent” as used interchangeably herein, refer to an agent that contains RNA as that term is defined herein, and which mediates the targeted cleavage of an RNA transcript via an RNA-induced silencing complex (RISC) pathway. iRNA directs the sequence-specific degradation of mRNA through a process known as RNA interference (RNAi). The iRNA modulates, e.g., inhibits, the expression of a KHK gene in a cell, e.g., a cell within a subject, such as a mammalian subject.

(37) In one embodiment, an RNAi agent of the invention includes a single stranded RNA that interacts with a target RNA sequence, e.g., a KHK target mRNA sequence, to direct the cleavage of the target RNA. Without wishing to be bound by theory it is believed that long double stranded RNA introduced into cells is broken down into siRNA by a Type III endonuclease known as Dicer

(Sharp et al. (2001) *Genes Dev.* 15:485). Dicer, a ribonuclease-III-like enzyme, processes the dsRNA into 19-23 base pair short interfering RNAs with characteristic two base 3' overhangs (Bernstein, et al., (2001) *Nature* 409:363). The siRNAs are then incorporated into an RNA-induced silencing complex (RISC) where one or more helicases unwind the siRNA duplex, enabling the complementary antisense strand to guide target recognition (Nykanen, et al., (2001) *Cell* 107:309). Upon binding to the appropriate target mRNA, one or more endonucleases within the RISC cleave the target to induce silencing (Elbashir, et al., (2001) *Genes Dev.* 15:188). Thus, in one aspect the invention relates to a single stranded RNA (siRNA) generated within a cell and which promotes the formation of a RISC complex to effect silencing of the target gene, i.e., a KHK gene. Accordingly, the term "siRNA" is also used herein to refer to an iRNA as described above.

(38) In certain embodiments, the RNAi agent may be a single-stranded siRNA (ssRNAi) that is introduced into a cell or organism to inhibit a target mRNA. Single-stranded RNAi agents bind to the RISC endonuclease, Argonaute 2, which then cleaves the target mRNA. The single-stranded siRNAs are generally 15-30 nucleotides and are chemically modified. The design and testing of single-stranded siRNAs are described in U.S. Pat. No. 8,101,348 and in Lima et al., (2012) *Cell* 150:883-894, the entire contents of each of which are hereby incorporated herein by reference. Any of the antisense nucleotide sequences described herein may be used as a single-stranded siRNA as described herein or as chemically modified by the methods described in Lima et al., (2012) *Cell* 150:883-894.

(39) In certain embodiments, an "iRNA" for use in the compositions, uses, and methods of the invention is a double stranded RNA and is referred to herein as a "double stranded RNA agent," "double stranded RNA (dsRNA) molecule," "dsRNA agent," or "dsRNA". The term "dsRNA", refers to a complex of ribonucleic acid molecules, having a duplex structure comprising two anti-parallel and substantially complementary nucleic acid strands, referred to as having "sense" and "antisense" orientations with respect to a target RNA, i.e., a KHK gene. In some embodiments of the invention, a double stranded RNA (dsRNA) triggers the degradation of a target RNA, e.g., an mRNA, through a post-transcriptional gene-silencing mechanism referred to herein as RNA interference or RNAi.

(40) In general, the majority of nucleotides of each strand of a dsRNA molecule are ribonucleotides, but as described in detail herein, each or both strands can also include one or more non-ribonucleotides, e.g., a deoxyribonucleotide or a modified nucleotide. In addition, as used in this specification, an "iRNA" may include ribonucleotides with chemical modifications; an iRNA may include substantial modifications at multiple nucleotides. As used herein, the term "modified nucleotide" refers to a nucleotide having, independently, a modified sugar moiety, a modified internucleotide linkage, or modified nucleobase, or any combination thereof. Thus, the term modified nucleotide encompasses substitutions, additions or removal of, e.g., a functional group or atom, to internucleoside linkages, sugar moieties, or nucleobases. The modifications suitable for use in the agents of the invention include all types of modifications disclosed herein or known in the art. Any such modifications, as used in a siRNA type molecule, are encompassed by "iRNA" or "RNAi agent" for the purposes of this specification and claims.

(41) In certain embodiments of the instant disclosure, inclusion of a deoxy-nucleotide if present within an RNAi agent can be considered to constitute a modified nucleotide.

(42) The duplex region may be of any length that permits specific degradation of a desired target RNA through a RISC pathway, and may range from about 19 to 36 base pairs in length, e.g., about 19-30 base pairs in length, for example, about 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, or 36 base pairs in length, such as about 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 base pairs in length. In certain embodiments, the duplex region is 19-21 base pairs in length, e.g., 21 base pairs in length. Ranges and lengths intermediate to the above recited ranges and

lengths are also contemplated to be part of the disclosure.

(43) The two strands forming the duplex structure may be different portions of one larger RNA molecule, or they may be separate RNA molecules. Where the two strands are part of one larger molecule, and therefore are connected by an uninterrupted chain of nucleotides between the 3'-end of one strand and the 5'-end of the respective other strand forming the duplex structure, the connecting RNA chain is referred to as a "hairpin loop." A hairpin loop can comprise at least one unpaired nucleotide. In some embodiments, the hairpin loop can comprise at least 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 23 or more unpaired nucleotides. In some embodiments, the hairpin loop can be 10 or fewer nucleotides. In some embodiments, the hairpin loop can be 8 or fewer unpaired nucleotides. In some embodiments, the hairpin loop can be 4-10 unpaired nucleotides. In some embodiments, the hairpin loop can be 4-8 nucleotides.

(44) Where the two substantially complementary strands of a dsRNA are comprised by separate RNA molecules, those molecules need not be, but can be covalently connected. Where the two strands are connected covalently by means other than an uninterrupted chain of nucleotides between the 3'-end of one strand and the 5'-end of the respective other strand forming the duplex structure, the connecting structure is referred to as a "linker." The RNA strands may have the same or a different number of nucleotides. The maximum number of base pairs is the number of nucleotides in the shortest strand of the dsRNA minus any overhangs that are present in the duplex. In addition to the duplex structure, an RNAi may comprise one or more nucleotide overhangs. In one embodiment of the RNAi agent, at least one strand comprises a 3' overhang of at least 1 nucleotide. In another embodiment, at least one strand comprises a 3' overhang of at least 2 nucleotides, e.g., 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, or 15 nucleotides. In other embodiments, at least one strand of the RNAi agent comprises a 5' overhang of at least 1 nucleotide. In certain embodiments, at least one strand comprises a 5' overhang of at least 2 nucleotides, e.g., 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, or 15 nucleotides. In still other embodiments, both the 3' and the 5' end of one strand of the RNAi agent comprise an overhang of at least 1 nucleotide.

(45) In certain embodiments, an iRNA agent of the invention is a dsRNA, each strand of which comprises 19-23 nucleotides, that interacts with a target RNA sequence, e.g., a KHK gene, to direct cleavage of the target RNA.

(46) In some embodiments, an iRNA of the invention is a dsRNA of 24-30 nucleotides that interacts with a target RNA sequence, e.g., a KHK target mRNA sequence, to direct the cleavage of the target RNA.

(47) As used herein, the term "nucleotide overhang" refers to at least one unpaired nucleotide that protrudes from the duplex structure of a double stranded iRNA. For example, when a 3'-end of one strand of a dsRNA extends beyond the 5'-end of the other strand, or vice versa, there is a nucleotide overhang. A dsRNA can comprise an overhang of at least one nucleotide; alternatively the overhang can comprise at least two nucleotides, at least three nucleotides, at least four nucleotides, at least five nucleotides or more. A nucleotide overhang can comprise or consist of a nucleotide/nucleoside analog, including a deoxynucleotide/nucleoside. The overhang(s) can be on the sense strand, the antisense strand, or any combination thereof. Furthermore, the nucleotide(s) of an overhang can be present on the 5'-end, 3'-end, or both ends of either an antisense or sense strand of a dsRNA.

(48) In one embodiment, the antisense strand of a dsRNA has a 1-10 nucleotide, e.g., a 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotide, overhang at the 3'-end or the 5'-end. In one embodiment, the sense strand of a dsRNA has a 1-10 nucleotide, e.g., a 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotide, overhang at the 3'-end or the 5'-end. In another embodiment, one or more of the nucleotides in the overhang is replaced with a nucleoside thiophosphate.

(49) In certain embodiments, the antisense strand of a dsRNA has a 1-10 nucleotide, e.g., 0-3, 1-3, 2-4, 2-5, 4-10, 5-10, e.g., a 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotide, overhang at the 3'-end or the 5'-end. In one embodiment, the sense strand of a dsRNA has a 1-10 nucleotide, e.g., a 1, 2, 3, 4, 5, 6,

7, 8, 9, or 10 nucleotide, overhang at the 3'-end or the 5'-end. In another embodiment, one or more of the nucleotides in the overhang is replaced with a nucleoside thiophosphate.

(50) In certain embodiments, the antisense strand of a dsRNA has a 1-10 nucleotides, e.g., a 1, 2, 3, 4, 5, 6, 7, 8, 9, or 10 nucleotide, overhang at the 3'-end or the 5'-end. In certain embodiments, the overhang on the sense strand or the antisense strand, or both, can include extended lengths longer than 10 nucleotides, e.g., 1-30 nucleotides, 2-30 nucleotides, 10-30 nucleotides, 10-25 nucleotides, 10-20 nucleotides, or 10-15 nucleotides in length. In certain embodiments, an extended overhang is on the sense strand of the duplex. In certain embodiments, an extended overhang is present on the 3' end of the sense strand of the duplex. In certain embodiments, an extended overhang is present on the 5' end of the sense strand of the duplex. In certain embodiments, an extended overhang is on the antisense strand of the duplex. In certain embodiments, an extended overhang is present on the 3' end of the antisense strand of the duplex. In certain embodiments, an extended overhang is present on the 5' end of the antisense strand of the duplex. In certain embodiments, one or more of the nucleotides in the extended overhang is replaced with a nucleoside thiophosphate. In certain embodiments, the overhang includes a self-complementary portion such that the overhang is capable of forming a hairpin structure that is stable under physiological conditions.

(51) "Blunt" or "blunt end" means that there are no unpaired nucleotides at that end of the double stranded RNA agent, i.e., no nucleotide overhang. A "blunt ended" double stranded RNA agent is double stranded over its entire length, i.e., no nucleotide overhang at either end of the molecule. The RNAi agents of the invention include RNAi agents with no nucleotide overhang at one end (i.e., agents with one overhang and one blunt end) or with no nucleotide overhangs at either end. Most often such a molecule will be double-stranded over its entire length.

(52) The term "antisense strand" or "guide strand" refers to the strand of an iRNA, e.g., a dsRNA, which includes a region that is substantially complementary to a target sequence, e.g., a KHK mRNA.

(53) As used herein, the term "region of complementarity" refers to the region on the antisense strand that is substantially complementary to a sequence, for example a target sequence, e.g., a KHK nucleotide sequence, as defined herein. Where the region of complementarity is not fully complementary to the target sequence, the mismatches can be in the internal or terminal regions of the molecule. Generally, the most tolerated mismatches are in the terminal regions, e.g., within 5, 4, or 3 nucleotides of the 5'- or 3'-end of the iRNA. In some embodiments, a double stranded RNA agent of the invention includes a nucleotide mismatch in the antisense strand. In some embodiments, the antisense strand of the double stranded RNA agent of the invention includes no more than 4 mismatches with the target mRNA, e.g., the antisense strand includes 4, 3, 2, 1, or 0 mismatches with the target mRNA. In some embodiments, the antisense strand double stranded RNA agent of the invention includes no more than 4 mismatches with the sense strand, e.g., the antisense strand includes 4, 3, 2, 1, or 0 mismatches with the sense strand. In some embodiments, a double stranded RNA agent of the invention includes a nucleotide mismatch in the sense strand. In some embodiments, the sense strand of the double stranded RNA agent of the invention includes no more than 4 mismatches with the antisense strand, e.g., the sense strand includes 4, 3, 2, 1, or 0 mismatches with the antisense strand. In some embodiments, the nucleotide mismatch is, for example, within 5, 4, 3 nucleotides from the 3'-end of the iRNA. In another embodiment, the nucleotide mismatch is, for example, in the 3'-terminal nucleotide of the iRNA agent. In some embodiments, the mismatch(s) is not in the seed region.

(54) Thus, an RNAi agent as described herein can contain one or more mismatches to the target sequence. In one embodiment, an RNAi agent as described herein contains no more than 3 mismatches (i.e., 3, 2, 1, or 0 mismatches). In one embodiment, an RNAi agent as described herein contains no more than 2 mismatches. In one embodiment, an RNAi agent as described herein contains no more than 1 mismatch. In one embodiment, an RNAi agent as described herein contains 0 mismatches. In certain embodiments, if the antisense strand of the RNAi agent contains

mismatches to the target sequence, the mismatch can optionally be restricted to be within the last 5 nucleotides from either the 5'- or 3'-end of the region of complementarity. For example, in such embodiments, for a 23 nucleotide RNAi agent, the strand which is complementary to a region of a KHK gene, generally does not contain any mismatch within the central 13 nucleotides. The methods described herein or methods known in the art can be used to determine whether an RNAi agent containing a mismatch to a target sequence is effective in inhibiting the expression of a KHK gene. Consideration of the efficacy of RNAi agents with mismatches in inhibiting expression of a KHK gene is important, especially if the particular region of complementarity in a KHK gene is known to have polymorphic sequence variation within the population.

(55) The term “sense strand” or “passenger strand” as used herein, refers to the strand of an iRNA that includes a region that is substantially complementary to a region of the antisense strand as that term is defined herein.

(56) As used herein, “substantially all of the nucleotides are modified” are largely but not wholly modified and can include not more than 5, 4, 3, 2, or 1 unmodified nucleotides.

(57) As used herein, the term “cleavage region” refers to a region that is located immediately adjacent to the cleavage site. The cleavage site is the site on the target at which cleavage occurs. In some embodiments, the cleavage region comprises three bases on either end of, and immediately adjacent to, the cleavage site. In some embodiments, the cleavage region comprises two bases on either end of, and immediately adjacent to, the cleavage site. In some embodiments, the cleavage site specifically occurs at the site bound by nucleotides 10 and 11 of the antisense strand, and the cleavage region comprises nucleotides 11, 12 and 13.

(58) As used herein, and unless otherwise indicated, the term “complementary,” when used to describe a first nucleotide sequence in relation to a second nucleotide sequence, refers to the ability of an oligonucleotide or polynucleotide comprising the first nucleotide sequence to hybridize and form a duplex structure under certain conditions with an oligonucleotide or polynucleotide comprising the second nucleotide sequence, as will be understood by the skilled person. Such conditions can, for example, be stringent conditions, where stringent conditions can include: 400 mM NaCl, 40 mM PIPES pH 6.4, 1 mM EDTA, 50° C. or 70° C. for 12-16 hours followed by washing (see, e.g., “Molecular Cloning: A Laboratory Manual, Sambrook, et al. (1989) Cold Spring Harbor Laboratory Press). Such conditions can be, for example, “stringent conditions”, where stringent conditions can include: 400 mM NaCl, 40 mM PIPES pH 6.4, 1 mM EDTA, 50° C. or 70° C. for 12-16 hours followed by washing (see, e.g., “Molecular Cloning: A Laboratory Manual, Sambrook, et al. (1989) Cold Spring Harbor Laboratory Press). Other conditions, such as physiologically relevant conditions as can be encountered inside an organism, can apply. The skilled person will be able to determine the set of conditions most appropriate for a test of complementarity of two sequences in accordance with the ultimate application of the hybridized nucleotides.

(59) Complementary sequences within an iRNA, e.g., within a dsRNA as described herein, include base-pairing of the oligonucleotide or polynucleotide comprising a first nucleotide sequence to an oligonucleotide or polynucleotide comprising a second nucleotide sequence over the entire length of one or both nucleotide sequences. Such sequences can be referred to as “fully complementary” with respect to each other herein. However, where a first sequence is referred to as “substantially complementary” with respect to a second sequence herein, the two sequences can be fully complementary, or they can form one or more, but generally not more than 5, 4, 3, or 2 mismatched base pairs upon hybridization for a duplex up to 30 base pairs, while retaining the ability to hybridize under the conditions most relevant to their ultimate application, e.g., inhibition of gene expression, in vitro or in vivo. However, where two oligonucleotides are designed to form, upon hybridization, one or more single stranded overhangs, such overhangs shall not be regarded as mismatches with regard to the determination of complementarity. For example, a dsRNA comprising one oligonucleotide 21 nucleotides in length and another oligonucleotide 23

nucleotides in length, wherein the longer oligonucleotide comprises a sequence of 21 nucleotides that is fully complementary to the shorter oligonucleotide, can yet be referred to as “fully complementary” for the purposes described herein.

(60) “Complementary” sequences, as used herein, can also include, or be formed entirely from, non-Watson-Crick base pairs or base pairs formed from non-natural and modified nucleotides, in so far as the above requirements with respect to their ability to hybridize are fulfilled. Such non-Watson-Crick base pairs include, but are not limited to, G:U Wobble or Hoogsteen base pairing.

(61) The terms “complementary,” “fully complementary” and “substantially complementary” herein can be used with respect to the base matching between the sense strand and the antisense strand of a dsRNA, or between two oligonucleotides or polynucleotides, such as the antisense strand of a double stranded RNA agent and a target sequence, as will be understood from the context of their use.

(62) As used herein, a polynucleotide that is “substantially complementary to at least part of” a messenger RNA (mRNA) refers to a polynucleotide that is substantially complementary to a contiguous portion of the mRNA of interest (e.g., an mRNA encoding a KHK gene). For example, a polynucleotide is complementary to at least a part of a KHK mRNA if the sequence is substantially complementary to a non-interrupted portion of an mRNA encoding a KHK gene.

(63) Accordingly, in some embodiments, the antisense polynucleotides disclosed herein are fully complementary to the target KHK sequence. In other embodiments, the antisense polynucleotides disclosed herein are substantially complementary to the target KHK sequence and comprise a contiguous nucleotide sequence which is at least 80% complementary over its entire length to the equivalent region of the nucleotide sequence of any one of SEQ ID NOs:1, 3, 5, 7, 9, or 11, or a fragment of any one of SEQ ID NOs:1, 3, 5, 7, 9, or 11, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary.

(64) In some embodiments, the antisense polynucleotides disclosed herein are substantially complementary to a fragment of a target KHK sequence and comprise a contiguous nucleotide sequence which is at least 80% complementary over its entire length to a fragment of SEQ ID NO: 1 selected from the group of nucleotides 120-162; 164-188; 181-207; 193-217; 209-231; 283-306; 508-546; 568-603; 596-632; 640-674; 746-806; 806-835; 917-942; 936-084; 1016-1041; 1100-1123; 1149-1175; 1160-1193; 1205-1229; 1252-1283; 1334-1356; 1407-1429; 1472-1497; 1506-1533; 1539-1561; 1704-1727; 1747-1787; 1850-1873; 1936-1964; 1960-1990; 2015-2048; 2060-2095; 2090-2118; 2124-2160; 2181-2200; 2221-2262 of SEQ ID NO: 1, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary.

(65) In some embodiments, the antisense polynucleotides disclosed herein are substantially complementary to a fragment of a target KHK sequence and comprise a contiguous nucleotide sequence which is at least 80% complementary over its entire length to a fragment of SEQ ID NO: 1 selected from the group of nucleotides 517-539; 521-543; 524-546; 517-546; 581-603; 610-632; 747-769; 749-771; 752-774; 755-777; 757-779; 758-780; 764-786; 776-798; 781-803; 747-803; 920-942; 941-963; 944-966; 950-972; 962-984; 920-984; 1149-1171; 1161-1183; 1165-1187; 1171-1193; 1149-1193; 1205-1227; 1206-1228; 1205-1228; 1334-1356; 1472-1494; 1475-1497; 1472-1497 of SEQ ID NO: 1, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary.

(66) In some embodiments, the antisense polynucleotides disclosed herein are substantially complementary to a fragment of a target KHK sequence and comprise a contiguous nucleotide sequence which is at least 80% complementary over its entire length to a fragment of SEQ ID NO: 1 selected from the group of nucleotides 517-539; 524-546; 517-546; 753-775; 757-779; 753-779; 764-786; 767-789; 768-790; 769-791; 764-791; 773-795; 781-803; 773-803; 753-803; 808-830; 937-959; 941-963; 944-966; 941-966; 948-970; 950-972; 948-972; 1160-1182; 1161-1183; 1160-

1183; and 1207-1229 of SEQ ID NO: 1, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary.

(67) In some embodiments, the antisense polynucleotides disclosed herein are substantially complementary to a fragment of a target KHK sequence and comprise a contiguous nucleotide sequence which is at least 80% complementary over its entire length to a fragment of SEQ ID NO: 1 selected from the group of nucleotides 495-517, 492-517, 500-529, 514-551, 517-539, 524-546, 517-548, 614-643, 625-647, 625-660, 642-664, 642-672, 753-811, 754-780, 762-791, 764-786, 772-800, 781-803, 805-827, 808-830, 809-831, 792-838, 931-982, 944-966, 947-969, 948-970, 948-982, 1011-1035, 1021-1043, 1019-1050, 1063-1091, 1150-1192, 1152-1176, 1160-1192, 1160-1182, 1162-1184, 1198-1230, 1198-1221, and 1202-1230 of SEQ ID NO: 1, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary.

(68) In some embodiments, the antisense polynucleotides disclosed herein are substantially complementary to a fragment of a target KHK sequence and comprise a contiguous nucleotide sequence which is at least 80% complementary over its entire length to a fragment of SEQ ID NO: 1 selected from the group of nucleotides 513-556, 753-813, 936-981, 1155-1193, 1200-1229, 1704-1727, 1747-1787, 1850-1873, 1936-1964, 1960-1990, 1936-1990, 2015-2048, 2060-2095, 2090-2118, 2060-2118, 2124-2160, 2181-2220, 2221-2249, 2181-2249, 2240-2262, 2221-2262, and 2181-2262 of SEQ ID NO: 1, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, or about 99% complementary.

(69) In other embodiments, the antisense polynucleotides disclosed herein are substantially complementary to the target KHK sequence and comprise a contiguous nucleotide sequence which is at least about 80% complementary over its entire length to any one of the sense strand nucleotide sequences in any one of any one of Tables 2, 3, 5, 6, and 8-13, or a fragment of any one of the sense strand nucleotide sequences in any one of Tables 2, 3, 5, 6, and 8-13, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or 100% complementary.

(70) In one embodiment, an RNAi agent of the disclosure includes a sense strand that is substantially complementary to an antisense polynucleotide which, in turn, is the same as a target KHK sequence, and wherein the sense strand polynucleotide comprises a contiguous nucleotide sequence which is at least about 80% complementary over its entire length to the equivalent region of the nucleotide sequence of SEQ ID NOs: 2, 4, 6, 8, 10, or 12, or a fragment of any one of SEQ ID NOs: 2, 4, 6, 8, 10, or 12, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or 100% complementary.

(71) In some embodiments, an iRNA of the invention includes a sense strand that is substantially complementary to an antisense polynucleotide which, in turn, is complementary to a target KHK sequence, and wherein the sense strand polynucleotide comprises a contiguous nucleotide sequence which is at least about 80% complementary over its entire length to any one of the antisense strand nucleotide sequences in any one of any one of Tables 2, 3, 5, 6, and 8-13, or a fragment of any one of the antisense strand nucleotide sequences in any one of Tables 2, 3, 5, 6, and 8-13, such as about 85%, about 90%, about 91%, about 92%, about 93%, about 94%, about 95%, about 96%, about 97%, about 98%, about 99%, or 100% complementary.

(72) In certain embodiments, the sense and antisense strands are selected from any one of duplexes AD-1613400; AD-1613243; AD-1290757.3; AD-1290878.3; AD-1290969.3; AD-1423317.2; AD-1423327.2; AD-1423336.2; AD-1290599.3; AD-1523172.1; AD-1290837.3; AD-1523173.1; AD-1290884.3; AD-1523174.1; AD-1290959.3; AD-1523175.1; AD-1423311.2; AD-1423324.2; AD-1523176.1; AD-1423329.2; AD-1423333.2; AD-1423330.2; AD-1523177.1; AD-1290885.3; AD-1523178.1; AD-1423334.2; AD-1523179.1; AD-1523180.1; AD-1290539.3; and AD-1523181.1.

(73) In general, an "iRNA" includes ribonucleotides with chemical modifications. Such

modifications may include all types of modifications disclosed herein or known in the art. Any such modifications, as used in a dsRNA molecule, are encompassed by “iRNA” for the purposes of this specification and claims.

(74) In certain embodiments of the instant disclosure, inclusion of a deoxy-nucleotide if present within an RNAi agent can be considered to constitute a modified nucleotide.

(75) In an aspect of the invention, an agent for use in the methods and compositions of the invention is a single-stranded antisense oligonucleotide molecule that inhibits a target mRNA via an antisense inhibition mechanism. The single-stranded antisense oligonucleotide molecule is complementary to a sequence within the target mRNA. The single-stranded antisense oligonucleotides can inhibit translation in a stoichiometric manner by base pairing to the mRNA and physically obstructing the translation machinery, see Dias, N. et al., (2002) *Mol Cancer Ther* 1:347-355. The single-stranded antisense oligonucleotide molecule may be about 14 to about 30 nucleotides in length and have a sequence that is complementary to a target sequence. For example, the single-stranded antisense oligonucleotide molecule may comprise a sequence that is at least about 14, 15, 16, 17, 18, 19, 20, or more contiguous nucleotides from any one of the antisense sequences described herein.

(76) The phrase “contacting a cell with an iRNA,” such as a dsRNA, as used herein, includes contacting a cell by any possible means. Contacting a cell with an iRNA includes contacting a cell in vitro with the iRNA or contacting a cell in vivo with the iRNA. The contacting may be done directly or indirectly. Thus, for example, the iRNA may be put into physical contact with the cell by the individual performing the method, or alternatively, the iRNA may be put into a situation that will permit or cause it to subsequently come into contact with the cell.

(77) Contacting a cell in vitro may be done, for example, by incubating the cell with the iRNA. Contacting a cell in vivo may be done, for example, by injecting the iRNA into or near the tissue where the cell is located, or by injecting the iRNA into another area, e.g., the bloodstream or the subcutaneous space, such that the agent will subsequently reach the tissue where the cell to be contacted is located. For example, the iRNA may contain or be coupled to a ligand, e.g., GalNAc, that directs the iRNA to a site of interest, e.g., the liver. Combinations of in vitro and in vivo methods of contacting are also possible. For example, a cell may also be contacted in vitro with an iRNA and subsequently transplanted into a subject.

(78) In certain embodiments, contacting a cell with an iRNA includes “introducing” or “delivering the iRNA into the cell” by facilitating or effecting uptake or absorption into the cell. Absorption or uptake of an iRNA can occur through unaided diffusion or active cellular processes, or by auxiliary agents or devices. Introducing an iRNA into a cell may be in vitro or in vivo. For example, for in vivo introduction, iRNA can be injected into a tissue site or administered systemically. In vitro introduction into a cell includes methods known in the art such as electroporation and lipofection. Further approaches are described herein below or are known in the art.

(79) The term “lipid nanoparticle” or “LNP” is a vesicle comprising a lipid layer encapsulating a pharmaceutically active molecule, such as a nucleic acid molecule, e.g., an iRNA or a plasmid from which an iRNA is transcribed. LNPs are described in, for example, U.S. Pat. Nos. 6,858,225, 6,815,432, 8,158,601, and 8,058,069, the entire contents of which are hereby incorporated herein by reference.

(80) As used herein, a “subject” is an animal, such as a mammal, including a primate (such as a human, a non-human primate, e.g., a monkey, and a chimpanzee), a non-primate (such as a cow, a pig, a horse, a goat, a rabbit, a sheep, a hamster, a guinea pig, a cat, a dog, a rat, or a mouse), or a bird that expresses the target gene, either endogenously or heterologously. In an embodiment, the subject is a human, such as a human being treated or assessed for a disease or disorder that would benefit from reduction in KHK expression; a human at risk for a disease or disorder that would benefit from reduction in KHK expression; a human having a disease or disorder that would benefit from reduction in KHK expression; or human being treated for a disease or disorder that would

benefit from reduction in KHK expression as described herein. In some embodiments, the subject is a female human. In other embodiments, the subject is a male human. In one embodiment, the subject is an adult subject. In another embodiment, the subject is a pediatric subject.

(81) As used herein, the terms “treating” or “treatment” refer to a beneficial or desired result, such as reducing at least one sign or symptom of a KHK-associated disorder in a subject. Treatment also includes a reduction of one or more sign or symptoms associated with unwanted KHK expression; diminishing the extent of unwanted KHK activation or stabilization; amelioration or palliation of unwanted KHK activation or stabilization. “Treatment” can also mean prolonging survival as compared to expected survival in the absence of treatment. The term “lower” in the context of the level of KHK in a subject or a disease marker or symptom refers to a statistically significant decrease in such level. The decrease can be, for example, at least 10%, 15%, 20%, 25%, 30%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, 95%, or more. In certain embodiments, a decrease is at least 20%. In certain embodiments, the decrease is at least 50% in a disease marker, e.g., protein or gene expression level. “Lower” in the context of the level of KHK in a subject is a decrease to a level accepted as within the range of normal for an individual without such disorder. In certain embodiments, “lower” is the decrease in the difference between the level of a marker or symptom for a subject suffering from a disease and a level accepted within the range of normal for an individual, e.g., the level of decrease in bodyweight between an obese individual and an individual having a weight accepted within the range of normal.

(82) As used herein, “prevention” or “preventing,” when used in reference to a disease, disorder or condition thereof, may be treated or ameliorated by a reduction in expression of a KHK gene, refers to a reduction in the likelihood that a subject will develop a symptom associated with such a disease, disorder, or condition, e.g., a symptom of unwanted or excessive KHK expression and/or activity, e.g., increased fructose metabolism, elevated uric acid and lipid levels. Without being bound by mechanism, it is known that fructose phosphorylation catalyzed by KHK to form fructose-1-phosphate is not regulated by feedback inhibition which can result in depletion of ATP and intracellular phosphate, and increased AMP levels, which results in the production of uric acid. Further, the fructose-1-phosphate is metabolized to glyceraldehyde which feeds into the citric acid cycle increasing the production of acetyl Co-A stimulating fatty acid synthesis. Diseases and conditions associated with elevated uric acid and fatty acid synthesis include, e.g., liver disease (e.g., fatty liver, steatohepatitis including non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance not related to immune response to insulin, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, disease of lipid deposition or dysfunction (e.g., adipocyte dysfunction, visceral adipose deposition, obesity), disease of elevated uric acid (e.g., hyperuricemia, gout), and eating disorders such as excessive sugar craving. The failure to develop a disease, disorder or condition, or the reduction in the development of a symptom or comorbidity associated with such a disease, disorder or condition (e.g., by at least about 10% on a clinically accepted scale for that disease or disorder), or the exhibition of delayed signs or symptoms or disease progression by days, weeks, months or years is considered effective prevention.

(83) As used herein, the term “ketohexokinase (KHK)-associated disease” or “KHK-associated disorder,” is a disease or disorder that is caused by, or associated with KHK gene expression or KHK protein production. The term “KHK-associated disease” includes a disease, disorder or condition that would benefit from a decrease in KHK gene expression, replication, or protein activity. Non-limiting examples of KHK-associated diseases include, for example, liver disease (e.g., fatty liver, steatohepatitis including non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia,

postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance not related to immune response to insulin, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, disease of lipid deposition or dysfunction (e.g., adipocyte dysfunction, visceral adipose deposition, obesity), disease of elevated uric acid (e.g., hyperuricemia, gout), and eating disorders such as excessive sugar craving.

(84) In certain embodiments, a KHK-associated disease is associated with elevated uric acid (e.g. hyperuricemia, gout).

(85) In certain embodiments, a KHK-associated disease is associated with elevated lipid levels (e.g., fatty liver, steatohepatitis including non-alcoholic steatohepatitis (NASH), dyslipidemia).

(86) “Therapeutically effective amount,” as used herein, is intended to include the amount of an RNAi agent that, when administered to a subject having a KHK-associated disease, is sufficient to effect treatment of the disease (e.g., by diminishing, ameliorating, or maintaining the existing disease or one or more symptoms of disease). The “therapeutically effective amount” may vary depending on the RNAi agent, how the agent is administered, the disease and its severity and the history, age, weight, family history, genetic makeup, the types of preceding or concomitant treatments, if any, and other individual characteristics of the subject to be treated.

(87) “Prophylactically effective amount,” as used herein, is intended to include the amount of an RNAi agent that, when administered to a subject having a KHK-associated disorder, is sufficient to prevent or ameliorate the disease or one or more symptoms of the disease. Ameliorating the disease includes slowing the course of the disease or reducing the severity of later-developing disease. The “prophylactically effective amount” may vary depending on the RNAi agent, how the agent is administered, the degree of risk of disease, and the history, age, weight, family history, genetic makeup, the types of preceding or concomitant treatments, if any, and other individual characteristics of the patient to be treated.

(88) A “therapeutically-effective amount” or “prophylactically effective amount” also includes an amount of an RNAi agent that produces some desired effect at a reasonable benefit/risk ratio applicable to any treatment. The iRNA employed in the methods of the present invention may be administered in a sufficient amount to produce a reasonable benefit/risk ratio applicable to such treatment.

(89) The phrase “pharmaceutically acceptable” is employed herein to refer to those compounds (including salts), materials, compositions, or dosage forms which are, within the scope of sound medical judgment, suitable for use in contact with the tissues of human subjects and animal subjects without excessive toxicity, irritation, allergic response, or other problem or complication, commensurate with a reasonable benefit/risk ratio.

(90) The phrase “pharmaceutically-acceptable carrier” as used herein means a pharmaceutically-acceptable material, composition, or vehicle, such as a liquid or solid filler, diluent, excipient, manufacturing aid (e.g., lubricant, talc magnesium, calcium or zinc stearate, or steric acid), or solvent encapsulating material, involved in carrying or transporting the subject compound from one organ, or portion of the body, to another organ, or portion of the body. Each carrier must be “acceptable” in the sense of being compatible with the other ingredients of the formulation and not injurious to the subject being treated. Such carriers are known in the art. Pharmaceutically acceptable carriers include carriers for administration by injection.

(91) The term “lower” in the context of the level of KHK gene expression or KHK protein production in a subject, or a disease marker or symptom refers to a statistically significant decrease in such level. The decrease can be, for example, at least 20%, 25%, 30%, 35%, 40%, 45%, 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95%, or below the level of detection for the detection method. In certain embodiments, the expression of the target is normalized, i.e., decreased towards or to a level accepted as within the range of normal for an individual without

such disorder, e.g., normalization of body weight, blood pressure, or a serum lipid level. As used here, “lower” in a subject can refer to lowering of gene expression or protein production in a cell in a subject does not require lowering of expression in all cells or tissues of a subject. For example, as used herein, lowering in a subject can include lowering of gene expression or protein production in the liver of a subject.

(92) The term “lower” can also be used in association with normalizing a symptom of a disease or condition, i.e. decreasing the difference between a level in a subject suffering from a KHK-associated disease towards or to a level in a normal subject not suffering from a KHK-associated disease. For example, if a subject with a normal weight of 70 kg weighs 90 kg prior to treatment (20 kg overweight) and 80 kg after treatment (10 kg overweight), the subject's weight is lowered towards a normal weight by 50% ($10/20 \times 100\%$). Similarly, if the HDL level of a woman is increased from 50 mg/dL (poor) to 57 mg/dL, with a normal level being 60 mg/dL, the difference between the prior level of the subject and the normal level is decreased by 70% (difference of 10 mg/dL between subject level and normal is decreased by 7 mg/dL, $7/10 \times 100\%$). As used herein, if a disease is associated with an elevated value for a symptom, “normal” is considered to be the upper limit of normal. If a disease is associated with a decreased value for a symptom, “normal” is considered to be the lower limit of normal.

(93) The term “sample,” as used herein, includes a collection of similar fluids, cells, or tissues isolated from a subject, as well as fluids, cells, or tissues present within a subject. Examples of biological fluids include blood, serum and serosal fluids, plasma, cerebrospinal fluid, ocular fluids, lymph, urine, saliva, and the like. Tissue samples may include samples from tissues, organs, or localized regions. For example, samples may be derived from particular organs, parts of organs, or fluids or cells within those organs. In certain embodiments, samples may be derived from the liver (e.g., whole liver or certain segments of liver or certain types of cells in the liver, such as, e.g., hepatocytes). In some embodiments, a “sample derived from a subject” refers to urine obtained from the subject. A “sample derived from a subject” can refer to blood (which can be readily converted to plasma or serum) drawn from the subject.

II. iRNAs of the Invention

(94) The present invention provides iRNAs which inhibit the expression of a KHK gene. In certain embodiments, the iRNA includes double stranded ribonucleic acid (dsRNA) molecules for inhibiting the expression of a KHK gene in a cell, such as a cell within a subject, e.g., a mammal, such as a human having or susceptible to developing a KHK-associated disease. The dsRNAi agent includes an antisense strand having a region of complementarity which is complementary to at least a part of an mRNA formed in the expression of a KHK gene. The region of complementarity is about 30 nucleotides or less in length (e.g., about 30, 29, 28, 27, 26, 25, 24, 23, 22, 21, 20, 19, or 18 nucleotides or less in length). Upon contact with a cell expressing the KHK gene, the iRNA inhibits the expression of the KHK gene (e.g., a human, a primate, a non-primate, or a rat KHK gene) by at least about 50% as assayed by, for example, a PCR or branched DNA (bDNA)-based method, or by a protein-based method, such as by immunofluorescence analysis, using, for example, western blotting or flow cytometric techniques. In certain embodiments, inhibition of expression is determined by the qPCR method provided in the examples herein with the siRNA at, e.g., a 10 nM concentration, in an appropriate organism cell line provided therein. In certain embodiments, inhibition of expression in vivo is determined by knockdown of the human gene in a rodent expressing the human gene, e.g., a mouse or an AAV-infected mouse expressing the human target gene, e.g., when administered as single dose, e.g., at 3 mg/kg at the nadir of RNA expression.

(95) A dsRNA includes two RNA strands that are complementary and hybridize to form a duplex structure under conditions in which the dsRNA will be used. One strand of a dsRNA (the antisense strand) includes a region of complementarity that is substantially complementary, and generally fully complementary, to a target sequence. The target sequence can be derived from the sequence of an mRNA formed during the expression of a KHK gene. The other strand (the sense strand)

includes a region that is complementary to the antisense strand, such that the two strands hybridize and form a duplex structure when combined under suitable conditions. As described elsewhere herein and as known in the art, the complementary sequences of a dsRNA can also be contained as self-complementary regions of a single nucleic acid molecule, as opposed to being on separate oligonucleotides.

(96) Generally, the duplex structure is 15 to 30 base pairs in length, e.g., 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 base pairs in length. In certain embodiments, the duplex structure is 18 to 25 base pairs in length, e.g., 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-25, 20-24, 20-23, 20-22, 20-21, 21-25, 21-24, 21-23, 21-22, 22-25, 22-24, 22-23, 23-25, 23-24 or 24-25 base pairs in length, for example, 19-21 basepairs in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the disclosure.

(97) Similarly, the region of complementarity to the target sequence is 15 to 30 nucleotides in length, e.g., 15-29, 15-28, 15-27, 15-26, 15-25, 15-24, 15-23, 15-22, 15-21, 15-20, 15-19, 15-18, 15-17, 18-30, 18-29, 18-28, 18-27, 18-26, 18-25, 18-24, 18-23, 18-22, 18-21, 18-20, 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 nucleotides in length, for example 19-23 nucleotides in length or 21-23 nucleotides in length. Ranges and lengths intermediate to the above recited ranges and lengths are also contemplated to be part of the disclosure. In some embodiments, the duplex structure is 19 to 30 base pairs in length. Similarly, the region of complementarity to the target sequence is 19 to 30 nucleotides in length.

(98) In some embodiments, the dsRNA is about 19 to about 23 nucleotides in length, or about 25 to about 30 nucleotides in length. In general, the dsRNA is long enough to serve as a substrate for the Dicer enzyme. For example, it is well-known in the art that dsRNAs longer than about 21-23 nucleotides in length may serve as substrates for Dicer. As the ordinarily skilled person will also recognize, the region of an RNA targeted for cleavage will most often be part of a larger RNA molecule, often an mRNA molecule. Where relevant, a “part” of an mRNA target is a contiguous sequence of an mRNA target of sufficient length to allow it to be a substrate for RNAi-directed cleavage (i.e., cleavage through a RISC pathway).

(99) One of skill in the art will also recognize that the duplex region is a primary functional portion of a dsRNA, e.g., a duplex region of about 19 to about 30 base pairs, e.g., about 19-30, 19-29, 19-28, 19-27, 19-26, 19-25, 19-24, 19-23, 19-22, 19-21, 19-20, 20-30, 20-29, 20-28, 20-27, 20-26, 20-25, 20-24, 20-23, 20-22, 20-21, 21-30, 21-29, 21-28, 21-27, 21-26, 21-25, 21-24, 21-23, or 21-22 base pairs. Thus, in one embodiment, to the extent that it becomes processed to a functional duplex, of e.g., 15-30 base pairs, that targets a desired RNA for cleavage, an RNA molecule or complex of RNA molecules having a duplex region greater than 30 base pairs is a dsRNA. Thus, an ordinarily skilled artisan will recognize that in one embodiment, a miRNA is a dsRNA. In another embodiment, a dsRNA is not a naturally occurring miRNA. In another embodiment, an iRNA agent useful to target KHK gene expression is not generated in the target cell by cleavage of a larger dsRNA.

(100) A dsRNA as described herein can further include one or more single-stranded nucleotide overhangs e.g., 1-4, 2-4, 1-3, 2-3, 1, 2, 3, or 4 nucleotides. dsRNAs having at least one nucleotide overhang can have superior inhibitory properties relative to their blunt-ended counterparts. A nucleotide overhang can comprise or consist of a nucleotide/nucleoside analog, including a deoxynucleotide/nucleoside. The overhang(s) can be on the sense strand, the antisense strand, or any combination thereof. Furthermore, the nucleotide(s) of an overhang can be present on the 5'-

end, 3'-end, or both ends of an antisense or sense strand of a dsRNA.

(101) A dsRNA can be synthesized by standard methods known in the art. Double stranded RNAi compounds of the invention may be prepared using a two-step procedure. First, the individual strands of the double stranded RNA molecule are prepared separately. Then, the component strands are annealed. The individual strands of the siRNA compound can be prepared using solution-phase or solid-phase organic synthesis or both. Organic synthesis offers the advantage that the oligonucleotide strands comprising unnatural or modified nucleotides can be easily prepared. Similarly, single-stranded oligonucleotides of the invention can be prepared using solution-phase or solid-phase organic synthesis or both.

(102) In an aspect, a dsRNA of the invention includes at least two nucleotide sequences, a sense sequence and an anti-sense sequence. The sense strand is selected from the group of sequences provided in any one of Tables 2, 3, 5, 6, and 8-13, and the corresponding antisense strand of the sense strand is selected from the group of sequences of any one of Tables 2, 3, 5, 6, and 8-13. In this aspect, one of the two sequences is complementary to the other of the two sequences, with one of the sequences being substantially complementary to a sequence of an mRNA generated in the expression of a KHK gene. As such, in this aspect, a dsRNA will include two oligonucleotides, where one oligonucleotide is described as the sense strand in any one of Tables 2, 3, 5, 6, and 8-13, and the second oligonucleotide is described as the corresponding antisense strand of the sense strand in any one of Tables 2, 3, 5, 6, and 8-13.

(103) In certain embodiments, the substantially complementary sequences of the dsRNA are contained on separate oligonucleotides. In other embodiments, the substantially complementary sequences of the dsRNA are contained on a single oligonucleotide.

(104) In certain embodiments, the sense or antisense strand is selected from the sense or antisense strand of any one of duplexes AD-517197.2; AD-517258.2; AD-516748.2; AD-516851.2; AD-519351.2; AD-519754.2; AD-519828.2; AD-520018.2; AD-520035.2; AD-520062.2; AD-520064.2; AD-520065.2; AD-520067.2; AD-75289.2; AD-520069.2; AD-520099.2; AD-67575.7; AD-520101.2; AD-1193323.1; AD-1193344.1; AD-1193350.1; AD-1193365.1; AD-1193379.1; AD-1193407.1; AD-1193421.1; AD-1193422.1; AD-1193429.1; AD-1193437.1; AD-1193443.1; AD-1193471.1; AD-1193481.1 or AD-67605.7.

(105) In some embodiments, the sense or antisense strand is selected from the sense or antisense strand of any one of duplexes AD-519345.1, AD-519346.1, AD-519347.1, AD-67554.7, AD-519752.3, AD-1010731.1, AD-1010732.1, AD-519343.1, AD-519344.1, AD-519349.1, AD-519350.1, AD-519753.2, AD-519932.1, AD-519935.2, AD-520018.6, AD-517837.2, AD-805635.2, AD-519329.2, AD-520063.2, AD-519757.2, AD-805631.2, AD-516917.2, AD-516828.2, AD-518983.2, AD-805636.2, AD-519754.7, AD-520062.2, AD-67575.9, AD-518923.3, AD-520053.4, AD-519667.2, AD-519773.2, AD-519354.2, AD-520060.4, AD-520061.4, AD-1010733.2, AD-1010735.2, AD-1193323.1; AD-1193344.1; AD-1193350.1; AD-1193365.1; AD-1193379.1; AD-1193407.1; AD-1193421.1; AD-1193422.1; AD-1193429.1; AD-1193437.1; AD-1193443.1; AD-1193471.1; or AD-1193481.1.

(106) In some embodiments, the sense or antisense strand is selected from the sense or antisense strand of duplex AD-519351.

(107) It will be understood that, although the sequences in, for example, Tables 2, 5, 8 and 10 are not described as modified or conjugated sequences, the RNA of the iRNA of the invention e.g., a dsRNA of the invention, may comprise any one of the sequences set forth in any one of Tables 2, 3, 5, 6, and 8-13 that is un-modified, un-conjugated, or modified or conjugated differently than described therein. In other words, the invention encompasses dsRNA of Tables 2, 3, 5, 6, and 8-13 which are un-modified, un-conjugated, modified, or conjugated, as described herein.

(108) The skilled person is well aware that dsRNAs having a duplex structure of about 20 to 23 base pairs, e.g., 21, base pairs have been hailed as particularly effective in inducing RNA interference (Elbashir et al., EMBO 2001, 20:6877-6888). However, others have found that shorter

or longer RNA duplex structures can also be effective (Chu and Rana (2007) RNA 14:1714-1719; Kim et al. (2005) Nat Biotech 23:222-226). In the embodiments described above, by virtue of the nature of the oligonucleotide sequences provided in any one of Tables 2, 3, 5, 6, and 8-13, dsRNAs described herein can include at least one strand of a length of minimally 21 nucleotides. It can be reasonably expected that shorter duplexes having any one of the sequences in any one of Tables 2, 3, 5, 6, and 8-13 minus only a few nucleotides on one or both ends can be similarly effective as compared to the dsRNAs described above. Hence, dsRNAs having a sequence of at least 19, 20, 21, 22, 23 or more contiguous nucleotides derived from any one of the sequences of any one of Tables 2, 3, 5, 6, and 8-13, and differing in their ability to inhibit the expression of a KHK gene by not more than about 5, 10, 15, 20, 25, or 30% inhibition from a dsRNA comprising the full sequence, are contemplated to be within the scope of the present invention.

(109) In addition, the RNAs provided in Tables 2, 3, 5, 6, and 8-13 identify a site(s) in a KHK transcript that is susceptible to RISC-mediated cleavage. As such, the present invention further features iRNAs that target within one of these sites. As used herein, an iRNA is said to target within a particular site of an RNA transcript if the iRNA promotes cleavage of the transcript anywhere within that particular site. Such an iRNA will generally include at least about 19 contiguous nucleotides from any one of the sequences provided in any one of Tables 2, 3, 5, 6, and 8-13 coupled to additional nucleotide sequences taken from the region contiguous to the selected sequence in a KHK gene.

III. Modified iRNAs of the Invention

(110) In certain embodiments, the RNA of the iRNA of the invention e.g., a dsRNA, is unmodified, and does not comprise, e.g., chemical modifications or conjugations known in the art and described herein. In other embodiments, the RNA of an iRNA of the invention, e.g., a dsRNA, is chemically modified to enhance stability or other beneficial characteristics. In certain embodiments of the invention, substantially all of the nucleotides of an iRNA of the invention are modified. In other embodiments of the invention, all of the nucleotides of an iRNA or substantially all of the nucleotides of an iRNA are modified, i.e., not more than 5, 4, 3, 2, or 1 unmodified nucleotides are present in a strand of the iRNA.

(111) The nucleic acids featured in the invention can be synthesized or modified by methods well established in the art, such as those described in "Current protocols in nucleic acid chemistry," Beaucage, S. L. et al. (Eds.), John Wiley & Sons, Inc., New York, NY, USA, which is hereby incorporated herein by reference. Modifications include, for example, end modifications, e.g., 5'-end modifications (phosphorylation, conjugation, inverted linkages) or 3'-end modifications (conjugation, DNA nucleotides, inverted linkages, etc.); base modifications, e.g., replacement with stabilizing bases, destabilizing bases, or bases that base pair with an expanded repertoire of partners, removal of bases (abasic nucleotides), or conjugated bases; sugar modifications (e.g., at the 2'-position or 4'-position) or replacement of the sugar; or backbone modifications, including modification or replacement of the phosphodiester linkages. Specific examples of iRNA compounds useful in the embodiments described herein include, but are not limited to RNAs containing modified backbones or no natural internucleoside linkages. RNAs having modified backbones include, among others, those that do not have a phosphorus atom in the backbone. For the purposes of this specification, and as sometimes referenced in the art, modified RNAs that do not have a phosphorus atom in their internucleoside backbone can also be considered to be oligonucleosides. In some embodiments, a modified iRNA will have a phosphorus atom in its internucleoside backbone.

(112) Modified RNA backbones include, for example, phosphorothioates, chiral phosphorothioates, phosphorodithioates, phosphotriesters, aminoalkylphosphotriesters, methyl and other alkyl phosphonates including 3'-alkylene phosphonates and chiral phosphonates, phosphinates, phosphoramidates including 3'-amino phosphoramidate and aminoalkylphosphoramidates, thionophosphoramidates, thionoalkylphosphonates, thionoalkylphosphotriesters, and

boranophosphates having normal 3'-5' linkages, 2'-5'-linked analogs of these, and those having inverted polarity wherein the adjacent pairs of nucleoside units are linked 3'-5' to 5'-3' or 2'-5' to 5'-2'. Various salts, mixed salts and free acid forms are also included. In some embodiments of the invention, the dsRNA agents of the invention are in a free acid form. In other embodiments of the invention, the dsRNA agents of the invention are in a salt form. In one embodiment, the dsRNA agents of the invention are in a sodium salt form. In certain embodiments, when the dsRNA agents of the invention are in the sodium salt form, sodium ions are present in the agent as counterions for substantially all of the phosphodiester and/or phosphorothioate groups present in the agent. Agents in which substantially all of the phosphodiester and/or phosphorothioate linkages have a sodium counterion include not more than 5, 4, 3, 2, or 1 phosphodiester and/or phosphorothioate linkages without a sodium counterion. In some embodiments, when the dsRNA agents of the invention are in the sodium salt form, sodium ions are present in the agent as counterions for all of the phosphodiester and/or phosphorothioate groups present in the agent.

(113) Representative U.S. patents that teach the preparation of the above phosphorus-containing linkages include, but are not limited to, U.S. Pat. Nos. 3,687,808; 4,469,863; 4,476,301; 5,023,243; 5,177,195; 5,188,897; 5,264,423; 5,276,019; 5,278,302; 5,286,717; 5,321,131; 5,399,676; 5,405,939; 5,453,496; 5,455,233; 5,466,677; 5,476,925; 5,519,126; 5,536,821; 5,541,316; 5,550,111; 5,563,253; 5,571,799; 5,587,361; 5,625,050; 6,028,188; 6,124,445; 6,160,109; 6,169,170; 6,172,209; 6,239,265; 6,277,603; 6,326,199; 6,346,614; 6,444,423; 6,531,590; 6,534,639; 6,608,035; 6,683,167; 6,858,715; 6,867,294; 6,878,805; 7,015,315; 7,041,816; 7,273,933; 7,321,029; and U.S. Pat. RE39464, the entire contents of each of which are hereby incorporated herein by reference.

(114) Modified RNA backbones that do not include a phosphorus atom therein have backbones that are formed by short chain alkyl or cycloalkyl internucleoside linkages, mixed heteroatoms and alkyl or cycloalkyl internucleoside linkages, or one or more short chain heteroatomic or heterocyclic internucleoside linkages. These include those having morpholino linkages (formed in part from the sugar portion of a nucleoside); siloxane backbones; sulfide, sulfoxide and sulfone backbones; formacetyl and thioformacetyl backbones; methylene formacetyl and thioformacetyl backbones; alkene containing backbones; sulfamate backbones; methyleneimino and methylenehydrazino backbones; sulfonate and sulfonamide backbones; amide backbones; and others having mixed N, O, S, and CH₂ component parts.

(115) Representative U.S. patents that teach the preparation of the above oligonucleosides include, but are not limited to, U.S. Pat. Nos. 5,034,506; 5,166,315; 5,185,444; 5,214,134; 5,216,141; 5,235,033; 5,64,562; 5,264,564; 5,405,938; 5,434,257; 5,466,677; 5,470,967; 5,489,677; 5,541,307; 5,561,225; 5,596,086; 5,602,240; 5,608,046; 5,610,289; 5,618,704; 5,623,070; 5,663,312; 5,633,360; 5,677,437; and 5,677,439, the entire contents of each of which are hereby incorporated herein by reference.

(116) Suitable RNA mimetics are contemplated for use in iRNAs provided herein, in which both the sugar and the internucleoside linkage, i.e., the backbone, of the nucleotide units are replaced with novel groups. The base units are maintained for hybridization with an appropriate nucleic acid target compound. One such oligomeric compound in which an RNA mimetic that has been shown to have excellent hybridization properties is referred to as a peptide nucleic acid (PNA). In PNA compounds, the sugar backbone of an RNA is replaced with an amide containing backbone, in particular an aminoethylglycine backbone. The nucleobases are retained and are bound directly or indirectly to aza nitrogen atoms of the amide portion of the backbone. Representative US patents that teach the preparation of PNA compounds include, but are not limited to, U.S. Pat. Nos. 5,539,082; 5,714,331; and 5,719,262, the entire contents of each of which are hereby incorporated herein by reference. Additional PNA compounds suitable for use in the iRNAs of the invention are described in, for example, in Nielsen et al., *Science*, 1991, 254, 1497-1500.

(117) Some embodiments featured in the invention include RNAs with phosphorothioate

backbones and oligonucleosides with heteroatom backbones, and in particular —CH.sub.2—NH—CH.sub.2—, —CH.sub.2—N(CH.sub.3)—O—CH.sub.2— [known as a methylene (methylimino) or MMI backbone], —CH.sub.2—O—N(CH.sub.3)—CH.sub.2—, —CH.sub.2—N(CH.sub.3)—N(CH.sub.3)—CH.sub.2— and —N(CH.sub.3)—CH.sub.2—CH.sub.2— of the above-referenced U.S. Pat. No. 5,489,677, and the amide backbones of the above-referenced U.S. Pat. No. 5,602,240. In some embodiments, the RNAs featured herein have morpholino backbone structures of the above-referenced U.S. Pat. No. 5,034,506. The native phosphodiester backbone can be represented as O—P(O)(OH)—OCH.sub.2—.

(118) Modified RNAs can also contain one or more substituted sugar moieties. The iRNAs, e.g., dsRNAs, featured herein can include one of the following at the 2'-position: OH; F; O-, S-, or N-alkyl; O-, S-, or N-alkenyl; O-, S- or N-alkynyl; or O-alkyl-O-alkyl, wherein the alkyl, alkenyl and alkynyl can be substituted or unsubstituted C.sub.1 to C.sub.10 alkyl or C.sub.2 to C.sub.10 alkenyl and alkynyl. Exemplary suitable modifications include O[(CH.sub.2).sub.nO].sub.mCH.sub.3, O(CH.sub.2).sub.nOCH.sub.3, O(CH.sub.2).sub.nNH.sub.2, O(CH.sub.2).sub.nCH.sub.3, O(CH.sub.2).sub.nONH.sub.2, and O(CH.sub.2).sub.nON[(CH.sub.2).sub.nCH.sub.3].sub.2, where n and m are from 1 to about 10. In other embodiments, dsRNAs include one of the following at the 2' position: C.sub.1 to C.sub.10 lower alkyl, substituted lower alkyl, alkaryl, aralkyl, O-alkaryl or O-aralkyl, SH, SCH.sub.3, OCN, Cl, Br, CN, CF.sub.3, OCF.sub.3, SOCH.sub.3, SO.sub.2CH.sub.3, ONO.sub.2, NO.sub.2, N.sub.3, NH.sub.2, heterocycloalkyl, heterocycloalkaryl, aminoalkylamino, polyalkylamino, substituted silyl, an RNA cleaving group, a reporter group, an intercalator, a group for improving the pharmacokinetic properties of an iRNA, or a group for improving the pharmacodynamic properties of an iRNA, and other substituents having similar properties. In some embodiments, the modification includes a 2'-methoxyethoxy (2'-O—CH.sub.2CH.sub.2OCH.sub.3, also known as 2'-O-(2-methoxyethyl) or 2'-MOE) (Martin et al., *Helv. Chim. Acta*, 1995, 78:486-504) i.e., an alkoxy-alkoxy group. Another exemplary modification is 2'-dimethylaminoethoxy, i.e., a O(CH.sub.2).sub.2ON(CH.sub.3).sub.2 group, also known as 2'-DMAOE, as described in examples herein below, and 2'-dimethylaminoethoxyethoxy (also known in the art as 2'-O-dimethylaminoethoxyethyl or 2'-DMAEOE), i.e., 2'-O—CH.sub.2—O—CH.sub.2—N(CH.sub.3).sub.2. Further exemplary modifications include: 5'-Me-2'-F nucleotides, 5'-Me-2'-OMe nucleotides, 5'-Me-2'-deoxynucleotides, (both R and S isomers in these three families); 2'-alkoxyalkyl; and 2'-NMA (N-methylacetamide).

(119) Other modifications include 2'-methoxy (2'-OCH.sub.3), 2'-aminopropoxy (2'-OCH.sub.2CH.sub.2CH.sub.2NH.sub.2) and 2'-fluoro (2'-F). Similar modifications can also be made at other positions on the RNA of an iRNA, particularly the 3' position of the sugar on the 3' terminal nucleotide or in 2'-5' linked dsRNAs and the 5' position of 5' terminal nucleotide. iRNAs can also have sugar mimetics such as cyclobutyl moieties in place of the pentofuranosyl sugar. Representative US patents that teach the preparation of such modified sugar structures include, but are not limited to, U.S. Pat. Nos. 4,981,957; 5,118,800; 5,319,080; 5,359,044; 5,393,878; 5,446,137; 5,466,786; 5,514,785; 5,519,134; 5,567,811; 5,576,427; 5,591,722; 5,597,909; 5,610,300; 5,627,053; 5,639,873; 5,646,265; 5,658,873; 5,670,633; and 5,700,920, certain of which are commonly owned with the instant application. The entire contents of each of the foregoing are hereby incorporated herein by reference.

(120) An iRNA can also include nucleobase (often referred to in the art simply as “base”) modifications or substitutions. As used herein, “unmodified” or “natural” nucleobases include the purine bases adenine (A) and guanine (G), and the pyrimidine bases thymine (T), cytosine (C), and uracil (U). Modified nucleobases include other synthetic and natural nucleobases such as deoxythymidine (dT), 5-methylcytosine (5-me-C), 5-hydroxymethyl cytosine, xanthine, hypoxanthine, 2-aminoadenine, 6-methyl and other alkyl derivatives of adenine and guanine, 2-propyl and other alkyl derivatives of adenine and guanine, 2-thiouracil, 2-thiothymine and 2-

thiocytosine, 5-halouracil and cytosine, 5-propynyl uracil and cytosine, 6-azo uracil, cytosine and thymine, 5-uracil (pseudouracil), 4-thiouracil, 8-halo, 8-amino, 8-thiol, 8-thioalkyl, 8-hydroxyl and other 8-substituted adenines and guanines, 5-halo, particularly 5-bromo, 5-trifluoromethyl and other 5-substituted uracils and cytosines, 7-methylguanine and 7-methyladenine, 8-azaguanine and 8-azaadenine, 7-deazaguanine and 7-daazaadenine and 3-deazaguanine and 3-deazaadenine. Further nucleobases include those disclosed in U.S. Pat. No. 3,687,808, those disclosed in *Modified Nucleosides in Biochemistry, Biotechnology and Medicine*, Herdewijn, P. ed. Wiley-VCH, 2008; those disclosed in *The Concise Encyclopedia Of Polymer Science And Engineering*, pages 858-859, Kroschwitz, J. L., ed. John Wiley & Sons, 1990, those disclosed by Englisch et al., *Angewandte Chemie, International Edition*, 1991, 30, 613, and those disclosed by Sanghvi, Y S., Chapter 15, *dsRNA Research and Applications*, pages 289-302, Crooke, S. T. and Lebleu, B., Ed., CRC Press, 1993. Certain of these nucleobases are particularly useful for increasing the binding affinity of the oligomeric compounds featured in the invention. These include 5-substituted pyrimidines, 6-azapyrimidines and N-2, N-6 and O-6 substituted purines, including 2-aminopropyladenine, 5-propynyluracil and 5-propynylcytosine. 5-methylcytosine substitutions have been shown to increase nucleic acid duplex stability by 0.6-1.2° C. (Sanghvi, Y. S., Crooke, S. T. and Lebleu, B., Eds., *dsRNA Research and Applications*, CRC Press, Boca Raton, 1993, pp. 276-278) and are exemplary base substitutions, even more particularly when combined with 2'-O-methoxyethyl sugar modifications.

(121) Representative U.S. patents that teach the preparation of certain of the above noted modified nucleobases as well as other modified nucleobases include, but are not limited to, the above noted U.S. Pat. Nos. 3,687,808, 4,845,205; 5,130,30; 5,134,066; 5,175,273; 5,367,066; 5,432,272; 5,457,187; 5,459,255; 5,484,908; 5,502,177; 5,525,711; 5,552,540; 5,587,469; 5,594,121, 5,596,091; 5,614,617; 5,681,941; 5,750,692; 6,015,886; 6,147,200; 6,166,197; 6,222,025; 6,235,887; 6,380,368; 6,528,640; 6,639,062; 6,617,438; 7,045,610; 7,427,672; and 7,495,088, the entire contents of each of which are hereby incorporated herein by reference.

(122) An iRNA agent of the disclosure can also be modified to include one or more bicyclic sugar moieties. A "bicyclic sugar" is a furanosyl ring modified by a ring formed by the bridging of two carbons, whether adjacent or non-adjacent. A "bicyclic nucleoside" ("BNA") is a nucleoside having a sugar moiety comprising a ring formed by bridging two carbons, whether adjacent or non-adjacent, of the sugar ring, thereby forming a bicyclic ring system. In certain embodiments, the bridge connects the 4'-carbon and the 2'-carbon of the sugar ring, optionally, via the 2'-acyclic oxygen atom. Thus, in some embodiments an agent of the invention may include one or more locked nucleic acids (LNA). A locked nucleic acid is a nucleotide having a modified ribose moiety in which the ribose moiety comprises an extra bridge connecting the 2' and 4' carbons. In other words, an LNA is a nucleotide comprising a bicyclic sugar moiety comprising a 4'-CH.sub.2—O-2' bridge. This structure effectively "locks" the ribose in the 3'-endo structural conformation. The addition of locked nucleic acids to siRNAs has been shown to increase siRNA stability in serum, and to reduce off-target effects (Elmen, J. et al., (2005) *Nucleic Acids Research* 33(1):439-447; Mook, O R. et al., (2007) *Mol Canc Ther* 6(3):833-843; Grunweller, A. et al., (2003) *Nucleic Acids Research* 31(12):3185-3193). Examples of bicyclic nucleosides for use in the polynucleotides of the invention include without limitation nucleosides comprising a bridge between the 4' and the 2' ribosyl ring atoms. In certain embodiments, the antisense polynucleotide agents of the invention include one or more bicyclic nucleosides comprising a 4' to 2' bridge.

(123) A locked nucleoside can be represented by the structure (omitting stereochemistry),

(124) ##STR00005## wherein B is a nucleobase or modified nucleobase and L is the linking group that joins the 2'-carbon to the 4'-carbon of the ribose ring. Examples of such 4' to 2' bridged bicyclic nucleosides, include but are not limited to 4'-(CH.sub.2)—O-2' (LNA); 4'-(CH.sub.2)—S-2'; 4'-(CH.sub.2).sub.2-O-2' (ENA); 4'-CH(CH.sub.3)—O-2' (also referred to as "constrained ethyl" or "cEt") and 4'-CH(CH.sub.2OCH.sub.3)—O-2' (and analogs thereof; see, e.g., U.S. Pat.

No. 7,399,845); 4'-C(CH₂.sub.3)(CH₂.sub.3)—O-2' (and analogs thereof; see e.g., U.S. Pat. No. 8,278,283); 4'-CH₂.sub.2—N(OCH₂.sub.3)-2' (and analogs thereof; see e.g., U.S. Pat. No. 8,278,425); 4'-CH₂.sub.2—O—N(CH₂.sub.3)-2' (see, e.g., U.S. Patent Publication No. 2004/0171570); 4'-CH₂.sub.2—N(R)—O-2', wherein R is H, C₁-C₁₂ alkyl, or a nitrogen protecting group (see, e.g., U.S. Pat. No. 7,427,672); 4'-CH₂.sub.2—C(H)(CH₂.sub.3)-2' (see, e.g., Chattopadhyaya et al., *J. Org. Chem.*, 2009, 74, 118-134); and 4'-CH₂.sub.2—C(=CH₂.sub.2)-2' (and analogs thereof; see, e.g., U.S. Pat. No. 8,278,426). The entire contents of each of the foregoing are hereby incorporated herein by reference.

(125) Additional representative U.S. Patents and U.S. Patent Publications that teach the preparation of locked nucleic acid nucleotides include, but are not limited to, the following: U.S. Pat. Nos. 6,268,490; 6,525,191; 6,670,461; 6,770,748; 6,794,499; 6,998,484; 7,053,207; 7,034,133; 7,084,125; 7,399,845; 7,427,672; 7,569,686; 7,741,457; 8,022,193; 8,030,467; 8,278,425; 8,278,426; 8,278,283; US 2008/0039618; and US 2009/0012281, the entire contents of each of which are hereby incorporated herein by reference.

(126) Any of the foregoing bicyclic nucleosides can be prepared having one or more stereochemical sugar configurations including for example α -L-ribofuranose and β -D-ribofuranose (see WO 99/14226).

(127) The RNA of an iRNA can also be modified to include one or more constrained ethyl nucleotides. As used herein, a “constrained ethyl nucleotide” or “cEt” is a locked nucleic acid comprising a bicyclic sugar moiety comprising a 4'-CH(CH₂.sub.3)—O-2' bridge (i.e., L in the preceding structure). In one embodiment, a constrained ethyl nucleotide is in the S conformation referred to herein as “S-cEt.”

(128) An iRNA of the invention may also include one or more “conformationally restricted nucleotides” (“CRN”). CRN are nucleotide analogs with a linker connecting the C2' and C4' carbons of ribose or the C3 and —C5' carbons of ribose. CRN lock the ribose ring into a stable conformation and increase the hybridization affinity to mRNA. The linker is of sufficient length to place the oxygen in an optimal position for stability and affinity resulting in less ribose ring puckering.

(129) Representative publications that teach the preparation of certain of the above noted CRN include, but are not limited to, U.S. Patent Publication No. 2013/0190383; and PCT publication WO 2013/036868, the entire contents of each of which are hereby incorporated herein by reference.

(130) In some embodiments, an iRNA of the invention comprises one or more monomers that are UNA (unlocked nucleic acid) nucleotides. UNA is unlocked acyclic nucleic acid, wherein any of the bonds of the sugar has been removed, forming an unlocked “sugar” residue. In one example, UNA also encompasses monomer with bonds between C1'-C4' have been removed (i.e. the covalent carbon-oxygen-carbon bond between the C1' and C4' carbons). In another example, the C2'-C3' bond (i.e. the covalent carbon-carbon bond between the C2' and C3' carbons) of the sugar has been removed (see *Nuc. Acids Symp. Series*, 52, 133-134 (2008) and Fluiter et al., *Mol. Biosyst.*, 2009, 10, 1039 hereby incorporated by reference).

(131) Representative U.S. publications that teach the preparation of UNA include, but are not limited to, U.S. Pat. No. 8,314,227; and U.S. Patent Publication Nos. 2013/0096289; 2013/0011922; and 2011/0313020, the entire contents of each of which are hereby incorporated herein by reference.

(132) Potentially stabilizing modifications to the ends of RNA molecules can include N-(acetylaminocaproyl)-4-hydroxyprolinol (Hyp-C6-NHAc), N-(caproyl-4-hydroxyprolinol (Hyp-C6), N-(acetyl-4-hydroxyprolinol (Hyp-NHAc), thymidine-2'-O-deoxythymidine (ether), N-(aminocaproyl)-4-hydroxyprolinol (Hyp-C6-amino), 2-docosanoyl-uridine-3'-phosphate, inverted 2'-deoxy-modified ribonucleotide, such as inverted dT(idT), inverted dA (idA), and inverted abasic 2'-deoxyribonucleotide (iAb) and others. Disclosure of this modification can be found in WO 2011/005861.

(133) In one example, the 3' or 5' terminal end of a oligonucleotide is linked to an inverted 2'-deoxy-modified ribonucleotide, such as inverted dT(idT), inverted dA (idA), or a inverted abasic 2'-deoxyribonucleotide (iAb). In one particular example, the inverted 2'-deoxy-modified ribonucleotide is linked to the 3'end of an oligonucleotide, such as the 3'-end of a sense strand described herein, where the linking is via a 3'-3' phosphodiester linkage or a 3'-3'-phosphorothioate linkage.

(134) In another example, the 3'-end of a sense strand is linked via a 3'-3'-phosphorothioate linkage to an inverted abasic ribonucleotide (iAb). In another example, the 3'-end of a sense strand is linked via a 3'-3'-phosphorothioate linkage to an inverted dA (idA).

(135) In one particular example, the inverted 2'-deoxy-modified ribonucleotide is linked to the 3'end of an oligonucleotide, such as the 3'-end of a sense strand described herein, where the linking is via a 3'-3' phosphodiester linkage or a 3'-3'-phosphorothioate linkage.

(136) In another example, the 3'-terminal nucleotides of a sense strand is an inverted dA (idA) and is linked to the preceding nucleotide via a 3'-3'-linkage (e.g., 3'-3'-phosphorothioate linkage).

(137) Other modifications of the nucleotides of an iRNA of the invention include a 5' phosphate or 5' phosphate mimic, e.g., a 5'-terminal phosphate or phosphate mimic on the antisense strand of an iRNA. Suitable phosphate mimics are disclosed in, for example U.S. Patent Publication No. 2012/0157511, the entire contents of which are incorporated herein by reference.

(138) A. Modified iRNAs Comprising Motifs of the Invention

(139) In certain aspects of the invention, the double stranded RNA agents of the invention include agents with chemical modifications as disclosed, for example, in WO2013/075035, the entire contents of each of which are incorporated herein by reference. As shown herein and in WO2013/075035, one or more motifs of three identical modifications on three consecutive nucleotides may be introduced into a sense strand or antisense strand of a dsRNAi agent, particularly at or near the cleavage site. In some embodiments, the sense strand and antisense strand of the dsRNAi agent may otherwise be completely modified. The introduction of these motifs interrupts the modification pattern, if present, of the sense or antisense strand. The dsRNAi agent may be optionally conjugated with a GalNAc derivative ligand, for instance on the sense strand.

(140) More specifically, when the sense strand and antisense strand of the double stranded RNA agent are completely modified to have one or more motifs of three identical modifications on three consecutive nucleotides at or near the cleavage site of at least one strand of a dsRNAi agent, the gene silencing activity of the dsRNAi agent was observed.

(141) Accordingly, the invention provides double stranded RNA agents capable of inhibiting the expression of a target gene (i.e., KHK gene) in vivo. The RNAi agent comprises a sense strand and an antisense strand. Each strand of the RNAi agent may be, for example, 17-30 nucleotides in length, 25-30 nucleotides in length, 27-30 nucleotides in length, 19-25 nucleotides in length, 19-23 nucleotides in length, 19-21 nucleotides in length, 21-25 nucleotides in length, or 21-23 nucleotides in length.

(142) The sense strand and antisense strand typically form a duplex double stranded RNA ("dsRNA"), also referred to herein as "dsRNAi agent." The duplex region of a dsRNAi agent may be, for example, the duplex region can be 27-30 nucleotide pairs in length, 19-25 nucleotide pairs in length, 19-23 nucleotide pairs in length, 19-21 nucleotide pairs in length, 21-25 nucleotide pairs in length, or 21-23 nucleotide pairs in length. In another example, the duplex region is selected from 19, 20, 21, 22, 23, 24, 25, 26, and 27 nucleotides in length.

(143) In certain embodiments, the dsRNAi agent may contain one or more overhang regions or capping groups at the 3'-end, 5'-end, or both ends of one or both strands. The overhang can be, independently, 1-6 nucleotides in length, for instance 2-6 nucleotides in length, 1-5 nucleotides in length, 2-5 nucleotides in length, 1-4 nucleotides in length, 2-4 nucleotides in length, 1-3 nucleotides in length, 2-3 nucleotides in length, or 1-2 nucleotides in length. In certain

embodiments, the overhang regions can include extended overhang regions as provided above. The overhangs can be the result of one strand being longer than the other, or the result of two strands of the same length being staggered. The overhang can form a mismatch with the target mRNA or it can be complementary to the gene sequences being targeted or can be another sequence. The first and second strands can also be joined, e.g., by additional bases to form a hairpin, or by other non-base linkers.

(144) In certain embodiments, the nucleotides in the overhang region of the dsRNAi agent can each independently be a modified or unmodified nucleotide including, but not limited to 2'-sugar modified, such as, 2'-F, 2'-O-methyl, thymidine (T), 2'-O-methoxyethyl-5-methyluridine (Teo), 2'-O-methoxyethyladenosine (Aeo), 2'-O-methoxyethyl-5-methylcytidine (m5Ceo), and any combinations thereof.

(145) For example, TT can be an overhang sequence for either end on either strand. The overhang can form a mismatch with the target mRNA or it can be complementary to the gene sequences being targeted or can be another sequence.

(146) The 5'- or 3'-overhangs at the sense strand, antisense strand, or both strands of the dsRNAi agent may be phosphorylated. In some embodiments, the overhang region(s) contains two nucleotides having a phosphorothioate between the two nucleotides, where the two nucleotides can be the same or different. In some embodiments, the overhang is present at the 3'-end of the sense strand, antisense strand, or both strands. In some embodiments, this 3'-overhang is present in the antisense strand. In some embodiments, this 3'-overhang is present in the sense strand.

(147) The dsRNAi agent may contain only a single overhang, which can strengthen the interference activity of the RNAi, without affecting its overall stability. For example, the single-stranded overhang may be located at the 3'-end of the sense strand or, alternatively, at the 3'-end of the antisense strand. The RNAi may also have a blunt end, located at the 5'-end of the antisense strand (or the 3'-end of the sense strand) or vice versa. Generally, the antisense strand of the dsRNAi agent has a nucleotide overhang at the 3'-end, and the 5'-end is blunt. While not wishing to be bound by theory, the asymmetric blunt end at the 5'-end of the antisense strand and 3'-end overhang of the antisense strand favor the guide strand loading into RISC process.

(148) In certain embodiments, the dsRNAi agent is a double blunt-ended of 19 nucleotides in length, wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides at positions 7, 8, and 9 from the 5'end. The antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at positions 11, 12, and 13 from the 5'end.

(149) In other embodiments, the dsRNAi agent is a double blunt-ended of 20 nucleotides in length, wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides at positions 8, 9, and 10 from the 5'end. The antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at positions 11, 12, and 13 from the 5'end.

(150) In yet other embodiments, the dsRNAi agent is a double blunt-ended of 21 nucleotides in length, wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides at positions 9, 10, and 11 from the 5'end. The antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at positions 11, 12, and 13 from the 5'end.

(151) In certain embodiments, the dsRNAi agent comprises a 21 nucleotide sense strand and a 23 nucleotide antisense strand, wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides at positions 9, 10, and 11 from the 5'end; the antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at positions 11, 12, and 13 from the 5'end, wherein one end of the RNAi agent is blunt, while the other end comprises a two nucleotide overhang. In some embodiments, the two nucleotide overhang is at the 3'-end of the antisense strand.

(152) When the two nucleotide overhang is at the 3'-end of the antisense strand, there may be two phosphorothioate internucleotide linkages between the terminal three nucleotides, wherein two of the three nucleotides are the overhang nucleotides, and the third nucleotide is a paired nucleotide next to the overhang nucleotide. In one embodiment, the RNAi agent additionally has two phosphorothioate internucleotide linkages between the terminal three nucleotides at both the 5'-end of the sense strand and at the 5'-end of the antisense strand. In certain embodiments, every nucleotide in the sense strand and the antisense strand of the dsRNAi agent, including the nucleotides that are part of the motifs are modified nucleotides. In certain embodiments each residue is independently modified with a 2'-O-methyl or 2'-fluoro, e.g., in an alternating motif. Optionally, the dsRNAi agent further comprises a ligand (such as, GalNAc.sub.3).

(153) In certain embodiments, the dsRNAi agent comprises a sense and an antisense strand, wherein the sense strand is 25-30 nucleotide residues in length, wherein starting from the 5' terminal nucleotide (position 1) positions 1 to 23 of the first strand comprise at least 8 ribonucleotides; the antisense strand is 36-66 nucleotide residues in length and, starting from the 3' terminal nucleotide, comprises at least 8 ribonucleotides in the positions paired with positions 1-23 of sense strand to form a duplex; wherein at least the 3' terminal nucleotide of antisense strand is unpaired with sense strand, and up to 6 consecutive 3' terminal nucleotides are unpaired with sense strand, thereby forming a 3' single stranded overhang of 1-6 nucleotides; wherein the 5' terminus of antisense strand comprises from 10-30 consecutive nucleotides which are unpaired with sense strand, thereby forming a 10-30 nucleotide single stranded 5' overhang; wherein at least the sense strand 5' terminal and 3' terminal nucleotides are base paired with nucleotides of antisense strand when sense and antisense strands are aligned for maximum complementarity, thereby forming a substantially duplexed region between sense and antisense strands; and antisense strand is sufficiently complementary to a target RNA along at least 19 ribonucleotides of antisense strand length to reduce target gene expression when the double stranded nucleic acid is introduced into a mammalian cell; and wherein the sense strand contains at least one motif of three 2'-F modifications on three consecutive nucleotides, where at least one of the motifs occurs at or near the cleavage site. The antisense strand contains at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at or near the cleavage site.

(154) In certain embodiments, the dsRNAi agent comprises sense and antisense strands, wherein the dsRNAi agent comprises a first strand having a length which is at least 25 and at most 29 nucleotides and a second strand having a length which is at most 30 nucleotides with at least one motif of three 2'-O-methyl modifications on three consecutive nucleotides at position 11, 12, 13 from the 5' end; wherein the 3' end of the first strand and the 5' end of the second strand form a blunt end and the second strand is 1-4 nucleotides longer at its 3' end than the first strand, wherein the duplex region which is at least 25 nucleotides in length, and the second strand is sufficiently complementary to a target mRNA along at least 19 nucleotide of the second strand length to reduce target gene expression when the RNAi agent is introduced into a mammalian cell, and wherein Dicer cleavage of the dsRNAi agent results in an siRNA comprising the 3'-end of the second strand, thereby reducing expression of the target gene in the mammal. Optionally, the dsRNAi agent further comprises a ligand.

(155) In certain embodiments, the sense strand of the dsRNAi agent contains at least one motif of three identical modifications on three consecutive nucleotides, where one of the motifs occurs at the cleavage site in the sense strand.

(156) In certain embodiments, the antisense strand of the dsRNAi agent can also contain at least one motif of three identical modifications on three consecutive nucleotides, where one of the motifs occurs at or near the cleavage site in the antisense strand.

(157) For a dsRNAi agent having a duplex region of 19-23 nucleotides in length, the cleavage site of the antisense strand is typically around the 10, 11, and 12 positions from the 5'-end. Thus the motifs of three identical modifications may occur at the 9, 10, and 11 positions; the 10, 11, and 12

positions; the 11, 12, and 13 positions; the 12, 13, and 14 positions; or the 13, 14, and 15 positions of the antisense strand, the count starting from the first nucleotide from the 5'-end of the antisense strand, or, the count starting from the first paired nucleotide within the duplex region from the 5'-end of the antisense strand. The cleavage site in the antisense strand may also change according to the length of the duplex region of the dsRNAi agent from the 5'-end.

(158) The sense strand of the dsRNAi agent may contain at least one motif of three identical modifications on three consecutive nucleotides at the cleavage site of the strand; and the antisense strand may have at least one motif of three identical modifications on three consecutive nucleotides at or near the cleavage site of the strand. When the sense strand and the antisense strand form a dsRNA duplex, the sense strand and the antisense strand can be so aligned that one motif of the three nucleotides on the sense strand and one motif of the three nucleotides on the antisense strand have at least one nucleotide overlap, i.e., at least one of the three nucleotides of the motif in the sense strand forms a base pair with at least one of the three nucleotides of the motif in the antisense strand. Alternatively, at least two nucleotides may overlap, or all three nucleotides may overlap.

(159) In some embodiments, the sense strand of the dsRNAi agent may contain more than one motif of three identical modifications on three consecutive nucleotides. The first motif may occur at or near the cleavage site of the strand and the other motifs may be a wing modification. The term "wing modification" herein refers to a motif occurring at another portion of the strand that is separated from the motif at or near the cleavage site of the same strand. The wing modification is either adjacent to the first motif or is separated by at least one or more nucleotides. When the motifs are immediately adjacent to each other then the chemistries of the motifs are distinct from each other, and when the motifs are separated by one or more nucleotide then the chemistries can be the same or different. Two or more wing modifications may be present. For instance, when two wing modifications are present, each wing modification may occur at one end relative to the first motif which is at or near cleavage site or on either side of the lead motif.

(160) Like the sense strand, the antisense strand of the dsRNAi agent may contain more than one motif of three identical modifications on three consecutive nucleotides, with at least one of the motifs occurring at or near the cleavage site of the strand. This antisense strand may also contain one or more wing modifications in an alignment similar to the wing modifications that may be present on the sense strand.

(161) In some embodiments, the wing modification on the sense strand or antisense strand of the dsRNAi agent typically does not include the first one or two terminal nucleotides at the 3'-end, 5'-end, or both ends of the strand.

(162) In other embodiments, the wing modification on the sense strand or antisense strand of the dsRNAi agent typically does not include the first one or two paired nucleotides within the duplex region at the 3'-end, 5'-end, or both ends of the strand.

(163) When the sense strand and the antisense strand of the dsRNAi agent each contain at least one wing modification, the wing modifications may fall on the same end of the duplex region, and have an overlap of one, two, or three nucleotides.

(164) When the sense strand and the antisense strand of the dsRNAi agent each contain at least two wing modifications, the sense strand and the antisense strand can be so aligned that two modifications each from one strand fall on one end of the duplex region, having an overlap of one, two, or three nucleotides; two modifications each from one strand fall on the other end of the duplex region, having an overlap of one, two or three nucleotides; two modifications one strand fall on each side of the lead motif, having an overlap of one, two or three nucleotides in the duplex region.

(165) In some embodiments, every nucleotide in the sense strand and antisense strand of the dsRNAi agent, including the nucleotides that are part of the motifs, may be modified. Each nucleotide may be modified with the same or different modification which can include one or more alteration of one or both of the non-linking phosphate oxygens or of one or more of the linking

phosphate oxygens; alteration of a constituent of the ribose sugar, e.g., of the 2'-hydroxyl on the ribose sugar; wholesale replacement of the phosphate moiety with "dephospho" linkers; modification or replacement of a naturally occurring base; and replacement or modification of the ribose-phosphate backbone.

(166) As nucleic acids are polymers of subunits, many of the modifications occur at a position which is repeated within a nucleic acid, e.g., a modification of a base, or a phosphate moiety, or a non-linking O of a phosphate moiety. In some cases the modification will occur at all of the subject positions in the nucleic acid but in many cases it will not. By way of example, a modification may only occur at a 3'- or 5' terminal position, may only occur in a terminal region, e.g., at a position on a terminal nucleotide or in the last 2, 3, 4, 5, or 10 nucleotides of a strand. A modification may occur in a double strand region, a single strand region, or in both. A modification may occur only in the double strand region of an RNA or may only occur in a single strand region of a RNA. For example, a phosphorothioate modification at a non-linking O position may only occur at one or both termini, may only occur in a terminal region, e.g., at a position on a terminal nucleotide or in the last 2, 3, 4, 5, or 10 nucleotides of a strand, or may occur in double strand and single strand regions, particularly at termini. The 5'-end or ends can be phosphorylated.

(167) It may be possible, e.g., to enhance stability, to include particular bases in overhangs, or to include modified nucleotides or nucleotide surrogates, in single strand overhangs, e.g., in a 5'- or 3'-overhang, or in both. For example, it can be desirable to include purine nucleotides in overhangs. In some embodiments all or some of the bases in a 3'- or 5'-overhang may be modified, e.g., with a modification described herein. Modifications can include, e.g., the use of modifications at the 2' position of the ribose sugar with modifications that are known in the art, e.g., the use of deoxyribonucleotides, 2'-deoxy-2'-fluoro (2'-F) or 2'-O-methyl modified instead of the ribosugar of the nucleobase, and modifications in the phosphate group, e.g., phosphorothioate modifications. Overhangs need not be homologous with the target sequence.

(168) In some embodiments, each residue of the sense strand and antisense strand is independently modified with LNA, CRN, cET, UNA, HNA, CeNA, 2'-methoxyethyl, 2'-O-methyl, 2'-O-allyl, 2'-C-allyl, 2'-deoxy, 2'-hydroxyl, or 2'-fluoro. The strands can contain more than one modification. In one embodiment, each residue of the sense strand and antisense strand is independently modified with 2'-O-methyl or 2'-fluoro.

(169) At least two different modifications are typically present on the sense strand and antisense strand. Those two modifications may be the 2'-O-methyl or 2'-fluoro modifications, or others.

(170) In certain embodiments, the N.sub.a or N.sub.b comprise modifications of an alternating pattern. The term "alternating motif" as used herein refers to a motif having one or more modifications, each modification occurring on alternating nucleotides of one strand. The alternating nucleotide may refer to one per every other nucleotide or one per every three nucleotides, or a similar pattern. For example, if A, B and C each represent one type of modification to the nucleotide, the alternating motif can be "ABABABABABAB . . .," "AABBAABBAABB . . .," "AABAABAABAAB . . .," "AAABAAABAAAB . . .," "AAABBBAAABBB . . .," or "ABCABCABCABC . . .," etc.

(171) The type of modifications contained in the alternating motif may be the same or different. For example, if A, B, C, D each represent one type of modification on the nucleotide, the alternating pattern, i.e., modifications on every other nucleotide, may be the same, but each of the sense strand or antisense strand can be selected from several possibilities of modifications within the alternating motif such as "ABABAB . . .," "ACACAC . . .," "BDBDBD . . ." or "CDCDCD . . .," etc.

(172) In some embodiments, the dsRNAi agent of the invention comprises the modification pattern for the alternating motif on the sense strand relative to the modification pattern for the alternating motif on the antisense strand is shifted. The shift may be such that the modified group of nucleotides of the sense strand corresponds to a differently modified group of nucleotides of the antisense strand and vice versa. For example, the sense strand when paired with the antisense

strand in the dsRNA duplex, the alternating motif in the sense strand may start with “ABABAB” from 5' to 3' of the strand and the alternating motif in the antisense strand may start with “BABABA” from 5' to 3' of the strand within the duplex region. As another example, the alternating motif in the sense strand may start with “AABBAABB” from 5' to 3' of the strand and the alternating motif in the antisense strand may start with “BBAABBAA” from 5' to 3' of the strand within the duplex region, so that there is a complete or partial shift of the modification patterns between the sense strand and the antisense strand.

(173) In one particular example, the alternating motif in the sense strand is “ABABAB” from 5' 3' of the strand, where each A is an unmodified ribonucleotide and each B is a 2'-O-methyl modified nucleotide.

(174) In one particular example, the alternating motif in the sense strand is “ABABAB” from 5' 3' of the strand, where each A is an 2'-deoxy-2'-fluoro modified nucleotide and each B is a 2'-O-methyl modified nucleotide.

(175) In another particular example, the alternating motif in the antisense strand is “BABABA” from 3'-5' of the strand, where each A is a 2'-deoxy-2'-fluoro modified nucleotide and each B is a 2'-O-methyl modified nucleotide.

(176) In one particular example, the alternating motif in the sense strand is “ABABAB” from 5' 3' of the strand and the alternating motif in the antisense strand is “BABABA” from 3'-5' of the strand, where each A is an unmodified ribonucleotide and each B is a 2'-O-methyl modified nucleotide.

(177) In one particular example, the alternating motif in the sense strand is “ABABAB” from 5' 3' of the strand and the alternating motif in the antisense strand is “BABABA” from 3'-5' of the strand, where each A is a 2'-deoxy-2'-fluoro modified nucleotide and each B is a 2'-O-methyl modified nucleotide.

(178) In some embodiments, the dsRNAi agent comprises the pattern of the alternating motif of 2'-O-methyl modification and 2'-F modification on the sense strand initially has a shift relative to the pattern of the alternating motif of 2'-O-methyl modification and 2'-F modification on the antisense strand initially, i.e., the 2'-O-methyl modified nucleotide on the sense strand base pairs with a 2'-F modified nucleotide on the antisense strand and vice versa. The 1 position of the sense strand may start with the 2'-F modification, and the 1 position of the antisense strand may start with the 2'-O-methyl modification.

(179) The introduction of one or more motifs of three identical modifications on three consecutive nucleotides to the sense strand or antisense strand interrupts the initial modification pattern present in the sense strand or antisense strand. This interruption of the modification pattern of the sense or antisense strand by introducing one or more motifs of three identical modifications on three consecutive nucleotides to the sense or antisense strand may enhance the gene silencing activity against the target gene.

(180) In some embodiments, when the motif of three identical modifications on three consecutive nucleotides is introduced to any of the strands, the modification of the nucleotide next to the motif is a different modification than the modification of the motif. For example, the portion of the sequence containing the motif is “... N.sub.aYYYN.sub.b...,” where “Y” represents the modification of the motif of three identical modifications on three consecutive nucleotide, and “N.sub.a” and “N.sub.b” represent a modification to the nucleotide next to the motif “YYY” that is different than the modification of Y, and where N.sub.a and N.sub.b can be the same or different modifications. Alternatively, N.sub.a or N.sub.b may be present or absent when there is a wing modification present.

(181) The iRNA may further comprise at least one phosphorothioate or methylphosphonate internucleotide linkage. The phosphorothioate or methylphosphonate internucleotide linkage modification may occur on any nucleotide of the sense strand, antisense strand, or both strands in any position of the strand. For instance, the internucleotide linkage modification may occur on

every nucleotide on the sense strand or antisense strand; each internucleotide linkage modification may occur in an alternating pattern on the sense strand or antisense strand; or the sense strand or antisense strand may contain both internucleotide linkage modifications in an alternating pattern. The alternating pattern of the internucleotide linkage modification on the sense strand may be the same or different from the antisense strand, and the alternating pattern of the internucleotide linkage modification on the sense strand may have a shift relative to the alternating pattern of the internucleotide linkage modification on the antisense strand. In one embodiment, a double-stranded RNAi agent comprises 6-8 phosphorothioate internucleotide linkages. In some embodiments, the antisense strand comprises two phosphorothioate internucleotide linkages at the 5'-end and two phosphorothioate internucleotide linkages at the 3'-end, and the sense strand comprises at least two phosphorothioate internucleotide linkages at either the 5'-end or the 3'-end.

(182) In some embodiments, the dsRNAi agent comprises a phosphorothioate or methylphosphonate internucleotide linkage modification in the overhang region. For example, the overhang region may contain two nucleotides having a phosphorothioate or methylphosphonate internucleotide linkage between the two nucleotides. Internucleotide linkage modifications also may be made to link the overhang nucleotides with the terminal paired nucleotides within the duplex region. For example, at least 2, 3, 4, or all the overhang nucleotides may be linked through phosphorothioate or methylphosphonate internucleotide linkage, and optionally, there may be additional phosphorothioate or methylphosphonate internucleotide linkages linking the overhang nucleotide with a paired nucleotide that is next to the overhang nucleotide. For instance, there may be at least two phosphorothioate internucleotide linkages between the terminal three nucleotides, in which two of the three nucleotides are overhang nucleotides, and the third is a paired nucleotide next to the overhang nucleotide. These terminal three nucleotides may be at the 3'-end of the antisense strand, the 3'-end of the sense strand, the 5'-end of the antisense strand, or the 5'-end of the antisense strand.

(183) In some embodiments, the 2-nucleotide overhang is at the 3'-end of the antisense strand, and there are two phosphorothioate internucleotide linkages between the terminal three nucleotides, wherein two of the three nucleotides are the overhang nucleotides, and the third nucleotide is a paired nucleotide next to the overhang nucleotide. Optionally, the dsRNAi agent may additionally have two phosphorothioate internucleotide linkages between the terminal three nucleotides at both the 5'-end of the sense strand and at the 5'-end of the antisense strand.

(184) In one embodiment, the dsRNAi agent comprises mismatch(es) with the target, within the duplex, or combinations thereof. The mismatch may occur in the overhang region or the duplex region. The base pair may be ranked on the basis of their propensity to promote dissociation or melting (e.g., on the free energy of association or dissociation of a particular pairing, the simplest approach is to examine the pairs on an individual pair basis, though next neighbor or similar analysis can also be used). In terms of promoting dissociation: A:U is preferred over G:C; G:U is preferred over G:C; and I:C is preferred over G:C (I=inosine). Mismatches, e.g., non-canonical or other than canonical pairings (as described elsewhere herein) are preferred over canonical (A:T, A:U, G:C) pairings; and pairings which include a universal base are preferred over canonical pairings.

(185) In certain embodiments, the dsRNAi agent comprises at least one of the first 1, 2, 3, 4, or 5 base pairs within the duplex regions from the 5'-end of the antisense strand independently selected from the group of: A:U, G:U, I:C, and mismatched pairs, e.g., non-canonical or other than canonical pairings or pairings which include a universal base, to promote the dissociation of the antisense strand at the 5'-end of the duplex.

(186) In certain embodiments, the nucleotide at the 1 position within the duplex region from the 5'-end in the antisense strand is selected from A, dA, dU, U, and dT. Alternatively, at least one of the first 1, 2, or 3 base pair within the duplex region from the 5'-end of the antisense strand is an AU base pair. For example, the first base pair within the duplex region from the 5'-end of the antisense

strand is an AU base pair.

(187) In other embodiments, the nucleotide at the 3'-end of the sense strand is deoxythymidine (dT) or the nucleotide at the 3'-end of the antisense strand is deoxythymidine (dT). For example, there is a short sequence of deoxythymidine nucleotides, for example, two dT nucleotides on the 3'-end of the sense, antisense strand, or both strands.

(188) In certain embodiments, the sense strand sequence may be represented by formula (I):

(189) TABLE-US-00001 (I) 5' n.sub.p-N.sub.a-(X X X).sub.i-N.sub.b-Y Y Y-N.sub.b-(Z Z Z).sub.j-N.sub.a-n.sub.q 3' wherein: i and j are each independently 0 or 1; p and q are each independently 0-6; each N.sub.a independently represents an oligonucleotide sequence comprising 0-25 modified nucleotides, each sequence comprising at least two differently modified nucleotides; each N.sub.b independently represents an oligonucleotide sequence comprising 0-10 modified nucleotides; each n.sub.p and n.sub.q independently represent an overhang nucleotide; wherein Nb and Y do not have the same modification; and XXX, YYY, and ZZZ each independently represent one motif of three identical modifications on three consecutive nucleotides. In some embodiments, YYY is all 2'-F modified nucleotides.

(190) In some embodiments, the N.sub.a or N.sub.b comprises modifications of alternating pattern.

(191) In some embodiments, the YYY motif occurs at or near the cleavage site of the sense strand. For example, when the dsRNAi agent has a duplex region of 17-23 nucleotides in length, the YYY motif can occur at or the vicinity of the cleavage site (e.g.: can occur at positions 6, 7, 8; 7, 8, 9; 8, 9, 10; 9, 10, 11; 10, 11, 12; or 11, 12, 13) of the sense strand, the count starting from the first nucleotide, from the 5'-end; or optionally, the count starting at the first paired nucleotide within the duplex region, from the 5'-end.

(192) In one embodiment, i is 1 and j is 0, or i is 0 and j is 1, or both i and j are 1. The sense strand can therefore be represented by the following formulas:

(193) TABLE-US-00002 (Ib) 5' n.sub.p-N.sub.a-YYY-N.sub.b-ZZZ-N.sub.a-n.sub.q 3'; (Ic) 5' n.sub.p-N.sub.a-XXX-N.sub.b-YYY-N.sub.a-n.sub.q 3'; or (Id) 5' n.sub.p-N.sub.a-XXX-N.sub.b-YYY-N.sub.b-ZZZ-N.sub.a-n.sub.q 3'.

(194) When the sense strand is represented by formula (Ib), N.sub.b represents an oligonucleotide sequence comprising 0-10, 0-7, 0-5, 0-4, 0-2, or 0 modified nucleotides. Each N.sub.a independently can represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

(195) When the sense strand is represented as formula (Ic), N.sub.b represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2, or 0 modified nucleotides. Each N.sub.a can independently represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

(196) When the sense strand is represented as formula (Id), each N.sub.b independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-5, 0-4, 0-2, or 0 modified nucleotides. In some embodiments, N.sub.b is 0, 1, 2, 3, 4, 5, or 6 Each N.sub.a can independently represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

(197) Each of X, Y and Z may be the same or different from each other.

(198) In other embodiments, i is 0 and j is 0, and the sense strand may be represented by the formula:

5'n.sub.p-N.sub.a-YYY-N.sub.a-n.sub.q3' (Ia).

(199) When the sense strand is represented by formula (Ia), each N.sub.a independently can represent an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

(200) In one embodiment, the antisense strand sequence of the RNAi may be represented by formula (II):

(201) TABLE-US-00003 (II) 5' n.sub.q'-N.sub.a'-(Z'Z'Z').sub.k-N.sub.b'-Y'Y'Y'-N.sub.b'-(X'X'X').sub.l-N'.sub.a-n.sub.p' 3' wherein: k and l are each independently 0 or 1; p' and q' are each independently 0-6; each N.sub.a' independently represents an oligonucleotide sequence

comprising 0-25 modified nucleotides, each sequence comprising at least two differently modified nucleotides; each N.sub.b' independently represents an oligonucleotide sequence comprising 0-10 modified nucleotides; each n.sub.p' and n.sub.q' independently represent an overhang nucleotide; wherein N.sub.b' and Y' do not have the same modification; and X'X'X', Y'Y'Y', and Z'Z'Z' each independently represent one motif of three identical modifications on three consecutive nucleotides.

(202) In some embodiments, the N.sub.a' or N.sub.b' comprises modifications of alternating pattern.

(203) The Y'Y'Y' motif occurs at or near the cleavage site of the antisense strand. For example, when the dsRNAi agent has a duplex region of 17-23 nucleotides in length, the Y'Y'Y' motif can occur at positions 9, 10, 11; 10, 11, 12; 11, 12, 13; 12, 13, 14; or 13, 14, 15 of the antisense strand, with the count starting from the first nucleotide, from the 5'-end; or optionally, the count starting at the first paired nucleotide within the duplex region, from the 5'-end. In some embodiments, the Y'Y'Y' motif occurs at positions 11, 12, 13.

(204) In certain embodiments, Y'Y'Y' motif is all 2'-OMe modified nucleotides.

(205) In certain embodiments, k is 1 and l is 0, or k is 0 and l is 1, or both k and l are 1.

(206) The antisense strand can therefore be represented by the following formulas:

(207) TABLE-US-00004 (IIb) 5' n.sub.q'-N.sub.a'-Z'Z'Z'-N.sub.b'-Y'Y'Y'-N.sub.a'-n.sub.p' 3';

(IIc) 5' n.sub.q'-N.sub.a'-Y'Y'Y'-N.sub.b'-X'X'X'-n.sub.p' 3'; or (IId) 5' n.sub.q'-N.sub.a'-Z'Z'Z'-N.sub.b'-Y'Y'Y'-N.sub.b'-X'X'X'-N.sub.a'-n.sub.p' 3'.

(208) When the antisense strand is represented by formula (IIb), N.sub.b' represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2, or 0 modified nucleotides. Each N.sub.a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

(209) When the antisense strand is represented as formula (IIc), N.sub.b' represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2, or 0 modified nucleotides. Each N.sub.a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

(210) When the antisense strand is represented as formula (IId), each N.sub.b' independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2, or 0 modified nucleotides. Each N.sub.a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides. In some embodiments, N.sub.b is 0, 1, 2, 3, 4, 5, or 6.

(211) In other embodiments, k is 0 and l is 0 and the antisense strand may be represented by the formula:

(212) TABLE-US-00005 (Ia) 5' n.sub.p'-N.sub.a'-Y'Y'Y'-N.sub.a'-n.sub.q' 3'.

(213) When the antisense strand is represented as formula (IIa), each N.sub.a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

(214) Each of X', Y' and Z' may be the same or different from each other.

(215) Each nucleotide of the sense strand and antisense strand may be independently modified with LNA, CRN, UNA, cEt, HNA, CeNA, 2'-methoxyethyl, 2'-O-methyl, 2'-O-allyl, 2'-C-allyl, 2'-hydroxyl, or 2'-fluoro. For example, each nucleotide of the sense strand and antisense strand is independently modified with 2'-O-methyl or 2'-fluoro. Each X, Y, Z, X', Y', and Z', in particular, may represent a 2'-O-methyl modification or a 2'-fluoro modification.

(216) In some embodiments, the sense strand of the dsRNAi agent may contain YYY motif occurring at 9, 10, and 11 positions of the strand when the duplex region is 21 nt, the count starting from the first nucleotide from the 5'-end, or optionally, the count starting at the first paired nucleotide within the duplex region, from the 5'-end; and Y represents 2'-F modification. The sense strand may additionally contain XXX motif or ZZZ motifs as wing modifications at the opposite end of the duplex region; and XXX and ZZZ each independently represents a 2'-OMe modification or 2'-F modification.

(217) In some embodiments the antisense strand may contain Y'Y'Y' motif occurring at positions 11, 12, 13 of the strand, the count starting from the first nucleotide from the 5'-end, or optionally, the count starting at the first paired nucleotide within the duplex region, from the 5'-end; and Y' represents 2'-O-methyl modification. The antisense strand may additionally contain X'X'X' motif or Z'Z'Z' motifs as wing modifications at the opposite end of the duplex region; and X'X'X' and Z'Z'Z' each independently represents a 2'-OMe modification or 2'-F modification.

(218) The sense strand represented by any one of the above formulas (Ia), (Ib), (Ic), and (Id) forms a duplex with an antisense strand being represented by any one of formulas (IIa), (IIb), (IIc), and (IId), respectively.

(219) Accordingly, the dsRNAi agents for use in the methods of the invention may comprise a sense strand and an antisense strand, each strand having 14 to 30 nucleotides, the iRNA duplex represented by formula (III):

(220) TABLE-US-00006 (III) sense: 5' n.sub.p-N.sub.a-(X X X).sub.i-N.sub.b-Y Y Y-N.sub.b-(ZZZ).sub.j-N.sub.a-n.sub.q 3' antisense: 3' n.sub.p'-N.sub.a'-(X'X'X').sub.k-N.sub.b'-Y'Y'Y'-N.sub.b'-(Z'Z'Z').sub.l- N.sub.a'-n.sub.q'5' wherein: i, j, k, and l are each independently 0 or 1; p, p', q, and q' are each independently 0-6; each N.sub.a and N.sub.a' independently represents an oligonucleotide sequence comprising 0-25 modified nucleotides, each sequence comprising at least two differently modified nucleotides; each N.sub.b and N.sub.b' independently represents an oligonucleotide sequence comprising 0-10 modified nucleotides; wherein each n.sub.p', n.sub.p, n.sub.q', and n.sub.q, each of which may or may not be present, independently represents an overhang nucleotide; and XXX, YYY, ZZZ, X'X'X', Y'Y'Y', and Z'Z'Z' each independently represent one motif of three identical modifications on three consecutive nucleotides.

(221) In one embodiment, i is 0 and j is 0; or i is 1 and j is 0; or i is 0 and j is 1; or both i and j are 0; or both i and j are 1. In another embodiment, k is 0 and l is 0; or k is 1 and l is 0; k is 0 and l is 1; or both k and l are 0; or both k and l are 1.

(222) Exemplary combinations of the sense strand and antisense strand forming an iRNA duplex include the formulas below:

(223) TABLE-US-00007 (IIIa) 5' n.sub.p-N.sub.a-Y Y Y-N.sub.a-n.sub.q 3' 3' n.sub.p'-N.sub.a'-Y'Y'Y'-N.sub.a'n.sub.q' 5' (IIIb) 5' n.sub.p-N.sub.a-Y Y Y-N.sub.b-ZZZ-N.sub.a-n.sub.q 3' 3' n.sub.p'-N.sub.a'-Y'Y'Y'-N.sub.b'-Z'Z'Z'-N.sub.a'n.sub.q' 5' (IIIc) 5' n.sub.p-N.sub.a-X X X-N.sub.b-Y Y Y-N.sub.a-n.sub.q 3' 3' n.sub.p'-N.sub.a'-X'X'X'-N.sub.b'-Y'Y'Y'-N.sub.a-n.sub.q' 5' (IIId) 5' n.sub.p-N.sub.a-X X X-N.sub.b-Y Y Y-N.sub.b-ZZZ-N.sub.a-n.sub.q 3' 3' n.sub.p'-N.sub.a'-X'X'X'-N.sub.b'-Y'Y'Y'-N.sub.b'-Z'Z'Z'-N.sub.a-n.sub.q' 5'

(224) When the dsRNAi agent is represented by formula (IIIa), each N.sub.a independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

(225) When the dsRNAi agent is represented by formula (IIIb), each N.sub.b independently represents an oligonucleotide sequence comprising 1-10, 1-7, 1-5, or 1-4 modified nucleotides. Each N.sub.a independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

(226) When the dsRNAi agent is represented as formula (IIIc), each N.sub.b, N.sub.b' independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2, or 0 modified nucleotides. Each N.sub.a independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides.

(227) When the dsRNAi agent is represented as formula (IIId), each N.sub.b, N.sub.b' independently represents an oligonucleotide sequence comprising 0-10, 0-7, 0-10, 0-7, 0-5, 0-4, 0-2, or 0 modified nucleotides. Each N.sub.a, N.sub.a' independently represents an oligonucleotide sequence comprising 2-20, 2-15, or 2-10 modified nucleotides. Each of N.sub.a, N.sub.a', N.sub.b, and N.sub.b' independently comprises modifications of alternating pattern.

(228) Each of X, Y, and Z in formulas (III), (IIIa), (IIIb), (IIIc), and (IIId) may be the same or

different from each other.

(229) When the dsRNAi agent is represented by formula (III), (IIIa), (IIIb), (IIIc), and (IIId), at least one of the Y nucleotides may form a base pair with one of the Y' nucleotides. Alternatively, at least two of the Y nucleotides form base pairs with the corresponding Y' nucleotides; or all three of the Y nucleotides all form base pairs with the corresponding Y' nucleotides.

(230) When the dsRNAi agent is represented by formula (IIIb) or (IIId), at least one of the Z nucleotides may form a base pair with one of the Z' nucleotides. Alternatively, at least two of the Z nucleotides form base pairs with the corresponding Z' nucleotides; or all three of the Z nucleotides all form base pairs with the corresponding Z' nucleotides.

(231) When the dsRNAi agent is represented as formula (IIIc) or (IIId), at least one of the X nucleotides may form a base pair with one of the X' nucleotides. Alternatively, at least two of the X nucleotides form base pairs with the corresponding X' nucleotides; or all three of the X nucleotides all form base pairs with the corresponding X' nucleotides.

(232) In certain embodiments, the modification on the Y nucleotide is different than the modification on the Y' nucleotide, the modification on the Z nucleotide is different than the modification on the Z' nucleotide, or the modification on the X nucleotide is different than the modification on the X' nucleotide.

(233) In certain embodiments, when the dsRNAi agent is represented by formula (IIId), the N.sub.a modifications are 2'-O-methyl or 2'-fluoro modifications. In other embodiments, when the RNAi agent is represented by formula (IIId), the N.sub.a modifications are 2'-O-methyl or 2'-fluoro modifications and n.sub.p' > 0 and at least one n.sub.p' is linked to a neighboring nucleotide a via phosphorothioate linkage. In yet other embodiments, when the RNAi agent is represented by formula (IIId), the N.sub.a modifications are 2'-O-methyl or 2'-fluoro modifications, n.sub.p' > 0 and at least one n.sub.p' is linked to a neighboring nucleotide via phosphorothioate linkage, and the sense strand is conjugated to one or more GalNAc derivatives attached through a bivalent or trivalent branched linker (described below). In other embodiments, when the RNAi agent is represented by formula (IIId), the N.sub.a modifications are 2'-O-methyl or 2'-fluoro modifications, n.sub.p' > 0 and at least one n.sub.p' is linked to a neighboring nucleotide via phosphorothioate linkage, the sense strand comprises at least one phosphorothioate linkage, and the sense strand is conjugated to one or more GalNAc derivatives attached through a bivalent or trivalent branched linker.

(234) In some embodiments, when the dsRNAi agent is represented by formula (IIIa), the N.sub.a modifications are 2'-O-methyl or 2'-fluoro modifications, n.sub.p' > 0 and at least one n.sub.p' is linked to a neighboring nucleotide via phosphorothioate linkage, the sense strand comprises at least one phosphorothioate linkage, and the sense strand is conjugated to one or more GalNAc derivatives attached through a bivalent or trivalent branched linker.

(235) In some embodiments, the dsRNAi agent is a multimer containing at least two duplexes represented by formula (III), (IIIa), (IIIb), (IIIc), and (IIId), wherein the duplexes are connected by a linker. The linker can be cleavable or non-cleavable. Optionally, the multimer further comprises a ligand. Each of the duplexes can target the same gene or two different genes; or each of the duplexes can target same gene at two different target sites.

(236) In some embodiments, the dsRNAi agent is a multimer containing three, four, five, six, or more duplexes represented by formula (III), (IIIa), (IIIb), (IIIc), and (IIId), wherein the duplexes are connected by a linker. The linker can be cleavable or non-cleavable. Optionally, the multimer further comprises a ligand. Each of the duplexes can target the same gene or two different genes; or each of the duplexes can target same gene at two different target sites.

(237) In one embodiment, two dsRNAi agents represented by at least one of formulas (III), (IIIa), (IIIb), (IIIc), and (IIId) are linked to each other at the 5' end, and one or both of the 3' ends, and are optionally conjugated to a ligand. Each of the agents can target the same gene or two different genes; or each of the agents can target same gene at two different target sites.

(238) In certain embodiments, an RNAi agent of the invention may contain a low number of nucleotides containing a 2'-fluoro modification, e.g., 10 or fewer nucleotides with 2'-fluoro modification. For example, the RNAi agent may contain 10, 9, 8, 7, 6, 5, 4, 3, 2, 1 or 0 nucleotides with a 2'-fluoro modification. In a specific embodiment, the RNAi agent of the invention contains 10 nucleotides with a 2'-fluoro modification, e.g., 4 nucleotides with a 2'-fluoro modification in the sense strand and 6 nucleotides with a 2'-fluoro modification in the antisense strand. In another specific embodiment, the RNAi agent of the invention contains 6 nucleotides with a 2'-fluoro modification, e.g., 4 nucleotides with a 2'-fluoro modification in the sense strand and 2 nucleotides with a 2'-fluoro modification in the antisense strand.

(239) In other embodiments, an RNAi agent of the invention may contain an ultra low number of nucleotides containing a 2'-fluoro modification, e.g., 2 or fewer nucleotides containing a 2'-fluoro modification. For example, the RNAi agent may contain 2, 1 or 0 nucleotides with a 2'-fluoro modification. In a specific embodiment, the RNAi agent may contain 2 nucleotides with a 2'-fluoro modification, e.g., 0 nucleotides with a 2'-fluoro modification in the sense strand and 2 nucleotides with a 2'-fluoro modification in the antisense strand.

(240) Various publications describe multimeric iRNAs that can be used in the methods of the invention. Such publications include WO2007/091269, U.S. Pat. No. 7,858,769, WO2010/141511, WO2007/117686, WO2009/014887, and WO2011/031520 the entire contents of each of which are hereby incorporated herein by reference.

(241) In certain embodiments, the compositions and methods of the disclosure include a vinyl phosphonate (VP) modification of an RNAi agent as described herein. In exemplary embodiments, a 5'-vinyl phosphonate modified nucleotide of the disclosure has the structure:

(242) ##STR00006##

wherein X is O or S; R is hydrogen, hydroxy, fluoro, or C_{sub}.1-20 alkoxy (e.g., methoxy or n-hexadecyloxy); R_{sup}.5' is =C(H)—P(O)(OH)_{sub}.2 and the double bond between the C5' carbon and R_{sup}.5' is in the E or Z orientation (e.g., E orientation); and B is a nucleobase or a modified nucleobase, optionally where B is adenine, guanine, cytosine, thymine, or uracil.

(243) A vinyl phosphonate of the instant disclosure may be attached to either the antisense or the sense strand of a dsRNA of the disclosure. In certain embodiments, a vinyl phosphonate of the instant disclosure is attached to the antisense strand of a dsRNA, optionally at the 5' end of the antisense strand of the dsRNA.

(244) Vinyl phosphonate modifications are also contemplated for the compositions and methods of the instant disclosure. An exemplary vinyl phosphonate structure includes the preceding structure, where R_{5'} is =C(H)—OP(O)(OH)₂ and the double bond between the C5' carbon and R_{5'} is in the E or Z orientation (e.g., E orientation).

(245) As described in more detail below, the iRNA that contains conjugations of one or more carbohydrate moieties to an iRNA can optimize one or more properties of the iRNA. In many cases, the carbohydrate moiety will be attached to a modified subunit of the iRNA. For example, the ribose sugar of one or more ribonucleotide subunits of a iRNA can be replaced with another moiety, e.g., a non-carbohydrate (such as, cyclic) carrier to which is attached a carbohydrate ligand. A ribonucleotide subunit in which the ribose sugar of the subunit has been so replaced is referred to herein as a ribose replacement modification subunit (RRMS). A cyclic carrier may be a carbocyclic ring system, i.e., all ring atoms are carbon atoms, or a heterocyclic ring system, i.e., one or more ring atoms may be a heteroatom, e.g., nitrogen, oxygen, sulfur. The cyclic carrier may be a monocyclic ring system, or may contain two or more rings, e.g. fused rings. The cyclic carrier may be a fully saturated ring system, or it may contain one or more double bonds.

(246) The ligand may be attached to the polynucleotide via a carrier. The carriers include (i) at least one "backbone attachment point," such as, two "backbone attachment points" and (ii) at least one "tethering attachment point." A "backbone attachment point" as used herein refers to a functional group, e.g. a hydroxyl group, or generally, a bond available for, and that is suitable for

incorporation of the carrier into the backbone, e.g., the phosphate, or modified phosphate, e.g., sulfur containing, backbone, of a ribonucleic acid. A “tethering attachment point” (TAP) in some embodiments refers to a constituent ring atom of the cyclic carrier, e.g., a carbon atom or a heteroatom (distinct from an atom which provides a backbone attachment point), that connects a selected moiety. The moiety can be, e.g., a carbohydrate, e.g. monosaccharide, disaccharide, trisaccharide, tetrasaccharide, oligosaccharide, or polysaccharide. Optionally, the selected moiety is connected by an intervening tether to the cyclic carrier. Thus, the cyclic carrier will often include a functional group, e.g., an amino group, or generally, provide a bond, that is suitable for incorporation or tethering of another chemical entity, e.g., a ligand to the constituent ring.

(247) The iRNA may be conjugated to a ligand via a carrier, wherein the carrier can be cyclic group or acyclic group. In some embodiments, the cyclic group is selected from pyrrolidinyl, pyrazolinyl, pyrazolidinyl, imidazolinyl, imidazolidinyl, piperidinyl, piperazinyl, [1,3]dioxolane, oxazolidinyl, isoxazolidinyl, morpholinyl, thiazolidinyl, isothiazolidinyl, quinoxalinyl, pyridazinonyl, tetrahydrofuryl, and decalin. In some embodiments, the acyclic group is a serinol backbone or diethanolamine backbone.

(248) i. Thermally Destabilizing Modifications

(249) In certain embodiments, a dsRNA molecule can be optimized for RNA interference by incorporating thermally destabilizing modifications in the seed region of the antisense strand. As used herein “seed region” means at positions 2-9 of the 5'-end of the referenced strand or at positions 2-8 of the 5'-end of the referenced strand. For example, thermally destabilizing modifications can be incorporated in the seed region of the antisense strand to reduce or inhibit off-target gene silencing.

(250) The term “thermally destabilizing modification(s)” includes modification(s) that would result with a dsRNA with a lower overall melting temperature (T_m) than the T_m of the dsRNA without having such modification(s). For example, the thermally destabilizing modification(s) can decrease the T_m of the dsRNA by 1-4° C., such as one, two, three or four degrees Celcius. And, the term “thermally destabilizing nucleotide” refers to a nucleotide containing one or more thermally destabilizing modifications.

(251) It has been discovered that dsRNAs with an antisense strand comprising at least one thermally destabilizing modification of the duplex within the first 9 nucleotide positions, counting from the 5' end, of the antisense strand have reduced off-target gene silencing activity. Accordingly, in some embodiments, the antisense strand comprises at least one (e.g., one, two, three, four, five or more) thermally destabilizing modification of the duplex within the first 9 nucleotide positions of the 5' region of the antisense strand. In some embodiments, one or more thermally destabilizing modification(s) of the duplex is/are located in positions 2-9, such as, positions 4-8, from the 5'-end of the antisense strand. In some further embodiments, the thermally destabilizing modification(s) of the duplex is/are located at position 6, 7 or 8 from the 5'-end of the antisense strand. In still some further embodiments, the thermally destabilizing modification of the duplex is located at position 7 from the 5'-end of the antisense strand. In some embodiments, the thermally destabilizing modification of the duplex is located at position 2, 3, 4, 5 or 9 from the 5'-end of the antisense strand.

(252) An iRNA agent comprises a sense strand and an antisense strand, each strand having 14 to 40 nucleotides. The RNAi agent may be represented by formula (L):

(253) ##STR00007##

In formula (L), B1, B2, B3, B1', B2', B3', and B4' each are independently a nucleotide containing a modification selected from the group consisting of 2'-O-alkyl, 2'-substituted alkoxy, 2'-substituted alkyl, 2'-halo, ENA, and BNA/LNA. In one embodiment, B1, B2, B3, B1', B2', B3', and B4' each contain 2'-OMe modifications. In one embodiment, B1, B2, B3, B1', B2', B3', and B4' each contain 2'-OMe or 2'-F modifications. In one embodiment, at least one of B1, B2, B3, B1', B2', B3', and B4' contain 2'-O—N-methylacetamido (2'-O-NMA, 2'O—CH₂C(O)N(Me)H) modification.

(254) C1 is a thermally destabilizing nucleotide placed at a site opposite to the seed region of the antisense strand (i.e., at positions 2-8 of the 5'-end of the antisense strand or at positions 2-9 of the 5'-end of the referenced strand). For example, C1 is at a position of the sense strand that pairs with a nucleotide at positions 2-8 of the 5'-end of the antisense strand. In one example, C1 is at position 15 from the 5'-end of the sense strand. C1 nucleotide bears the thermally destabilizing modification which can include abasic modification; mismatch with the opposing nucleotide in the duplex; and sugar modification such as 2'-deoxy modification or acyclic nucleotide e.g., unlocked nucleic acids (UNA) or glycerol nucleic acid (GNA); and 2'-5'-linked ribonucleotides ("3'-RNA"). In one embodiment, C1 has thermally destabilizing modification selected from the group consisting of: i) mismatch with the opposing nucleotide in the antisense strand; ii) abasic modification selected from the group consisting of:

(255) ##STR00008##

and iii) sugar modification selected from the group consisting of:

(256) ##STR00009##

wherein B is a modified or unmodified nucleobase, R.sup.1 and R.sup.2 independently are H, halogen, OR.sub.3, or alkyl; and R.sub.3 is H, alkyl, cycloalkyl, aryl, aralkyl, heteroaryl or sugar. In one embodiment, the thermally destabilizing modification in C1 is a mismatch selected from the group consisting of G:G, G:A, G:U, G:T, A:A, A:C, C:C, C:U, C:T, U:U, T:T, and U:T; and optionally, at least one nucleobase in the mismatch pair is a 2'-deoxy nucleobase. In one example, the thermally destabilizing modification in C1 is GNA or

(257) ##STR00010##

T1, T1', T2', and T3' each independently represent a nucleotide comprising a modification providing the nucleotide a steric bulk that is less or equal to the steric bulk of a 2'-OMe modification. A steric bulk refers to the sum of steric effects of a modification. Methods for determining steric effects of a modification of a nucleotide are known to one skilled in the art. The modification can be at the 2' position of a ribose sugar of the nucleotide, or a modification to a non-ribose nucleotide, acyclic nucleotide, or the backbone of the nucleotide that is similar or equivalent to the 2' position of the ribose sugar, and provides the nucleotide a steric bulk that is less than or equal to the steric bulk of a 2'-OMe modification. For example, T1, T1', T2', and T3' are each independently selected from DNA, RNA, LNA, 2'-F, and 2'-F-5'-methyl. In one embodiment, T1 is DNA. In one embodiment, T1' is DNA, RNA or LNA. In one embodiment, T2' is DNA or RNA. In one embodiment, T3' is DNA or RNA. n.sup.1, n.sup.3, and q' are independently 4 to 15 nucleotides in length. n.sup.5, q.sup.3, and q.sup.7 are independently 1-6 nucleotide(s) in length. n.sup.4, q.sup.2, and q.sup.6 are independently 1-3 nucleotide(s) in length; alternatively, n.sup.4 is 0. q.sup.5 is independently 0-10 nucleotide(s) in length. n.sup.2 and q.sup.4 are independently 0-3 nucleotide(s) in length.

(258) Alternatively, n.sup.4 is 0-3 nucleotide(s) in length.

(259) In one embodiment, n.sup.4 can be 0. In one example, n.sup.4 is 0, and q.sup.2 and q.sup.6 are 1. In another example, n.sup.4 is 0, and q.sup.2 and q.sup.6 are 1, with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand).

(260) In one embodiment, n.sup.4, q.sup.2, and q.sup.6 are each 1.

(261) In one embodiment, n.sup.2, n.sup.4, q.sup.2, q.sup.4, and q.sup.6 are each 1.

(262) In one embodiment, C1 is at position 14-17 of the 5'-end of the sense strand, when the sense strand is 19-22 nucleotides in length, and n.sup.4 is 1. In one embodiment, C1 is at position 15 of the 5'-end of the sense strand

(263) In one embodiment, T3' starts at position 2 from the 5' end of the antisense strand. In one example, T3' is at position 2 from the 5' end of the antisense strand and q.sup.6 is equal to 1.

(264) In one embodiment, T1' starts at position 14 from the 5' end of the antisense strand. In one example, T1' is at position 14 from the 5' end of the antisense strand and q.sup.2 is equal to 1.

(265) In an exemplary embodiment, T3' starts from position 2 from the 5' end of the antisense strand and T1' starts from position 14 from the 5' end of the antisense strand. In one example, T3' starts from position 2 from the 5' end of the antisense strand and q.sup.6 is equal to 1 and T1' starts from position 14 from the 5' end of the antisense strand and q.sup.2 is equal to 1.

(266) In one embodiment, T1' and T3' are separated by 11 nucleotides in length (i.e. not counting the T1' and T3' nucleotides).

(267) In one embodiment, T1' is at position 14 from the 5' end of the antisense strand. In one example, T1' is at position 14 from the 5' end of the antisense strand and q.sup.2 is equal to 1, and the modification at the 2' position or positions in a non-ribose, acyclic or backbone that provide less steric bulk than a 2'-OMe ribose.

(268) In one embodiment, T3' is at position 2 from the 5' end of the antisense strand. In one example, T3' is at position 2 from the 5' end of the antisense strand and q.sup.6 is equal to 1, and the modification at the 2' position or positions in a non-ribose, acyclic or backbone that provide less than or equal to steric bulk than a 2'-OMe ribose.

(269) In one embodiment, T1 is at the cleavage site of the sense strand. In one example, T1 is at position 11 from the 5' end of the sense strand, when the sense strand is 19-22 nucleotides in length, and n.sup.2 is 1. In an exemplary embodiment, T1 is at the cleavage site of the sense strand at position 11 from the 5' end of the sense strand, when the sense strand is 19-22 nucleotides in length, and n.sup.2 is 1,

(270) In one embodiment, T2' starts at position 6 from the 5' end of the antisense strand. In one example, T2' is at positions 6-10 from the 5' end of the antisense strand, and q.sup.4 is 1.

(271) In an exemplary embodiment, T1 is at the cleavage site of the sense strand, for instance, at position 11 from the 5' end of the sense strand, when the sense strand is 19-22 nucleotides in length, and n.sup.2 is 1; T1' is at position 14 from the 5' end of the antisense strand, and q.sup.2 is equal to 1, and the modification to T1' is at the 2' position of a ribose sugar or at positions in a non-ribose, acyclic or backbone that provide less steric bulk than a 2'-OMe ribose; T2' is at positions 6-10 from the 5' end of the antisense strand, and q.sup.4 is 1; and T3' is at position 2 from the 5' end of the antisense strand, and q.sup.6 is equal to 1, and the modification to T3' is at the 2' position or at positions in a non-ribose, acyclic or backbone that provide less than or equal to steric bulk than a 2'-OMe ribose.

(272) In one embodiment, T2' starts at position 8 from the 5' end of the antisense strand. In one example, T2' starts at position 8 from the 5' end of the antisense strand, and q.sup.4 is 2.

(273) In one embodiment, T2' starts at position 9 from the 5' end of the antisense strand. In one example, T2' is at position 9 from the 5' end of the antisense strand, and q.sup.4 is 1.

(274) In one embodiment, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 1, B3' is 2'-OMe or 2'-F, q.sup.5 is 6, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within positions 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand).

(275) In one embodiment, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 1, B3' is 2'-OMe or 2'-F, q.sup.5 is 6, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within positions 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand).

(284) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1.

(285) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within positions 1-5 of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end).

(286) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1. In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within positions 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand).

(287) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1.

(288) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within positions 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand).

(289) The RNAi agent can comprise a phosphorus-containing group at the 5'-end of the sense strand or antisense strand. The 5'-end phosphorus-containing group can be 5'-end phosphate (5'-P), 5'-end phosphorothioate (5'-PS), 5'-end phosphorodithioate (5'-PS.sub.2), 5'-end vinylphosphonate (5'-VP), 5'-end methylphosphonate (MePhos), or 5'-deoxy-5'-C-malonyl

(290) ##STR00011##

When the 5'-end phosphorus-containing group is 5'-end vinylphosphonate (5'-VP), the 5'-VP can be either 5'-E-VP isomer (i.e., trans-vinylphosphonate,

(291) ##STR00012##

5'-Z-VP isomer (i.e., cis-vinylphosphonate,

(292) ##STR00013##

or mixtures thereof.

(293) In one embodiment, the RNAi agent comprises a phosphorus-containing group at the 5'-end of the sense strand. In one embodiment, the RNAi agent comprises a phosphorus-containing group at the 5'-end of the antisense strand.

(294) In one embodiment, the RNAi agent comprises a 5'-P. In one embodiment, the RNAi agent comprises a 5'-P in the antisense strand.

(295) In one embodiment, the RNAi agent comprises a 5'-PS. In one embodiment, the RNAi agent comprises a 5'-PS in the antisense strand.

(296) In one embodiment, the RNAi agent comprises a 5'-VP. In one embodiment, the RNAi agent comprises a 5'-VP in the antisense strand. In one embodiment, the RNAi agent comprises a 5'-E-VP in the antisense strand. In one embodiment, the RNAi agent comprises a 5'-Z-VP in the antisense strand.

(297) In one embodiment, the RNAi agent comprises a 5'-PS.sub.2. In one embodiment, the RNAi agent comprises a 5'-PS.sub.2 in the antisense strand.

(298) In one embodiment, the RNAi agent comprises a 5'-PS.sub.2. In one embodiment, the RNAi agent comprises a 5'-deoxy-5'-C-malonyl in the antisense strand.

(299) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1. The RNAi agent also comprises a 5'-PS.

(300) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1. The RNAi agent also comprises a 5'-P.

(301) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1. The RNAi agent also comprises a 5'-VP. The 5'-VP may be 5'-E-VP, 5'-Z-VP, or combination thereof.

(302) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1. The RNAi agent also comprises a 5'-PS.sub.2.

(303) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1. The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl.

(304) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-P.

(305) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate

internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS.

(306) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-VP. The 5'-VP may be 5'-E-VP, 5'-Z-VP, or combination thereof.

(307) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS.sub.2.

(308) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl.

(309) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1. The RNAi agent also comprises a 5'-P.

(310) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1. The dsRNA agent also comprises a 5'-PS.

(311) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1. The RNAi agent also comprises a 5'-VP. The 5'-VP may be 5'-E-VP, 5'-Z-VP, or combination thereof.

(312) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1. The RNAi agent also comprises a 5'-PS.sub.2.

(313) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe,

n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1. The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl.

(314) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end). The RNAi agent also comprises a 5'-P.

(315) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end). The RNAi agent also comprises a 5'-PS.

(316) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end). The RNAi agent also comprises a 5'-VP. The 5'-VP may be 5'-E-VP, 5'-Z-VP, or combination thereof.

(317) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end). The RNAi agent also comprises a 5'-PS.sub.2.

(318) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end). The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl.

(319) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1. The RNAi agent also comprises a 5'-P.

(320) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1. The RNAi agent also comprises a 5'-PS.

(321) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1. The RNAi agent also comprises a 5'-VP. The 5'-VP may be 5'-E-VP, 5'-Z-VP, or combination thereof.

(322) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1. The dsRNAi RNA agent also comprises a 5'-PS.sub.2.

(323) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1. The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl.

(324) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-P.

(325) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS.

(326) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-VP. The 5'-VP may be 5'-E-VP, 5'-Z-VP, or combination thereof.

(327) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F,

q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS.sub.2.

(328) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl.

(329) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1. The RNAi agent also comprises a 5'-P.

(330) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1. The RNAi agent also comprises a 5'-PS.

(331) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1. The RNAi agent also comprises a 5'-VP. The 5'-VP may be 5'-E-VP, 5'-Z-VP, or combination thereof.

(332) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1. The RNAi agent also comprises a 5'-PS.sub.2.

(333) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1. The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl.

(334) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-P.

(335) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand),

and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS. (336) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-VP. The 5'-VP may be 5'-E-VP, 5'-Z-VP, or combination thereof.

(337) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS.sub.2.

(338) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl.

(339) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-P and a targeting ligand. In one embodiment, the 5'-P is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(340) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS and a targeting ligand. In one embodiment, the 5'-PS is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(341) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-VP (e.g., a 5'-E-VP, 5'-Z-VP, or combination thereof), and a targeting ligand.

(342) In one embodiment, the 5'-VP is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(343) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS.sub.2 and a targeting ligand. In one embodiment, the 5'-PS.sub.2 is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(344) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl and a targeting ligand. In one embodiment, the 5'-deoxy-5'-C-malonyl is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(345) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end). The RNAi agent also comprises a 5'-P and a targeting ligand. In one embodiment, the 5'-P is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(346) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end). The RNAi agent also comprises a 5'-PS and a targeting ligand. In

one embodiment, the 5'-PS is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(347) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end). The RNAi agent also comprises a 5'-VP (e.g., a 5'-E-VP, 5'-Z-VP, or combination thereof) and a targeting ligand. In one embodiment, the 5'-VP is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(348) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end). The RNAi agent also comprises a 5'-PS.sub.2 and a targeting ligand. In one embodiment, the 5'-PS.sub.2 is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(349) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-OMe, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end). The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl and a targeting ligand. In one embodiment, the 5'-deoxy-5'-C-malonyl is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(350) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-P and a targeting ligand. In one embodiment, the 5'-P is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(351) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent

also comprises a 5'-PS and a targeting ligand. In one embodiment, the 5'-PS is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(352) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-VP (e.g., a 5'-E-VP, 5'-Z-VP, or combination thereof) and a targeting ligand. In one embodiment, the 5'-VP is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(353) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS.sub.2 and a targeting ligand. In one embodiment, the 5'-PS.sub.2 is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(354) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, T2' is 2'-F, q.sup.4 is 2, B3' is 2'-OMe or 2'-F, q.sup.5 is 5, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl and a targeting ligand. In one embodiment, the 5'-deoxy-5'-C-malonyl is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(355) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-P and a targeting ligand. In one embodiment, the 5'-P is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(356) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two

phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS and a targeting ligand. In one embodiment, the 5'-PS is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(357) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-VP (e.g., a 5'-E-VP, 5'-Z-VP, or combination thereof) and a targeting ligand. In one embodiment, the 5'-VP is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(358) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-PS.sub.2 and a targeting ligand. In one embodiment, the 5'-PS.sub.2 is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(359) In one embodiment, B1 is 2'-OMe or 2'-F, n.sup.1 is 8, T1 is 2'F, n.sup.2 is 3, B2 is 2'-OMe, n.sup.3 is 7, n.sup.4 is 0, B3 is 2'-OMe, n.sup.5 is 3, B1' is 2'-OMe or 2'-F, q.sup.1 is 9, T1' is 2'-F, q.sup.2 is 1, B2' is 2'-OMe or 2'-F, q.sup.3 is 4, q.sup.4 is 0, B3' is 2'-OMe or 2'-F, q.sup.5 is 7, T3' is 2'-F, q.sup.6 is 1, B4' is 2'-F, and q.sup.7 is 1; with two phosphorothioate internucleotide linkage modifications within position 1-5 of the sense strand (counting from the 5'-end of the sense strand), and two phosphorothioate internucleotide linkage modifications at positions 1 and 2 and two phosphorothioate internucleotide linkage modifications within positions 18-23 of the antisense strand (counting from the 5'-end of the antisense strand). The RNAi agent also comprises a 5'-deoxy-5'-C-malonyl and a targeting ligand. In one embodiment, the 5'-deoxy-5'-C-malonyl is at the 5'-end of the antisense strand, and the targeting ligand is at the 3'-end of the sense strand.

(360) In a particular embodiment, an RNAi agent of the present invention comprises: (a) a sense strand having: (i) a length of 21 nucleotides; (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker; and (iii) 2'-F modifications at positions 1, 3, 5, 7, 9 to 11, 13, 17, 19, and 21, and 2'-OMe modifications at positions 2, 4, 6, 8, 12, 14 to 16, 18, and 20 (counting from the 5' end); and (b) an antisense strand having: (i) a length of 23 nucleotides; (ii) 2'-OMe modifications at positions 1, 3, 5, 9, 11 to 13, 15, 17, 19, 21, and 23, and 2'F modifications at positions 2, 4, 6 to 8, 10, 14, 16, 18, 20, and 22 (counting from the 5' end); and (iii) phosphorothioate internucleotide linkages between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end); wherein the dsRNA agents have a two nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

(361) In another particular embodiment, an RNAi agent of the present invention comprises: (a) a sense strand having: (i) a length of 21 nucleotides; (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker; (iii) 2'-F modifications at positions 1, 3, 5, 7, 9 to 11, 13, 15, 17, 19, and 21, and

2'-OMe modifications at positions 2, 4, 6, 8, 12, 14, 16, 18, and 20 (counting from the 5' end); and (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end); and (b) an antisense strand having: (i) a length of 23 nucleotides; (ii) 2'-OMe modifications at positions 1, 3, 5, 7, 9, 11 to 13, 15, 17, 19, and 21 to 23, and 2'-F modifications at positions 2, 4, 6, 8, 10, 14, 16, 18, and 20 (counting from the 5' end); and (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end);

wherein the RNAi agents have a two nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

(362) In another particular embodiment, a RNAi agent of the present invention comprises: (a) a sense strand having: (i) a length of 21 nucleotides; (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker; (iii) 2'-OMe modifications at positions 1 to 6, 8, 10, and 12 to 21, 2'-F modifications at positions 7, and 9, and a deoxy-nucleotide (e.g. dT) at position 11 (counting from the 5' end); and (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end); and (b) an antisense strand having: (i) a length of 23 nucleotides; (ii) 2'-OMe modifications at positions 1, 3, 7, 9, 11, 13, 15, 17, and 19 to 23, and 2'-F modifications at positions 2, 4 to 6, 8, 10, 12, 14, 16, and 18 (counting from the 5' end); and (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end);

wherein the RNAi agents have a two nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

(363) In another particular embodiment, a RNAi agent of the present invention comprises: (a) a sense strand having: (i) a length of 21 nucleotides; (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker; (iii) 2'-OMe modifications at positions 1 to 6, 8, 10, 12, 14, and 16 to 21, and 2'-F modifications at positions 7, 9, 11, 13, and 15; and (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end); and (b) an antisense strand having: (i) a length of 23 nucleotides; (ii) 2'-OMe modifications at positions 1, 5, 7, 9, 11, 13, 15, 17, 19, and 21 to 23, and 2'-F modifications at positions 2 to 4, 6, 8, 10, 12, 14, 16, 18, and 20 (counting from the 5' end); and (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end);

wherein the RNAi agents have a two nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

(364) In another particular embodiment, a RNAi agent of the present invention comprises: (a) a sense strand having: (i) a length of 21 nucleotides; (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker; (iii) 2'-OMe modifications at positions 1 to 9, and 12 to 21, and 2'-F modifications at positions 10, and 11; and (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end); and (b) an antisense strand having: (i) a length of 23 nucleotides; (ii) 2'-OMe modifications at positions 1, 3, 5, 7, 9, 11 to 13, 15, 17, 19, and 21 to 23, and 2'-F modifications at positions 2, 4, 6, 8, 10, 14, 16, 18, and 20 (counting from the 5' end); and (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end);

wherein the RNAi agents have a two nucleotide overhang at the 3'-end of the antisense strand, and

a blunt end at the 5'-end of the antisense strand.

(365) In another particular embodiment, a RNAi agent of the present invention comprises: (a) a sense strand having: (i) a length of 21 nucleotides; (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker; (iii) 2'-F modifications at positions 1, 3, 5, 7, 9 to 11, and 13, and 2'-OMe modifications at positions 2, 4, 6, 8, 12, and 14 to 21; and (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end); and (b) an antisense strand having: (i) a length of 23 nucleotides; (ii) 2'-OMe modifications at positions 1, 3, 5 to 7, 9, 11 to 13, 15, 17 to 19, and 21 to 23, and 2'-F modifications at positions 2, 4, 8, 10, 14, 16, and 20 (counting from the 5' end); and (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end);

wherein the RNAi agents have a two nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

(366) In another particular embodiment, a RNAi agent of the present invention comprises: (a) a sense strand having: (i) a length of 21 nucleotides; (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker; (iii) 2'-OMe modifications at positions 1, 2, 4, 6, 8, 12, 14, 15, 17, and 19 to 21, and 2'-F modifications at positions 3, 5, 7, 9 to 11, 13, 16, and 18; and (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end); and (b) an antisense strand having: (i) a length of 25 nucleotides; (ii) 2'-OMe modifications at positions 1, 4, 6, 7, 9, 11 to 13, 15, 17, and 19 to 23, 2'-F modifications at positions 2, 3, 5, 8, 10, 14, 16, and 18, and desoxy-nucleotides (e.g. dT) at positions 24 and 25 (counting from the 5' end); and (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end); wherein the RNAi agents have a four nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

(367) In another particular embodiment, a RNAi agent of the present invention comprises: (a) a sense strand having: (i) a length of 21 nucleotides; (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker; (iii) 2'-OMe modifications at positions 1 to 6, 8, and 12 to 21, and 2'-F modifications at positions 7, and 9 to 11; and (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end); and (b) an antisense strand having: (i) a length of 23 nucleotides; (ii) 2'-OMe modifications at positions 1, 3 to 5, 7, 8, 10 to 13, 15, and 17 to 23, and 2'-F modifications at positions 2, 6, 9, 14, and 16 (counting from the 5' end); and (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end); wherein the RNAi agents have a two nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

(368) In another particular embodiment, a RNAi agent of the present invention comprises: (a) a sense strand having: (i) a length of 21 nucleotides; (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker; (iii) 2'-OMe modifications at positions 1 to 6, 8, and 12 to 21, and 2'-F modifications at positions 7, and 9 to 11; and (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end); and (b) an antisense strand having: (i) a length of 23 nucleotides; (ii) 2'-OMe modifications at positions 1, 3 to 5, 7, 10 to 13, 15, and 17 to 23, and 2'-F modifications at positions 2, 6, 8, 9, 14,

and 16 (counting from the 5' end); and (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 21 and 22, and between nucleotide positions 22 and 23 (counting from the 5' end); wherein the RNAi agents have a two nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

(369) In another particular embodiment, a RNAi agent of the present invention comprises: (a) a sense strand having: (i) a length of 19 nucleotides; (ii) an ASGPR ligand attached to the 3'-end, wherein said ASGPR ligand comprises three GalNAc derivatives attached through a trivalent branched linker; (iii) 2'-OMe modifications at positions 1 to 4, 6, and 10 to 19, and 2'-F modifications at positions 5, and 7 to 9; and (iv) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, and between nucleotide positions 2 and 3 (counting from the 5' end); and (b) an antisense strand having: (i) a length of 21 nucleotides; (ii) 2'-OMe modifications at positions 1, 3 to 5, 7, 10 to 13, 15, and 17 to 21, and 2'-F modifications at positions 2, 6, 8, 9, 14, and 16 (counting from the 5' end); and (iii) phosphorothioate internucleotide linkages between nucleotide positions 1 and 2, between nucleotide positions 2 and 3, between nucleotide positions 19 and 20, and between nucleotide positions 20 and 21 (counting from the 5' end); wherein the RNAi agents have a two nucleotide overhang at the 3'-end of the antisense strand, and a blunt end at the 5'-end of the antisense strand.

(370) In certain embodiments, the iRNA for use in the methods of the invention is an agent selected from agents listed in any one of Tables 2, 3, 5, 6, and 8-13. These agents may further comprise a ligand.

IV. iRNAs Conjugated to Ligands

(371) Another modification of the RNA of an iRNA of the invention involves chemically linking to the iRNA one or more ligands, moieties or conjugates that enhance the activity, cellular distribution, or cellular uptake of the iRNA e.g., into a cell. Such moieties include but are not limited to lipid moieties such as a cholesterol moiety (Letsinger et al., *Proc. Natl. Acad. Sci. USA*, 1989, 86: 6553-6556). In other embodiments, the ligand is cholic acid (Manoharan et al., *Biorg. Med. Chem. Lett.*, 1994, 4:1053-1060), a thioether, e.g., beryl-S-tritylthiol (Manoharan et al., *Ann. N.Y. Acad. Sci.*, 1992, 660:306-309; Manoharan et al., *Biorg. Med. Chem. Lett.*, 1993, 3:2765-2770), a thiocholesterol (Oberhauser et al., *Nucl. Acids Res.*, 1992, 20:533-538), an aliphatic chain, e.g., dodecandiol or undecyl residues (Saison-Behmoaras et al., *EMBO J*, 1991, 10:1111-1118; Kabanov et al., *FEBS Lett.*, 1990, 259:327-330; Svinarchuk et al., *Biochimie*, 1993, 75:49-54), a phospholipid, e.g., di-hexadecyl-rac-glycerol or triethyl-ammonium 1,2-di-O-hexadecyl-rac-glycero-3-phosphonate (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651-3654; Shea et al., *Nucl. Acids Res.*, 1990, 18:3777-3783), a polyamine or a polyethylene glycol chain (Manoharan et al., *Nucleosides & Nucleotides*, 1995, 14:969-973), or adamantane acetic acid (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651-3654), a palmityl moiety (Mishra et al., *Biochim. Biophys. Acta*, 1995, 1264:229-237), or an octadecylamine or hexylamino-carbonyloxycholesterol moiety (Crooke et al., *J. Pharmacol. Exp. Ther.*, 1996, 277:923-937).

(372) In certain embodiments, a ligand alters the distribution, targeting, or lifetime of an iRNA agent into which it is incorporated. In some embodiments a ligand provides an enhanced affinity for a selected target, e.g., molecule, cell or cell type, compartment, e.g., a cellular or organ compartment, tissue, organ or region of the body, as, e.g., compared to a species absent such a ligand. In some embodiments, ligands do not take part in duplex pairing in a duplexed nucleic acid.

(373) Ligands can include a naturally occurring substance, such as a protein (e.g., human serum albumin (HSA), low-density lipoprotein (LDL), or globulin); carbohydrate (e.g., a dextran, pullulan, chitin, chitosan, inulin, cyclodextrin, N-acetylglucosamine, N-acetylgalactosamine, or hyaluronic acid); or a lipid. The ligand can also be a recombinant or synthetic molecule, such as a synthetic polymer, e.g., a synthetic polyamino acid. Examples of polyamino acids include polyamino acid is a polylysine (PLL), poly L-aspartic acid, poly L-glutamic acid, styrene-maleic

acid anhydride copolymer, poly(L-lactide-co-glycolide) copolymer, divinyl ether-maleic anhydride copolymer, N-(2-hydroxypropyl)methacrylamide copolymer (HMPA), polyethylene glycol (PEG), polyvinyl alcohol (PVA), polyurethane, poly(2-ethylacrylic acid), N-isopropylacrylamide polymers, or polyphosphazine. Example of polyamines include: polyethylenimine, polylysine (PLL), spermine, spermidine, polyamine, pseudopeptide-polyamine, peptidomimetic polyamine, dendrimer polyamine, arginine, amidine, protamine, cationic lipid, cationic porphyrin, quaternary salt of a polyamine, or an alpha helical peptide.

(374) Ligands can also include targeting groups, e.g., a cell or tissue targeting agent, e.g., a lectin, glycoprotein, lipid or protein, e.g., an antibody, that binds to a specified cell type such as a kidney cell. A targeting group can be a thyrotropin, melanotropin, lectin, glycoprotein, surfactant protein A, Mucin carbohydrate, multivalent lactose, multivalent galactose, N-acetyl-galactosamine, N-acetyl-glucosamine multivalent mannose, multivalent fucose, glycosylated polyaminoacids, multivalent galactose, transferrin, bisphosphonate, polyglutamate, polyaspartate, a lipid, cholesterol, a steroid, bile acid, folate, vitamin B12, vitamin A, biotin, or an RGD peptide or RGD peptide mimetic. In certain embodiments, the ligand is a multivalent galactose, e.g., an N-acetyl-galactosamine.

(375) Other examples of ligands include dyes, intercalating agents (e.g. acridines), cross-linkers (e.g. psoralene, mitomycin C), porphyrins (TPPC4, texaphyrin, Sapphyrin), polycyclic aromatic hydrocarbons (e.g., phenazine, dihydrophenazine), artificial endonucleases (e.g. EDTA), lipophilic molecules, e.g., cholesterol, cholic acid, adamantane acetic acid, 1-pyrene butyric acid, dihydrotestosterone, 1,3-Bis-O(hexadecyl)glycerol, geranyloxyhexyl group, hexadecylglycerol, borneol, menthol, 1,3-propanediol, heptadecyl group, palmitic acid, myristic acid, O3-(oleoyl)lithocholic acid, O3-(oleoyl)cholenic acid, dimethoxytrityl, or phenoxazine) and peptide conjugates (e.g., antennapedia peptide, Tat peptide), alkylating agents, phosphate, amino, mercapto, PEG (e.g., PEG-40K), MPEG, [MPEG].sub.2, polyamino, alkyl, substituted alkyl, radiolabeled markers, enzymes, haptens (e.g. biotin), transport/absorption facilitators (e.g., aspirin, vitamin E, folic acid), synthetic ribonucleases (e.g., imidazole, bisimidazole, histamine, imidazole clusters, acridine-imidazole conjugates, Eu³⁺ complexes of tetraazamacrocycles), dinitrophenyl, HRP, or AP.

(376) Ligands can be proteins, e.g., glycoproteins, or peptides, e.g., molecules having a specific affinity for a co-ligand, or antibodies e.g., an antibody, that binds to a specified cell type such as a hepatic cell. Ligands can also include hormones and hormone receptors. They can also include non-peptidic species, such as lipids, lectins, carbohydrates, vitamins, cofactors, multivalent lactose, multivalent galactose, N-acetyl-galactosamine, N-acetyl-glucosamine multivalent mannose, or multivalent fucose. The ligand can be, for example, a lipopolysaccharide, an activator of p38 MAP kinase, or an activator of NF- κ B.

(377) The ligand can be a substance, e.g., a drug, which can increase the uptake of the iRNA agent into the cell, for example, by disrupting the cell's cytoskeleton, e.g., by disrupting the cell's microtubules, microfilaments, or intermediate filaments. The drug can be, for example, taxol, vincristine, vinblastine, cytochalasin, nocodazole, japlakinolide, latrunculin A, phalloidin, swinholide A, indanocine, or myoservin.

(378) In some embodiments, a ligand attached to an iRNA as described herein acts as a pharmacokinetic modulator (PK modulator). PK modulators include lipophiles, bile acids, steroids, phospholipid analogues, peptides, protein binding agents, PEG, vitamins, etc. Exemplary PK modulators include, but are not limited to, cholesterol, fatty acids, cholic acid, lithocholic acid, dialkylglycerides, diacylglyceride, phospholipids, sphingolipids, naproxen, ibuprofen, vitamin E, biotin. Oligonucleotides that comprise a number of phosphorothioate linkages are also known to bind to serum protein, thus short oligonucleotides, e.g., oligonucleotides of about 5 bases, 10 bases, 15 bases, or 20 bases, comprising multiple of phosphorothioate linkages in the backbone are also amenable to the present invention as ligands (e.g. as PK modulating ligands). In addition, aptamers

that bind serum components (e.g. serum proteins) are also suitable for use as PK modulating ligands in the embodiments described herein.

(379) Ligand-conjugated iRNAs of the invention may be synthesized by the use of an oligonucleotide that bears a pendant reactive functionality, such as that derived from the attachment of a linking molecule onto the oligonucleotide (described below). This reactive oligonucleotide may be reacted directly with commercially-available ligands, ligands that are synthesized bearing any of a variety of protecting groups, or ligands that have a linking moiety attached thereto.

(380) The oligonucleotides used in the conjugates of the present invention may be conveniently and routinely made through the well-known technique of solid-phase synthesis. Equipment for such synthesis is sold by several vendors including, for example, Applied Biosystems® (Foster City, Calif.). Any other methods for such synthesis known in the art may additionally or alternatively be employed. It is also known to use similar techniques to prepare other oligonucleotides, such as the phosphorothioates and alkylated derivatives.

(381) In the ligand-conjugated iRNAs and ligand-molecule bearing sequence-specific linked nucleosides of the present invention, the oligonucleotides and oligonucleosides may be assembled on a suitable DNA synthesizer utilizing standard nucleotide or nucleoside precursors, or nucleotide or nucleoside conjugate precursors that already bear the linking moiety, ligand-nucleotide or nucleoside-conjugate precursors that already bear the ligand molecule, or non-nucleoside ligand-bearing building blocks.

(382) When using nucleotide-conjugate precursors that already bear a linking moiety, the synthesis of the sequence-specific linked nucleosides is typically completed, and the ligand molecule is then reacted with the linking moiety to form the ligand-conjugated oligonucleotide. In some embodiments, the oligonucleotides or linked nucleosides of the present invention are synthesized by an automated synthesizer using phosphoramidites derived from ligand-nucleoside conjugates in addition to the standard phosphoramidites and non-standard phosphoramidites that are commercially available and routinely used in oligonucleotide synthesis.

(383) A. Lipid Conjugates

(384) In certain embodiments, the ligand or conjugate is a lipid or lipid-based molecule. Such a lipid or lipid-based molecule may bind a serum protein, e.g., human serum albumin (HSA). An HSA binding ligand allows for distribution of the conjugate to a target tissue, e.g., a non-kidney target tissue of the body. For example, the target tissue can be the liver, including parenchymal cells of the liver. Other molecules that can bind HSA can also be used as ligands. For example, naproxen or aspirin can be used. A lipid or lipid-based ligand can (a) increase resistance to degradation of the conjugate, (b) increase targeting or transport into a target cell or cell membrane, or (c) can be used to adjust binding to a serum protein, e.g., HSA.

(385) A lipid based ligand can be used to inhibit, e.g., control the binding of the conjugate to a target tissue. For example, a lipid or lipid-based ligand that binds to HSA more strongly will be less likely to be targeted to the kidney and therefore less likely to be cleared from the body. A lipid or lipid-based ligand that binds to HSA less strongly can be used to target the conjugate to the kidney.

(386) In certain embodiments, the lipid based ligand binds HSA. In some embodiments, it binds HSA with a sufficient affinity such that the conjugate will be distributed to a non-kidney tissue. However, it is preferred that the affinity not be so strong that the HSA-ligand binding cannot be reversed.

(387) In other embodiments, the lipid based ligand binds HSA weakly or not at all, such that the conjugate will be distributed to the kidney. Other moieties that target to kidney cells can also be used in place of, or in addition to, the lipid based ligand.

(388) In another aspect, the ligand is a moiety, e.g., a vitamin, which is taken up by a target cell, e.g., a proliferating cell. These are particularly useful for treating disorders characterized by unwanted cell proliferation, e.g., of the malignant or non-malignant type, e.g., cancer cells. Exemplary vitamins include vitamin A, E, and K. Other exemplary vitamins include B vitamin,

e.g., folic acid, B12, riboflavin, biotin, pyridoxal or other vitamins or nutrients taken up by target cells such as liver cells. Also included are HSA and low density lipoprotein (LDL).

(389) B. Cell Permeation Agents

(390) In another aspect, the ligand is a cell-permeation agent, such as, a helical cell-permeation agent. In some embodiments, the agent is amphipathic. An exemplary agent is a peptide such as tat or antennopodia. If the agent is a peptide, it can be modified, including a peptidylmimetic, invertomers, non-peptide or pseudo-peptide linkages, and use of D-amino acids. In some embodiments, the helical agent is an alpha-helical agent, which has a lipophilic and a lipophobic phase.

(391) The ligand can be a peptide or peptidomimetic. A peptidomimetic (also referred to herein as an oligopeptidomimetic) is a molecule capable of folding into a defined three-dimensional structure similar to a natural peptide. The attachment of peptide and peptidomimetics to iRNA agents can affect pharmacokinetic distribution of the iRNA, such as by enhancing cellular recognition and absorption. The peptide or peptidomimetic moiety can be about 5-50 amino acids long, e.g., about 5, 10, 15, 20, 25, 30, 35, 40, 45, or 50 amino acids long.

(392) A peptide or peptidomimetic can be, for example, a cell permeation peptide, cationic peptide, amphipathic peptide, or hydrophobic peptide (e.g., consisting primarily of Tyr, Trp, or Phe). The peptide moiety can be a dendrimer peptide, constrained peptide or crosslinked peptide. In another alternative, the peptide moiety can include a hydrophobic membrane translocation sequence (MTS). An exemplary hydrophobic MTS-containing peptide is RFGF having the amino acid sequence AAVALLPAVLLALLAP (SEQ ID NO: 14). An RFGF analogue (e.g., amino acid sequence AALLPVLLAAP (SEQ ID NO:15) containing a hydrophobic MTS can also be a targeting moiety. The peptide moiety can be a "delivery" peptide, which can carry large polar molecules including peptides, oligonucleotides, and protein across cell membranes. For example, sequences from the HIV Tat protein (GRKKRRQRRRPPQ (SEQ ID NO:16) and the *Drosophila* Antennapedia protein (RQIKIWFQNRRMKWKK (SEQ ID NO:17) have been found to be capable of functioning as delivery peptides. A peptide or peptidomimetic can be encoded by a random sequence of DNA, such as a peptide identified from a phage-display library, or one-bead-one-compound (OBOC) combinatorial library (Lam et al., Nature, 354:82-84, 1991). Examples of a peptide or peptidomimetic tethered to a dsRNA agent via an incorporated monomer unit for cell targeting purposes is an arginine-glycine-aspartic acid (RGD)-peptide, or RGD mimic. A peptide moiety can range in length from about 5 amino acids to about 40 amino acids. The peptide moieties can have a structural modification, such as to increase stability or direct conformational properties. Any of the structural modifications described below can be utilized.

(393) An RGD peptide for use in the compositions and methods of the invention may be linear or cyclic, and may be modified, e.g., glycosylated or methylated, to facilitate targeting to a specific tissue(s). RGD-containing peptides and peptidomimetics may include D-amino acids, as well as synthetic RGD mimics. In addition to RGD, one can use other moieties that target the integrin ligand, e.g., PECAM-1 or VEGF.

(394) A "cell permeation peptide" is capable of permeating a cell, e.g., a microbial cell, such as a bacterial or fungal cell, or a mammalian cell, such as a human cell. A microbial cell-permeating peptide can be, for example, an α -helical linear peptide (e.g., LL-37 or Ceropin P1), a disulfide bond-containing peptide (e.g., α -defensin, β -defensin or bactenecin), or a peptide containing only one or two dominating amino acids (e.g., PR-39 or indolicidin). A cell permeation peptide can also include a nuclear localization signal (NLS). For example, a cell permeation peptide can be a bipartite amphipathic peptide, such as MPG, which is derived from the fusion peptide domain of HIV-1 gp41 and the NLS of SV40 large T antigen (Simeoni et al., Nucl. Acids Res. 31:2717-2724, 2003).

(395) C. Carbohydrate Conjugates

(396) In some embodiments of the compositions and methods of the invention, an iRNA further

comprises a carbohydrate. The carbohydrate conjugated iRNA is advantageous for the in vivo delivery of nucleic acids, as well as compositions suitable for in vivo therapeutic use, as described herein. As used herein, "carbohydrate" refers to a compound which is either a carbohydrate per se made up of one or more monosaccharide units having at least 6 carbon atoms (which can be linear, branched or cyclic) with an oxygen, nitrogen or sulfur atom bonded to each carbon atom; or a compound having as a part thereof a carbohydrate moiety made up of one or more monosaccharide units each having at least six carbon atoms (which can be linear, branched or cyclic), with an oxygen, nitrogen or sulfur atom bonded to each carbon atom. Representative carbohydrates include the sugars (mono-, di-, tri-, and oligosaccharides containing from about 4, 5, 6, 7, 8, or 9 monosaccharide units), and polysaccharides such as starches, glycogen, cellulose and polysaccharide gums. Specific monosaccharides include C5 and above (e.g., C5, C6, C7, or C8) sugars; di- and trisaccharides include sugars having two or three monosaccharide units (e.g., C5, C6, C7, or C8).

(397) In certain embodiments, a carbohydrate conjugate for use in the compositions and methods of the invention is a monosaccharide.

(398) In certain embodiments, the monosaccharide is an N-acetylgalactosamine (GalNAc). GalNAc conjugates, which comprise one or more N-acetylgalactosamine (GalNAc) derivatives, are described, for example, in U.S. Pat. No. 8,106,022, the entire content of which is hereby incorporated herein by reference. In some embodiments, the GalNAc conjugate serves as a ligand that targets the iRNA to particular cells. In some embodiments, the GalNAc conjugate targets the iRNA to liver cells, e.g., by serving as a ligand for the asialoglycoprotein receptor of liver cells (e.g., hepatocytes).

(399) In some embodiments, the carbohydrate conjugate comprises one or more GalNAc derivatives. The GalNAc derivatives may be attached via a linker, e.g., a bivalent or trivalent branched linker. In some embodiments the GalNAc conjugate is conjugated to the 3' end of the sense strand. In some embodiments, the GalNAc conjugate is conjugated to the iRNA agent (e.g., to the 3' end of the sense strand) via a linker, e.g., a linker as described herein. In some embodiments the GalNAc conjugate is conjugated to the 5' end of the sense strand. In some embodiments, the GalNAc conjugate is conjugated to the iRNA agent (e.g., to the 5' end of the sense strand) via a linker, e.g., a linker as described herein.

(400) In certain embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a monovalent linker. In some embodiments, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a bivalent linker. In yet other embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a trivalent linker. In other embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a tetravalent linker.

(401) In certain embodiments, the double stranded RNAi agents of the invention comprise one GalNAc or GalNAc derivative attached to the iRNA agent. In certain embodiments, the double stranded RNAi agents of the invention comprise a plurality (e.g., 2, 3, 4, 5, or 6) GalNAc or GalNAc derivatives, each independently attached to a plurality of nucleotides of the double stranded RNAi agent through a plurality of monovalent linkers.

(402) In some embodiments, for example, when the two strands of an iRNA agent of the invention are part of one larger molecule connected by an uninterrupted chain of nucleotides between the 3'-end of one strand and the 5'-end of the respective other strand forming a hairpin loop comprising a plurality of unpaired nucleotides, each unpaired nucleotide within the hairpin loop may independently comprise a GalNAc or GalNAc derivative attached via a monovalent linker. The hairpin loop may also be formed by an extended overhang in one strand of the duplex.

(403) In some embodiments, for example, when the two strands of an iRNA agent of the invention are part of one larger molecule connected by an uninterrupted chain of nucleotides between the 3'-end of one strand and the 5'-end of the respective other strand forming a hairpin loop comprising a

plurality of unpaired nucleotides, each unpaired nucleotide within the hairpin loop may independently comprise a GalNAc or GalNAc derivative attached via a monovalent linker. The hairpin loop may also be formed by an extended overhang in one strand of the duplex.

(404) In one embodiment, a carbohydrate conjugate for use in the compositions and methods of the invention is selected from the group consisting of:

(405) ##STR00014## ##STR00015## ##STR00016## ##STR00017## ##STR00018##
##STR00019## ##STR00020##

(406) ##STR00021##
wherein Y is O or S and n is 3-6 (Formula XXIV);

(407) ##STR00022##
wherein Y is O or S and n is 3-6 (Formula XXV);

(408) ##STR00023##
wherein X is O or S (Formula XXVII);

(409) ##STR00024## ##STR00025## ##STR00026##

(410) In another embodiment, a carbohydrate conjugate for use in the compositions and methods of the invention is a monosaccharide. In one embodiment, the monosaccharide is an N-acetylgalactosamine, such as

(411) ##STR00027##

(412) In some embodiments, the RNAi agent is attached to the carbohydrate conjugate via a linker as shown in the following schematic, wherein X is O or S

(413) ##STR00028##

(414) In some embodiments, the RNAi agent is conjugated to L96 as defined in Table 1 and shown below:

(415) ##STR00029##

(416) Another representative carbohydrate conjugate for use in the embodiments described herein includes, but is not limited to

(417) ##STR00030##
(Formula XXXVI), when one of X or Y is an oligonucleotide, the other is a hydrogen.

(418) In some embodiments, a suitable ligand is a ligand disclosed in WO 2019/055633, the entire contents of which are incorporated herein by reference. In one embodiment the ligand comprises the structure below:

(419) ##STR00031##

(420) In certain embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a monovalent linker. In some embodiments, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a bivalent linker. In yet other embodiments of the invention, the GalNAc or GalNAc derivative is attached to an iRNA agent of the invention via a trivalent linker.

(421) In one embodiment, the double stranded RNAi agents of the invention comprise one or more GalNAc or GalNAc derivative attached to the iRNA agent. The GalNAc may be attached to any nucleotide via a linker on the sense strand or antisense strand. The GalNAc may be attached to the 5'-end of the sense strand, the 3' end of the sense strand, the 5'-end of the antisense strand, or the 3'-end of the antisense strand. In one embodiment, the GalNAc is attached to the 3' end of the sense strand, e.g., via a trivalent linker.

(422) In other embodiments, the double stranded RNAi agents of the invention comprise a plurality (e.g., 2, 3, 4, 5, or 6) GalNAc or GalNAc derivatives, each independently attached to a plurality of nucleotides of the double stranded RNAi agent through a plurality of linkers, e.g., monovalent linkers.

(423) In some embodiments, for example, when the two strands of an iRNA agent of the invention is part of one larger molecule connected by an uninterrupted chain of nucleotides between the 3'-end of one strand and the 5'-end of the respective other strand forming a hairpin loop comprising, a

plurality of unpaired nucleotides, each unpaired nucleotide within the hairpin loop may independently comprise a GalNAc or GalNAc derivative attached via a monovalent linker. (424) In some embodiments, the carbohydrate conjugate further comprises one or more additional ligands as described above, such as, but not limited to, a PK modulator or a cell permeation peptide.

(425) Additional carbohydrate conjugates and linkers suitable for use in the present invention include those described in PCT Publication Nos. WO 2014/179620 and WO 2014/179627, the entire contents of each of which are incorporated herein by reference.

(426) D. Linkers

(427) In some embodiments, the conjugate or ligand described herein can be attached to an iRNA oligonucleotide with various linkers that can be cleavable or non-cleavable.

(428) The term “linker” or “linking group” means an organic moiety that connects two parts of a compound, e.g., covalently attaches two parts of a compound. Linkers typically comprise a direct bond or an atom such as oxygen or sulfur, a unit such as NR₈, C(O), C(O)NH, SO, SO.sub.2, SO.sub.2NH or a chain of atoms, such as, but not limited to, substituted or unsubstituted alkyl, substituted or unsubstituted alkenyl, substituted or unsubstituted alkynyl, arylalkyl, arylalkenyl, arylalkynyl, heteroarylalkyl, heteroarylalkenyl, heteroarylalkynyl, heterocyclalkyl, heterocyclalkenyl, heterocyclalkynyl, aryl, heteroaryl, heterocycl, cycloalkyl, cycloalkenyl, alkylarylalkyl, alkylarylalkenyl, alkylarylalkynyl, alkenylarylalkyl, alkenylarylalkenyl, alkenylarylalkynyl, alkynylarylalkyl, alkynylarylalkenyl, alkynylarylalkynyl, alkylheteroarylalkyl, alkylheteroarylalkenyl, alkylheteroarylalkynyl, alkenylheteroarylalkyl, alkenylheteroarylalkenyl, alkenylheteroarylalkynyl, alkynylheteroarylalkyl, alkynylheteroarylalkenyl, alkynylheteroarylalkynyl, alkylheterocyclalkyl, alkylheterocyclalkenyl, alkylheterocyclalkynyl, alkenylheterocyclalkyl, alkenylheterocyclalkenyl, alkenylheterocyclalkynyl, alkynylheterocyclalkyl, alkynylheterocyclalkenyl, alkynylheterocyclalkynyl, alkylaryl, alkenylaryl, alkynylaryl, alkylheteroaryl, alkenylheteroaryl, alkynylheteroaryl, which one or more methylenes can be interrupted or terminated by O, S, S(O), SO.sub.2, N(R₈), C(O), substituted or unsubstituted aryl, substituted or unsubstituted heteroaryl, or substituted or unsubstituted heterocyclic; where R₈ is hydrogen, acyl, aliphatic, or substituted aliphatic. In one embodiment, the linker is about 1-24 atoms, 2-24, 3-24, 4-24, 5-24, 6-24, 6-18, 7-18, 8-18, 7-17, 8-17, 6-16, 7-17, or 8-16 atoms.

(429) A cleavable linking group is one which is sufficiently stable outside the cell, but which upon entry into a target cell is cleaved to release the two parts the linker is holding together. In one embodiment, the cleavable linking group is cleaved at least about 10 times, 20, times, 30 times, 40 times, 50 times, 60 times, 70 times, 80 times, 90 times, or more, or at least 100 times faster in a target cell or under a first reference condition (which can, e.g., be selected to mimic or represent intracellular conditions) than in the blood of a subject, or under a second reference condition (which can, e.g., be selected to mimic or represent conditions found in the blood or serum).

(430) Cleavable linking groups are susceptible to cleavage agents, e.g., pH, redox potential, or the presence of degradative molecules. Generally, cleavage agents are more prevalent or found at higher levels or activities inside cells than in serum or blood. Examples of such degradative agents include: redox agents which are selected for particular substrates or which have no substrate specificity, including, e.g., oxidative or reductive enzymes or reductive agents such as mercaptans, present in cells, that can degrade a redox cleavable linking group by reduction; esterases; endosomes or agents that can create an acidic environment, e.g., those that result in a pH of five or lower; enzymes that can hydrolyze or degrade an acid cleavable linking group by acting as a general acid, peptidases (which can be substrate specific), and phosphatases.

(431) A cleavable linkage group, such as a disulfide bond can be susceptible to pH. The pH of human serum is 7.4, while the average intracellular pH is slightly lower, ranging from about 7.1-7.3. Endosomes have a more acidic pH, in the range of 5.5-6.0, and lysosomes have an even more

acidic pH at around 5.0. Some linkers will have a cleavable linking group that is cleaved at a selected pH, thereby releasing a cationic lipid from the ligand inside the cell, or into the desired compartment of the cell.

(432) A linker can include a cleavable linking group that is cleavable by a particular enzyme. The type of cleavable linking group incorporated into a linker can depend on the cell to be targeted. For example, a liver-targeting ligand can be linked to a cationic lipid through a linker that includes an ester group. Liver cells are rich in esterases, and therefore the linker will be cleaved more efficiently in liver cells than in cell types that are not esterase-rich. Other cell-types rich in esterases include cells of the lung, renal cortex, and testis.

(433) Linkers that contain peptide bonds can be used when targeting cell types rich in peptidases, such as liver cells and synoviocytes.

(434) In general, the suitability of a candidate cleavable linking group can be evaluated by testing the ability of a degradative agent (or condition) to cleave the candidate linking group. It will also be desirable to also test the candidate cleavable linking group for the ability to resist cleavage in the blood or when in contact with other non-target tissue. Thus, one can determine the relative susceptibility to cleavage between a first and a second condition, where the first is selected to be indicative of cleavage in a target cell and the second is selected to be indicative of cleavage in other tissues or biological fluids, e.g., blood or serum. The evaluations can be carried out in cell free systems, in cells, in cell culture, in organ or tissue culture, or in whole animals. It can be useful to make initial evaluations in cell-free or culture conditions and to confirm by further evaluations in whole animals. In some embodiments, useful candidate compounds are cleaved at least about 2, 4, 10, 20, 30, 40, 50, 60, 70, 80, 90, or 100 times faster in the cell (or under in vitro conditions selected to mimic intracellular conditions) as compared to blood or serum (or under in vitro conditions selected to mimic extracellular conditions).

(435) i. Redox Cleavable Linking Groups

(436) In certain embodiments, a cleavable linking group is a redox cleavable linking group that is cleaved upon reduction or oxidation. An example of reductively cleavable linking group is a disulphide linking group (—S—S—). To determine if a candidate cleavable linking group is a suitable “reductively cleavable linking group,” or for example is suitable for use with a particular iRNA moiety and particular targeting agent one can look to methods described herein. For example, a candidate can be evaluated by incubation with dithiothreitol (DTT), or other reducing agent using reagents known in the art, which mimic the rate of cleavage which would be observed in a cell, e.g., a target cell. The candidates can also be evaluated under conditions which are selected to mimic blood or serum conditions. In one, candidate compounds are cleaved by at most about 10% in the blood. In other embodiments, useful candidate compounds are degraded at least about 2, 4, 10, 20, 30, 40, 50, 60, 70, 80, 90, or about 100 times faster in the cell (or under in vitro conditions selected to mimic intracellular conditions) as compared to blood (or under in vitro conditions selected to mimic extracellular conditions). The rate of cleavage of candidate compounds can be determined using standard enzyme kinetics assays under conditions chosen to mimic intracellular media and compared to conditions chosen to mimic extracellular media.

(437) ii. Phosphate-Based Cleavable Linking Groups

(438) In other embodiments, a cleavable linker comprises a phosphate-based cleavable linking group. A phosphate-based cleavable linking group is cleaved by agents that degrade or hydrolyze the phosphate group. An example of an agent that cleaves phosphate groups in cells are enzymes such as phosphatases in cells. Examples of phosphate-based linking groups are —O—P(O)(ORk)—O— , —O—P(S)(ORk)—O— , —O—P(S)(SRk)—O— , —S—P(O)(ORk)—O— , —O—P(O)(ORk)—S— , —S—P(O)(ORk)—S— , —O—P(S)(ORk)—S— , —S—P(S)(ORk)—O— , —O—P(O)(Rk)—O— , —O—P(S)(Rk)—O— , —S—P(O)(Rk)—O— , —S—P(S)(Rk)—O— , —S—P(O)(Rk)—S— , —O—P(S)(Rk)—S— , wherein Rk at each occurrence can be, independently, C1-C20 alkyl, C1-C20 haloalkyl, C6-C10 aryl, or C7-C12 aralkyl. Exemplary embodiments include —O—P(O)(OH)—O— , —O—

$\text{P(S)}(\text{OH})\text{—O—}$, $\text{—O—P(S)}(\text{OH})\text{—O—}$, $\text{—S—P(O)}(\text{OH})\text{—O—}$, $\text{—O—P(O)}(\text{OH})\text{—S—}$, $\text{—S—P(O)}(\text{OH})\text{—S—}$, $\text{—O—P(S)}(\text{OH})\text{—S—}$, $\text{—S—P(S)}(\text{OH})\text{—O—}$, $\text{—O—P(O)}(\text{H})\text{—O—}$, $\text{—O—P(S)}(\text{H})\text{—O—}$, $\text{—S—P(O)}(\text{H})\text{—O—}$, $\text{—S—P(S)}(\text{H})\text{—O—}$, $\text{—S—P(O)}(\text{H})\text{—S—}$, and $\text{—O—P(S)}(\text{H})\text{—S—}$. In certain embodiments, a phosphate-based linking group is $\text{—O—P(O)}(\text{OH})\text{—O—}$. These candidates can be evaluated using methods analogous to those described above.

(439) iii. Acid Cleavable Linking Groups

(440) In other embodiments, a cleavable linker comprises an acid cleavable linking group. An acid cleavable linking group is a linking group that is cleaved under acidic conditions. In some embodiments acid cleavable linking groups are cleaved in an acidic environment with a pH of about 6.5 or lower (e.g., about 6.0, 5.5, 5.0, or lower), or by agents such as enzymes that can act as a general acid. In a cell, specific low pH organelles, such as endosomes and lysosomes can provide a cleaving environment for acid cleavable linking groups. Examples of acid cleavable linking groups include but are not limited to hydrazones, esters, and esters of amino acids. Acid cleavable groups can have the general formula —C=NN— , C(O)O , or —OC(O) . An exemplary embodiment is when the carbon attached to the oxygen of the ester (the alkoxy group) is an aryl group, substituted alkyl group, or tertiary alkyl group such as dimethyl pentyl or t-butyl. These candidates can be evaluated using methods analogous to those described above.

(441) iv. Ester-Based Linking Groups

(442) In other embodiments, a cleavable linker comprises an ester-based cleavable linking group. An ester-based cleavable linking group is cleaved by enzymes such as esterases and amidases in cells. Examples of ester-based cleavable linking groups include, but are not limited to, esters of alkylene, alkenylene and alkynylene groups. Ester cleavable linking groups have the general formula —C(O)O— , or —OC(O)— . These candidates can be evaluated using methods analogous to those described above.

(443) v. Peptide-Based Cleaving Groups

(444) In yet other embodiments, a cleavable linker comprises a peptide-based cleavable linking group. A peptide-based cleavable linking group is cleaved by enzymes such as peptidases and proteases in cells. Peptide-based cleavable linking groups are peptide bonds formed between amino acids to yield oligopeptides (e.g., dipeptides, tripeptides etc.) and polypeptides. Peptide-based cleavable groups do not include the amide group (—C(O)NH—). The amide group can be formed between any alkylene, alkenylene or alkynylene. A peptide bond is a special type of amide bond formed between amino acids to yield peptides and proteins. The peptide based cleavage group is generally limited to the peptide bond (i.e., the amide bond) formed between amino acids yielding peptides and proteins and does not include the entire amide functional group. Peptide-based cleavable linking groups have the general formula $\text{—NHCHRAC(O)NHCHRBC(O)—}$, where RA and RB are the R groups of the two adjacent amino acids. These candidates can be evaluated using methods analogous to those described above.

(445) In some embodiments, an iRNA of the invention is conjugated to a carbohydrate through a linker. Non-limiting examples of iRNA carbohydrate conjugates with linkers of the compositions and methods of the invention include, but are not limited to,

(446) ##STR00032## ##STR00033## ##STR00034## ##STR00035##

when one of X or Y is an oligonucleotide, the other is a hydrogen.

(447) In certain embodiments of the compositions and methods of the invention, a ligand is one or more “GalNAc” (N-acetylgalactosamine) derivatives attached through a bivalent or trivalent branched linker.

(448) In one embodiment, a dsRNA of the invention is conjugated to a bivalent or trivalent branched linker selected from the group of structures shown in any of formula (XLV)-(XLVI):

(449) ##STR00036##

wherein: q2A, q2B, q3A, q3B, q4A, q4B, q5A, q5B and q5C represent independently for each occurrence 0-20 and wherein the repeating unit can be the same or different; P^{sup}.2A, P^{sup}.2B,

P.sup.3A, P.sup.3B, P.sup.4A, P.sup.4B, P.sup.5A, P.sup.5B, P.sup.5C, T.sup.2A, T.sup.2B, T.sup.3A, T.sup.3B, T.sup.4A, T.sup.4B, T.sup.4A, T.sup.5B, T.sup.5C are each independently for each occurrence absent, CO, NH, O, S, OC(O), NHC(O), CH.sub.2, CH.sub.2NH or CH.sub.2O; Q.sup.2A, Q.sup.2B, Q.sup.3A, Q.sup.3B, Q.sup.4A, Q.sup.4B, Q.sup.5A, Q.sup.5B, Q.sup.5C are independently for each occurrence absent, alkylene, substituted alkylene wherein one or more methylenes can be interrupted or terminated by one or more of O, S, S(O), SO.sub.2, N(RN), C(R')=C(R''), C≡C or C(O); R.sup.2A, R.sup.2B, R.sup.3A, R.sup.3B, R.sup.4A, R.sup.4B, R.sup.5A, R.sup.5B, R.sup.5C are each independently for each occurrence absent, NH, O, S, CH.sub.2, C(O)O, C(O)NH, NHCH(R.sup.a)C(O), —C(O)CH(R.sup.a)—NH—, CO, CH=N—O (450) ##STR00037## or heterocyclyl; L.sup.2A, L.sup.2B, L.sup.3A, L.sup.3B, L.sup.4A, L.sup.4B, L.sup.5A, L.sup.5B and L.sup.5C represent the ligand; i.e. each independently for each occurrence a monosaccharide (such as GalNAc), disaccharide, trisaccharide, tetrasaccharide, oligosaccharide, or polysaccharide; and R.sup.a is H or amino acid side chain. Trivalent conjugating GalNAc derivatives are particularly useful for use with RNAi agents for inhibiting the expression of a target gene, such as those of formula (XLIX):

(451) ##STR00038##

wherein L.sup.5A, L.sup.5B and L.sup.5C represent a monosaccharide, such as GalNAc derivative.

(452) Examples of suitable bivalent and trivalent branched linker groups conjugating GalNAc derivatives include, but are not limited to, the structures recited above as formulas II, VII, XI, X, and XIII.

(453) Representative U.S. patents that teach the preparation of RNA conjugates include, but are not limited to, U.S. Pat. Nos. 4,828,979; 4,948,882; 5,218,105; 5,525,465; 5,541,313; 5,545,730; 5,552,538; 5,578,717; 5,580,731; 5,591,584; 5,109,124; 5,118,802; 5,138,045; 5,414,077; 5,486,603; 5,512,439; 5,578,718; 5,608,046; 4,587,044; 4,605,735; 4,667,025; 4,762,779; 4,789,737; 4,824,941; 4,835,263; 4,876,335; 4,904,582; 4,958,013; 5,082,830; 5,112,963; 5,214,136; 5,082,830; 5,112,963; 5,214,136; 5,245,022; 5,254,469; 5,258,506; 5,262,536; 5,272,250; 5,292,873; 5,317,098; 5,371,241; 5,391,723; 5,416,203; 5,451,463; 5,510,475; 5,512,667; 5,514,785; 5,565,552; 5,567,810; 5,574,142; 5,585,481; 5,587,371; 5,595,726; 5,597,696; 5,599,923; 5,599,928; 5,688,941; 6,294,664; 6,320,017; 6,576,752; 6,783,931; 6,900,297; 7,037,646; and 8,106,022, the entire contents of each of which are hereby incorporated herein by reference.

(454) It is not necessary for all positions in a given compound to be uniformly modified, and in fact more than one of the aforementioned modifications can be incorporated in a single compound or even at a single nucleoside within an iRNA. The present invention also includes iRNA compounds that are chimeric compounds.

(455) “Chimeric” iRNA compounds or “chimeras,” in the context of this invention, are iRNA compounds, such as, dsRNAi agents, that contain two or more chemically distinct regions, each made up of at least one monomer unit, i.e., a nucleotide in the case of a dsRNA compound. These iRNAs typically contain at least one region wherein the RNA is modified so as to confer upon the iRNA increased resistance to nuclease degradation, increased cellular uptake, or increased binding affinity for the target nucleic acid. An additional region of the iRNA can serve as a substrate for enzymes capable of cleaving RNA:DNA or RNA:RNA hybrids. By way of example, RNase H is a cellular endonuclease which cleaves the RNA strand of an RNA:DNA duplex. Activation of RNase H, therefore, results in cleavage of the RNA target, thereby greatly enhancing the efficiency of iRNA inhibition of gene expression. Consequently, comparable results can often be obtained with shorter iRNAs when chimeric dsRNAs are used, compared to phosphorothioate deoxy dsRNAs hybridizing to the same target region. Cleavage of the RNA target can be routinely detected by gel electrophoresis and, if necessary, associated nucleic acid hybridization techniques known in the art.

(456) In certain instances, the RNA of an iRNA can be modified by a non-ligand group. A number of non-ligand molecules have been conjugated to iRNAs in order to enhance the activity, cellular

distribution or cellular uptake of the iRNA, and procedures for performing such conjugations are available in the scientific literature. Such non-ligand moieties have included lipid moieties, such as cholesterol (Kubo, T. et al., *Biochem. Biophys. Res. Comm.*, 2007, 365(1):54-61; Letsinger et al., *Proc. Natl. Acad. Sci. USA*, 1989, 86:6553), cholic acid (Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1994, 4:1053), a thioether, e.g., hexyl-S-tritylthiol (Manoharan et al., *Ann. N.Y. Acad. Sci.*, 1992, 660:306; Manoharan et al., *Bioorg. Med. Chem. Lett.*, 1993, 3:2765), a thiocholesterol (Oberhauser et al., *Nucl. Acids Res.*, 1992, 20:533), an aliphatic chain, e.g., dodecandiol or undecyl residues (Saison-Behmoaras et al., *EMBO J.*, 1991, 10:111; Kabanov et al., *FEBS Lett.*, 1990, 259:327; Svinarchuk et al., *Biochimie*, 1993, 75:49), a phospholipid, e.g., di-hexadecyl-rac-glycerol or triethylammonium 1,2-di-O-hexadecyl-rac-glycero-3-H-phosphonate (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651; Shea et al., *Nucl. Acids Res.*, 1990, 18:3777), a polyamine or a polyethylene glycol chain (Manoharan et al., *Nucleosides & Nucleotides*, 1995, 14:969), or adamantane acetic acid (Manoharan et al., *Tetrahedron Lett.*, 1995, 36:3651), a palmityl moiety (Mishra et al., *Biochim. Biophys. Acta*, 1995, 1264:229), or an octadecylamine or hexylamino-carbonyl-oxycholesterol moiety (Crooke et al., *J. Pharmacol. Exp. Ther.*, 1996, 277:923).

Representative United States patents that teach the preparation of such RNA conjugates have been listed above. Typical conjugation protocols involve the synthesis of RNAs bearing an aminolinker at one or more positions of the sequence. The amino group is then reacted with the molecule being conjugated using appropriate coupling or activating reagents. The conjugation reaction can be performed either with the RNA still bound to the solid support or following cleavage of the RNA, in solution phase. Purification of the RNA conjugate by HPLC typically affords the pure conjugate.

V. Delivery of an iRNA of the Invention

(457) The delivery of an iRNA of the invention to a cell e.g., a cell within a subject, such as a human subject (e.g., a subject in need thereof, such as a subject susceptible to or diagnosed with a KHK-associated disorder, as described herein) can be achieved in a number of different ways. For example, delivery may be performed by contacting a cell with an iRNA of the invention either in vitro or in vivo. In vivo delivery may also be performed directly by administering a composition comprising an iRNA, e.g., a dsRNA, to a subject. Alternatively, in vivo delivery may be performed indirectly by administering one or more vectors that encode and direct the expression of the iRNA. These alternatives are discussed further below.

(458) In general, any method of delivering a nucleic acid molecule (in vitro or in vivo) can be adapted for use with an iRNA of the invention (see e.g., Akhtar S. and Julian R L. (1992) *Trends Cell. Biol.* 2(5):139-144 and WO94/02595, which are incorporated herein by reference in their entireties). For in vivo delivery, factors to consider in order to deliver an iRNA molecule include, for example, biological stability of the delivered molecule, prevention of non-specific effects, and accumulation of the delivered molecule in the target tissue. RNA interference has also shown success with local delivery to the CNS by direct injection (Dorn, G., et al. (2004) *Nucleic Acids* 32:e49; Tan, P H., et al (2005) *Gene Ther.* 12:59-66; Makimura, H., et al (2002) *BMC Neurosci.* 3:18; Shishkina, G T., et al (2004) *Neuroscience* 129:521-528; Thakker, E R., et al (2004) *Proc. Natl. Acad. Sci. U.S.A.* 101:17270-17275; Akaneya, Y., et al (2005) *J. Neurophysiol.* 93:594-602). Modification of the RNA or the pharmaceutical carrier can also permit targeting of the iRNA to the target tissue and avoid undesirable off-target effects. iRNA molecules can be modified by chemical conjugation to lipophilic groups such as cholesterol to enhance cellular uptake and prevent degradation. For example, an iRNA directed against ApoB conjugated to a lipophilic cholesterol moiety was injected systemically into mice and resulted in knockdown of apoB mRNA in both the liver and jejunum (Soutschek, J., et al (2004) *Nature* 432:173-178).

(459) In an alternative embodiment, the iRNA can be delivered using drug delivery systems such as a nanoparticle, a dendrimer, a polymer, liposomes, or a cationic delivery system. Positively charged cationic delivery systems facilitate binding of an iRNA molecule (negatively charged) and also enhance interactions at the negatively charged cell membrane to permit efficient uptake of an iRNA

by the cell. Cationic lipids, dendrimers, or polymers can either be bound to an iRNA, or induced to form a vesicle or micelle (see e.g., Kim S H, et al (2008) *Journal of Controlled Release* 129(2):107-116) that encases an iRNA. The formation of vesicles or micelles further prevents degradation of the iRNA when administered systemically. Methods for making and administering cationic-iRNA complexes are well within the abilities of one skilled in the art (see e.g., Sorensen, D R, et al (2003) *J. Mol. Biol* 327:761-766; Verma, U N, et al (2003) *Clin. Cancer Res.* 9:1291-1300; Arnold, A S et al (2007) *J. Hypertens.* 25:197-205, which are incorporated herein by reference in their entirety). Some non-limiting examples of drug delivery systems useful for systemic delivery of iRNAs include DOTAP (Sorensen, D R., et al (2003), supra; Verma, U N, et al (2003), supra), "solid nucleic acid lipid particles" (Zimmermann, T S, et al (2006) *Nature* 441:111-114), cardiolipin (Chien, P Y, et al (2005) *Cancer Gene Ther.* 12:321-328; Pal, A, et al (2005) *Int J. Oncol.* 26:1087-1091), polyethyleneimine (Bonnet M E, et al (2008) *Pharm. Res.* August 16 Epub ahead of print; Aigner, A. (2006) *J. Biomed. Biotechnol.* 71659), Arg-Gly-Asp (RGD) peptides (Liu, S. (2006) *Mol. Pharm.* 3:472-487), and polyamidoamines (Tomalia, D A, et al (2007) *Biochem. Soc. Trans.* 35:61-67; Yoo, H., et al (1999) *Pharm. Res.* 16:1799-1804). In some embodiments, an iRNA forms a complex with cyclodextrin for systemic administration. Methods for administration and pharmaceutical compositions of iRNAs and cyclodextrins can be found in U.S. Pat. No. 7,427,605, which is herein incorporated by reference in its entirety.

(460) A. Vector Encoded iRNAs of the Invention

(461) iRNA targeting the KHK gene can be expressed from transcription units inserted into DNA or RNA vectors (see, e.g., Couture, A, et al., *TIG.* (1996), 12:5-10; Skillern, A, et al., International PCT Publication No. WO 00/22113, Conrad, International PCT Publication No. WO 00/22114, and Conrad, U.S. Pat. No. 6,054,299). Expression can be transient (on the order of hours to weeks) or sustained (weeks to months or longer), depending upon the specific construct used and the target tissue or cell type. These transgenes can be introduced as a linear construct, a circular plasmid, or a viral vector, which can be an integrating or non-integrating vector. The transgene can also be constructed to permit it to be inherited as an extrachromosomal plasmid (Gassmann, et al., *Proc. Natl. Acad. Sci. USA* (1995) 92:1292).

(462) Viral vector systems which can be utilized with the methods and compositions described herein include, but are not limited to, (a) adenovirus vectors; (b) retrovirus vectors, including but not limited to lentiviral vectors, moloney murine leukemia virus, etc.; (c) adeno-associated virus vectors; (d) herpes simplex virus vectors; (e) SV 40 vectors; (f) polyoma virus vectors; (g) papilloma virus vectors; (h) picornavirus vectors; (i) pox virus vectors such as an orthopox, e.g., vaccinia virus vectors or avipox, e.g. canary pox or fowl pox; and (j) a helper-dependent or gutless adenovirus. Replication-defective viruses can also be advantageous. Different vectors will or will not become incorporated into the cells' genome. The constructs can include viral sequences for transfection, if desired. Alternatively, the construct can be incorporated into vectors capable of episomal replication, e.g. EPV and EBV vectors. Constructs for the recombinant expression of an iRNA will generally require regulatory elements, e.g., promoters, enhancers, etc., to ensure the expression of the iRNA in target cells. Other aspects to consider for vectors and constructs are known in the art.

VI. Pharmaceutical Compositions of the Invention

(463) The present invention also includes pharmaceutical compositions and formulations which include the iRNAs of the invention. In one embodiment, provided herein are pharmaceutical compositions containing an iRNA, as described herein, and a pharmaceutically acceptable carrier. The pharmaceutical compositions containing the iRNA are useful for treating a disease or disorder associated with the expression or activity of a KHK gene. Such pharmaceutical compositions are formulated based on the mode of delivery. One example is compositions that are formulated for systemic administration via parenteral delivery, e.g., by subcutaneous (SC) or intravenous (IV) delivery. The pharmaceutical compositions of the invention may be administered in dosages

sufficient to inhibit expression of a KHK gene.

(464) In some embodiments, the pharmaceutical compositions of the invention are sterile. In another embodiment, the pharmaceutical compositions of the invention are pyrogen free.

(465) The pharmaceutical compositions of the invention may be administered in dosages sufficient to inhibit expression of a KHK gene. In general, a suitable dose of an iRNA of the invention will be in the range of about 0.001 to about 200.0 milligrams per kilogram body weight of the recipient per day, generally in the range of about 1 to 50 mg per kilogram body weight per day. Typically, a suitable dose of an iRNA of the invention will be in the range of about 0.1 mg/kg to about 5.0 mg/kg, about 0.3 mg/kg and about 3.0 mg/kg. A repeat-dose regimen may include administration of a therapeutic amount of iRNA on a regular basis, such as every month, once every 3-6 months, or once a year. In certain embodiments, the iRNA is administered about once per month to about once per six months.

(466) After an initial treatment regimen, the treatments can be administered on a less frequent basis. Duration of treatment can be determined based on the severity of disease.

(467) In other embodiments, a single dose of the pharmaceutical compositions can be long lasting, such that doses are administered at not more than 1, 2, 3, or 4 month intervals. In some embodiments of the invention, a single dose of the pharmaceutical compositions of the invention is administered about once per month. In other embodiments of the invention, a single dose of the pharmaceutical compositions of the invention is administered quarterly (i.e., about every three months). In other embodiments of the invention, a single dose of the pharmaceutical compositions of the invention is administered twice per year (i.e., about once every six months).

(468) The skilled artisan will appreciate that certain factors can influence the dosage and timing required to effectively treat a subject, including but not limited to mutations present in the subject, previous treatments, the general health or age of the subject, and other diseases present. Moreover, treatment of a subject with a prophylactically or therapeutically effective amount, as appropriate, of a composition can include a single treatment or a series of treatments.

(469) The iRNA can be delivered in a manner to target a particular tissue (e.g., hepatocytes).

(470) Pharmaceutical compositions of the present invention include, but are not limited to, solutions, emulsions, and liposome-containing formulations. These compositions can be generated from a variety of components that include, but are not limited to, preformed liquids, self-emulsifying solids, and self-emulsifying semisolids. Formulations include those that target the liver.

(471) The pharmaceutical formulations of the present invention, which can conveniently be presented in unit dosage form, can be prepared according to conventional techniques well known in the pharmaceutical industry. Such techniques include the step of bringing into association the active ingredients with the pharmaceutical carrier(s) or excipient(s). In general, the formulations are prepared by uniformly and intimately bringing into association the active ingredients with liquid carriers.

(472) A. Additional Formulations

(473) i. Emulsions

(474) The compositions of the present invention can be prepared and formulated as emulsions. Emulsions are typically heterogeneous systems of one liquid dispersed in another in the form of droplets usually exceeding 0.1 μm in diameter (see e.g., Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems, Allen, L V, Popovich N G, and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Idson, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199; Rosoff, in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., Volume 1, p. 245; Block in Pharmaceutical Dosage Forms, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 2, p. 335; Higuchi et al., in Remington's Pharmaceutical Sciences, Mack Publishing Co., Easton, Pa., 1985, p. 301). Emulsions

are often biphasic systems comprising two immiscible liquid phases intimately mixed and dispersed with each other. In general, emulsions can be of either the water-in-oil (w/o) or the oil-in-water (o/w) variety. When an aqueous phase is finely divided into and dispersed as minute droplets into a bulk oily phase, the resulting composition is called a water-in-oil (w/o) emulsion. Alternatively, when an oily phase is finely divided into and dispersed as minute droplets into a bulk aqueous phase, the resulting composition is called an oil-in-water (o/w) emulsion. Emulsions can contain additional components in addition to the dispersed phases, and the active drug which can be present as a solution either in the aqueous phase, oily phase or itself as a separate phase. Pharmaceutical excipients such as emulsifiers, stabilizers, dyes, and anti-oxidants can also be present in emulsions as needed. Pharmaceutical emulsions can also be multiple emulsions that are comprised of more than two phases such as, for example, in the case of oil-in-water-in-oil (o/w/o) and water-in-oil-in-water (w/o/w) emulsions. Such complex formulations often provide certain advantages that simple binary emulsions do not. Multiple emulsions in which individual oil droplets of an o/w emulsion enclose small water droplets constitute a w/o/w emulsion. Likewise a system of oil droplets enclosed in globules of water stabilized in an oily continuous phase provides an o/w/o emulsion.

(475) Emulsions are characterized by little or no thermodynamic stability. Often, the dispersed or discontinuous phase of the emulsion is well dispersed into the external or continuous phase and maintained in this form through the means of emulsifiers or the viscosity of the formulation. Other means of stabilizing emulsions entail the use of emulsifiers that can be incorporated into either phase of the emulsion. Emulsifiers can broadly be classified into four categories: synthetic surfactants, naturally occurring emulsifiers, absorption bases, and finely dispersed solids (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, L V, Popovich N G, and Ansel H C, 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Idson, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199).

(476) Synthetic surfactants, also known as surface active agents, have found wide applicability in the formulation of emulsions and have been reviewed in the literature (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, L V, Popovich N G, and Ansel H C, 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rieger, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 285; Idson, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), Marcel Dekker, Inc., New York, N.Y., 1988, volume 1, p. 199). Surfactants are typically amphiphilic and comprise a hydrophilic and a hydrophobic portion. The ratio of the hydrophilic to the hydrophobic nature of the surfactant has been termed the hydrophile/lipophile balance (HLB) and is a valuable tool in categorizing and selecting surfactants in the preparation of formulations. Surfactants can be classified into different classes based on the nature of the hydrophilic group: nonionic, anionic, cationic, and amphoteric (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, L V, Popovich N G, and Ansel H C, 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rieger, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 285).

(477) A large variety of non-emulsifying materials are also included in emulsion formulations and contribute to the properties of emulsions. These include fats, oils, waxes, fatty acids, fatty alcohols, fatty esters, humectants, hydrophilic colloids, preservatives, and antioxidants (Block, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 335; Idson, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199).

(478) The application of emulsion formulations via dermatological, oral, and parenteral routes, and methods for their manufacture have been reviewed in the literature (see e.g., Ansel's *Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, L V, Popovich N G, and Ansel H

C, 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Idson, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 199).

(479) ii. Microemulsions

(480) In one embodiment of the present invention, the compositions of iRNAs and nucleic acids are formulated as microemulsions. A microemulsion can be defined as a system of water, oil, and amphiphile which is a single optically isotropic and thermodynamically stable liquid solution (see e.g., *Ansel's Pharmaceutical Dosage Forms and Drug Delivery Systems*, Allen, L V, Popovich N G, and Ansel H C., 2004, Lippincott Williams & Wilkins (8th ed.), New York, NY; Rosoff, in *Pharmaceutical Dosage Forms*, Lieberman, Rieger and Banker (Eds.), 1988, Marcel Dekker, Inc., New York, N.Y., volume 1, p. 245). Typically microemulsions are systems that are prepared by first dispersing an oil in an aqueous surfactant solution and then adding a sufficient amount of a fourth component, generally an intermediate chain-length alcohol to form a transparent system. Therefore, microemulsions have also been described as thermodynamically stable, isotropically clear dispersions of two immiscible liquids that are stabilized by interfacial films of surface-active molecules (Leung and Shah, in: *Controlled Release of Drugs: Polymers and Aggregate Systems*, Rosoff, M., Ed., 1989, VCH Publishers, New York, pages 185-215).

(481) iii. Microparticles

(482) An iRNA of the invention may be incorporated into a particle, e.g., a microparticle. Microparticles can be produced by spray-drying, but may also be produced by other methods including lyophilization, evaporation, fluid bed drying, vacuum drying, or a combination of these techniques.

(483) iv. Penetration Enhancers

(484) In one embodiment, the present invention employs various penetration enhancers to effect the efficient delivery of nucleic acids, particularly iRNAs, to the skin of animals. Most drugs are present in solution in both ionized and nonionized forms. However, usually only lipid soluble or lipophilic drugs readily cross cell membranes. It has been discovered that even non-lipophilic drugs can cross cell membranes if the membrane to be crossed is treated with a penetration enhancer. In addition to aiding the diffusion of non-lipophilic drugs across cell membranes, penetration enhancers also enhance the permeability of lipophilic drugs.

(485) Penetration enhancers can be classified as belonging to one of five broad categories, i.e., surfactants, fatty acids, bile salts, chelating agents, and non-chelating non-surfactants (see e.g., Malmsten, M. *Surfactants and polymers in drug delivery*, Informa Health Care, New York, NY, 2002; Lee et al., *Critical Reviews in Therapeutic Drug Carrier Systems*, 1991, p. 92). Each of the above mentioned classes of penetration enhancers and their use in manufacture of pharmaceutical compositions and delivery of pharmaceutical agents are well known in the art.

(486) v. Excipients

(487) In contrast to a carrier compound, a "pharmaceutical carrier" or "excipient" is a pharmaceutically acceptable solvent, suspending agent, or any other pharmacologically inert vehicle for delivering one or more nucleic acids to an animal. The excipient can be liquid or solid and is selected, with the planned manner of administration in mind, so as to provide for the desired bulk, consistency, etc., when combined with a nucleic acid and the other components of a given pharmaceutical composition. Such agent are well known in the art.

(488) vi. Other Components

(489) The compositions of the present invention can additionally contain other adjunct components conventionally found in pharmaceutical compositions, at their art-established usage levels. Thus, for example, the compositions can contain additional, compatible, pharmaceutically-active materials such as, for example, antipruritics, astringents, local anesthetics or anti-inflammatory agents, or can contain additional materials useful in physically formulating various dosage forms of the compositions of the present invention, such as dyes, flavoring agents, preservatives,

antioxidants, opacifiers, thickening agents and stabilizers. However, such materials, when added, should not unduly interfere with the biological activities of the components of the compositions of the present invention. The formulations can be sterilized and, if desired, mixed with auxiliary agents, e.g., lubricants, preservatives, stabilizers, wetting agents, emulsifiers, salts for influencing osmotic pressure, buffers, colorings, flavorings, or aromatic substances, and the like which do not deleteriously interact with the nucleic acid(s) of the formulation.

(490) Aqueous suspensions can contain substances which increase the viscosity of the suspension including, for example, sodium carboxymethylcellulose, sorbitol, or dextran. The suspension can also contain stabilizers.

(491) In some embodiments, pharmaceutical compositions featured in the invention include (a) one or more iRNA and (b) one or more agents which function by a non-iRNA mechanism and which are useful in treating a KHK-associated disorder, e.g., liver disease (e.g., fatty liver, steatohepatitis, non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, hyperuricemia, gout, eating disorders, and excessive sugar craving.

(492) Toxicity and prophylactic efficacy of such compounds can be determined by standard pharmaceutical procedures in cell cultures or experimental animals, e.g., for determining the LD50 (the dose lethal to 50% of the population) and the ED50 (the dose prophylactically effective in 50% of the population). The dose ratio between toxic and therapeutic effects is the therapeutic index and it can be expressed as the ratio LD50/ED50. Compounds that exhibit high therapeutic indices are preferred.

(493) The data obtained from cell culture assays and animal studies can be used in formulating a range of dosage for use in humans. The dosage of compositions featured herein in the invention lies generally within a range of circulating concentrations that include the ED50, such as, an ED80 or ED90, with little or no toxicity. The dosage can vary within this range depending upon the dosage form employed and the route of administration utilized. For any compound used in the methods featured in the invention, the prophylactically effective dose can be estimated initially from cell culture assays. A dose can be formulated in animal models to achieve a circulating plasma concentration range of the compound or, when appropriate, of the polypeptide product of a target sequence (e.g., achieving a decreased concentration of the polypeptide) that includes the IC50 (i.e., the concentration of the test compound which achieves a half-maximal inhibition of symptoms) or higher levels of inhibition as determined in cell culture. Such information can be used to more accurately determine useful doses in humans. Levels in plasma can be measured, for example, by high performance liquid chromatography.

(494) In addition to their administration, as discussed above, the iRNAs featured in the invention can be administered in combination with other known agents used for the prevention or treatment of a KHK-associated disorder, e.g., liver disease (e.g., fatty liver, steatohepatitis, non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, hyperuricemia, gout, eating disorders, and excessive sugar craving. In any event, the administering physician can adjust the amount and timing of iRNA administration on the basis of results observed using standard measures of efficacy known in the art or described herein.

VII. Methods For Inhibiting KHK Expression

(495) The present invention also provides methods of inhibiting expression of a KHK gene in a cell. The methods include contacting a cell with an RNAi agent, e.g., double stranded RNAi agent, in an amount effective to inhibit expression of KHK in the cell, thereby inhibiting expression of KHK in the cell.

(496) Contacting of a cell with an iRNA, e.g., a double stranded RNA agent, may be done in vitro or in vivo. Contacting a cell in vivo with the iRNA includes contacting a cell or group of cells within a subject, e.g., a human subject, with the iRNA. Combinations of in vitro and in vivo methods of contacting a cell are also possible. Contacting a cell may be direct or indirect, as discussed above. Furthermore, contacting a cell may be accomplished via a targeting ligand, including any ligand described herein or known in the art. In some embodiments, the targeting ligand is a carbohydrate moiety, e.g., a GalNAc.sub.3 ligand, or any other ligand that directs the RNAi agent to a site of interest.

(497) The term “inhibiting,” as used herein, is used interchangeably with “reducing,” “silencing,” “downregulating”, “suppressing”, and other similar terms, and includes any level of inhibition.

(498) The phrase “inhibiting expression of a KHK” is intended to refer to inhibition of expression of any KHK gene (such as, e.g., a mouse KHK 3 gene, a rat KHK gene, a monkey KHK gene, or a human KHK gene) as well as variants or mutants of a KHK gene. Thus, the KHK gene may be a wild-type KHK gene, a mutant KHK gene, or a transgenic KHK gene in the context of a genetically manipulated cell, group of cells, or organism.

(499) “Inhibiting expression of a KHK gene” includes any level of inhibition of a KHK gene, e.g., at least partial suppression of the expression of a KHK gene. The expression of the KHK gene may be assessed based on the level, or the change in the level, of any variable associated with KHK gene expression, e.g., KHK mRNA level or KHK protein level. This level may be assessed in an individual cell or in a group of cells, including, for example, a sample derived from a subject.

(500) Inhibition may be assessed by a decrease in an absolute or relative level of one or more variables that are associated with KHK expression compared with a control level. The control level may be any type of control level that is utilized in the art, e.g., a pre-dose baseline level, or a level determined from a similar subject, cell, or sample that is untreated or treated with a control (such as, e.g., buffer only control or inactive agent control).

(501) It is understood that the degree and duration of elevation of a sign of a KHK-associated disease will vary depending upon the sign. For example, lipid signs, e.g., fasting lipid levels, NAFLD, NASH, obesity; signs of liver and kidney function, and glucose or insulin response, are durable signs that will not vary in a clinically significant manner within a day or even within a week. Other markers, e.g., serum uric acid and glucose levels, and urine fructose levels, will vary within and likely between days. Blood pressure can be elevated transiently and durably in response to fructose. As fructose likely results in weight gain at least in part by reducing satiety, fructose consumption in conjunction with caloric limitation may not result in weight gain.

(502) Further, depending on the disease state in the subject, as many as one third of adults and two thirds of children malabsorb fructose (Johnson et al. (2013) Diabetes. 62:3307-3315), e.g., due to variations in expression of the GLUTS transporter in the gut. However, repeated exposure to fructose can increase fructose absorption. Fructose metabolism has demonstrated to be different depending on the source of fructose, e.g., in high fructose corn syrup vs. in natural fruit, and at high concentrations, such as those provided by soft drinks, glucose can be converted to fructose by the polyol pathway. However, fructose will have more metabolic effects than glucose. Body composition, e.g., lean body mass, has also been demonstrated to affect fructose metabolism. Therefore, both the timing of testing and controls must be carefully selected.

(503) In some embodiments of the methods of the invention, expression of a KHK gene is inhibited by at least 50%, 55%, 60%, 65%, 70%, 75%, 80%, 85%, 90%, or 95%, or to below the level of detection of the assay. In some embodiments, expression of a KHK gene is inhibited by at least

70%. In some embodiments, expression level is determined using the assay method provided in Example 2 with a 10 nM siRNA concentration in the appropriate species matched cell line.

(504) In certain embodiments, inhibition of expression in vivo is determined by knockdown of the human gene in a rodent expressing the human gene, e.g., an AAV-infected mouse expressing the human target gene (i.e., KHK), e.g., when administered as a single dose, e.g., at 3 mg/kg at the nadir of RNA expression. Knockdown of expression of an endogenous gene in a model animal system can also be determined, e.g., after administration of a single dose at, e.g., 3 mg/kg at the nadir of RNA expression. Such systems are useful when the nucleic acid sequence of the human gene and the model animal gene are sufficiently close such that the human iRNA provides effective knockdown of the model animal gene. RNA expression in liver is determined using the PCR methods provided in Example 2.

(505) Inhibition of the expression of a KHK gene may be manifested by a reduction of the amount of mRNA expressed by a first cell or group of cells (such cells may be present, for example, in a sample derived from a subject) in which a KHK gene is transcribed and which has or have been treated (e.g., by contacting the cell or cells with an iRNA of the invention, or by administering an iRNA of the invention to a subject in which the cells are or were present) such that the expression of a KHK gene is inhibited, as compared to a second cell or group of cells substantially identical to the first cell or group of cells but which has not or have not been so treated (control cell(s) not treated with an iRNA or not treated with an iRNA targeted to the gene of interest). In some embodiments, the inhibition is assessed by the method provided in Example 2 using a 10 nM siRNA concentration in the species matched cell line and expressing the level of mRNA in treated cells as a percentage of the level of mRNA in control cells, using the following formula:

$$(506) \frac{(\text{mRNA}_{\text{in control cells}}) - (\text{mRNA}_{\text{in treated cells}})}{(\text{mRNA}_{\text{in control cells}})} \cdot \text{Math. } 100\%$$

(507) In other embodiments, inhibition of the expression of a KHK gene may be assessed in terms of a reduction of a parameter that is functionally linked to KHK gene expression, e.g., KHK protein level in blood or serum from a subject. KHK gene silencing may be determined in any cell expressing KHK, either endogenous or heterologous from an expression construct, and by any assay known in the art.

(508) Inhibition of the expression of a KHK protein may be manifested by a reduction in the level of the KHK protein that is expressed by a cell or group of cells or in a subject sample (e.g., the level of protein in a blood sample derived from a subject). As explained above, for the assessment of mRNA suppression, the inhibition of protein expression levels in a treated cell or group of cells may similarly be expressed as a percentage of the level of protein in a control cell or group of cells, or the change in the level of protein in a subject sample, e.g., blood or serum derived therefrom.

(509) A control cell, a group of cells, or subject sample that may be used to assess the inhibition of the expression of a KHK gene includes a cell, group of cells, or subject sample that has not yet been contacted with an RNAi agent of the invention. For example, the control cell, group of cells, or subject sample may be derived from an individual subject (e.g., a human or animal subject) prior to treatment of the subject with an RNAi agent or an appropriately matched population control.

(510) The level of KHK mRNA that is expressed by a cell or group of cells may be determined using any method known in the art for assessing mRNA expression. In one embodiment, the level of expression of KHK in a sample is determined by detecting a transcribed polynucleotide, or portion thereof, e.g., mRNA of the KHK gene. RNA may be extracted from cells using RNA extraction techniques including, for example, using acid phenol/guanidine isothiocyanate extraction (RNAzol B; Biogenesis), RNeasy™ RNA preparation kits (Qiagen®) or PAXgene™ (PreAnalytix™, Switzerland). Typical assay formats utilizing ribonucleic acid hybridization include nuclear run-on assays, RT-PCR, RNase protection assays, northern blotting, in situ hybridization, and microarray analysis.

(511) In some embodiments, the level of expression of KHK is determined using a nucleic acid probe. The term “probe”, as used herein, refers to any molecule that is capable of selectively

binding to a specific KHK. Probes can be synthesized by one of skill in the art, or derived from appropriate biological preparations. Probes may be specifically designed to be labeled. Examples of molecules that can be utilized as probes include, but are not limited to, RNA, DNA, proteins, antibodies, and organic molecules.

(512) Isolated mRNA can be used in hybridization or amplification assays that include, but are not limited to, Southern or northern analyses, polymerase chain reaction (PCR) analyses and probe arrays. One method for the determination of mRNA levels involves contacting the isolated mRNA with a nucleic acid molecule (probe) that can hybridize to KHK mRNA. In one embodiment, the mRNA is immobilized on a solid surface and contacted with a probe, for example by running the isolated mRNA on an agarose gel and transferring the mRNA from the gel to a membrane, such as nitrocellulose. In an alternative embodiment, the probe(s) are immobilized on a solid surface and the mRNA is contacted with the probe(s), for example, in an Affymetrix® gene chip array. A skilled artisan can readily adapt known mRNA detection methods for use in determining the level of KHK mRNA.

(513) An alternative method for determining the level of expression of KHK in a sample involves the process of nucleic acid amplification or reverse transcriptase (to prepare cDNA) of for example mRNA in the sample, e.g., by RT-PCR (the experimental embodiment set forth in Mullis, 1987, U.S. Pat. No. 4,683,202), ligase chain reaction (Barany (1991) *Proc. Natl. Acad. Sci. USA* 88:189-193), self sustained sequence replication (Guatelli et al. (1990) *Proc. Natl. Acad. Sci. USA* 87:1874-1878), transcriptional amplification system (Kwoh et al. (1989) *Proc. Natl. Acad. Sci. USA* 86:1173-1177), Q-Beta Replicase (Lizardi et al. (1988) *Bio/Technology* 6:1197), rolling circle replication (Lizardi et al., U.S. Pat. No. 5,854,033) or any other nucleic acid amplification method, followed by the detection of the amplified molecules using techniques well known to those of skill in the art. These detection schemes are especially useful for the detection of nucleic acid molecules if such molecules are present in very low numbers. In particular aspects of the invention, the level of expression of KHK is determined by quantitative fluorogenic RT-PCR (i.e., the TaqMan™ System). In some embodiments, expression level is determined by the method provided in Example 2 using, e.g., a 10 nM siRNA concentration, in the species matched cell line.

(514) The expression levels of KHK mRNA may be monitored using a membrane blot (such as used in hybridization analysis such as northern, Southern, dot, and the like), or microwells, sample tubes, gels, beads or fibers (or any solid support comprising bound nucleic acids). See U.S. Pat. Nos. 5,770,722, 5,874,219, 5,744,305, 5,677,195 and 5,445,934, which are incorporated herein by reference. The determination of KHK expression level may also comprise using nucleic acid probes in solution.

(515) In some embodiments, the level of mRNA expression is assessed using branched DNA (bDNA) assays or real time PCR (qPCR). The use of these methods is described and exemplified in the Examples presented herein. In some embodiments, expression level is determined by the method provided in Example 2 using a 10 nM siRNA concentration in the species matched cell line.

(516) The level of KHK protein expression may be determined using any method known in the art for the measurement of protein levels. Such methods include, for example, electrophoresis, capillary electrophoresis, high performance liquid chromatography (HPLC), thin layer chromatography (TLC), hyperdiffusion chromatography, fluid or gel precipitin reactions, absorption spectroscopy, a colorimetric assays, spectrophotometric assays, flow cytometry, immunodiffusion (single or double), immunoelectrophoresis, western blotting, radioimmunoassay (RIA), enzyme-linked immunosorbent assays (ELISAs), immunofluorescent assays, electrochemiluminescence assays, and the like.

(517) In some embodiments, the efficacy of the methods of the invention are assessed by a decrease in KHK mRNA or protein level (e.g., in a liver biopsy).

(518) In some embodiments of the methods of the invention, the iRNA is administered to a subject

such that the iRNA is delivered to a specific site within the subject. The inhibition of expression of KHK may be assessed using measurements of the level or change in the level of KHK mRNA or KHK protein in a sample derived from fluid or tissue from the specific site within the subject (e.g., liver or blood).

(519) As used herein, the terms detecting or determining a level of an analyte are understood to mean performing the steps to determine if a material, e.g., protein, RNA, is present. As used herein, methods of detecting or determining include detection or determination of an analyte level that is below the level of detection for the method used.

VIII. Prophylactic and Treatment Methods of the Invention

(520) The present invention also provides methods of using an iRNA of the invention or a composition containing an iRNA of the invention to inhibit expression of KHK, thereby preventing or treating an KHK-associated disorder, e.g., liver disease (e.g., fatty liver, steatohepatitis, non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, hyperuricemia, gout, eating disorders, and excessive sugar craving. In the methods of the invention the cell may be contacted with the siRNA in vitro or in vivo, i.e., the cell may be within a subject.

(521) A cell suitable for treatment using the methods of the invention may be any cell that expresses a KHK gene, e.g., a liver cell. A cell suitable for use in the methods of the invention may be a mammalian cell, e.g., a primate cell (such as a human cell, including human cell in a chimeric non-human animal, or a non-human primate cell, e.g., a monkey cell or a chimpanzee cell), or a non-primate cell. In certain embodiments, the cell is a human cell, e.g., a human liver cell. In the methods of the invention, KHK expression is inhibited in the cell by at least 50, 55, 60, 65, 70, 75, 80, 85, 90, or 95, or to a level below the level of detection of the assay.

(522) The in vivo methods of the invention may include administering to a subject a composition containing an iRNA, where the iRNA includes a nucleotide sequence that is complementary to at least a part of an RNA transcript of the KHK gene of the mammal to which the RNAi agent is to be administered. The composition can be administered by any means known in the art including, but not limited to oral, intraperitoneal, or parenteral routes, including intracranial (e.g., intraventricular, intraparenchymal, and intrathecal), intravenous, intramuscular, subcutaneous, transdermal, airway (aerosol), nasal, rectal, and topical (including buccal and sublingual) administration. In certain embodiments, the compositions are administered by intravenous infusion or injection. In certain embodiments, the compositions are administered by subcutaneous injection. In certain embodiments, the compositions are administered by intramuscular injection.

(523) In one aspect, the present invention also provides methods for inhibiting the expression of an KHK gene in a mammal. The methods include administering to the mammal a composition comprising a dsRNA that targets a KHK gene in a cell of the mammal and maintaining the mammal for a time sufficient to obtain degradation of the mRNA transcript of the KHK gene, thereby inhibiting expression of the KHK gene in the cell. Reduction in gene expression can be assessed by any methods known in the art and by methods, e.g. qRT-PCR, described herein, e.g., in Example 2. Reduction in protein production can be assessed by any methods known in the art, e.g. ELISA. In certain embodiments, a puncture liver biopsy sample serves as the tissue material for monitoring the reduction in the KHK gene or protein expression. In other embodiments, a blood sample serves as the subject sample for monitoring the reduction in the KHK protein expression. A reduction in the expression of KHK may also be assessed indirectly by measuring a decrease in fructose metabolism by detecting one or more indicators of fructose metabolism, e.g., the presence of fructose in the urine indicating lack of fructose metabolism.

(524) The present invention further provides methods of treatment in a subject in need thereof, e.g., a subject diagnosed with a KHK-associated disorder, such as, liver disease (e.g., fatty liver, steatohepatitis, non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, hyperuricemia, gout, eating disorders, and excessive sugar craving.

(525) The present invention further provides methods of prophylaxis in a subject in need thereof. The treatment methods of the invention include administering an iRNA of the invention to a subject, e.g., a subject that would benefit from a reduction of KHK expression, in a prophylactically effective amount of an iRNA targeting a KHK gene or a pharmaceutical composition comprising an iRNA targeting a KHK gene.

(526) In one aspect, the present invention provides methods of treating a subject having a disorder that would benefit from reduction in KHK expression, e.g., a KHK-associated disease, such as liver disease (e.g., fatty liver, steatohepatitis, non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, hyperuricemia, gout, eating disorders, and excessive sugar craving.

(527) In certain embodiments, the KHK-associated disorder is a liver disease, e.g., fatty liver disease such as NAFLD or NASH. In certain embodiments, the KHK-associated disorder is dyslipidemia, e.g., elevated serum triglycerides, elevated serum LDL, elevated serum cholesterol, lowered serum HDL, postprandial hypertriglyceridemia. In another embodiment, the KHK-associated disorder is a disorder of glycemic control, e.g., insulin resistance not resulting from an immune response against insulin, glucose resistance, type 2 diabetes. In certain embodiments, the KHK-associated disorder is a cardiovascular disease, e.g., hypertension, endothelial cell dysfunction. In certain embodiments, the KHK-associated disorder is a kidney disease, e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease. In certain embodiments, the disease is metabolic syndrome. In certain embodiments, the KHK-associated disorder is a disease of lipid deposition or dysfunction, e.g., visceral adipose deposition, fatty liver, obesity. In certain embodiments, the KHK-associated disorder is a disease of elevated uric acid, e.g., gout, hyperuricemia. In certain embodiments the KHK-associated disorder is an eating disorder such as excessive sugar craving.

(528) An iRNA of the invention may be administered as a “free iRNA.” A free iRNA is administered in the absence of a pharmaceutical composition. The naked iRNA may be in a suitable buffer solution. The buffer solution may comprise acetate, citrate, prolamine, carbonate, or phosphate, or any combination thereof. In one embodiment, the buffer solution is phosphate buffered saline (PBS). The pH and osmolarity of the buffer solution containing the iRNA can be adjusted such that it is suitable for administering to a subject.

(529) Alternatively, an iRNA of the invention may be administered as a pharmaceutical composition, such as a dsRNA liposomal formulation.

(530) Subjects that would benefit from an inhibition of KHK gene expression are subjects susceptible to or diagnosed with an KHK-associated disorder, such as liver disease (e.g., fatty liver, steatohepatitis, non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), cardiovascular disease (e.g.,

hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, hyperuricemia, gout, eating disorders, and excessive sugar craving.

(531) In an embodiment, the method includes administering a composition featured herein such that expression of the target a KHK gene is decreased, such as for about 1, 2, 3, 4, 5, 6, 1-6, 1-3, or 3-6 months per dose. In certain embodiments, the composition is administered once every 3-6 months.

(532) In some embodiments, the iRNAs useful for the methods and compositions featured herein specifically target RNAs (primary or processed) of the target a KHK gene. Compositions and methods for inhibiting the expression of these genes using iRNAs can be prepared and performed as described herein.

(533) Administration of the iRNA according to the methods of the invention may result prevention or treatment of a KHK-associated disorder, e.g., liver disease (e.g., fatty liver, steatohepatitis, non-alcoholic steatohepatitis (NASH)), dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), cardiovascular disease (e.g., hypertension, endothelial cell dysfunction), kidney disease (e.g., acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, chronic kidney disease), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, hyperuricemia, gout, eating disorders, and excessive sugar craving.

(534) Subjects can be administered a therapeutic amount of iRNA, such as about 0.01 mg/kg to about 200 mg/kg.

(535) In some embodiments, the iRNA is administered subcutaneously, i.e., by subcutaneous injection. One or more injections may be used to deliver the desired dose of iRNA to a subject. The injections may be repeated over a period of time.

(536) The administration may be repeated on a regular basis. In certain embodiments, after an initial treatment regimen, the treatments can be administered on a less frequent basis. A repeat-dose regimen may include administration of a therapeutic amount of iRNA on a regular basis, such as once per month to once a year. In certain embodiments, the iRNA is administered about once per month to about once every three months, or about once every three months to about once every six months.

(537) The invention further provides methods and uses of an iRNA agent or a pharmaceutical composition thereof for treating a subject that would benefit from reduction and/or inhibition of KHK gene expression, e.g., a subject having an KHK-associated disease, in combination with other pharmaceuticals and/or other therapeutic methods, e.g., with known pharmaceuticals and/or known therapeutic methods, such as, for example, those which are currently employed for treating these disorders.

(538) Accordingly, in some aspects of the invention, the methods which include either a single iRNA agent of the invention, further include administering to the subject one or more additional therapeutic agents.

(539) The iRNA agent and an additional therapeutic agent and/or treatment may be administered at the same time and/or in the same combination, e.g., parenterally, or the additional therapeutic agent can be administered as part of a separate composition or at separate times and/or by another method known in the art or described herein.

IX. Diagnostic Criteria and Treatment for KHK-Associated Diseases

(540) Diagnostic criteria, therapeutic agents, and considerations for treatment for various KHK-associated diseases are provided below.

(541) A. Hyperuricemia

(542) Serum uric acid levels are not routinely obtained as clinical lab values. However,

hyperuricemia (elevated uric acid) is associated with a number of diseases and conditions including gout, NAFLD, NASH, metabolic disorder, insulin resistance (not resulting from an immune response to insulin), cardiovascular disease, hypertension, and type 2 diabetes. It is expected that decreasing KHK expression can be useful in the prevention or treatment of one or more conditions associated with elevated serum uric acid levels. Further, it is expected that a subject would derive clinical benefit from normalization of serum uric acid levels towards or to a normal serum uric acid level, e.g., no more than 6.8 mg/dl, such as, no more than 6 mg/dl, even in the absence of overt signs or symptoms of one or more conditions associated with elevated uric acid.

(543) Animal models of hyperuricemia include, for example, high fructose diet, e.g., in rats and mice, which can induce one or more of fat accumulation including fatty liver, insulin resistance, type 2 diabetes, obesity including visceral obesity, metabolic syndrome, decreased adiponectin secretion, reduced renal function, and inflammation (see, e.g., Johnson et al. (2013) *Diabetes*. 62:3307-3315). Administration of oxonic acid, a uricase inhibitor, can also be used to induce hyperuricemia (see, e.g., Mazalli et al. (2001) *Hypertens.* 38:1101-1106). Genetic models of hyperuricemia include the B6; 129S7-Uox.sup.tm1Bay/J mouse available from Jackson Laboratory (jaxmice.jax.org/strain/002223.html) which develops hyperuricemia, with 10-fold higher levels of serum uric acid levels.

(544) Various treatments for hyperuricemia are known in the art. However, some of the agents can only be used in limited populations. For example, allopurinol is a xanthine oxidase inhibitor that is used to reduce serum uric acid levels for the treatment of a number of conditions, e.g., gout, cardiovascular disease including ischemia-reperfusion injury, hypertension, atherosclerosis, and stroke, and inflammatory diseases (Pacher et al., (2006) *Pharma. Rev.* 58:87-114). However, the use of allopurinol is contraindicated in subjects with impaired renal function, e.g., chronic kidney disease, hypothyroidism, hyperinsulinemia, or insulin resistance; or in subjects predisposed to kidney disease or impaired renal function, e.g., subjects with hypertension, metabolic disorder, diabetes, and the elderly. Further, allopurinol should not be taken by subjects taking oral coagulants or probenecid as well as subjects taking diuretics, especially thiazide diuretics or other drugs that can reduce kidney function or have potential kidney toxicity.

(545) In certain embodiments, the compositions and methods of the invention are used in combination with other compositions and methods to treat hyperuricemia, e.g., allopurinol, oxypurinol, febuxostat. In certain embodiments, the compositions and methods of the invention are used for treatment of subjects with reduced kidney function or susceptible to reduced kidney function, e.g., due to age, comorbidities, or drug interactions.

(546) B. Gout

(547) Gout affects approximately 1 in 40 adults, most commonly men between 30-60 years of age. Gout less commonly affects women. Gout is one of a few types of arthritis where future damage to joints can be avoided by treatment. Gout is characterized by recurrent attacks of acute inflammatory arthritis caused by an inflammatory reaction to uric acid crystals in the joint due to hyperuricemia resulting from insufficient renal clearance of uric acid or excessive uric acid production. Fructose associated gout is sometimes associated with variants of transporters expressed in the kidney, intestine, and liver. Gout is characterized by the formation and deposition of tophi, monosodium urate (MSU) crystals, in the joints and subcutaneously. Pain associated with gout is not related to the size of the tophi, but is a result of an immune response against the MSU crystals. There is a linear inverse relation between serum uric acid and the rate of decrease in tophus size. For example, in one study of 18 patients with non-tophaceous gout, serum uric acid declined to 2.7-5.4 mg/dL (0.16-0.32 mM) in all subjects within 3 months of starting urate lowering therapy (Pascual and Sivera (2007) *Ann. Rheum. Dis.* 66:1056-1058). However, it took 12 months with normalized serum uric acid for MSU crystals to disappear from asymptomatic knee or first MTP joints in patients who had gout for less than 10 years, vs. 18 months in those with gout for more than 10 years. Therefore, effective treatment of gout does not require complete clearance of

tophi or resolution of all symptoms, e.g., joint pain and swelling, inflammation, but simply a reduction in at least one sign or symptom of gout, e.g., reduction in severity or frequency of gout attacks, in conjunction with a reduction in serum urate levels.

(548) Animal models of gout include oxonic acid-induced hyperuricemia (see, e.g., Jang et al. (2014) *Mycobiology*. 42:296-300).

(549) Currently available treatments for gout are contraindicated or ineffective in a number of subjects. Allopurinol, a common first line treatment to reduce uric acid levels in subjects with gout, is contraindicated in a number of populations, especially those with compromised renal function, as discussed above. Further, a number of subjects fail treatment with allopurinol, e.g., subjects who suffer gout flares despite treatment, or subjects who suffer from rashes or hypersensitivity reactions associated with allopurinol.

(550) In certain embodiments, the compositions and methods of the invention are used in combination with other agents to reduce serum uric acid. In certain embodiments, the compositions and methods of the invention are used in combination with agents for treatment of symptoms of gout, e.g., analgesic or anti-inflammatory agents, e.g., NSAIDs. In certain embodiments, the compositions and methods of the invention are used for treatment of subjects with reduced kidney function or susceptible to reduced kidney function, e.g., due to age, comorbidities, or drug interactions.

(551) C. Liver disease

(552) NAFLD is associated with hyperuricemia (Xu et al. (2015) *J. Hepatol.* 62:1412-1419) which, in turn, is associated with elevated fructose metabolism. The definition of nonalcoholic fatty liver disease (NAFLD) requires that (a) there is evidence of hepatic steatosis, either by imaging or by histology and (b) there are no causes for secondary hepatic fat accumulation such as significant alcohol consumption, use of steatogenic medication or hereditary disorders. In the majority of patients, NAFLD is associated with metabolic risk factors such as obesity, diabetes mellitus, and dyslipidemia. NAFLD is histologically further categorized into nonalcoholic fatty liver (NAFL) and nonalcoholic steatohepatitis (NASH). NAFL is defined as the presence of hepatic steatosis with no evidence of hepatocellular injury in the form of ballooning of the hepatocytes. NASH is defined as the presence of hepatic steatosis and inflammation with hepatocyte injury (ballooning) with or without fibrosis (Chalasani et al. (2012) *Hepatol.* 55:2005-2023). It is generally agreed that patients with simple steatosis have very slow, if any, histological progression, while patients with NASH can exhibit histological progression to cirrhotic-stage disease. The long term outcomes of patients with NAFLD and NASH have been reported in several studies. Their findings can be summarized as follows; (a) patients with NAFLD have increased overall mortality compared to matched control populations, (b) the most common cause of death in patients with NAFLD, NAFL, and NASH is cardiovascular disease, and (c) patients with NASH (but not NAFL) have an increased liver-related mortality rate.

(553) Animal models of NAFLD include various high fat- or high fructose-fed animal models. Genetic models of NAFLD include the B6.129S7-Ld1r.sup.tm1Her/J and the B6.129S4-Pten.sup.tm1Hwu/J mice available from The Jackson Laboratory.

(554) Treatment of NAFLD is typically to manage the conditions that resulted in development of NAFLD. For example, patients with dyslipidemia are treated with agents to normalize cholesterol or triglycerides, as needed, to treat or prevent further progression of NAFLD. Patients with type 2 diabetes are treated with agents to normalize glucose or insulin sensitivity. Lifestyle changes, e.g., changes in diet and exercise, are also used to treat NAFLD. In a mouse model of NAFLD, treatment with allopurinol both prevented the development of hepatic steatosis, but also significantly ameliorated established hepatic steatosis in mice (Xu et al., *J. Hepatol.* 62:1412-1419, 2015).

(555) In certain embodiments, the compositions and methods of the invention are used in combination with other agents to reduce serum uric acid. In certain embodiments, the compositions

and methods of the invention are used in combination with agents for treatment of symptoms of NAFLD. In certain embodiments, the compositions and methods of the invention are used for treatment of subjects with reduced kidney function or susceptible to reduced kidney function, e.g., due to age, comorbidities, or drug interactions.

(556) D. Dyslipidemia, Disorders of Glycemic Control, Metabolic Syndrome, and Obesity

(557) Dyslipidemia (e.g., hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia), disorders of glycemic control (e.g., insulin resistance, type 2 diabetes), metabolic syndrome, adipocyte dysfunction, visceral adipose deposition, obesity, and excessive sugar craving are associated with elevated fructose metabolism. Characteristics or diagnostic criteria for the conditions are provided below. Animal models of metabolic disorder and the component features include various high fat- or high fructose-fed animal models. Genetic models include leptin deficient B6.Cg-Lep.sup.ob/J, commonly known as ob or ob/ob mice, which are available from The Jackson Laboratory.

(558) Normal and abnormal fasting levels of the lipids are provided in the table below.

(559) TABLE-US-00008 Lipid Value Interpretation Total Cholesterol Below 200 mg/dL Desirable 200-239 mg/dL Borderline high 240 mg/dL and above High LDL Below 70 mg/dL Best for people who have heart cholesterol disease or diabetes. Below 100 mg/dL Optimal for people at risk of heart disease. 100-129 mg/dL Near optimal if there is no heart disease. High if there is heart disease. 130-159 mg/dL Borderline high if there is no heart disease. High if there is heart disease. 160-189 mg/dL High if there is no heart disease. Very high if there is heart disease. 190 mg/dL and above Very high HDL Below 40 mg/dL (men) Poor cholesterol Below 50 mg/dL (women) 50-59 mg/dL Moderate 60 mg/dL and above Normal Triglycerides Below 150 mg/dL Desirable 150-199 mg/dL Borderline high 200-499 mg/dL High 500 mg/dL and above Very High

(560) Postprandial hypertriglyceridemia is principally initiated by overproduction or decreased catabolism of triglyceride-rich lipoproteins (TRLs) and is a consequence of predisposing genetic variations and medical conditions such as obesity and insulin resistance.

(561) Insulin resistance is characterized by the presence of at least one of: 1. A fasting blood glucose level of 100-125 mg/dL taken at two different times; or 2. An oral glucose tolerance test with a result of a glucose level of 140-199 mg/dL at 2 hours after glucose consumption.

(562) As used herein, insulin resistance does not include a lack of response to insulin as a result of an immune response to administered insulin as often occurs in late stages of insulin dependent diabetes, especially type 1 diabetes.

(563) Type 2 diabetes is characterized by at least one of: 1. A fasting blood glucose level ≥ 126 mg/dL taken at two different times; 2. A hemoglobin A1c (A1C) test with a result of $\geq 6.5\%$ or higher; or 3. An oral glucose tolerance test with a result of a glucose level ≥ 200 mg/dL at 2 hours after glucose consumption.

(564) Pharmacological treatments for type 2 diabetes and insulin resistance include treatment with agents to normalize blood sugar such as metformin (e.g., glucophage, glumetza), sulfonylureas (e.g., glyburide, glipizide, glimepiride), meglitinides (e.g., repaglinide, nateglinide), thiazolidinediones (rosiglitazone, pioglitazone), DPP-4 inhibitors (sitagliptin, saxagliptin, linagliptin), GLP-1 receptor antagonists (exenatide, liraglutide), and SGLT2 inhibitors (e.g., canagliflozin, dapagliflozin).

(565) Obesity is characterized as disease of excess body fat. Body mass index (BMI), which is calculated by dividing body weight in kilograms (kg) by height in meters (m) squared, provides a reasonable estimate of body fat for most, but not all, people. Generally, a BMI below 18.5 is characterized as underweight, 18.5 to 24.9 is normal, 25.0-29.9 is overweight, 30.0-34.9 is obese (class I), 35-39.9 is obese (class II), and 40.0 and higher is extremely obese (class III).

(566) Methods for assessment of subcutaneous vs. visceral fat are provided, for example, in Wajchenberg (2000) Subcutaneous and visceral adipose tissue: their relation to the metabolic syndrome, Endocr Rev. 21:697-738, which is incorporated herein by reference.

(567) Metabolic syndrome is characterized by a cluster of conditions defined as at least three of the five following metabolic risk factors: 1. Large waistline (≥ 35 inches for women or ≥ 40 inches for men); 2. High triglyceride level (≥ 150 mg/dl); 3. Low HDL cholesterol (≤ 50 mg/dl for women or ≤ 40 mg/dl for men); 4. Elevated blood pressure ($\geq 130/85$) or on medicine to treat high blood pressure; and 5. High fasting blood sugar (≥ 100 mg/dl) or being in medicine to treat high blood sugar.

(568) As with NAFLD, the agents for treatment of metabolic syndrome depend on the specific risk factors present, e.g., normalize lipids when lipids are abnormal, normalize glucose or insulin sensitivity when they are abnormal.

(569) Metabolic syndrome, insulin resistance, and type 2 diabetes are often associated with decreased renal function or the potential for decreased renal function.

(570) In certain embodiments, the compositions and methods of the invention are for use in treatment of subjects with dyslipidemia, disorders of glycemic control, metabolic syndrome, and obesity. For example, in certain embodiments, the compositions and methods of the invention are for use in subjects with metabolic syndrome, insulin resistance, or type 2 diabetes and chronic kidney disease. In certain embodiments, the compositions and methods are for use in subjects with metabolic syndrome, insulin resistance, or type 2 diabetes who are suffering from one or more of cardiovascular disease, hypothyroidism, or inflammatory disease; or elderly subjects (e.g., over 65). In certain embodiments, the compositions and methods are for use in subjects with metabolic syndrome, insulin resistance, or type 2 diabetes who are also taking a drug that can reduce kidney function as demonstrated by the drug label. For example, in certain embodiments the compositions and methods of the invention are for use in subjects with metabolic syndrome, insulin resistance, or type 2 diabetes who are being treated with oral coagulants or probenecid. For example, in certain embodiments the compositions and methods of the invention are for use in subjects with metabolic syndrome, insulin resistance, or type 2 diabetes who are being treated with diuretics, especially thiazide diuretics.

(571) In certain embodiments, the compositions and methods of the invention are used in combination with other agents to reduce serum uric acid. In certain embodiments, the compositions and methods of the invention are used in combination with agents for treatment of symptoms of metabolic syndrome, insulin resistance, or type 2 diabetes. In certain embodiments, subjects are treated with e.g., agents to decrease blood pressure, e.g., diuretics, beta-blockers, ACE inhibitors, angiotensin II receptor blockers, calcium channel blockers, alpha blockers, alpha-2 receptor antagonists, combined alpha- and beta-blockers, central agonists, peripheral adrenergic inhibitors, and blood vessel dilators; agents to decrease cholesterol, e.g., statins, selective cholesterol absorption inhibitors, resins, or lipid lowering therapies; or agents to normalize blood sugar, e.g., metformin, sulfonylureas, meglitinides, thiazolidinediones, DPP-4 inhibitors, GLP-1 receptor antagonists, and SGLT2 inhibitors.

(572) In certain embodiments, the compositions and methods of the invention are used for treatment of subjects with reduced kidney function or susceptible to reduced kidney function, e.g., due to age, comorbidities, or drug interactions.

(573) The iRNA and additional therapeutic agents may be administered at the same time or in the same combination, e.g., parenterally, or the additional therapeutic agent can be administered as part of a separate composition or at separate times or by another method known in the art or described herein.

(574) E. Cardiovascular Disease

(575) In certain embodiments, the compositions and methods of the invention are for use in treatment of subjects with cardiovascular disease. For example, in certain embodiments, the compositions and methods of the invention are for use in subjects with cardiovascular disease and chronic kidney disease. In certain embodiments, the compositions and methods are for use in subjects with cardiovascular disease who are suffering from one or more of metabolic disorder,

insulin resistance, hyperinsulinemia, diabetes, hypothyroidism, or inflammatory disease. In certain embodiments, the compositions and methods are for use in subjects with cardiovascular disease who are also taking a drug that can reduce kidney function as demonstrated by the drug label. For example, in certain embodiments the compositions and methods of the invention are for use in subjects with cardiovascular disease who are being treated with oral coagulants or probenecid. For example, in certain embodiments the compositions and methods of the invention are for use in subjects with cardiovascular disease who are being treated with diuretics, especially thiazide diuretics. For example, in certain embodiments the compositions and methods of the invention are for use in subjects with cardiovascular disease who have failed treatment with allopurinol.

(576) In certain embodiments, the compositions and methods of the invention are used in combination with other agents to reduce serum uric acid. In certain embodiments, the compositions and methods of the invention are used in combination with agents for treatment of symptoms of cardiovascular disease, e.g., agents to decrease blood pressure, e.g., diuretics, beta-blockers, ACE inhibitors, angiotensin II receptor blockers, calcium channel blockers, alpha blockers, alpha-2 receptor antagonists, combined alpha- and beta-blockers, central agonists, peripheral adrenergic inhibitors, and blood vessel dilators; or agents to decrease cholesterol, e.g., statins, selective cholesterol absorption inhibitors, resins, or lipid lowering therapies.

(577) F. Kidney Disease

(578) Kidney disease includes, for example, acute kidney disorder, tubular dysfunction, proinflammatory changes to the proximal tubules, and chronic kidney disease.

(579) Acute kidney (renal) failure occurs when the kidneys suddenly become unable to filter waste products from the blood resulting in accumulation of dangerous levels of wastes in serum and systemic chemical imbalance. Acute kidney failure can develop rapidly over a few hours or a few days, and is most common in individuals who are already hospitalized, particularly in critically ill individuals who need intensive care. Acute kidney failure can be fatal and requires intensive treatment. However, acute kidney failure may be reversible. If you're otherwise in good health, you may recover normal or nearly normal kidney function.

(580) Chronic kidney disease, also called chronic kidney failure, describes the gradual loss of kidney function. When chronic kidney disease reaches an advanced stage, dangerous levels of fluid, electrolytes and wastes can accumulate in the body. Signs and symptoms of kidney disease may include nausea, vomiting, loss of appetite, fatigue and weakness, sleep problems, changes in urine output, decreased mental sharpness, muscle twitches and cramps, hiccups, swelling of feet and ankles, persistent itching, chest pain, if fluid builds up around the lining of the heart, shortness of breath, if fluid builds up in the lungs, high blood pressure (hypertension) that's difficult to control. Signs and symptoms of chronic kidney disease are often nonspecific and can develop slowly, and may not appear until irreversible damage has occurred.

(581) Kidney disease is treated by removing the damaging agent or condition that is causing kidney damage, e.g. normalize blood pressure to improve kidney function, end treatment with agents that can induce kidney damage, reduce inflammation that is causing kidney damage, or by providing renal support (e.g., renal dialysis) to assist kidney function.

(582) Renal function is typically determined using one or more routine laboratory tests, BUN (blood urea nitrogen), creatinine (blood), creatinine (urine), or creatinine clearance (see, e.g., www.nlm.nih.gov/medlineplus/ency/article/003435.htm). The tests may also be diagnostic of conditions in other organs.

(583) Generally, a BUN level of 6 to 20 mg/dL is considered normal, although normal values may vary among different laboratories. Elevated BUN level can be indicative of kidney disease, including glomerulonephritis, pyelonephritis, and acute tubular necrosis, or kidney failure.

(584) A normal result for blood creatinine is 0.7 to 1.3 mg/dL for men and 0.6 to 1.1 mg/dL for women. Elevated blood creatinine can be indicative of compromised kidney function due to kidney damage or failure, infection, or reduced blood flow.

(585) Urine creatinine (24-hour sample) values can range from 500 to 2000 mg/day. Results depend on age and amount of lean body mass. Normal results are 14 to 26 mg per kg of body mass per day for men and 11 to 20 mg per kg of body mass per day for women. Abnormal results can be indicative of kidney damage, such as damage to the tubule cells, kidney failure, decreased blood flow to the kidneys, or kidney infection (pyelonephritis).

(586) The creatinine clearance test helps provide information regarding kidney function by comparing the creatinine level in urine with the creatinine level in blood. Clearance is often measured as milliliters per minute (ml/min). Normal values are 97 to 137 ml/min. for men and 88 to 128 ml/min. for women. Lower than normal creatinine clearance can be indicative of kidney damage, such as damage to the tubule cells, kidney failure, decreased blood flow to the kidneys, or reduced glomerular filtration in the kidneys.

(587) In certain embodiments, the compositions and methods of the invention can be used for the treatment of kidney disease. It is expected that such agents would not cause damage to the kidney.

X. Kits

(588) In certain aspects, the instant disclosure provides kits that include a suitable container containing a pharmaceutical formulation of a siRNA compound, e.g., a double-stranded siRNA compound, or siRNA compound, (e.g., a precursor, e.g., a larger siRNA compound which can be processed into a siRNA compound, or a DNA which encodes an siRNA compound, e.g., a double-stranded siRNA compound, or ssiRNA compound, or precursor thereof).

(589) Such kits include one or more dsRNA agent(s) and instructions for use, e.g., instructions for administering a prophylactically or therapeutically effective amount of a dsRNA agent(s). The dsRNA agent may be in a vial or a pre-filled syringe. The kits may optionally further comprise means for administering the dsRNA agent (e.g., an injection device, such as a pre-filled syringe), or means for measuring the inhibition of KHK (e.g., means for measuring the inhibition of KHK mRNA, KHK protein, and/or KHK activity). Such means for measuring the inhibition of KHK may comprise a means for obtaining a sample from a subject, such as, e.g., a plasma sample. The kits of the invention may optionally further comprise means for determining the therapeutically effective or prophylactically effective amount.

(590) In certain embodiments the individual components of the pharmaceutical formulation may be provided in one container, e.g., a vial or a pre-filled syringe. Alternatively, it may be desirable to provide the components of the pharmaceutical formulation separately in two or more containers, e.g., one container for a siRNA compound preparation, and at least another for a carrier compound. The kit may be packaged in a number of different configurations such as one or more containers in a single box. The different components can be combined, e.g., according to instructions provided with the kit. The components can be combined according to a method described herein, e.g., to prepare and administer a pharmaceutical composition. The kit can also include a delivery device.

(591) This invention is further illustrated by the following examples which should not be construed as limiting. The entire contents of all references, patents and published patent applications cited throughout this application, as well as the informal Sequence Listing, are hereby incorporated herein by reference.

EXAMPLES

Example 1. iRNA Synthesis

(592) Source of Reagents

(593) Where the source of a reagent is not specifically given herein, such reagent can be obtained from any supplier of reagents for molecular biology at a quality/purity standard for application in molecular biology.

(594) siRNA Design

(595) siRNAs targeting the human KHK (human NCBI refseqID: XM_017004061.1; NCBI GeneID: 3795) were designed using custom R and Python scripts. The human KHK REFSEQ mRNA has a length of 2283 bases.

(596) Detailed lists of the unmodified KHK sense and antisense strand nucleotide sequences are shown in Tables 2, 5 and 8. Detailed lists of the modified KHK sense and antisense strand nucleotide sequences are shown in Tables 3, 6 and 9.

(597) It is to be understood that, throughout the application, a duplex name without a decimal is equivalent to a duplex name with a decimal which merely references the batch number of the duplex. For example, AD-959917 is equivalent to AD-959917.1.

(598) siRNA Synthesis

(599) siRNAs were designed, synthesized, and prepared using methods known in the art.

(600) Briefly, siRNA sequences were synthesized on a 1 μ mol scale using a Mermade 192 synthesizer (BioAutomation) with phosphoramidite chemistry on solid supports. The solid support was controlled pore glass (500-1000 Å) loaded with a custom GalNAc ligand (3'-GalNAc conjugates), universal solid support (AM Chemicals), or the first nucleotide of interest. Ancillary synthesis reagents and standard 2-cyanoethyl phosphoramidite monomers (2'-deoxy-2'-fluoro, 2'-O-methyl, RNA, DNA) were obtained from Thermo-Fisher (Milwaukee, WI), Hongene (China), or Chemgenes (Wilmington, MA, USA). Additional phosphoramidite monomers were procured from commercial suppliers, prepared in-house, or procured using custom synthesis from various CMOs. Phosphoramidites were prepared at a concentration of 100 mM in either acetonitrile or 9:1 acetonitrile:DMF and were coupled using 5-Ethylthio-1H-tetrazole (ETT, 0.25 M in acetonitrile) with a reaction time of 400 s. Phosphorothioate linkages were generated using a 100 mM solution of 3-((Dimethylamino-methylidene) amino)-3H-1,2,4-dithiazole-3-thione (DDTT, obtained from Chemgenes (Wilmington, MA, USA)) in anhydrous acetonitrile/pyridine (9:1 v/v). Oxidation time was 5 minutes. All sequences were synthesized with final removal of the DMT group ("DMT-Off").

(601) Upon completion of the solid phase synthesis, solid-supported oligoribonucleotides were treated with 300 μ L of Methylamine (40% aqueous) at room temperature in 96 well plates for approximately 2 hours to afford cleavage from the solid support and subsequent removal of all additional base-labile protecting groups. For sequences containing any natural ribonucleotide linkages (2'-OH) protected with a tert-butyl dimethyl silyl (TBDMS) group, a second deprotection step was performed using TEA.Math.3HF (triethylamine trihydrofluoride). To each oligonucleotide solution in aqueous methylamine was added 200 μ L of dimethyl sulfoxide (DMSO) and 300 μ L TEA.Math.3HF and the solution was incubated for approximately 30 mins at 60° C. After incubation, the plate was allowed to come to room temperature and crude oligonucleotides were precipitated by the addition of 1 mL of 9:1 acetonitrile:ethanol or 1:1 ethanol:isopropanol. The plates were then centrifuged at 4° C. for 45 mins and the supernatant carefully decanted with the aid of a multichannel pipette. The oligonucleotide pellet was resuspended in 20 mM NaOAc and subsequently desalted using a HiTrap size exclusion column (5 mL, GE Healthcare) on an Agilent LC system equipped with an autosampler, UV detector, conductivity meter, and fraction collector. Desalted samples were collected in 96 well plates and then analyzed by LC-MS and UV spectrometry to confirm identity and quantify the amount of material, respectively.

(602) Duplexing of single strands was performed on a Tecan liquid handling robot. Sense and antisense single strands were combined in an equimolar ratio to a final concentration of 10 μ M in 1 $\times$ PBS in 96 well plates, the plate sealed, incubated at 100° C. for 10 minutes, and subsequently allowed to return slowly to room temperature over a period of 2-3 hours. The concentration and identity of each duplex was confirmed and then subsequently utilized for in vitro screening assays. Example 2. In Vitro Screening Methods

(603) HepG2 Cell Culture and 96-Well Transfections

(604) HepG2 cells were grown to near confluence at 37° C. in an atmosphere of 5% CO₂ in Eagle's Minimum Essential Medium (Gibco) supplemented with 10% FBS (ATCC) before being released from the plate by trypsinization. Transfection was carried out by adding 18.5 μ L of Opti-MEM plus 0.25 μ L of Lipofectamine RNAiMax per well (Invitrogen, Carlsbad CA. cat #13778-

150) to 5 µl of each siRNA duplex to each individual well in a 96-well plate. The mixture was then incubated at room temperature for 15 minutes. Eighty µl of complete growth media without antibiotic containing $\sim 2 \times 10^4$ HepG2 cells were then added to the siRNA mixture. Cells were incubated for 24 hours prior to RNA purification. Dose experiments were performed at 10 nM, 1 nM and 0.1 nM final duplex concentration.

(605) Total RNA Isolation Using DYNABEADS mRNA Isolation Kit (Invitrogen™, Part #: 610-12)

(606) Cells were lysed in 75 µl of Lysis/Binding Buffer containing 3 µL of beads per well and mixed for 10 minutes on an electrostatic shaker. The washing steps were automated on a Biotek EL406, using a magnetic plate support. Beads were washed (in 90 µL) once in Buffer A, once in Buffer B, and twice in Buffer E, with aspiration steps in between. Following a final aspiration, complete 10 µL RT mixture was added to each well, as described below.

(607) cDNA Synthesis Using ABI High Capacity cDNA Reverse Transcription Kit (Applied Biosystems, Foster City, CA, Cat #4368813)

(608) A master mix of 1 µl 10× Buffer, 0.4 µl 25× dNTPs, 1 µl Random primers, 0.5 µl Reverse Transcriptase, 0.5 µl RNase inhibitor and 6.6 µl of H.sub.2O per reaction were added per well. Plates were sealed, agitated for 10 minutes on an electrostatic shaker, and then incubated at 37 degrees C. for 2 hours. Following this, the plates were agitated at 80 degrees C. for 8 minutes.

(609) Real Time PCR

(610) Two microlitre (µl) of cDNA were added to a master mix containing 0.5 µl of human GAPDH TaqMan Probe (4326317E), 0.5 µl human KHK, 2 µl nuclease-free water and 5 µl Lightcycler 480 probe master mix (Roche Cat #04887301001) per well in a 384 well plates (Roche cat #04887301001). Real time PCR was done in a LightCycler480 Real Time PCR system (Roche).

(611) To calculate relative fold change, data were analyzed using the $\Delta\Delta C_t$ method and normalized to assays performed with cells transfected with 10 nM AD-1955, or mock transfected cells. IC50s were calculated using a 4 parameter fit model using XLFit and normalized to cells transfected with AD-1955 or mock-transfected. The sense and antisense sequences of AD-1955 are: sense:

cuuAcGcuGAGuAcuucGAdTsdT (SEQ ID NO: 18) and antisense

UCGAAGuACUcAGCGuAAGdTsdT (SEQ ID NO: 19).

(612) The results of the transfection assays of the dsRNA agents listed in Tables 2 and 3 in HepG2 cells are shown in Table 4.

(613) TABLE-US-00009 TABLE 1 Abbreviations of nucleotide monomers used in nucleic acid sequence representation. It will be understood that these monomers, when present in an oligonucleotide, are mutually linked by 5'-3'- phosphodiester bonds; and it is understood that when the nucleotide contains a 2'-fluoro modification, then the fluoro replaces the hydroxy at that position in the parent nucleotide (i.e., it is a 2'-deoxy-2'- fluoronucleotide). Abbreviation Nucleotide(s) A Adenosine-3'-phosphate Ab beta-L-adenosine-3'-phosphate Abs beta-L-adenosine-3'-phosphorothioate Af 2'-fluoroadenosine-3'-phosphate Afs 2'-fluoroadenosine-3'-phosphorothioate As adenosine-3'-phosphorothioate C cytidine-3'-phosphate Cb beta-L-cytidine-3'-phosphate Cbs beta-L-cytidine-3'-phosphorothioate Cf 2'-fluorocytidine-3'-phosphate Cfs 2'-fluorocytidine-3'-phosphorothioate Cs cytidine-3'-phosphorothioate G guanosine-3'-phosphate Gb beta-L-guanosine-3'-phosphate Gbs beta-L-guanosine-3'-phosphorothioate Gf 2'-fluoroguanosine-3'-phosphate Gfs 2'-fluoroguanosine-3'-phosphorothioate Gs guanosine-3'-phosphorothioate T 5'-methyluridine-3'-phosphate Tf 2'-fluoro-5-methyluridine-3'-phosphate Tfs 2'-fluoro-5-methyluridine-3'-phosphorothioate Ts 5-methyluridine-3'-phosphorothioate U Uridine-3'-phosphate Uf 2'-fluorouridine-3'-phosphate Ufs 2'-fluorouridine-3'-phosphorothioate Us uridine-3'-phosphorothioate N any nucleotide, modified or unmodified a 2'-O-methyladenosine-3'-phosphate as 2'-O-methyladenosine-3'-phosphorothioate c 2'-O-methylcytidine-3'-phosphate cs 2'-O-methylcytidine-3'-phosphorothioate g 2'-O-methylguanosine-3'-phosphate gs 2'-O-methylguanosine-3'-phosphorothioate t 2'-O-methyl-5-methyluridine-3'-phosphate ts 2'-O-methyl-

5-methyluridine-3'-phosphorothioate u 2'-O-methyluridine-3'-phosphorothioate s phosphorothioate linkage L10 N-(cholesterylcarboxamidocaproyl)-4-hydroxyprolinol (Hyp-C6-Chol) L96 N-[tris(GalNAc-alkyl)-amidodecanoyl]-4-hydroxyprolinol (Hyp-(GalNAc-alkyl)3)  Y34 2-hydroxymethyl-tetrahydrofuran-4-methoxy-3-phosphate (abasic 2'-OMe furanose) Y44 inverted abasic DNA (2-hydroxymethyl-tetrahydrofuran-5-phosphate) (Agn) Adenosine-glycol nucleic acid (GNA) S-Isomer (Cgn) Cytidine-glycol nucleic acid (GNA) S-Isomer (Ggn) Guanosine-glycol nucleic acid (GNA) S-Isomer (Tgn) Thymidine-glycol nucleic acid (GNA) S-Isomer P Phosphate VP Vinyl-phosphonate dA 2'-deoxyadenosine-3'-phosphate dAs 2'-deoxyadenosine-3'-phosphorothioate dC 2'-deoxycytidine-3'-phosphate dCs 2'-deoxycytidine-3'-phosphorothioate dG 2'-deoxyguanosine-3'-phosphate dGs 2'-deoxyguanosine-3'-phosphorothioate dT 2'-deoxythymidine-3'-phosphate dTs 2'-deoxythymidine-3'-phosphorothioate dU 2'-deoxyuridine dUs 2'-deoxyuridine-3'-phosphorothioate (C2p) cytidine-2'-phosphate (G2p) guanosine-2'-phosphate (U2p) uridine-2'-phosphate (A2p) adenosine-2'-phosphate (Chd) 2'-O-hexadecyl-cytidine-3'-phosphate (Ahd) 2'-O-hexadecyl-adenosine-3'-phosphate (Ghd) 2'-O-hexadecyl-guanosine-3'-phosphate (Uhd) 2'-O-hexadecyl-uridine-3'-phosphate s phosphorothioate

(614) TABLE-US-00010 TABLE 2 Unmodified Sense and Antisense Strand Sequences of KHK dsRNAs SEQ SEQ Duplex Sense ID Range in Antisense ID Range in Name Sequence 5' to 3' NO: XM_017004061.1 sequence 5' to 3' NO: XM_017004061.1 AD-1290652 GUGGUGUUUGUCAGCAAAGAU 20 810-830 AUCUUUGCUGACAAACACCACGU 244 808-830 AD-1290731 ACGUGGUGUUUGUCAGCAAU 21 808-828 AUUUGCUGACAAACACCACGU 245 806-828 AD-1290560 UCAGAGCAAUAUAAUCUCCU 22 1336-1356 AGGAAGAUUUUUUGCUCUGAGG 246 1334-1356 AD-1290854 CAGUUCAAGUGGAUCCACAUU 23 642-662 AAUGUGGAUCCACUUGAACUGGG 247 640-662 AD-1290629 AGUAGCGCAUUUUCUCUUUGU 24 122-142 ACAAAGAGAAAAUGCGCUACUUG 248 120-142 AD-1290517 GCGCAUUUUCUCUUUGCAUUU 25 126-146 AAAUGCAAAGAGAAAAUGCGCUA 249 124-146 AD-1290526 CGCAUUUUCUCUUUGCAUUCU 26 127-147 AGAAUGCAAAGAGAAAAUGCGCU 250 125-147 AD-1290559 GCAUUUUCUCUUUGCAUUCUU 27 128-148 AAGAAUGCAAAGAGAAAAUGCGC 251 126-148 AD-1290548 CAUUUUCUCUUUGCAUUCUCU 28 129-149 AGAGAAUGCAAAGAGAAAAUGCG 252 127-149 AD-1290569 UUUUCUCUUUGCAUUCUCGAU 29 131-151 AUCGAGAAUGCAAAGAGAAAAUG 253 129-151 AD-1290547 UUCUCUUUGCAUUCUCGAGAU 30 133-153 AUCUCGAGAAUGCAAAGAGAAAA 254 131-153 AD-1290544 UCUCUUUGCAUUCUCGAGAUU 31 134-154 AAUCUCGAGAAUGCAAAGAGAAA 255 132-154 AD-1290667 CUCUUUGCAUUCUCGAGAU 32 135-155 AGAUCUCGAGAAUGCAAAGAGAA 256 133-155 AD-1290685 UCUUUGCAUUCUCGAGAU 33 136-156 ACGAUCUCGAGAAUGCAAAGAGA 257 134-156 AD-1290695 UUUGCAUUCUCGAGAU 34 138-158 AAGCGAUCUCGAGAAUGCAAAGA 258 136-158 AD-1290653 UGCAUUCUCGAGAU 35 140-160 AUAAGCGAUCUCGAGAAUGCAA 259 138-160 AD-1290853 GCAUUCUCGAGAU 36 141-161 ACUAAGCGAUCUCGAGAAUGCAA 260 139-161 AD-1290990 CAUUCUCGAGAU 37 142-162 AGCUAAGCGAUCUCGAGAAUGCA 261 140-162 AD-1290540 CUUUAAAAAGGUUUGCAUCAU 38 166-186 AUGAUGCAAACCUUUUUAAAGCG 262 164-186 AD-1290580 UUUAAAAAGGUUUGCAUCAGU 39 167-187 ACUGAUGCAAACCUUUUUAAAGC 263 165-187 AD-1290664 UUUAAAAAGGUUUGCAUCAGCU 40 168-188

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181-203 AD-1290938 GCUGUGAGUCCAUCUGACAAU 42 186-206
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185-207 AD-1290914 CCAUCUGACAAGCGAGGAAAU 44 195-215
AUUUCUCGCUUGUCAGAUGGAC 268 193-215 AD-1290982
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194-216 AD-1290708 GGAAACUAAGGCUGAGAAGUU 46 210-230
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209-231 AD-1290942 AGACCUCUGGGUUGGCUUUCU 48 286-306
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512-534 AD-1290881 CCUCAUGGAAGAGAAGCAGAU 50 519-539
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518-540 AD-1290814 AUGGAAGAGAAGCAGAUCCU 52 523-543
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523-545 AD-1290964 GAAGAGAAGCAGAUCCUGUGU 54 526-546
ACACAGGAUCUGCUUCUCUCCA 278 524-546 AD-1290802
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569-591 AD-1290816 GUGGACAAGUACCCUAAGGAU 56 580-600
AUCCUUAGGGUACUUGUCCACCA 280 578-600 AD-1290821
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581-603 AD-1290870 GAGGACUCGGAGAUAAAGGUGU 58 598-618
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609-631 AD-1290682 AAGGUGUUUGUCCCAGAGAUU 60 612-632
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754-776 AD-1290530 GACUUUGAGAAGGUUGAUCUU 70 757-777
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1018-1040 AD-1290890 CUGGAGACACCUUCAAUUGCCU 102 1021-1041
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2226-2248 AD-1290600 GCUUGGCUACAGAAUUAUUGU 190 2229-2249
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2126-2148 AD-1290542 UCAAUUCUUCAUCUGUCAAU 194 2195-2215
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2197-2219 AD-1290557 GGCCUUAUAAUGUAAAGAGCU 196 2074-2094
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UAUAAUGUAAAGGGCUUUAGU 197 2095-2115 ACUAAAGCCCUUACAUAUAUG 421
2093-2115 AD-1290561 AUUUUCUCUUUGCAUUCUCGU 198 130-150
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2191-2213 AD-1290565 CUCAAUUCUUCAUCUGUCAU 200 2194-2214
AUGACAGAUGAAGAAAUUGAGGC 424 2192-2214 AD-1290574
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2069-2091 AD-1290592 CAUAUAUGUAAAGGGCUUUU 202 2093-2113
AAAAGCCCUUACAUAUAUGCU 426 2091-2113 AD-1290609
UGCCAUAUAAUAGCUGCAU 203 2140-2160 AAUGCAGCUAAUAAAUGGCAGA 427
2138-2160 AD-1290624 AUUAUUGUGAGGAUAAAUCU 204 2242-2262
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2067-2089 AD-1290635 UUCUGCUUGGCUACAGAAUU 206 2225-2245
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2021-2043 AD-1290657 UUUCUCUUGCAUUCUCGAGU 210 132-152
ACUCGAGAAUGCAAAGAGAAAAU 434 130-152 AD-1290659
CAAUCUGCCUCAAUUUCUUCU 211 2186-2206 AGAAGAAAUUGAGGCAGAUUGCG 435
2184-2206 AD-1290665 AGAGCAAUAAAUUCUUCCUCU 212 1477-1497
AGAGGAAGAUUUAUUUGCUCUGA 436 1475-1497 AD-1290666
ACUUUGAGAAGGUUGAUCUGU 213 758-778 ACAGAUCAACCUUCUCAAGUCU 437
756-778 AD-1290680 UACAGACUUUGAGAAGGUUGU 214 753-773
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AAAGCGAUCUCGAGAAUGCAAAG 440 137-159 AD-1290684
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AGCAGCUAAUAAAUGGCAGAUU 442 2136-2158 AD-1290718
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946-968 AD-1290746 UUGUAUGGUCGUGUGAGGAAU 220 1018-1038
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1704-1726 AD-1290765 GACGUGGUGUUUGUCAGCAU 222 946-966
AUUGCUGACAAACACCACGUCUC 446 944-966 AD-1290778
UCA AUGCCUCCGUCAUCUUCU 223 1166-1186 AGAAGAUGACGGAGGCAUUGAAG 447
1164-1186 AD-1290795 UUGCUCACUCGGAUGCUUUU 224 1102-1122
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1937-1959 AD-1290803 AUCUGACAAGCGAGGAAACUU 226 197-217
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UGGAGCCCACCUUGGAAUUAU 227 1940-1960 AUAAUCCAAGGUGGGCUCCAAG 451
1938-1960 AD-1290835 GCUCUCCAGCUGUUUGGCUU 228 921-941
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1850-1872 AD-1290863 AAGACCUCUGGGUUGGCUUUU 230 285-305
AAAAGCCAACCCAGAGGUCUUGG 454 283-305 AD-1290875
GCCCACCAGCCUGUGAUUUGU 231 1853-1873 ACAAUUCACAGGCUGGUGGGCAG 455
1851-1873 AD-1290891 ACGGAGACGUGGUGUUUGUCU 232 941-961
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2015-2037 AD-1290903 GAGUAGCCUCAUGGAAGAGAU 234 513-533
AUCUCUCCAUGAGGCUACUCCC 458 511-533 AD-1290908
CCAGUGAACCUGCCAAAGAAU 235 1707-1727 AUUCUUUGGCAGGUUCACUGGGU 459
1705-1727 AD-1290910 UAGGGUGGGUAAGGCCUUAUU 236 2062-2082
AAUAAGGCCUUAACCCACCCUAU 460 2060-2082 AD-1290924
UGGGAGUAGCCUCAUGGAAGU 237 510-530 ACUCCAUGAGGCUACUCCCAGA 461
508-530 AD-1290939 AGGGUGGGUAAGGCCUUAUAU 238 2063-2083
AUUAAGGCCUUAACCCACCCUAU 462 2061-2083 AD-1290946
GGUUGAUCUGACCCAGUUCAU 239 768-788 AUGAACUGGGUCAGAUAACCUU 463
766-788 AD-1290950 CAUCAGCCUGGUGGACAAGUU 240 570-590

AACUUGUCCACCAUGAC 464 568-590 AD-1290956
 AGCUGUGAGUCCAUCUGACAU 241 185-205 AUGUCAGAUGGACUCACAGCUGA 465
 183-205 AD-1290971 GCUACGGAGACGUGGUGUUUU 242 938-958
 AAAACACCACGUCUCCGUAGCCA 466 936-958 AD-1290973
 GUUGUGCAGACUCUAUUCCCU 243 1753-1773 AGGGAAUAGAGUCUGCACAACGC 467
 1751-1773
 (615) TABLE-US-00011 TABLE 3 Modified Sense and Antisense Strand Sequences
 of KHK dsRNAs Duplex Name Sense Sequence 5' to 3' SEQ ID NO: Antisense
 Sequence 5' to 3' SEQ ID NO: mRNA target sequence SEQ ID NO: AD-1290652
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 469 asUfsuugCfugacaaaCfaCfcacguscsu 693 AGACGUGGUGUUUGUCAGCAAAG 917 AD-
 1290560 uscsagagCfaAfAfUfaaaucuncuL96 470 asGfsgaaGfauuuuuUfgCfucugasgsg 694
 CCUCAGAGCAAUAAAUCUUCCU 918 AD-1290854 csasguucAfaGfUfGfgauccacauL96
 471 asAfsuguGfgauccacUfuGfaacugsgsg 695 CCCAGUUCAAGUGGAUCCACAUAU 919 AD-
 1290629 asgsuagcGfcAfUfUfuucucuuguL96 472 asCfsaaaGfagaaaauGfcGfcuacusug 696
 CAAGUAGCGCAUUUUCUCUUUGC 920 AD-1290517 gscsgcauUfuUfCfUfcuuugcauuL96
 473 asAfsaugCfaaagagaAfaAfugcgcsusa 697 UAGCGCAUUUUCUCUUUGCAUUC 921 AD-
 1290526 csgscauuUfuCfUfCfuugcauucL96 474 asGfsaaUGfcaaagagAfaAfaugcgscsu 698
 AGCGCAUUUUCUCUUUGCAUUCU 922 AD-1290559 gscsaaUUfcUfCfUfuugcauucL96
 475 asAfsгааUfgcaaagaGfaAfaaugcsgsc 699 GCGCAUUUUCUCUUUGCAUUCUC 923 AD-
 1290548 csasuuuuCfuCfUfUfugcauucL96 476 asGfsagaAfugcaaagAfgAfaaugsccsg 700
 CGCAUUUUCUCUUUGCAUUCUCG 924 AD-1290569 ususuucuCfuUfUfGfcauucucgauL96
 477 asUfscgaGfaaugcaaAfgAfgaaaasug 701 CAUUUUCUCUUUGCAUUCUCGAG 925 AD-
 1290547 ususcucuUfuGfCfAfuucucgagauL96 478 asUfscucGfagaauGcAfaAfgagaasasa 702
 UUUUCUCUUUGCAUUCUCGAGAU 926 AD-1290544 uscsucuuUfgCfAfUfucucgagauL96
 479 asAfsucuCfagaaugCfaAfagagasasa 703 UUUCUCUUUGCAUUCUCGAGAU 927 AD-
 1290667 csuscuuUfcAfUfUfcucgagauL96 480 asGfsaucUfcgagaauGfcAfaagagsasa 704
 UUCUCUUUGCAUUCUCGAGAU 928 AD-1290685 uscsuuugCfaUfUfCfucgagauL96
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 1290695 ususugcaUfuCfUfCfagauL96 482 asAfsugcAfucucgagAfaUfgcaaasgsa 706
 UCUUUGCAUUCUCGAGAU 930 AD-1290653 usgscauuCfuCfGfAfgaucgcuauL96
 483 asUfsaagCfgaucucgAfgAfaugcasasa 707 UUUGCAUUCUCGAGAU 931 AD-
 1290853 gscsaaUfcGfAfGfaucgcuuaguL96 484 asCfsuaaGfcgaucucGfaGfaaugcsasa 708
 UUGCAUUCUCGAGAU 932 AD-1290990 csasuucuCfGfAfGfAfuL96 485 asGfscuaAfgcgaucCfGfAfgaaugscsa 709 UGCAUUCUCGAGAU 933 AD-
 1290540 csusuuaaAfaAfGfGfuugcauL96 486 asUfsgauGfcaaaccUfuUfuuaaagscsg 710
 CGCUUUAAAAGGUUUGCAUCAG 934 AD-1290580 ususuuaaAfaGfGfUfuugcauL96 487 asCfsugaUfgcaaaccUfuUfuuaaasgsc 711 GCUUUAAAAGGUUUGCAUCAG 935 AD-
 1290664 usuaaaaAfgGfUfUfugcauL96 488 asGfscugAfugcaaacCfuUfuuaaasag 712
 CUUUAAAAGGUUUGCAUCAG 936 AD-1290916 uscsagcuGfuGfAfGfuccaucuL96 489 asUfscagAfuggacucAfcAfgcugasusg 713 CAUCAGCUGUGAGUCCAUCUGAC 937 AD-
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 CAGCUGUGAGUCCAUCUGACAAG 938 AD-1290896 csusgugaGfuCfCfAfuL96 491 asCfsuugUfcagauggAfcUfcacagcsu 715 AGCUGUGAGUCCAUCUGACAAGC 939 AD-
 1290914 cscsaucuGfaCfAfAfgcgaggaaauL96 492 asUfsuucCfucgcuugUfcAfgauggsasc 716
 GUCCAUCUGACAAGCGAGGAAAC 940 AD-1290982 csasucugAfcAfAfGfcgaggaaacuL96 493 asGfsuuuUfcucgcuuGfuCfagaugsgsa 717 UCCAUCUGACAAGCGAGGAAACU 941 AD-
 1290708 gsgsaaacUfaAfGfGfcugagaaguL96 494 asAfscuuCfucagccuUfaGfuuuccsusc 718
 GAGGAAACUAAGGCUGAGAAGUG 942 AD-1290693 gsasaacuAfaGfGfCfugagaaguL96

495 asCfsacuUfcuagaccUfuAfguucscsu 719 AGGAUAGCCUGAAGGUAAGUG 943 AD-
1290942 asgsaccuCfuGfGfGfuuggcuuucuL96 496 asGfsaaaGfccaaaccAfgAfggucususg 720
CAAGACCUCUGGGUUGGCUUUC 944 AD-1290807 asgsuagcCfuCfAfUfggaagagaauL96
497 asUfsucuCfuuccaugAfgGfcuacuscsc 721 GGAGUAGCCUCAUGGAAGAGAAG 945 AD-
1290881 cscsucauGfgAfAfGfagaagcagauL96 498 asUfscugCfuucucuCfcAfugaggscsu 722
AGCCUCAUGGAAGAGAAGCAGAU 946 AD-1290745 csuscaugGfaAfGfAfgaagcagauuL96
499 asAfsucuGfcuucucuUfcCfaugagsgsc 723 GCCUCAUGGAAGAGAAGCAGAU 947 AD-
1290814 asusggaaGfaGfAfAfgcagauccuuL96 500 asAfsaggUfcugcuucUfcUfuccausgsa 724
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501 asAfsagGfaucugcuUfcUfcuuccsas 725 AUGGAAGAGAAGCAGAUCCUGUG 949 AD-
1290964 gsasagagAfaGfCfAfgauccuguguL96 502 asCfsacaGfgaucugcUfuCfucuucscsa 726
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1290816 gsusggacAfaGfUfAfccuaaggauL96 504 asUfscuUfagggucUfuGfuccacscsa 728
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505 asUfscuCfcuuaaggUfaCfuugucscsa 729 UGACAAGUACCCUAAGGAGGAC 953 AD-
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GUGCAGGAAGCACUGAGAUUCGG 1012 AD-1290977 usgsgccuGfcAfGfGfgcuuugauguL96
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1290961 csusgcagGfgCfUfUfugauggcauuL96 566 asAfsugcCfaucaaagCfcCfugcagsgsc 790
GCCUGCAGGGCUUUGAUGGCAUC 1014 AD-1290834 csasgggcUfuUfGfAfuggcaucguuL96
567 asAfsccaUfgccaucaAfaGfccccugscsa 791 UGCAGGGCUUUGAUGGCAUCGUG 1015 AD-
1290867 gsgsgcuUfgAfUfGfgcaucguguuL96 568 asAfsccGfaugccauCfaAfagcccsusg 792
CAGGGCUUUGAUGGCAUCGUGUG 1016 AD-1290879 csuscugcCfuGfUfGfuccuguguuuL96
569 asAfsacaCfaggacacAfgGfcagagsusc 793 GACUCUGCCUGUGUCCUGUGUUC 1017 AD-
1290549 csuscagaGfcAfAfAfuaaaucuuL96 570 asGfsaagAfuuuuuuGfcUfcugagsgsc 794
GCCUCAGAGCAAUAUAUCUUC 1018 AD-1290525 csasgagcAfaAfUfAfaaucuuccuuL96
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1290747 gsgsagccCfaCfCfUfuggaauuaauL96 586 asUfsuaUfuccaaggUfgGfgcuccsasa 810
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1290911 asgscccaCfcUfUfGfgaauuaagguL96 588 asCfscuuAfauuccaaGfgUfgggcuscsc 812
GGAGCCCACCUUGGAAUUAAGG 1036 AD-1290926 gscscacCfuUfGfGfaauuaagguL96
589 asCfscuuUfaauuccaAfgGfugggcsusc 813 GAGCCCACCUUGGAAUUAAGGGC 1037 AD-
1291003 gsgscgugCfcUfCfAfgccacaaauL96 590 asAfsuuGfuggcugaGfgCfacgccscsu 814
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591 asUfsgggUfcacauuuGfuGfgcugasgs 815 CCUCAGCCACAAUUGUGACCCAG 1039 AD-
1290764 gsgsuccgAfuCfUfGfgaacacauauL96 592 asUfsaugUfguuccagAfuCfaggaccsusc 816
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1290670 uscscgauCfuGfGfAfacacauuuuL96 594 asAfsauUfguguuccAfgAfucggascsc 818
GGUCCGAUCUGGAACACAUUAUG 1042 AD-1290712 cscsgaucUfgGfAfAfcacauuuuL96

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1290755 gsgsugggUfaAfGfGfccuuauaaL96 602 asAfsuaaUfaaggccuUfaCfccaccscsu 826
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691 asGfsggaAfuagagucUfcCfacaacscsc 915 GCGUUGUGCAGACUCUAUUCCCA 1139

(616) TABLE-US-00012 TABLE 4 KHK Dose Screen in HepG2 cells KHK/gapdh 10 nM 1 nM
0.1 nM Duplex Name mean SD mean SD mean SD AD-1290887.1 0.404 0.066 0.946 0.145 1.155
0.187 AD-1290629.1 0.613 0.010 1.098 0.229 1.020 0.166 AD-1290807.1 0.247 0.007 0.663 0.138
1.226 0.074 AD-1290681.1 0.990 0.171 1.179 0.170 1.183 0.178 AD-1290639.1 0.927 0.044 1.302
0.254 1.104 0.178 AD-1290597.1 0.904 0.040 1.425 0.206 1.156 0.123 AD-1290775.1 0.870 0.026

1.498 0.159 1.210 0.162 AD-1290795.1 0.182 0.023 0.526 0.119 1.005 0.106 AD-1291003.1 0.835
0.113 1.353 0.222 1.296 0.081 AD-1290903.1 0.419 0.021 1.151 0.233 0.887 0.055 AD-1290896.1
0.807 0.065 1.389 0.334 0.970 0.054 AD-1290605.1 0.675 0.077 1.474 0.345 0.989 0.083 AD-
1290742.1 0.991 0.143 1.355 0.379 1.054 0.145 AD-1290554.1 0.924 0.069 1.578 0.256 1.118
0.082 AD-1290509.1 0.865 0.048 1.256 0.129 1.118 0.165 AD-1290938.1 0.482 0.044 0.772 0.164
0.990 0.327 AD-1290750.1 0.990 0.051 1.077 0.059 1.058 0.116 AD-1290659.1 0.690 0.031 0.937
0.194 0.971 0.100 AD-1290755.1 0.916 0.058 1.122 0.098 1.100 0.119 AD-1290558.1 0.888 0.034
1.064 0.214 0.850 0.035 AD-1290516.1 0.777 0.044 1.046 0.236 0.864 0.132 AD-1290621.1 0.893
0.039 0.990 0.210 0.910 0.148 AD-1290956.1 0.498 0.040 0.582 0.063 0.762 0.089 AD-1290672.1
0.814 0.034 0.828 0.073 0.826 0.210 AD-1290800.1 0.859 0.116 0.898 0.082 0.764 0.096 AD-
1290555.1 0.781 0.061 0.843 0.036 0.805 0.143 AD-1290527.1 0.912 0.114 0.908 0.056 0.664
0.037 AD-1290921.1 0.920 0.045 0.896 0.102 0.741 0.082 AD-1290664.1 0.693 0.071 0.956 0.095
0.806 0.035 AD-1290924.1 0.282 0.025 0.576 0.045 0.630 0.011 AD-1290650.1 0.756 0.035 0.914
0.130 0.830 0.081 AD-1290939.1 0.829 0.027 0.873 0.030 0.804 0.128 AD-1290592.1 0.784 0.055
0.894 0.136 0.775 0.069 AD-1290551.1 0.724 0.024 0.891 0.189 0.687 0.009 AD-1290638.1 0.921
0.052 0.976 0.230 0.758 0.071 AD-1290580.1 0.410 0.017 0.702 0.116 0.741 0.112 AD-1290916.1
0.552 0.026 0.694 0.024 0.723 0.105 AD-1290741.1 0.799 0.057 0.864 0.034 0.731 0.075 AD-
1290910.1 0.945 0.078 0.948 0.020 0.665 0.030 AD-1290661.1 0.985 0.065 0.937 0.064 0.665
0.011 AD-1290660.1 0.901 0.159 0.936 0.022 0.912 0.032 AD-1290622.1 1.020 0.207 0.940 0.040
0.932 0.174 AD-1290540.1 0.474 0.033 0.698 0.062 0.978 0.109 AD-1290618.1 1.075 0.102 0.930
0.116 0.786 0.032 AD-1290600.1 1.072 0.213 0.901 0.055 0.953 0.133 AD-1290908.1 0.812 0.052
0.810 0.019 0.802 0.172 AD-1290665.1 0.191 0.030 0.216 0.116 0.656 0.117 AD-1290870.1 0.831
0.092 0.916 0.071 0.731 0.047 AD-1290990.1 0.698 0.020 0.687 0.036 0.785 0.075 AD-1290989.1
0.829 0.167 0.862 0.101 0.607 0.241 AD-1290987.1 0.376 0.042 0.581 0.024 0.985 0.177 AD-
1290983.1 0.987 0.249 0.993 0.176 0.883 0.125 AD-1290982.1 0.949 0.138 0.846 0.054 0.866
0.052 AD-1290973.1 0.884 0.229 0.954 0.070 0.813 0.127 AD-1290971.1 0.135 0.043 0.289 0.029
0.699 0.071 AD-1290970.1 0.153 0.039 0.207 0.011 0.660 0.127 AD-1290969.1 0.140 0.034 0.221
0.017 0.608 0.113 AD-1290964.1 0.265 0.038 0.360 0.026 0.709 0.100 AD-1290963.1 0.158 0.045
0.311 0.063 0.739 0.176 AD-1290962.1 0.317 0.114 0.436 0.042 0.730 0.095 AD-1290959.1 0.157
0.044 0.295 0.014 0.835 0.044 AD-1290950.1 0.282 0.044 0.624 0.049 0.896 0.080 AD-1290946.1
0.656 0.156 0.866 0.031 0.883 0.042 AD-1290915.1 0.117 0.040 0.396 0.024 0.882 0.121 AD-
1290914.1 0.676 0.046 0.689 0.038 0.862 0.102 AD-1290911.1 1.052 0.136 0.914 0.028 0.930
0.143 AD-1290900.1 0.230 0.034 0.512 0.013 0.805 0.026 AD-1290897.1 1.054 0.014 0.957 0.027
0.867 0.065 AD-1290891.1 0.534 0.134 0.675 0.016 0.853 0.068 AD-1290890.1 1.094 0.160 0.905
0.025 0.945 0.138 AD-1290886.1 0.626 0.146 0.764 0.044 0.953 0.112 AD-1290885.1 0.115 0.017
0.217 0.066 0.770 0.030 AD-1290884.1 0.133 0.025 0.264 0.014 0.783 0.120 AD-1290882.1 0.265
0.032 0.659 0.013 0.897 0.032 AD-1290881.1 0.156 0.052 0.393 0.024 0.952 0.137 AD-1290878.1
0.156 0.026 0.381 0.011 0.933 0.201 AD-1290874.1 0.135 0.006 0.371 0.010 0.827 0.127 AD-
1290872.1 0.269 0.055 0.256 0.013 0.606 0.098 AD-1290865.1 0.680 0.264 0.838 0.020 0.895
0.202 AD-1290861.1 0.814 0.323 0.885 0.022 0.961 0.088 AD-1290860.1 0.169 0.022 0.641 0.031
0.957 0.032 AD-1290857.1 0.656 0.179 0.921 0.068 0.990 0.068 AD-1290854.1 0.532 0.083 0.644
0.032 0.994 0.041 AD-1290853.1 0.550 0.135 0.732 0.018 0.966 0.051 AD-1290852.1 0.750 0.108
0.762 0.161 1.074 0.156 AD-1290842.1 0.887 0.182 0.748 0.049 0.982 0.039 AD-1290837.1 0.409
0.365 0.260 0.016 0.919 0.135 AD-1290828.1 0.370 0.070 0.565 0.059 0.904 0.028 AD-1290823.1
0.213 0.072 0.329 0.034 0.916 0.051 AD-1290821.1 0.198 0.043 0.363 0.012 1.036 0.111 AD-
1290818.1 0.053 0.009 0.162 0.011 0.699 0.039 AD-1290816.1 0.208 0.040 0.472 0.026 0.916
0.037 AD-1290814.1 0.116 0.037 0.221 0.012 0.625 0.025 AD-1290811.1 0.120 0.031 0.291 0.023
0.830 0.034 AD-1290810.1 0.194 0.025 0.572 0.033 0.949 0.018 AD-1290802.1 0.332 0.110 0.511
0.004 0.983 0.032 AD-1290778.1 0.146 0.019 0.440 0.037 0.961 0.045 AD-1290765.1 0.086 0.017
0.141 0.010 0.647 0.020 AD-1290759.1 0.262 0.082 0.432 0.008 0.948 0.065 AD-1290757.1 0.136

0.041 0.242 0.011 0.838 0.035 AD-1290752.1 0.155 0.018 0.423 0.074 0.893 0.042 AD-1290746.1
0.123 0.013 0.598 0.099 0.970 0.125 AD-1290745.1 0.199 0.018 0.614 0.068 0.964 0.081 AD-
1290740.1 0.138 0.016 0.620 0.120 0.985 0.031 AD-1290739.1 0.379 0.035 1.002 0.121 0.992
0.049 AD-1290736.1 0.525 0.046 1.239 0.176 1.017 0.173 AD-1290734.1 0.512 0.029 1.320 0.191
1.066 0.083 AD-1290731.1 0.154 0.004 0.536 0.100 1.089 0.077 AD-1290718.1 0.128 0.051 0.699
0.158 1.227 0.081 AD-1290710.1 0.237 0.082 0.841 0.138 1.133 0.125 AD-1290702.1 0.771 0.024
1.217 0.214 1.069 0.019 AD-1290695.1 0.379 0.050 0.758 0.127 0.997 0.026 AD-1290685.1 0.373
0.009 0.887 0.154 1.003 0.045 AD-1290683.1 0.363 0.013 0.741 0.104 1.030 0.056 AD-1290668.1
0.096 0.008 0.285 0.042 0.874 0.040 AD-1290667.1 0.549 0.036 0.987 0.122 1.200 0.067 AD-
1290657.1 0.459 0.045 1.116 0.220 1.181 0.044 AD-1290656.1 0.591 0.104 1.197 0.190 1.231
0.021 AD-1290653.1 0.325 0.008 0.636 0.092 1.087 0.022 AD-1290652.1 0.106 0.008 0.382 0.063
1.143 0.023 AD-1290651.1 0.232 0.008 0.415 0.053 0.983 0.091 AD-1290633.1 0.943 0.033 0.882
0.084 0.898 0.126 AD-1290632.1 0.406 0.154 0.773 0.099 0.866 0.099 AD-1290604.1 1.000 0.035
0.737 0.043 0.934 0.078 AD-1290599.1 0.128 0.003 0.200 0.016 0.749 0.027 AD-1290589.1 0.992
0.050 0.731 0.038 0.839 0.047 AD-1290584.1 0.222 0.012 0.362 0.031 0.830 0.037 AD-1290576.1
0.140 0.009 0.218 0.020 0.846 0.118 AD-1290569.1 0.531 0.139 0.575 0.056 0.814 0.192 AD-
1290565.1 1.031 0.070 0.703 0.049 0.911 0.086 AD-1290561.1 0.400 0.039 0.640 0.106 1.109
0.180 AD-1290559.1 0.429 0.037 0.573 0.086 1.094 0.053 AD-1290557.1 0.861 0.121 0.917 0.251
1.148 0.214 AD-1290552.1 0.866 0.110 0.759 0.128 0.739 0.317 AD-1290548.1 0.427 0.040 0.436
0.041 0.854 0.037 AD-1290547.1 0.534 0.102 0.569 0.065 1.079 0.218 AD-1290546.1 0.081 0.017
0.127 0.070 0.689 0.115 AD-1290544.1 0.420 0.176 0.533 0.047 0.964 0.045 AD-1290543.1 0.798
0.127 0.855 0.085 0.973 0.049 AD-1290542.1 0.807 0.059 0.730 0.024 0.901 0.187 AD-1290528.1
0.638 0.045 0.777 0.049 1.080 0.063 AD-1290526.1 0.496 0.040 0.680 0.045 1.027 0.164 AD-
1290524.1 0.886 0.029 1.005 0.038 0.998 0.132 AD-1290522.1 0.825 0.036 0.937 0.042 0.863
0.034 AD-1290515.1 0.925 0.061 0.886 0.060 0.802 0.049 AD-1290514.1 1.056 0.046 1.130 0.080
0.899 0.030 AD-1290510.1 0.794 0.058 0.921 0.052 0.869 0.073 AD-1290980.1 0.309 0.016 0.756
0.022 0.804 0.042 AD-1290933.1 0.444 0.029 0.747 0.039 0.862 0.125 AD-1290834.1 0.296 0.017
0.611 0.064 0.904 0.119 AD-1290747.1 0.839 0.068 0.831 0.135 1.002 0.093 AD-1290666.1 0.265
0.003 0.457 0.036 0.988 0.117 AD-1290612.1 0.776 0.084 0.811 0.050 0.828 0.027 AD-1290570.1
0.811 0.035 0.909 0.096 0.844 0.016 AD-1290564.1 0.956 0.026 0.910 0.053 1.112 0.182 AD-
1290535.1 0.916 0.048 0.820 0.086 0.772 0.175 AD-1290517.1 0.443 0.038 0.508 0.035 0.908
0.157 AD-1291001.1 0.214 0.024 0.423 0.052 0.874 0.134 AD-1290805.1 0.927 0.033 0.863 0.094
0.948 0.152 AD-1290655.1 0.926 0.058 0.999 0.085 0.902 0.068 AD-1290574.1 0.978 0.105 0.874
0.046 0.826 0.069 AD-1290533.1 0.937 0.105 0.788 0.035 0.885 0.090 AD-1290530.1 0.127 0.007
0.230 0.022 0.731 0.057 AD-1290523.1 0.997 0.162 0.713 0.015 0.937 0.168 AD-1290904.1 0.465
0.035 0.660 0.132 0.973 0.114 AD-1290796.1 0.944 0.075 0.734 0.131 0.936 0.147 AD-1290684.1
1.012 0.060 0.678 0.064 0.957 0.197 AD-1290611.1 0.201 0.009 0.224 0.023 0.781 0.084 AD-
1290602.1 0.927 0.141 0.648 0.089 1.032 0.068 AD-1290563.1 0.839 0.111 0.650 0.045 1.135
0.189 AD-1290961.1 0.289 0.029 0.473 0.028 0.923 0.133 AD-1290923.1 0.137 0.005 0.280 0.038
0.771 0.071 AD-1290880.1 0.913 0.055 0.859 0.098 0.855 0.084 AD-1290712.1 0.942 0.014 0.729
0.038 0.935 0.078 AD-1290626.1 0.730 0.044 0.705 0.068 0.991 0.097 AD-1290539.1 0.121 0.012
0.166 0.041 0.609 0.026 AD-1290531.1 0.909 0.182 0.781 0.132 1.007 0.135 AD-1290670.1 0.973
0.038 0.790 0.179 1.075 0.159 AD-1290654.1 0.921 0.046 0.798 0.151 1.175 0.185 AD-1290615.1
0.918 0.026 0.772 0.081 1.243 0.213 AD-1290591.1 0.096 0.006 0.163 0.012 0.767 0.159 AD-
1290879.1 0.209 0.012 0.478 0.021 0.920 0.147 AD-1290763.1 0.665 0.047 0.752 0.030 0.810
0.057 AD-1290680.1 0.092 0.013 0.198 0.018 0.759 0.050 AD-1290643.1 0.708 0.049 0.770 0.033
0.873 0.029 AD-1290573.1 0.585 0.040 0.744 0.034 0.909 0.065 AD-1290764.1 0.607 0.049 0.788
0.031 1.020 0.087 AD-1290687.1 0.646 0.032 0.894 0.092 1.038 0.155 AD-1290579.1 0.118 0.007
0.456 0.015 1.022 0.022 AD-1290977.1 0.688 0.059 0.950 0.061 1.176 0.120 AD-1290955.1 0.172
0.011 0.269 0.025 0.963 0.125 AD-1290894.1 0.864 0.038 0.804 0.051 0.997 0.068 AD-1290722.1

0.822 0.068 0.839 0.044 0.907 0.040 AD-1290692.1 0.106 0.011 0.192 0.022 0.840 0.025 AD-1290835.1 0.384 0.030 0.692 0.019 0.975 0.135 AD-1290730.1 0.246 0.020 0.491 0.022 0.990 0.131 AD-1290693.1 0.553 0.051 0.740 0.120 1.105 0.191 AD-1290682.1 0.250 0.020 0.348 0.036 0.884 0.105 AD-1290635.1 0.765 0.068 0.800 0.075 1.070 0.110 AD-1290525.1 0.128 0.019 0.203 0.019 0.958 0.069 AD-1290984.1 0.411 0.044 0.685 0.045 0.754 0.043 AD-1290942.1 0.474 0.024 0.586 0.037 0.673 0.103 AD-1290841.1 0.995 0.059 0.960 0.049 0.767 0.029 AD-1290719.1 0.887 0.047 0.789 0.043 0.913 0.113 AD-1290708.1 0.751 0.032 0.794 0.046 0.980 0.136 AD-1290627.1 0.222 0.018 0.309 0.132 0.986 0.113 AD-1290624.1 0.828 0.027 0.737 0.053 0.899 0.140 AD-1290560.1 0.145 0.005 0.213 0.021 0.854 0.151 AD-1290993.1 1.013 0.140 0.825 0.053 1.298 0.137 AD-1290988.1 0.433 0.023 0.650 0.031 1.274 0.288 AD-1290863.1 0.856 0.088 0.914 0.050 0.916 0.041 AD-1290836.1 0.818 0.080 0.906 0.072 0.922 0.124 AD-1290663.1 0.247 0.019 0.496 0.025 1.019 0.180 AD-1290549.1 0.126 0.009 0.263 0.019 0.885 0.056 AD-1290931.1 1.106 0.034 0.850 0.043 1.057 0.104 AD-1290926.1 1.096 0.033 0.849 0.016 0.934 0.044 AD-1290909.1 1.023 0.042 0.776 0.065 0.935 0.089 AD-1290875.1 0.982 0.074 0.704 0.047 0.943 0.096 AD-1290867.1 1.168 0.044 0.798 0.033 1.048 0.189 AD-1290803.1 0.646 0.007 0.569 0.021 0.852 0.263 AD-1290748.1 0.830 0.098 0.901 0.126 1.229 0.126 AD-1290609.1 0.822 0.087 0.808 0.259 1.227 0.116 AD-1290556.1 0.915 0.043 0.826 0.240 1.251 0.124 AD-1290507.1 0.927 0.080 0.691 0.130 1.342 0.077 Neg ctrl 1.130 0.343 0.981 0.222 0.863 0.237 Neg ctrl 0.897 0.088 1.089 0.141 1.102 0.113 Pos ctrl 0.091 0.013 0.101 0.008 0.196 0.037

Example 3. Design, Synthesis and In Vitro Screening of Additional dsRNA Duplexes

(617) Additional siRNAs were designed, synthesized, and prepared using methods known in the art and described above in Example 1.

(618) Detailed lists of the additional unmodified KHK sense and antisense strand nucleotide sequences are shown in Table 5. Detailed lists of the modified KHK sense and antisense strand nucleotide sequences are shown in Table 6.

(619) For transfections, Hep3b cells (ATCC, Manassas, VA) were grown to near confluence at 37° C. in an atmosphere of 5% CO₂ in Eagle's Minimum Essential Medium (Gibco) supplemented with 10% FBS (ATCC) before being released from the plate by trypsinization. Transfection was carried out by adding 7.5 µl of Opti-MEM plus 0.1 µl of Lipofectamine RNAiMax per well (Invitrogen, Carlsbad CA. cat #13778-150) to 2.5 µl of each siRNA duplex to an individual well in a 384-well plate. The mixture was then incubated at room temperature for 15 minutes. Forty µl of complete growth media without antibiotic containing ~1.5×10⁴ Hep3b cells were then added to the siRNA mixture. Cells were incubated for 24 hours prior to RNA purification. Single dose experiments were performed at 10 nM final duplex concentration.

(620) Total RNA isolation was performed using DYNABEADS. Briefly, cells were lysed in 10 µl of Lysis/Binding Buffer containing 3 µL of beads per well and mixed for 10 minutes on an electrostatic shaker. The washing steps were automated on a Biotek EL406, using a magnetic plate support. Beads were washed (in 3 L) once in Buffer A, once in Buffer B, and twice in Buffer E, with aspiration steps in between. Following a final aspiration, complete 12 µL RT mixture was added to each well, as described below.

(621) For cDNA synthesis, a master mix of 10.5 µl 10× Buffer, 0.6 µl 10× dNTPs, 1.5 µl Random primers, 0.75 µl Reverse Transcriptase, 0.75 µl RNase inhibitor and 9.9 µl of H₂O per reaction were added per well. Plates were sealed, agitated for 10 minutes on an electrostatic shaker, and then incubated at 37 degrees C. for 2 hours. Following this, the plates were agitated at 80 degrees C. for 8 minutes.

(622) RT-qPCR was performed as described above and relative fold change was calculated as described above.

(623) The results of the transfection assays of the dsRNA agents listed in Tables 5 and 6 in Hep3b cells are shown in Table 7.

(624) Table 8 provides a list of the unmodified KHK sense and antisense strand nucleotide

sequences of selected dsRNA agents. Table 9 provides a list of the modified KHK sense and antisense strand nucleotide sequences of selected dsRNA agents.

(625) TABLE-US-00013 TABLE 5 Unmodified Sense and Antisense Strand Sequences of KHK dsRNAs									
Duplex	Name	Sequence 5' to 3'	NO: XM_017004061.1	Sequence 5' to 3'	NO: XM_017004061.1	AD-1423310.1	UUUGAGAAGGUUGAUCUGACU	1140	760-780
		AGUCAGAUCAACCUUCUCAAGU	296	758-780	AD-1423311.1				
		GACGUGGUGUUUGUCAGCAAU	222	946-966	AUUGCUGACAAACACCACGUCUC	446			
		944-966	AD-1423312.1	UGCAGGAAGCACUGAGAUUCU	115	1208-1228			
		AGAATCTCAGUGCUUCCUGCACG	1141	1206-1228	AD-1423313.1				
		ACAGACUUUGAGAAGGUUGAU	67	754-774	AUCAACCUUCUCAAGUCUGUAG	291			
		752-774	AD-1423314.1	GCUACAGACUUUGAGAAGGUU	65	751-771			
		AACCTUCUCAAGUCUGUAGCAG	1142	749-771	AD-1423315.1				
		ACCCAGUUCAAGUGGAUCCAU	79	778-798	AUGGAUCCACUUGAACUGGGUCA	303			
		776-798	AD-1423316.1	UCAGAGCAAUAAAUCUCCU	22	1336-1356			
		AGGAAGAUUUUUUGCUCUGAGG	246	1334-1356	AD-1290665.2				
		AGAGCAAUAAAUCUUCCUCU	212	1477-1497	AGAGGAAGAUUUUUUGCUCUGA	436			
		1475-1497	AD-1423317.1	CUUUGAGAAGGUUGAUCUGAU	71	759-779			
		AUCAGATCAACCUUCUCAAGUC	1143	757-779	AD-1423318.1				
		AUGGAAGAGAAGCAGAUCCUU	52	523-543	AAGGAUCUGCUUCUCUCCAUGA	276			
		521-543	AD-1423319.1	GACUUUGAGAAGGUUGAUCUU	70	757-777			
		AAGATCAACCUUCUCAAGUCUG	1144	755-777	AD-1423320.1				
		CUCAGAGCAAUAAAUCUUCU	122	1474-1494	AGAAGATUUUUUGCUCUGAGGC	1145			
		1472-1494	AD-1423321.1	CUCUCCAGCUGUUUGGCUAU	88	922-942			
		AUAGCCAAACAGCUGGAAGAGCU	312	920-942	AD-1423322.1				
		GAGCUGGAGACACCUUCAAUU	104	1151-1171	AAUUGAAGGUGUCUCCAGCUCCC	328			
		1149-1171	AD-1423323.1	CAAUGCCUCCGUCAUCUUCAU	111	1167-1187			
		AUGAAGAUGACGGAGGCAUUGAA	335	1165-1187	AD-1423324.1				
		CCUUCA AUGCCUCCGUCAUCU	109	1163-1183	AGAUGACGGAGGCAUUGAAGGUG	333			
		1161-1183	AD-1423325.1	CUGCUACAGACUUUGAGAAGU	63	749-769			
		ACUUCUCAAGUCUGUAGCAGAC	287	747-769	AD-1423326.1				
		AAAGAUGUGGCCAAGCACUUU	99	964-984	AAAGTGCUUGGCCACAUCUUUGC	1146			
		962-984	AD-1423327.1	AAGGUUGAUCUGACCCAGUUU	73	766-786			
		AAACTGGGUCAGAUCAACCUUCU	1147	764-786	AD-1423328.1				
		AAGGUGUUUGUCCCAGAGAUU	60	612-632	AAUCTCTGGGACAAACACCUUUAU	1148			
		610-632	AD-1423329.1	GAAGAGAAGCAGAUCCUGUGU	54	526-546			
		ACACAGGAUCUGCUUCUCUCCA	278	524-546	AD-1423330.1				
		GUGUUUGUCAGCAAAGAUGUU	95	952-972	AACATCTUUGCUGACAAACACCA	1149			
		950-972	AD-1423331.1	GACAAGUACCCUAAGGAGGAU	57	583-603			
		AUCCTCCUAGGGUACUUGUCCA	1150	581-603	AD-1423332.1				
		GUGCAGGAAGCACUGAGAUUU	114	1207-1227	AAUCUCAGUGCUUCCUGCACGC	338			
		1205-1227	AD-1290878.2	GGAGACGUGGUGUUUGUCAGU	92	943-963			
		ACUGACAAACACCACGUCUCCGU	316	941-963	AD-1423334.1				
		CCUCAUGGAAGAGAAGCAGAU	50	519-539	AUCUGCTUCUCUCCAUGAGGCU	1151			
		517-539	AD-1423335.1	CUCCGUCAUCUUCAGCCUCUU	113	1173-1193			
		AAGAGGCUGAAGAUGACGGAGGC	337	1171-1193	AD-1423336.1				
		GUUCAAGUGGAUCCACAUGU	207	783-803	ACAATGTGGAUCCACUUGAACUG	1152			
		781-803							

(626) TABLE-US-00014 TABLE 6 Modified Sense and Antisense Strand Sequences of KHK dsRNAs

SEQ	SEQ	SEQ	Sense ID	Antisense ID	ID	Duplex	Name	Sequence	5'
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to 3' NO: Sequence 5' to 3' NO mRNA target sequence NO AD-1423310.1
ususugagAfaGfGfU 1153 asGfsucdAg(Agn)uca 1154 ACUUUGAGAAGGUUGAUCUGACC 968
fugaucugacuL96 accUfuCfucaaaasgsu AD-1423311.1 gsascgugGfuGfUfu 670
asUfsugdCu(G2p)aca 1155 GAGACGUGGUGUUUGUCAGCAA 1118 fugucagcaauL96
aacAfcCfacgucsusc AD-1423312.1 usgscaggAfaGfCfA 563 asGfsaadTc(Tgn)cag 1156
CGUGCAGGAAGCACUGAGAUUCG 1011 fcugagauucuL96 ugcUfuCfcugcascsg AD-
1423313.1 ascsagacUfuUfGfA 515 asUfscadAc(C2p)uuc 1157
CUACAGACUUUGAGAAGGUUGAU 963 fgaagguugauL96 ucaAfaGfucugusasg AD-1423314.1
gscsuacaGfaCfUfu 513 asAfscddTu(C2p)uca 1158 CUGCUACAGACUUUGAGAAGGUU 961
fugagaagguuL96 aagUfcUfguagcsasg AD-1423315.1 ascscagUfuCfAfA 527
asUfsggdAu(C2p)cac 1159 UGACCCAGUUCAAGUGGAUCCAC 975 fguggauccauL96
uugAfaCfuggguscsa AD-1423316.1 uscsagagCfaAfAfU 470 asGfsgadAg(Agn)uuu 1160
CCUCAGAGCAAUAAAUCUUCCU 918 faaauucuuccuL96 auuUfgCfucugasgsg AD-1290665.2
asgsagcaAfaUfAfA 660 asGfsaggAfagauuuuU 884 UCAGAGCAAUAAAUCUUCCUCA 1108
faucuuccucuL96 fuUfgcucusgsa AD-1423317.1 csusuugaGfaAfGfG 519 asUfscadGa(Tgn)caa
1161 GACUUUGAGAAGGUUGAUCUGAC 967 fuugaucugauL96 ccuUfcUfcaaagsusc AD-
1423318.1 asusggaaGfaGfAfA 500 asAfsggdAu(C2p)ugc 1162
UCAUGGAAGAGAAGCAGAUCCUG 948 fgcagaucuuL96 uucUfcUfuccausgsa AD-1423319.1
gsascuuuGfaGfAfA 518 asAfsgadTc(Agn)acc 1163 CAGACUUUGAGAAGGUUGAUCUG 966
fgguugaucuuL96 uucUfcAfaagucsusg AD-1423320.1 csuscagaGfcAfAfA 570
asGfsaadGa(Tgn)uuu 1164 GCCUCAGAGCAAUAAAUCUUCC 1018 fuaaaucuucuL96
uuuGfcUfcugagsgsc AD-1423321.1 csuscuucCfaGfCfU 536 asUfsagdCc(Agn)aac 1165
AGCUCUUCCAGCUGUUUGGCUAC 984 fguuuggcuauL96 agcUfgGfaagagscsu AD-1423322.1
gsasgcugGfaGfAfC 552 asAfsuudGa(Agn)ggu 1166 GGGAGCUGGAGACACCUUCAUG 1000
faccuucuuuL96 gucUfcCfagcuscsc AD-1423323.1 csasaugcCfuCfCfG 559 asUfsgadAg(Agn)uga
1167 UUCA AUGCCUCCGUCAUCUUCAG 1007 fucaucuucuuL96 cggAfgGfcauugsasa AD-
1423324.1 cscsuucaAfuGfCfC 557 asGfsaudGa(C2p)gga 1168
CACCUUCA AUGCCUCCGUCAUCU 1005 fuccgucaucuL96 ggcAfuUfgaaggsusg AD-1423325.1
csusgcuaCfaGfAfC 511 asCfsuudCu(C2p)aaa 1169 GUCUGCUACAGACUUUGAGAAGG 959
fuuugagaaguL96 gucUfgUfagcagsasc AD-1423326.1 asasagauGfuGfGfC 547
asAfsagdTg(C2p)uug 1170 GCAAAGAUGUGGCCAAGCACUUG 995 fcaagcacuuuL96
gccAfcAfucuuusgsc AD-1423327.1 asasgguuGfaUfCfU 521 asAfsacdTg(G2p)guc 1171
AGAAGGUUGAUCUGACCCAGUUC 969 fgaccaguuuL96 agaUfcAfaccuuscsu AD-1423328.1
asasggugUfuUfGfU 508 asAfsucdTc(Tgn)ggg 1172 AUAAGGUGUUUGUCCCAGAGAUG 956
fccagagauuL96 acaAfaCfaccuusasu AD-1423329.1 gsasagagAfaGfCfA 502 asCfsacdAg(G2p)auc
1173 UGGAAGAGAAGCAGAUCUGUGC 950 fgauccuguguL96 ugcUfuCfucuuuscsa AD-
1423330.1 gsusguuuGfuCfAfG 543 asAfscadTc(Tgn)uug 1174
UGGUGUUUGUCAGCAAAGAUGUG 991 fcaaagauguuL96 cugAfcAfaacacscsa AD-1423331.1
gsascaagUfaCfCfC 505 asUfscddTc(C2p)uuu 1175 UGGACAAGUACCCUAAGGAGGAC 953
fuaaggaggauL96 gggUfaCfuugucsca AD-1423332.1 gsusgcagGfaAfGfC 562
asAfsaudCu(C2p)agu 1176 GCGUGCAGGAAGCACUGAGAUUC 1010 facugagauuuL96
gcuUfcCfugcacsgsc AD-1290878.2 gsgsagacGfuGfGfU 540 asCfsugaCfaaacaccA 764
ACGGAGACGUGGUGUUUGUCAGC 988 fguuugucaguL96 fcGfucuccsgsu AD-1423334.1
cscsucauGfgAfAfG 498 asUfscudGc(Tgn)ucu 1177 AGCCUCAUGGAAGAGAAGCAGAU 946
fagaagcagauL96 cuuCfcAfugaggscsu AD-1423335.1 csusccguCfaUfCfU 561
asAfsgadGg(C2p)uga 1178 GCCUCCGUCAUCUUCAGCCUCUC 1009 fucagccucuuL96
agaUfgAfcggagsgsc AD-1423336.1 gsusucaaGfuGfGfA 655 asCfsaadTg(Tgn)gga 1179
CAGUUCAAGUGGAUCCACAUUGA 1103 fuccacauuguL96 uccAfcUfugaacsusg
(627) TABLE-US-00015 TABLE 7 KHK Single Dose Screen in Hep3b cells % Message Duplex
Name Remaining (qPCR) AD-1423317 9.38 AD-1423327 7.75 AD-1423336 7.70 AD-1423311

6.06 AD-1423320 21.59 AD-1423324 10.44 AD-1423329 12.28 AD-1423333 9.09 AD-1423330
 8.78 AD-1423310 6.66 AD-1423314 8.66 AD-1423316 10.59 AD-1423322 18.97 AD-1423325
 16.15 AD-1423334 10.71 AD-1423312 12.77 AD-1423313 9.33 AD-1423315 7.66 AD-1423318
 13.43 AD-1423319 8.34 AD-1423321 11.82 AD-1423323 9.65 AD-1423326 5.15 AD-1423328
 22.84 AD-1423331 13.21 AD-1423332 10.54 AD-1423335 8.60
 (628) TABLE-US-00016 TABLE 8 Unmodified Sense and Antisense Strand
 Sequences of KHK dsRNAs Range in Range in Duplex SEQ ID GenBank SEQ ID
 GenBank Name Sense sequence 5' to 3' NO: Source Name Source Name Antisense
 sequence 5' to 3' NO: Source Name AD-1290757.3 GUUGAUCUGACCCAGUUCAAU
 1180 XM_017004061.1_769- 769-789 AUUGAACUGGGUCAGAUCAACCU 298 767-789
 789_G21U_s AD-1290878.3 GGAGACGUGGUGUUUGUCAGU 92 XM_017004061.1_943-
 943-963 ACUGACAAACACCACGUCUCCGU 316 941-963 963_C21U_s AD-1290969.3
 ACCUUCA AUGCCUCCGUCAUU 108 XM_017004061.1_1162- 1162-1182
 AAUGACGGAGGCAUUGAAGGUGU 332 1160-1182 1182_C21U_s AD-1423317.2
 CUUUGAGAAGGUUGAUCUGAU 71 XM_017004061.1_759- 759-779
 AUCAGATCAACCUUCUCAAGUC 1143 757-779 779_C21U_s AD-1423327.2
 AAGGUUGAUCUGACCCAGUUU 73 XM_017004061.1_766- 766-786
 AAAGTGGGUCAGAUCAACCUUCU 1147 764-786 786_C21U_s AD-1423336.2
 GUUCAAGUGGAUCCACAUGU 207 783-803 ACAATGTGGAUCCACUUGAACUG 1152
 781-803 AD-1290599.3 UGGUGUUUGUCAGCAAAGAUU 93 XM_017004061.1_950- 950-970
 AAUCUUUGCUGACAAACACCACG 317 948-970 970_G21U_s AD-1523172.1
 UGGUGUUUGUCAGCAAAGAUU 93 XM_017004061.1_950- 950-970
 AAUCTUUGCUGACAAACACCACG 1182 948-970 970_G21U_s AD-1290837.3
 UGAUCUGACCCAGUUCAAGUU 76 XM_017004061.1_771- 771-791
 AACUUGAACUGGGUCAGAUCAAC 300 769-791 791_G21U_s AD-1523173.1
 UGAUCUGACCCAGUUCAAGUU 76 XM_017004061.1_771- 771-791
 AACUTGAACUGGGUCAGAUCAAC 1183 769-791 791_G21U_s AD-1290884.3
 GCAGGAAGCACUGAGAUUCGU 116 XM_017004061.1_1209- 1209-1229
 ACGAAUCUCAGUGCUUCCUGCAC 340 1207-1229 1229_G21U_s AD-1523174.1
 GCAGGAAGCACUGAGAUUCGU 116 XM_017004061.1_1209- 1209-1229
 ACGAAUCUCAGUGCUUCCUGCAC 340 1207-1229 1229_G21U_s AD-1290959.3
 UUGAUCUGACCCAGUUCAAGU 75 XM_017004061.1_770-790_s 770-790
 ACUUGAACUGGGUCAGAUCAACC 299 768-790 AD-1523175.1
 UUGAUCUGACCCAGUUCAAGU 75 XM_017004061.1_770-790_s 770-790
 ACUUGAACUGGGUCAGAUCAACC 299 768-790 AD-1423311.2
 GACGUGGUGUUUGUCAGCAAU 222 946-966 AUUGCUGACAAACACCACGUCUC 446
 944-966 AD-1423324.2 CCUUCA AUGCCUCCGUCAUCU 109 XM_017004061.1_1163- 1163-
 1183 AGAUGACGGAGGCAUUGAAGGUG 333 1161-1183 1183_s AD-1523176.1
 CCUUCA AUGCCUCCGUCAUCU 109 XM_017004061.1_1163- 1163-1183
 AGAUGACGGAGGCAUUGAAGGUG 333 1161-1183 1183_s AD-1423329.2
 GAAGAGAAGCAGAUCCUGUGU 54 XM_017004061.1_526-546_C21U_s 526-546
 ACACAGGAUCUGCUUCUCUCCA 278 524-546 AD-1423333.2
 GUGGUGUUUGUCAGCAAAGAU 1181 XM_005264298.1_812-830_s 810-830
 AUCUTUGCUGACAAACACCACGU 1184 808-830 AD-1423330.2
 GUGUUUGUCAGCAAAGAUGUU 95 XM_017004061.1_952-972_G21U_s 952-972
 AACATCTUUGCUGACAAACACCA 1149 950-972 AD-1523177.1
 GUGUUUGUCAGCAAAGAUGUU 95 XM_017004061.1_952-972 G21U s 952-972
 AACATCUUUGCUGACAAACACCA 1185 950-972 AD-1290885.3
 CUGACCCAGUUCAAGUGGAUU 78 XM_017004061.1_775-795_C21U_s 775-795
 AAUCCACUUGAACUGGGUCAGAU 302 773-795 AD-1523178.1

CUGACCUUGAUCUUGAUU 78 XM_017004061.1_775-795_C21U_s 775-795
 AAUCCACUUGAACUGGGUCAGAU 302 773-795 AD-1423334.2
 CCUCAUGGAAGAGAAGCAGAU 50 XM_017004061.1_519-539_s 519-539
 AUCUGCTUCUCUCCAUGAGGCU 1151 517-539 AD-1523179.1
 CCUCAUGGAAGAGAAGCAGAU 50 XM_017004061.1_519-539_s 519-539
 AUCUGCUUCUCUCCAUGAGGCU 274 517-539 AD-1523180.1
 CCUCAUGGAAGAGAAGCAGAU 50 XM_017004061.1_519-539_s 519-539
 AUCUGCTUCUCUCCAUGAGGCU 1151 517-539 AD-1290539.3
 CAGACUUUGAGAAGGUUGAUU 68 XM_017004061.1_755-775_C21U_s 755-775
 AAUCAACCUUCUCAAGUCUGUA 292 753-775 AD-1523181.1
 CAGACUUUGAGAAGGUUGAUU 68 XM_017004061.1_755-775_C21U_s 755-775
 AAUCAACCUUCUCAAGUCUGUA 292 753-775
 (629) TABLE-US-00017 TABLE 9 Modified Sense and Antisense Strand Sequences
 of KHK dsRNAs Sense SEQ SEQ SEQ Duplex Sequence ID ID ID Name 5' to 3' NO:
 Antisense Sequence 5' to 3' NO: mRNA Target Sequence NO: AD-1290757.3
 gsusugauCfuGfAfC 1186 asUfsugaAfcugggucAfgAfucaacscsu 746
 AGGUUGAUCUGACCCAGUUCAAG 970 fccaguucaauL96 AD-1290878.3 gsgsagacGfuGfGfU
 540 asCfsugaCfaaacaccAfcGfucuccsgsu 764 ACGGAGACGUGGUGUUUGUCAGC 988
 fguuugucaguL96 AD-1290969.3 ascscuucAfaUfGfC 556 asAfsugaCfaggaggcaUfuGfaaggusgsu 780
 ACACCUUCAAUGCCUCCGUCAUC 1004 fcuccgucauuL96 AD-1423317.2 csusuugaGfaAfGfG
 519 asUfscadGa(Tgn)caaccuUfcUfcaaag 1161 GACUUUGAGAAGGUUGAUCUGAC 967
 fuugaucugauL96 susc AD-1423327.2 asasgguuGfaUfCfU 521
 asAfsacdTg(G2p)gucagaUfcAfaccuu 1171 AGAAGGUUGAUCUGACCCAGUUC 969
 fgaccaguuuL96 scsu AD-1423336.2 gsusucaaGfuGfGfA 655 asCfsaadTg(Tgn)ggauccAfcUfugaac
 1179 CAGUUCAAGUGGAUCCACAUGA 1103 fuccacauuguL96 susg AD-1290599.3
 usgsguguUfuGfUfC 541 asAfsucuUfugcugacAfaAfcaccascsg 765
 CGUGGUGUUUGUCAGCAAAGAUG 989 fagcaaagauuL96 AD-1523172.1 usgsguguUfuGfUfC
 541 asAfsucdTu(U2p)gcugacAfaAfcacca 1188 CGUGGUGUUUGUCAGCAAAGAUG 989
 fagcaaagauuL96 scsg AD-1290837.3 usgsaucuGfaCfCfC 524 asAfscuuGfaacugggUfcAfgaucasasc
 748 GUUGAUCUGACCCAGUUCAAGUG 972 faguucaaguuL96 AD-1523173.1
 usgsaucuGfaCfCfC 524 asAfscudTg(A2p)acugggUfcAfgauca 1189
 GUUGAUCUGACCCAGUUCAAGUG 972 faguucaaguuL96 sasc AD-1290884.3
 gscsaggaAfgCfAfC 564 asCfsgaaUfcucagugCfuUfccugcsasc 788
 GUGCAGGAAGCACUGAGAUUCGG 1012 fugagauucguL96 AD-1523174.1
 gscsaggaAfgCfAfC 564 asCfsgadAu(C2p)ucagugCfuUfccugc 1190
 GUGCAGGAAGCACUGAGAUUCGG 1012 fugagauucguL96 sasc AD-1290959.3
 ususgaucUfgAfCfC 523 asCfsuugAfacuggguCfaGfaucaascsc 747
 GGUUGAUCUGACCCAGUUCAAGU 971 fcaguucaaguL96 AD-1523175.1 ususgaucUfgAfCfC
 523 asCfsuudGa(A2p)cuggguCfaGfaucaa 1191 GGUUGAUCUGACCCAGUUCAAGU 971
 fcaguucaaguL96 scSc AD-1423311.2 gsascgugGfuGfUfU 670
 asUfsugdCu(G2p)acaaacAfcCfacguc 1155 GAGACGUGGUGUUUGUCAGCAA 1118
 fugucagcaauL96 susc AD-1423324.2 cscsuucaAfuGfCfC 557 asGfsaudGa(C2p)ggaggcAfuUfgaag
 1168 CACCUUCAAUGCCUCCGUCAUCU 1005 fuccgucaucuL96 gsusg AD-1523176.1
 cscsuucaAfuGfCfC 557 asGfsaudGa(Cgn)ggaggcAfuUfgaag 1192
 CACCUUCAAUGCCUCCGUCAUCU 1005 fuccgucaucuL96 gsusg AD-1423329.2
 gsasagagAfaGfCfA 502 asCfsacdAg(G2p)aucugcUfuCfucuuc 1173
 UGGAAGAGAAGCAGAUCCUGUGC 950 fgauccuguguL96 scsa AD-1423333.2
 gsusggugUfuUfGfU 468 asUfscudTu(G2p)cugacaAfaCfaccac 1193
 ACGUGGUGUUUGUCAGCAAAGAU 916 fcagcaaagauL96 sgsu AD-1423330.2
 gsusguuuGfuCfAfG 543 asAfscadTc(Tgn)uugcugAfcAfaacac 1174

UGGUGUUUGUCAGCAAAGAUGUG 991 fcaaagauuuL96 scsa AD-1523177.1
 gsusguuuGfuCfAfG 543 asAfscaTc(U2p)uugcugAfcAfaacac 1194
 UGGUGUUUGUCAGCAAAGAUGUG 991 fcaaagauuuL96 scsa AD-1290885.3
 csusgaccCfaGfUfU 526 asAfsuccAfcuugaacUfgGfgucagsasu 750
 AUCUGACCCAGUUCAAGUGGAUC 974 fcaaguggauuL96 AD-1523178.1 csusgaccCfaGfUfU
 526 asAfsucdCa(C2p)uugaacUfgGfgucag 1195 AUCUGACCCAGUUCAAGUGGAUC 974
 fcaaguggauuL96 AD-1423334.2 cscsucauGfgAfAfG 498 sUfscudGc(Tgn)ucucuuCfcAfugagg 1177
 AGCCUCAUGGAAGAGAAGCAGAU 946 fagaagcagauL96 scsu AD-1523179.1
 cscsucauGfgAfAfG 498 asUfscudGc(U2p)ucucuuCfcAfugag 1196
 AGCCUCAUGGAAGAGAAGCAGAU 946 fagaagcagauL96 gscsu AD-1523180.1
 cscsucauGfgAfAfG 498 asUfscudGcdTucucuuCfcAfugaggsc 1197
 AGCCUCAUGGAAGAGAAGCAGAU 946 fagaagcagauL96 su AD-1290539.3
 csasgacuUfuGfAfG 516 asAfsucaAfcuucucAfaAfgucugsusa 740
 UACAGACUUUGAGAAGGUUGAUC 964 faagguugauuL96 AD-1523181.1 csasgacuUfuGfAfG
 516 asAfsucdAa(C2p)cuucucAfaAfgucug 1198 UACAGACUUUGAGAAGGUUGAUC 964
 faagguugauuL96 susa

Example 4. Design, Synthesis and In Vitro Screening of Additional dsRNA Duplexes

(630) Additional siRNAs were designed, synthesized, and prepared using the methods described above in Example 1.

(631) Detailed lists of the additional unmodified KHK sense and antisense strand nucleotide sequences are shown in Table 10. Detailed lists of the modified KHK sense and antisense strand nucleotide sequences are shown in Table 11.

(632) TABLE-US-00018 TABLE 10 Unmodified Sense and Antisense Strand Sequences of KHK dsRNA SEQ SEQ Duplex Sense ID Range in Antisense ID Range in Name Sequence 5' to 3' NO: XM_017004061.1 Sequence 5' to 3' NO:

Name	Sequence	5' to 3' NO:	XM_017004061.1	Sequence	5' to 3' NO:
AD-1290969	ACCUUCA AUGCCUCCGUCAUU	1199	1162-1182		
332	AAUGACGGAGGCAUUGAAGGUGU	1160-1182	AD-1423310		
72	UUUGAGAAGGUUGAUCUGACU	760-780	AGUCAGAUCAACCUUCUCAAAGU	296	
758-780	AD-1423311	GACGUGGUGUUUGUCAGCAAU	222	946-966	
446	AUUGCUGACAAACACCACGUCUC	944-966	AD-1423312		
115	UGCAGGAAGCACUGAGAUCU	1208-1228	AGAATCTCAGUGCUUCCUGCACG	1141	
1206-1228	AD-1423317	CUUUGAGAAGGUUGAUCUGAU	71	759-779	
1143	AUCAGATCAACCUUCUCAAAGUC	757-779	AD-1423319		
70	GACUUUGAGAAGGUUGAUCUU	757-777	AAGATCAACCUUCUCAAAGUCUG	1144	
755-777	AD-1423323	CAAUGCCUCCGUCAUCUUCAU	111	1167-1187	
335	AUGAAGAUGACGGAGGCAUUGAA	1165-1187	AD-1423327		
73	AAGGUUGAUCUGACCCAGUUU	766-786	AAACTGGGUCAGAUCAACCUUCU	1147	
764-786	AD-1423329	GAAGAGAAGCAGAUCCUGUGU	54	526-546	
278	ACACAGGAUCUGCUUCUCUCCA	524-546	AD-1423330		
95	GUGUUUGUCAGCAAAGAUGUU	952-972	AACATCTUUGCUGACAAACACCA	1149	
950-972	AD-1423333	GUGGUGUUUGUCAGCAAAGAU	20	949-969	
1184	AUCUTUGCUGACAAACACCACGU	947-969	AD-1423334		
50	CCUCAUGGAAGAGAAGCAGAU	519-539	AUCUGCTUCUCUCCAUGAGGCU	1151	
517-539	AD-1423336	GUUCAAGUGGAUCCACAUGU	207	783-803	
1152	ACAATGTGGAUCCACUUGAACUG	781-803	AD-1523180		
50	CCUCAUGGAAGAGAAGCAGAU	519-539	AUCUGCTUCUCUCCAUGAGGCU	1151	
517-539	AD-1548743	UGGUGUUUGUCAGCAAAGAUU	93	950-970	
1269	AAUCTUTGCUGACAAACACCACG	948-970	AD-1612957		
1200	GCCUCAUGGAAGAGAAGCAGU	518-538	ACUGCUTCUCUTCCAUGAGGCUA	1270	
516-538	AD-1612958	CCUCAUGGAAGAGAAGCAGAU	50	519-539	

ATCUGCTUACCTUCCUAGGCU 1271 517-539 AD-1612963
UGGAAGAGAAGCAGA UCCUGU 1201 524-544 ACAGGATCUGCTUCUCUCCAUG 1272
522-544 AD-1612967 AGAGAAGCAGA UCCUGUGCGU 1202 528-548
ACGCACAGGAUCUGCUUCUCUUC 1273 526-548 AD-1612969
AGAAGCAGA UCCUGUGCGUGU 1203 530-550 ACACGCACAGGAUCUGCUUCUCU 1274
528-550 AD-1612970 GAAGCAGA UCCUGUGCGUGGU 1204 531-551
ACCACGCACAGGAUCUGCUUCUC 1275 529-551 AD-1613059
CAGACUUUGAGAAGGUUGAUU 68 755-775 AAUCAACCUUCTCAAAGUCUGUA 1276
753-775 AD-1613060 AGACUUUGAGAAGGUUGAUCU 69 756-776
AGAUCAACCUUCUCAAGUCUGU 293 754-776 AD-1613062
ACUUUGAGAAGGUUGAUCUGU 213 758-778 ACAGAUCAACCTUCUCAAGUCU 1277
756-778 AD-1613070 AAGGUUGAUCUGACCCAGUUU 73 766-786
AAACTGGGUCAGA UCAACCUUCU 1147 764-786 AD-1613073
GUUGAUCUGACCCAGUUCAAU 74 769-789 ATUGAACUGGGTCAGA UCAACCU 1278
767-789 AD-1613074 UUGAUCUGACCCAGUUCAAGU 75 770-790
ACUUGAACUGGGUCAGA UCAACC 299 768-790 AD-1613075
UGAUCUGACCCAGUUCAAGUU 76 771-791 AACUTGAACUGGGUCAGA UCAAC 1183
769-791 AD-1613076 GAUCUGACCCAGUUCAAGUGU 77 772-792
ACACTUGAACUGGGUCAGA UCAA 1279 770-792 AD-1613077
AUCUGACCCAGUUCAAGUGGU 1205 773-793 ACCACUTGAACTGGGUCAGA UCA 1280
771-793 AD-1613079 CUGACCCAGUUCAAGUGGAUU 78 775-795
AAUCCACUUGAACUGGGUCAGAU 302 773-795 AD-1613088
UUCAAGUGGAUCCACA UUGAU 82 784-804 ATCAAUGUGGATCCACUUGAACU 1281
782-804 AD-1613089 UCAAGUGGAUCCACA UUGAGU 83 785-805
ACUCAATGUGGAUCCACUUGAAC 1282 783-805 AD-1613090
CAAGUGGAUCCACA UUGAGGU 84 786-806 ACCUCA AUGUGGAUCCACUUGAA 308
784-806 AD-1613091 AAGUGGAUCCACA UUGAGGGU 1206 787-807
ACCCTCAAUGUGGAUCCACUUGA 1283 785-807 AD-1613094
UGGAUCCACA UUGAGGGCCGU 1207 790-810 ACGGCCCUCAATGUGGAUCCACU 1284
788-810 AD-1613095 GGAUCCACA UUGAGGGCCGGU 1208 791-811
ACCGGCCCUCAAUGUGGAUCCAC 1285 789-811 AD-1613237
GUUUGGCUACGGAGACGUGGU 1209 933-953 ACCACGTCUCCGUAGCCAAACAG 1286
931-953 AD-1613238 UUUGGCUACGGAGACGUGGUU 1210 934-954
AACCACGUCUCCGUAGCCAAACA 1287 932-954 AD-1613239
UUGGCUACGGAGACGUGGUGU 1211 935-955 ACACCACGUCUCCGUAGCCAAAC 1288
933-955 AD-1613240 UGGCUACGGAGACGUGGUGUU 1212 936-956
AACACCACGUCTCCGUAGCCAAA 1289 934-956 AD-1613241
GGCUACGGAGACGUGGUGUUU 1213 937-957 AACACCACGUCUCCGUAGCCAA 1290
935-957 AD-1613242 GCUACGGAGACGUGGUGUUUU 242 938-958
AAAACACCACGTCUCCGUAGCCA 1291 936-958 AD-1613243
CUACGGAGACGUGGUGUUUGU 89 939-959 ACAAACACCACGUCUCCGUAGCC 313 937-
959 AD-1613244 UACGGAGACGUGGUGUUUGUU 90 940-960
AACAAACACCACGUCUCCGUAGC 314 938-960 AD-1613245
ACGGAGACGUGGUGUUUGUCU 232 941-961 AGACAAACACCACGUCUCCGUAG 456
939-961 AD-1613246 CGGAGACGUGGUGUUUGUCAU 91 942-962
ATGACAAACACCACGUCUCCGUA 1292 940-962 AD-1613247
GGAGACGUGGUGUUUGUCAGU 92 943-963 ACUGACAAACACCACGUCUCCGU 316
941-963 AD-1613254 UGGUGUUUGUCAGCAAAGAUU 93 950-970
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949-971 AD-1613256 GUGUUGUCAGCAAAGAUGUU 95 952-972
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UGUUUGUCAGCAAAGAUGUGU 96 953-973 ACACAUCUUUGCUGACAAACACC 320
951-973 AD-1613258 GUUUGUCAGCAAAGAUGUGGU 97 954-974
ACCACATCUUUGCUGACAAACAC 1295 952-974 AD-1613259
UUUGUCAGCAAAGAUGUGGCU 98 955-975 AGCCACAUCUUTGCUGACAAACA 1296
953-975 AD-1613361 AGCUGGAGACACCUUCA AUGU 105 1152-1172
ACAUTGAAGGUGUCUCCAGCUCC 1297 1150-1172 AD-1613362
GCUGGAGACACCUUCA AUGCU 1214 1153-1173 AGCATUGAAGGTGUCUCCAGCUC
1298 1151-1173 AD-1613363 CUGGAGACACCUUCA AUGCCU 106 1154-1174
AGGCAUTGAAGGUGUCUCCAGCU 1299 1152-1174 AD-1613369
ACACCUUCA AUGCCUCCGUCU 1215 1160-1180 AGACGGAGGCATUGAAGGUGUCU
1300 1158-1180 AD-1613370 CACCUUCA AUGCCUCCGUCAU 1216 1161-1181
ATGACGGAGGCAUUGAAGGUGUC 1301 1159-1181 AD-1613371
ACCUUCA AUGCCUCCGUCAU 108 1162-1182 AAUGACGGAGGCAUUGAAGGUGU 332
1160-1182 AD-1613374 UUCA AUGCCUCCGUCAU CUUU 110 1165-1185
AAAGAUGACGGAGGCAUUGAAGG 334 1163-1185 AD-1613377
AAUGCCUCCGUCAU CUUCAGU 112 1168-1188 ACUGAAGAUGACGGAGGCAUUGA 336
1166-1188 AD-1613378 AUGCCUCCGUCAU CUUCAGCU 1217 1169-1189
AGCUGAAGAUGACGGAGGCAUUG 1302 1167-1189 AD-1613395
AGCGUGCAGGAAGCACUGAGU 1218 1204-1224 ACUCAGTGCUUCCUGCACGCUCC
1303 1202-1224 AD-1613400 GCAGGAAGCACUGAGAUUCGU 116 1209-1229
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CAGGAAGCACUGAGAUUCGGU 1219 1210-1230 ACCGAATCUCAGUGCUUCCUGCA
1305 1208-1230 AD-1634353 UAGCCUCAUGGAAGAGAAGCU 1220 516-536
AGCUTCTCUUCCAUGAGGCUACU 1306 514-536 AD-1634354
GCCUCAUGGAAGAGAAGCAGU 1200 518-538 ACUGCUTCUCUCCAUGAGGCCUA 1307
516-538 AD-1634355 CAUGGAAGAGAAGCAGAUCCU 1221 522-542
AGGATCTGCUUCUCUCCAUGAG 1308 520-542 AD-1634356
GGAAGAGAAGCAGAUCCUGUU 53 525-545 AACAGGAUCUGCUUCUCUCCAU 277
523-545 AD-1634357 AGAGAAGCAGAUCCUGUGCGU 1202 528-548
ACGCACAGGAUCUGCUUCUCUUC 1273 526-548 AD-1634358
AGACUUUGAGAAGGUUGAUCU 69 756-776 AGAUCAACCUUCUCA AAGUCUGU 293
754-776 AD-1634359 UGAGAAGGUUGAUCUGACCCU 1222 762-782
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AGAAGGUUGAUCUGACCCAGU 1223 764-784 ACUGGGTCAGAUCAACCUUCUCA 1310
762-784 AD-1634361 GGUUGAUCUGACCCAGUUCAU 239 768-788
AUGAACTGGGUCAGAUCAACCUU 1311 766-788 AD-1634362
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769-791 AD-1634363 UCUGACCCAGUUCAAGUGGAU 1224 774-794
AUCCACTUGAACUGGGUCAGAU 1312 772-794 AD-1634364
UGACCCAGUUCAAGUGGAUCU 1225 776-796 AGAUCCACUUGAACUGGGUCAGA 1313
774-796 AD-1634365 CCCAGUUCAAGUGGAUCCACU 1226 779-799
AGUGGATCCACUUGAACUGGGUC 1314 777-799 AD-1634366
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778-800 AD-1634367 GUGGAUCCACAUUGAGGGCCU 1227 789-809
AGGCCCTCAAUGUGGAUCCACU 1315 787-809 AD-1634368
AGACGUGGUGUUUGUCAGCAU 1228 945-965 AUGCTGACAAACACCACGUCUCC 1316
943-965 AD-1634369 ACGUGGUGUUUGUCAGCAAU 21 947-967
AUUUGCTGACAAACACCACGUCU 1317 945-967 AD-1634370

UUUGCAGCAAGAUGCCUG 98 955-975 AGCCACAUCUUGACAAACA 322
953-975 AD-1634371 UGUCAGCAAAGAUGUGGCCAU 1229 957-977
AUGGCCACAUCUUUGCUGACAAA 1318 955-977 AD-1634372
GCAAAGAUGUGGCCAAGCACU 1230 962-982 AGUGCUTGGCCACAUCUUUGCUG 1319
960-982 AD-1634373 CUGGAGACACCUUCAUGCCU 106 1154-1174
AGGCAUTGAAGGUGUCUCCAGCU 1299 1152-1174 AD-1634374
UGGAGACACCUUCAUGCCUU 107 1155-1175 AAGGCATUGAAGGUGUCUCCAGC 1320
1153-1175 AD-1634375 GGAGACACCUUCAUGCCUCU 1231 1156-1176
AGAGGCAUUGAAGGUGUCUCCAG 1321 1154-1176 AD-1634376
ACACCUUCAUGCCUCCGUCU 1215 1160-1180 AGACGGAGGCAUUGAAGGUGUCU
1322 1158-1180 AD-1634377 CUUCAUGCCUCCGUCAUCUU 1232 1164-1184
AAGATGACGGAGGCAUUGAAGGU 1323 1162-1184 AD-1634378
UCA AUGCCUCCGUCAUCUUCU 223 1166-1186 AGAAGATGACGGAGGCAUUGAAG 1324
1164-1186 AD-1634379 AUGCCUCCGUCAUCUUCAGCU 1217 1169-1189
AGCUGAAGAUGACGGAGGCAUUG 1302 1167-1189 AD-1634380
UGCCUCCGUCAUCUUCAGCCU 1233 1170-1190 AGGCTGAAGAUGACGGAGGCAUU
1325 1168-1190 AD-1634381 CCUCCGUCAUCUUCAGCCUCU 1234 1172-1192
AGAGGCTGAAGAUGACGGAGGCA 1326 1170-1192 AD-1634382
GAGGAGCGUGCAGGAAGCACU 1235 1200-1220 AGUGCUTCCUGCACGCUCCUCCC
1327 1198-1220 AD-1634383 AGGAGCGUGCAGGAAGCACUU 1236 1201-1221
AAGUGCTUCCUGCACGCUCCUCC 1328 1199-1221 AD-1634384
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1303 1202-1224 AD-1634385 CGUGCAGGAAGCACUGAGAUU 1237 1206-1226
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CUUCA AUGCCUCCGUCAUU 1238 1164-1182 AAUGACGGAGGCAUUGAAGGU 1330
1162-1182 AD-1634387 ACCUACAAUGCCUCCGUCAUU 1239 1162-1182
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1332 1160-1182 AD-1634389 ACGUUCA AUGCCUCCGUCAUU 1241 1162-1182
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ACAUUCA AUGCCUCCGUCAUU 1242 1162-1182 AAUGACGGAGGCAUUGAAUGUGU
1334 1160-1182 AD-1634391 CUUCA AUGCCUCCGUCAUU 1238 1164-1182
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ACCUACAAUGCCUCCGUCAUU 1239 1162-1182 AAUGACGGAGGCAUUGUAGGUGU
1331 1160-1182 AD-1634393 ACCAUCA AUGCCUCCGUCAUU 1240 1162-1182
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1333 1160-1182 AD-1634395 ACAUUCA AUGCCUCCGUCAUU 1242 1162-1182
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GUUCAAGUGGAUCCACAUGU 207 783-803 ACAATGTGGAUCCACUUGAACUG 1152
781-803 AD-1634397 GUUCAAGUGGAUCCACAUGU 207 783-803
ACAATGTGGAUCCACUUGAACUG 1152 781-803 AD-1634398
GUUCUAGUGGAUCCACAUGU 1243 783-803 ACAATGTGGAUCCACUAGAACUG 1335
781-803 AD-1634399 GUUGAAGUGGAUCCACAUGU 1244 783-803
ACAATGTGGAUCCACUUGAACUG 1336 781-803 AD-1634400
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781-803 AD-1634401 GUACAAGUGGAUCCACAUGU 1246 783-803
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764-786 AD-1634403 AAGGUUGAUCUGACUCAGUUU 1248 766-786

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764-786 AD-1634405 AAGUUUGAUCUGACCUAGUUU 1250 766-786
AAAGTAGGUCAGAAACUUCU 1342 764-786 AD-1634406
AAGUUUGAUCUGACUCAGUUU 1251 766-786 AAAGTGAGUCAGAAACUUCU 1343
764-786 AD-1634407 AAGUUUGAUCUGACUCAGUUU 1252 766-786
AAAGTGAGUCAGAAACUUCU 1344 764-786 AD-1634408
UGGUUUUUGUCAGCAAAGAUU 1253 950-970 AAAGTUTGCUGACAAAACACACG 1345
948-970 AD-1634409 UGGUCUUUGUCAGCAAAGAUU 1254 950-970
AAAGTUTGCUGACAAAAGACACACG 1346 948-970 AD-1634410
UGGAGUUUGUCAGCAAAGAUU 1255 950-970 AAAGTUTGCUGACAAACUCCACG 1347
948-970 AD-1634411 UGUUGUUUGUCAGCAAAGAUU 1256 950-970
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UGCUGUUUGUCAGCAAAGAUU 1257 950-970 AAAGTUTGCUGACAAACAGCACG 1349
948-970 AD-1634413 UGGUGUUUGUCAGCAAAGAUU 93 950-970
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UGGUGUUUGUCAGCAAAGAUU 93 950-970 AAAGTUTGCUGACAAACACACACG 1269
948-970 AD-1634415 CCUCAUGGAAGAGAAUCAGAU 1258 519-539
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CCUCUUGGAAGAGAAUCAGAU 1259 519-539 ATCUGATUCUCTUCCAAGAGGCU 1352
517-539 AD-1634417 CCUGAUGGAAGAGAAUCAGAU 1260 519-539
ATCUGATUCUCTUCCAUCAGGCU 1353 517-539 AD-1634418
CCUAAUGGAAGAGAAUCAGAU 1261 519-539 ATCUGATUCUCTUCCAUUAGGCU 1354
517-539 AD-1634419 CCACAUGGAAGAGAAUCAGAU 1262 519-539
ATCUGATUCUCTUCCAUGUGGCU 1355 517-539 AD-1634420
GACGUGGUGUUUGUCAGCAAU 222 946-966 AUUGCUGACAAACACACGUCUC 446
944-966 AD-1634421 GACGUGGUGUUUGUCAGCAAU 222 946-966
AUUGCTGACAAACACACGUCUC 1356 944-966 AD-1634422
GACGUGGUGUUUGUCAGCAAU 222 946-966 AUUGCTGACAAACACACGUCUC 1356
944-966 AD-1634423 GUGGUGUUUGUCAGCAAAGAU 20 949-969
AUCUTUGCUGACAAACACACGUC 1184 947-969 AD-1634424
GUGGUGUUUGUCAGUAAAGAU 1263 949-969 AUCUTUACUGACAAACACACGUC 1357
947-969 AD-1634425 GUGGUGUUUGUCAGCAAAGAU 20 949-969
AUCTUTGCUGACAAACACACGUC 1358 947-969 AD-1634426
GUGGAGUUUGUCAGUAAAGAU 1264 949-969 AUCUTUACUGACAAACUCCACGUC 1359
947-969 AD-1634427 GUGCUGUUUGUCAGUAAAGAU 1265 949-969
AUCUTUACUGACAAACAGCACGUC 1360 947-969 AD-1634428
GUGUUGUUUGUCAGUAAAGAU 1266 949-969 AUCUTUACUGACAAACAACACGUC 1361
947-969 AD-1634429 GUCGUGUUUGUCAGUAAAGAU 1267 949-969
AUCUTUACUGACAAACACGACGUC 1362 947-969 AD-1634430
GUUGUGUUUGUCAGUAAAGAU 1268 949-969 AUCUTUACUGACAAACACAACGUC 1363
947-969 AD-1634431 GUGGAGUUUGUCAGUAAAGAU 1264 949-969
AUCTUTGCUGACAAACUCCACGUC 1364 947-969 AD-1634432
GUGCUGUUUGUCAGUAAAGAU 1265 949-969 AUCTUTGCUGACAAACAGCACGUC 1365
947-969 AD-1634433 GUGUUGUUUGUCAGUAAAGAU 1266 949-969
AUCTUTGCUGACAAACAACACGUC 1366 947-969 AD-1634434
GUCGUGUUUGUCAGUAAAGAU 1267 949-969 AUCTUTGCUGACAAACACGACGUC 1367
947-969 AD-1634435 GUUGUGUUUGUCAGUAAAGAU 1268 949-969
AUCTUTGCUGACAAACAACACGUC 1368 947-969 AD-1634436
GAAGAGAAGCAGAUCCUGUGU 54 526-546 ACACAGGAUCUGCUUCUCUUCU 1369

524-546 AD-1634AGAGAAAGCAGCAUCCUGUGU 54 526-546

ACACAGGAUCUGCUUCUCUCCU 1369 524-546

(633) TABLE-US-00019 TABLE 11 Modified Sense and Antisense Strand Sequences and KHK dsRNA SEQ SEQ Duplex ID ID Name Sense Sequence 5' to 3' NO:

Antisense Sequence 5' to 3' NO: AD- ascscuucAfaUfGfCfcuccgucauuL96 1370

asAfsugaCfaggaggcaUfuGfaaggusgsu 780 1290969 AD- ususugagAfaGfGfUfugaucugacuL96 1153

asGfsucdAg(Agn)ucaaccUfuCfucaaasgs 1154 1423310 u AD- gsascgugGfuGfUfUfugucagcaauL96

670 asUfsugdCu(G2p)acaaacAfcCfacgucsusc 1155 1423311 AD-

usgscaggAfaGfCfAfcugagauucuL96 563 asGfsaadTc(Tgn)cagugcUfuCfcugcascsg 1156 1423312

AD- csusuugaGfaAfGfGfuugaucugauL96 519 asUfscadGa(Tgn)caaccUfcUfcaaagsusc 1161

1423317 AD- gsascuuuGfaGfAfAfgguugaucuuL96 518 asAfsgadTc(Agn)accuucUfcAfaagucsus

1163 1423319 g AD- csasaugcCfuCfCfGfucaucuucL96 559

asUfsgadAg(Agn)ugacggAfgGfcauugsas 1167 1423323 a AD- asasgguuGfaUfCfUfgaccaguuuL96

521 asAfsacdTg(G2p)gucagaUfcAfaccuuscs 1171 1423327 u AD-

gsasagagAfaGfCfAfgauccuguguL96 502 asCfsacdAg(G2p)aucugcUfuCfucuucscs 1173 1423329 a

AD- gsusguuuGfuCfAfGfcaaagauguuL96 543 asAfsacdTc(Tgn)uugcugAfcAfaacacscsa 1174

1423330 AD- gsusggugUfuUfGfUfcagcaaagauL96 468 asUfscudTu(G2p)cugacaAfaCfaccacsgsu

1193 1423333 AD- cscsucauGfgAfAfGfagaagcagauL96 498

asUfscudGc(Tgn)ucucuUfcAfugaggscs 1177 1423334 u AD- gsusucaaGfuGfGfAfuccacauuguL96

655 asCfsaadTg(Tgn)ggauccAfcUfugaacsusg 1179 1423336 AD-

cscsucauGfgAfAfGfagaagcagauL96 498 asUfscudGcdTucucuUfcAfugaggscsu 1197 1523180

AD- usgsguguUfuGfUfCfagcaaagauuL96 541 asAfsucdTu(Tgn)gcugacAfaAfcaccascsg 1482

1548743 AD- gscscucaugGfAfAfgagaagcaguL96 1371 asdCsugdCudTcucudTcCfaugaggcsusa

1483 1612957 AD- cscsucauggAfAfGfagaagcagauL96 1372 asdTscudGcdTucudTuCfcaugaggscsu

1484 1612958 AD- usgsgaagagAfAfGfcagauccuguL96 1373

asdCsagdGadTcugcdTuCfucuuccasusg 1485 1612963 AD- asgsagaagcAfGfAfuccugugcguL96

1374 asdCsgcdAcdAggaudCuGfcuucucususc 1486 1612967 AD-

asgsaagcagAfUfCfcugugcguguL96 1375 asdCsacdGcdAcaggdAuCfugcuucuscscu 1487 1612969

AD- gsasagcagaUfCfCfugugcgugguL96 1376 asdCscadCgdCacagdGaUfcugcuucscusc 1488

1612970 AD- csasgacuuuGfAfGfaagguugauuL96 1377 asdAsucdAadCcuudTcAfaagucugsusa

1489 1613059 AD- asgsacuugAfGfAfagguugaucuL96 1378

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1379 asdCsagdAudCaacdTuCfucaaaguscsu 1491 1613062 AD-

asasgguugaUfCfUfgaccaguuuL96 1380 asdAsacdTgdGgucadGaUfcaaccuuscsu 1492 1613070

AD- gsusugaucuGfAfCfccaguucaauL96 1381 asdTsgudAadCugggdTcAfgaucaacscsu 1493

1613073 AD- ususgaucugAfCfCfcaguucaaguL96 1382 asdCsuudGadAcuggdGuCfagaucascsc

1494 1613074 AD- usgsaucugaCfCfCfaguucaaguL96 1383

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1384 asdCsacdTudGaacudGgGfucagaucsasa 1496 1613076 AD-

asuscugaccCfAfGfuucaagugguL96 1385 asdCscadCudTgaacdTgGfgucagauscscsa 1497 1613077

AD- csusgaccaGfUfUfcaaguggauuL96 1386 asdAsucdCadCuugadAcUfgggucagsasu 1498

1613079 AD- ususcaagugGfAfUfccacauugauL96 1387 asdTscadAudGuggadTcCfacuugaascscu

1499 1613088 AD- uscsaaguggAfUfCfcacauugaguL96 1388

asdCsucdAadTguggdAuCfcacuugasasc 1500 1613089 AD- csasaguggaUfCfCfacaugagguL96

1389 asdCscudCadAugugdGaUfccacuugsasa 1501 1613090 AD-

asasguggauCfCfAfcauugagggguL96 1390 asdCscddTcdAaugudGgAfuccacuusgsa 1502 1613091

AD- usgsgauccaCfAfUfugagggccguL96 1391 asdCsggdCcdCucaadTgUfggauccascscu 1503

1613094 AD- gsgsauccacAfUfUfgagggccguL96 1392 asdCscgdGcdCcucadAuGfuggauccascsc

1504 1613095 AD- gsusuuggcuAfCfGfgagacgugguL96 1393

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1394 asdAscdAcCdCgUfagccaaasc 1506 1613238 AD-
ususggcuacGfGfAfgacgugguguL96 1395 asdCsacdCadCgucudCcGfuagccaasasc 1507 1613239
AD- usgsgcuacgGfAfGfacgugguguL96 1396 asdAscadCcdAcgucdTcCfaguagccasasa 1508
1613240 AD- gsgscuacggAfGfAfcgugguguuuL96 1397 asdAsacdAcCdCagudCuCfcguagccsasa
1509 1613241 AD- gscsuacggaGfAfCfugguguuuuL96 1398
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1399 asdCsaadAcCdAccacdGuCfuccguagcsc 1511 1613243 AD-
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1613245 AD- csgsgagacGfUfGfGfuguuuugucauL96 1402 asdTsgadCadAacacdCaCfuguccgsusa
1514 1613246 AD- gsgsagacguGfGfUfguuugucaguL96 1403
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1404 asdAsudTudTgcugdAcAfaacaccascsg 1516 1613254 AD-
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AD- gsusguuuGfUfCfAfgcaaagauuL96 1406 asdAscadTcdTuugcdTgAfcaaaccascsa 1518
1613256 AD- usgsuuugucAfGfCfaaagauguL96 1407 asdCsacdAudCuuugdCuGfacaacascsc
1519 1613257 AD- gsusuugucaGfCfAfaagaugugguL96 1408
asdCscadCadTcuuudGcUfgacaaacsasc 1520 1613258 AD- ususugucagCfAfAfagauguggcuL96
1409 asdGscddAcCdAucudTgCfugacaaascsa 1521 1613259 AD-
asgscuggagAfCfAfccucaauguL96 1410 asdCsaudTgdAaggudGuCfuccagcuscsc 1522 1613361
AD- gscsuggagaCfAfCfcucaaugcuL96 1411 asdGscadTudGaaggdTgUfcuccagcsusc 1523
1613362 AD- csusggagacAfCfCfucaauggcuL96 1412 asdGsgcdAudTgaagdGuGfuccagcscsu
1524 1613363 AD- ascsaccuucAfAfUfgccuccgucuL96 1413
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1414 asdTsgadCgdGaggcdAuUfgaaggugsusc 1526 1613370 AD-
ascscuucaaUfGfCfuccgucauuL96 1415 asdAsugdAcCdGaggdCaUfugaaggusgsu 1527 1613371
AD- ususcaaugcCfUfCfcgucaucuuuL96 1416 asdAsagdAudGacggdAgGfcauugaasgsg 1528
1613374 AD- asasugccucCfGfUfcaucucaguL96 1417 asdCsugdAadGaugadCgGfaggcauusgsa
1529 1613377 AD- asusgccuccGfUfCfaucucagcuL96 1418
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1419 asdCsudAgdTgcuudCcUfgcacgcuscsc 1531 1613395 AD-
gscsaggaagCfAfCfugagauucguL96 1420 asdCsgadAudCucagdTgCfuuccugcsasc 1532 1613400
AD- csasggaagcAfCfUfgagauucgguL96 1421 asdCscgdAadTcucadGuGfcuuccugscsa 1533
1613401 AD- usasgccuCfaUfGfGfaagagaagcuL96 1422 asGfscudTc(Tgn)cuuccaUfgAfggcuascsu
1534 1634353 AD- gscscucaUfgGfAfAfgagaagcaguL96 1423
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1424 asGfsgadTc(Tgn)gcuucCfuUfccaugsag 1536 1634355 AD-
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AD- asgsagaaGfcAfGfAfuccugugcguL96 1425 asCfsgcdAc(Agn)ggaucuGfcUfucucusus 1538
1634357 c AD- asgsacuuUfgAfGfAfagguugaucuL96 517 asGfsaudCa(Agn)ccuucCfaAfagucusgs
1539 1634358 u AD- usgsagaaGfgUfUfGfaucugacccuL96 1426
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AD- usgsaucuGfaCfCfCfaguucaaguL96 524 asAfscudTg(Agn)acugggUfcAfgaucasas 1543
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1544 1634363 c AD- usgsacccAfgUfUfCfaaguggaucuL96 1429
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1430 asGfsugdGa(Tgn)ccacuGfaAfcugggsus 1546 1634365 c AD-
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AD- gsusggauCfCfAfCfAfugaggccuL96 1431 asGfsgcdCc(Tgn)cauguGfgAfuccacsus 1548
1634367 u AD- asgsacguGfgUfGfUfuugucagcauL96 1432
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1553 1634372 g AD- csusggagAfcAfCfCfuucaaugccuL96 554
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AD- gsasagagAfaGfCfAfgauccuguguL96 502 asdCsacdAgdGaucudGcUfucucuucscsu 1617
1634436 AD- gsasagagAfaGfCfAfgauccuguguL96 502 asCfsacadGg(Agn)ucugcUfuCfucuucscs
1618 1634437 u SEQ Duplex ID Name mRNA Target Sequence NO: AD-
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GACUUUGAGAAGGUUGAUCUGA 967 1423317 C AD-
CAGACUUUGAGAAGGUUGAUCU 966 1423319 G AD- UUCAUUGCCUCCGUCAUCUUA
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UGGAAGAGAAGCAGAUCCUGUG 950 1634437 C

Example 5. Design, Synthesis and In Vitro Screening of Additional dsRNA Duplexes

(634) Based on the in vitro analyses, structure-active relationship (SAR) analyses were performed. In particular, additional duplexes were designed, synthesized, and assayed.

(635) siRNAs were designed, synthesized, and prepared using the methods described above. In vitro screening assays in Hep3B and PCH cells with these siRNAs were performed as described above.

(636) Detailed lists of the unmodified KHK sense and antisense strand nucleotide sequences are shown in Table 12. Detailed lists of the modified KHK sense and antisense strand nucleotide sequences are shown in Table 13.

(637) For transfections, cells (ATCC, Manassas, VA) were grown to near confluence at 37° C. in an atmosphere of 5% CO₂ in Eagle's Minimum Essential Medium (Gibco) supplemented with 10% FBS (ATCC) before being released from the plate by trypsinization. Transfection was carried out by adding 7.5 µl of Opti-MEM plus 0.1 µl of Lipofectamine RNAiMax per well (Invitrogen, Carlsbad CA. cat #13778-150) to 2.5 µl of each siRNA duplex to an individual well in a 384-well plate. The mixture was then incubated at room temperature for 15 minutes. Forty µl of complete growth media without antibiotic containing $\sim 1.5 \times 10^4$ cells were then added to the siRNA mixture. Cells were incubated for 24 hours prior to RNA purification. Single dose experiments were performed at 10, 1 and 0.1 nM final duplex concentration.

(638) Total RNA isolation was performed using DYNABEADS. Briefly, cells were lysed in 10 µl of Lysis/Binding Buffer containing 3 µL of beads per well and mixed for 10 minutes on an electrostatic shaker. The washing steps were automated on a Biotek EL406, using a magnetic plate support. Beads were washed (in 3 L) once in Buffer A, once in Buffer B, and twice in Buffer E, with aspiration steps in between. Following a final aspiration, complete 12 µL RT mixture was added to each well, as described below.

(639) For cDNA synthesis, a master mix of 10.5 µl 10× Buffer, 0.6 µl 10× dNTPs, 1.5 µl Random primers, 0.75 µl Reverse Transcriptase, 0.75 µl RNase inhibitor and 9.9 µl of H₂O per reaction were added per well. Plates were sealed, agitated for 10 minutes on an electrostatic shaker, and then incubated at 37 degrees C. for 2 hours. Following this, the plates were agitated at 80 degrees C. for 8 minutes.

(640) RT-qPCR was performed as described above and relative fold change was calculated as described above. The results of the transfection assays of the dsRNA agents listed in Tables 12 and 13 in Hep3B cells are shown in Table 14. The results of the transfection assays of the dsRNA agents listed in Tables 12 and 13 in primary cynomolgus hepatocytes (PCH) are shown in Table 15.

(641) For Dual-Glo® Luciferase assay, cells (ATCC, Manassas, VA) were grown to near confluence at 37° C. in an atmosphere of 5% CO₂ in Eagle's Minimum Essential Medium (Gibco) supplemented with 10% FBS (ATCC) before being released from the plate by trypsinization. Dual-Glo® Luciferase constructs were generated in the psiCHECK2 plasmid containing the human KHK genomic sequence. Each dual-luciferase plasmid was co-transfected with siRNA (Tables 12 and 13) into approximately 2×10⁵ cells using Lipofectamine 2000 (Invitrogen, Carlsbad CA. cat #11668-019). For each well of a 96 well plate, 0.5 µl of Lipofectamine was added to 100 ng of plasmid vector and a single siRNA (Tables 12 and 13) in 14.8 µl of Opti-MEM and allowed to complex at room temperature for 15 minutes. The mixture was then added to the cells which were resuspended in 80 µl of fresh complete media. Cells were incubated for 24 hours before luciferase was measured. Single dose experiments were performed at 10, 1 and 0.1 nM final duplex concentration.

(642) Forty-eight hours after the siRNAs were transfected, Firefly (transfection control) and *Renella* (fused to KHK target sequence) luciferase were measured. First, media was removed from cells. Then Firefly luciferase activity was measured by adding 75 µl of Dual-Glo® Luciferase Reagent equal to the culture medium volume to each well and mix. The mixture was incubated at room temperature for 30 minutes before luminescence (500 nm) was measured on a Spectramax (Molecular Devices) to detect the Firefly luciferase signal. *Renilla* luciferase activity was measured by adding 75 µl of room temperature Dual-Glo® Stop & Glo® Reagent to each well and the plates were incubated for 10-15 minutes before luminescence was again measured to determine the *Renilla* luciferase signal. The Dual-Glo® Stop & Glo® Reagent, quenches the firefly luciferase signal and sustains luminescence for the *Renilla* luciferase reaction. siRNA activity was determined by normalizing the *Renilla* (KHK) signal to the Firefly (control) signal within each well. The magnitude of siRNA activity was then assessed relative to cells that were transfected with the same vector but were not treated with siRNA or were treated with a non-targeting siRNA. All transfections were done in quadruplicates.

(643) Table 16 shows the Dual-Glo® Luciferase assay results of a single dose screen in cells transfected with the indicated agents in Tables 12 and 13.

(644) TABLE-US-00020

TABLE 12 Unmodified Sense and Antisense Strand Sequences of KHK dsRNA		Sense Strand SEQ ID Range		Antisense Strand SEQ ID Range	
ID	Range	in Duplex	Name Sequence 5' to 3'	NO. XM_017004061.1	Sequence 5' to 3' NO. XM_017004061.1
AD-1290507.2	UUCUUCAUCUGUCAAAUGGAU	1658	2200-2220	AD-1290509.2	AUCCAUUUGACAGAUGAAGAAAU
415	2198-2220	AD-1290509.2	GAUUAAAAUCUGCCAUUUAAU	167	2130-2150
AUUAUAAUGGCAGAUUUUAAUCCA	391	2128-2150	AD-1290510.2	AAUUUCUUCAUCUGUCAAAUU	183
2197-2217	AAUUUGACAGAUGAAGAAAUUGA	407	2195-2217	AD-1290514.2	CAAUUUCUUCAUCUGUCAAAU
182	2196-2216	AUUUGACAGAUGAAGAAAUUGAG	406	2194-2216	AD-1290515.2
AGGCCUUAUAAUGUAAAGAGU	158	2073-2093	ACUCUUUACAUAUAAGGCCUUA	382	2071-2093
AD-1290516.2	GGAUUAAAAUCUGCCAUUUUAU	192	2129-2149	AUAAAUGGCAGAUUUUAAUCCAG	416
2127-2149	AD-1290522.2	UCUGGAACACAUAUUGGAAUU	149	2025-2045	AAUUCCAUAUGUGUCCAGAU
373	2023-2045	AD-1290523.2	GCCUCAAUUUCUUCAUCUGUU	181	2192-2212
AACAGAUGAAGAAAUUGAGGCAG	405	2190-2212	AD-1290524.2	AUUUCUUCAUCUGUCAAAUGU	184
2198-2218	ACAUUUGACAGAUGAAGAAAUUG	408	2196-2218	AD-1290527.2	UGGAUUAAAAUCUGCCAUUUU
193	2128-2148	AAAUGGCAGAUUUUAAUCCAGG	417	2126-2148	AD-1290528.2
CUGGAACACAUAUUGGAAUUU	150	2026-2046	AAAUUCCAUAUGUGUCCAGAU	374	2024-2046
AD-1290531.2	CUGCCUCAAUUUCUUCAUCUU	179	2190-2210	AAGAUGAAGAAAUUGAGGCAGAU	403
2188-2210	AD-1290533.2	AAUCUGCCAUUUAAUAGCU	169	2135-2155	

AGCUAAUUGGAGUUA 393 2133-2155 AD-1290535.2
AAUCUGCCAUAUUAAUAGCUU 170 2136-2156 AAGCUAAUAAAUGGCAGAUUUU 394
2134-2156 AD-1290539.5 CAGACUUUGAGAAGGUUGAUU 68 755-775
AAUCAACCUUCUCAAGUCUGUA 292 753-775 AD-1290542.2
UCAAUUUCUUCAUCUGUCAU 194 2195-2215 AUUGACAGAUGAAGAAAUUGAGG 418
2193-2215 AD-1290543.2 UGGAACACAUAUUGGAAUUGU 151 2027-2047
ACAAUUCCAAUAUGUGUUCAG 375 2025-2047 AD-1290551.2
CUGGAUUA AAAAUCUGCCAUAU 166 2127-2147 AAAUGGCAGAUUUUAAUCCAGGU 390
2125-2147 AD-1290552.2 UUUCUUCAUCUGUCAAAUGGU 195 2199-2219
ACCAUUUGACAGAUGAAGAAAUAU 419 2197-2219 AD-1290554.2
AUAAUGUAAAGGGCUUUAGAU 162 2096-2116 AUCUAAAGCCCUUUACAUAUAU 386
2094-2116 AD-1290555.2 AUAUAAUGUAAAGGGCUUUUAU 161 2094-2114
AUAAAGCCCUUUACAUAUAUAGC 385 2092-2114 AD-1290556.2
GCCUUAUA AUGUAAAGAGCAU 159 2075-2095 AUGCUCUUUACAUAUAAGGCCU 383
2073-2095 AD-1290557.2 GGCCUUAUA AUGUAAAGAGCU 196 2074-2094
AGCUCUUUACAUAUAAGGCCU 420 2072-2094 AD-1290558.2
UAUAAUGUAAAGGGCUUUAGU 197 2095-2115 ACUAAAGCCCUUUACAUAUAUG 421
2093-2115 AD-1290563.2 GUAAGGCCUUAUA AUGUAAAU 156 2070-2090
AUUUACAUAUAAGGCCUUAACC 380 2068-2090 AD-1290564.2
CCUCAAUUUCUUCAUCUGUCU 199 2193-2213 AGACAGAUGAAGAAAUUGAGGCA 423
2191-2213 AD-1290565.2 CUCAAUUUCUUCAUCUGUCAU 200 2194-2214
AUGACAGAUGAAGAAAUUGAGGC 424 2192-2214 AD-1290570.2
AAGGCCUUAUA AUGUAAAGAU 157 2072-2092 AUCUUUACAUAUAAGGCCUUAAC 381
2070-2092 AD-1290573.2 AUCUGCCUCAAUUUCUUCAU 177 2188-2208
AAUGAAGAAAUUGAGGCAGAUUG 401 2186-2208 AD-1290574.2
UAAGGCCUUAUA AUGUAAAGU 201 2071-2091 ACUUUACAUAUAAGGCCUUAAC 425
2069-2091 AD-1290584.2 GUGUUUGUCAGCAAAGAUGUU 95 952-972
AACAUUUUGCUGACAAACACCA 319 950-972 AD-1290589.2
GGAACACAUAUUGGAAUUGGU 152 2028-2048 ACCAAUUCCAAUAUGUGUUCAG 376
2026-2048 AD-1290592.2 CAUAUA AUGUAAAGGGCUUUU 202 2093-2113
AAAAGCCCUUUACAUAUAUGCU 426 2091-2113 AD-1290597.2
AUUAAAAUCUGCCAUAUUAU 168 2131-2151 AAUUA AUGGCAGAUUUUAAUCC 392
2129-2151 AD-1290599.7 UGGUGUUUGUCAGCAAAGAUU 93 950-970
AAUCUUUGCUGACAAACACCACG 317 948-970 AD-1290600.2
GCUUGGCUACAGAAUUAUUGU 190 2229-2249 ACAUA AUUCUGUAGCCAAGCAG 414
2227-2249 AD-1290602.2 UGCCUCAAUUUCUUCAUCUGU 180 2191-2211
ACAGAUGAAGAAAUUGAGGCAGA 404 2189-2211 AD-1290604.2
AUCUGCCAUAUUAUAGCUGU 171 2137-2157 ACAGCUAAUUA AUGGCAGAUUU 395
2135-2157 AD-1290605.2 AAUCUGCCUCAAUUUCUUCAU 176 2187-2207
AUGAAGAAAUUGAGGCAGAUUGC 400 2185-2207 AD-1290609.2
UGCCAUAUUAUAGCUGCAU 203 2140-2160 AAUGCAGCUAAUUA AUGGCAGA 427
2138-2160 AD-1290611.3 AGACUUUGAGAAGGUUGAUCU 59 756-776
AGAUCAACCUUCUCAAGUCUGU 293 754-776 AD-1290612.2
AUCUGGAACACAUAUUGGAAU 148 2024-2044 AUUCCAAUAUGUGUUCAGAUUCG 372
2022-2044 AD-1290615.2 UCUGCCUCAAUUUCUUCAUCU 178 2189-2209
AGAUGAAGAAAUUGAGGCAGAUU 402 2187-2209 AD-1290618.2
AAUGUAAAGGGCUUUAGAGUU 164 2098-2118 AACUCUAAAGCCCUUUACAUAU 388
2096-2118 AD-1290624.2 AUUAUUGUGAGGAUAAAUCU 204 2242-2262
AGAUUUUAUCCUCACAAUAUUC 428 2240-2262 AD-1290626.2
GGUAAGGCCUUAUA AUGUAAU 205 2069-2089 AUUACAUAUAAGGCCUUAACCA 429

2067-2089 AD-1290633.2 CUGCCAAUUAUAGCUGCAU 172 2139-2159
AUGCAGCUAAUUAUAAUGGCAGAU 396 2137-2159 AD-1290635.2
UUCUGCUUGGCUACAGAAUUU 206 2225-2245 AAUUCUGUAGCCAAGCAGAAUU 430
2223-2245 AD-1290639.2 UAAUGUAAAGGGCUUUAGAGU 163 2097-2117
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2226-2248 AD-1290650.2 CGCAAUCUGCCUCAAUUUCUU 174 2184-2204
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AUACAUUAUAAGGCCUUAACCCAC 432 2066-2088 AD-1290655.2
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1939-1961 AD-1290750.2 CCCAGUGAACCUGCCAAAGAU 221 1706-1726
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2063-2085 AD-1290763.2 GUCCGAUCUGGAACACAUAUU 145 2019-2039
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2062-2084 AD-1290805.2 UGGAGCCCACCUUGGAAUUAU 227 1940-1960
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2060-2082 AD-1290911.2 AGCCACCUUGGAAUUAAGGU 140 1943-1963
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GCCCACCUUGGAAUUAAGGGU 141 1944-1964 ACCCUUAAUCCAAGGUGGGCUC 365
1942-1964 AD-1290931.2 UCAGCCACAAUGUGACCTAU 143 1970-1990
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2061-2083 AD-1290969.7 ACCUUCAAUGCCUCCGUCAUU 108 1162-1182
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1750-1772 AD-1290989.2 GCGUUGUGCAGACUCUAUUCU 132 1751-1771
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1764-1786 AD-1291003.2 GGCGUGCCUCAGCCACAAAUU 142 1962-1982
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515-537 AD-1684594.1 CUCAUGGAAGAGAAGCAGAUU 51 520-540
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519-541 AD-1684596.1 AUGGAAGAGAAGCAGAUCCUU 52 523-543
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525-547 AD-1684598.1 GAGAAGCAGAUCCUGUGCGUU 1663 529-549
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532-554 AD-1684604.1 CAGAUCCUGUGCGUGGGGCUU 1667 535-555
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763-785 AD-1684616.1 AGGUUGAUCUGACCCAGUUCU 1676 767-787
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769-791 AD-1684620.1 AUCUGACCCAGUUCAAGUU 1678 773-791
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775-797 AD-1684622.1 ACCCAGUUCAAGUGGAUCCAU 79 778-798
AUGGAUCCACUUGAACUGGGUCA 303 776-798 AD-1684623.1
CAGUUCAAGUGGAUCCACAUU 23 781-801 AAUGTGGAUCCACUUGAACUGGG 1724
779-801 AD-1684624.1 AGUUCAAGUGGAUCCACAUUU 81 782-802
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786-808 AD-1684626.1 UGGAUCCACAUUGAGGGCCGU 1207 790-810
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790-812 AD-1684628.1 AUCCACAUUGAGGGCCGGAU 1682 793-813
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942-964 AD-1684630.1 CGUGGUGUUUGUCAGCAAAGU 219 948-968
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954-976 AD-1684632.1 GUCAGCAAAGAUGUGGCCAAU 1685 958-978
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957-979 AD-1684634.1 CAGCAAAGAUGUGGCCAAGCU 1687 960-980
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959-981 AD-1684636.1 GAGACACCUUCAUGCCUCCU 1689 1157-1177
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1737 1156-1178 AD-1684638.1 GACACCUUCAUGCCUCCGUU 1691 1159-1179
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1161-1183 AD-1684640.1 GCCUCCGUCAUCUUCAGCCUU 1692 1171-1191
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CUCCGUCAUCUUCAGCCUCUU 113 1173-1193 AAGAGGCUGAAGAUGACGGAGGC 337
1171-1193 AD-1684642.1 GGAGCGUGCAGGAAGCACUGU 1693 1202-1222
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1741 1201-1223 AD-1684644.1 GCGUGCAGGAAGCACUGAGAU 1695 1205-1225
AUCUCAGUGCUUCCUGCACGCUC 1742 1203-1225 AD-1684645.1
GUGCAGGAAGCACUGAGAUUU 114 1207-1227 AAAUCUCAGUGCUUCCUGCACGC 338
1205-1227 AD-1684646.1 CCCAGUGAACCUGCCAAAGAU 221 1706-1726
ATCUTUGGCAGGUUCACUGGGUG 1743 1704-1726 AD-1684647.1

CCAGCUAAGCCAAAGAAU 235 1707-1727 ATUCTTGGCAGUUCACUGGGU 1744
1705-1727 AD-1684648.1 CUGCGUUGUGCAGACUCUAUU 130 1749-1769
AAUAGAGUCUGCACAACGCAGGG 354 1747-1769 AD-1684649.1
UGCGUUGUGCAGACUCUAUUU 131 1750-1770 AAUAGAGUCUGCACAACGCAGG 355
1748-1770 AD-1684650.1 GCGUUGUGCAGACUCUAUUCU 132 1751-1771
AGAATAGAGUCTGCACAACGCAG 1745 1749-1771 AD-1684651.1
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1750-1772 AD-1684652.1 GUUGUGCAGACUCUAUUCU 243 1753-1773
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1752-1774 AD-1684654.1 UAUUCCACAGCUCAGAAGCU 135 1766-1786
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1765-1787 AD-1684656.1 UGCCCACCAGCCUGUGAUUUU 229 1852-1872
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1851-1873 AD-1684658.1 CUUGGAGCCCACCUUGGAAU 137 1938-1958
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1937-1959 AD-1684660.1 UGGAGCCCACCUUGGAAUUAU 227 1940-1960
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1939-1961 AD-1684662.1 GAGCCCACCUUGGAAUUAAGU 139 1942-1962
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1941-1963 AD-1684664.1 GCCCACCUCUGGAAUUAAGGGU 141 1944-1964
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1960-1982 AD-1684666.1 UCAGCCACAAUGUGACCCAU 143 1970-1990
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2015-2037 AD-1684668.1 GUCCGAUCUGGAACACAUAU 144 2018-2038
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2021-2043 AD-1684674.1 AUCUGGAACACAUAUUGGAU 148 2024-2044
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2025-2047 AD-1684678.1 GGAACACAUAUUGGAUUUGGU 152 2028-2048
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2062-2084 AD-1684682.1 GGUGGGUAAGGCCUUAUAAUU 154 2065-2085
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2064-2086 AD-1684684.1 UGGGUAAGGCCUUAUAAUGUU 215 2067-2087
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GGGUAAGGCCUUAUAAUGUAU 208 2068-2088 ATACAUTAUAAGGCCUUAACCCAC 1773
2066-2088 AD-1684686.1 GGUAAGGCCUUAUAAUGUAAU 205 2069-2089
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GUAAGGCCUUAUAAUGUAAAU 156 2070-2090 ATUUACAUAUAAGGCCUUAACCC 1775
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ACAGCUAAUUAUAAUGGCAGAUUU 395 2135-2157 AD-1684709.1
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2136-2158 AD-1684710.1 CUGCCAUAUUAAUUAGCUGCAU 172 2139-2159
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 2184-2206 AD-1684716.1 AAUCUGCCUCAAUUUCUUCAU 176 2187-2207
 ATGAAGAAAUUGAGGCAGAUUGC 1797 2185-2207 AD-1684717.1
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 1798 2186-2208 AD-1684718.1 UCUGCCUCAAUUUCUUCAUCU 178 2189-2209
 AGAUGAAGAAATUGAGGCAGAUU 1799 2187-2209 AD-1684719.1
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 2190-2212 AD-1684722.1 CCUCAAUUUCUUCAUCUGUCU 199 2193-2213
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 CUCAAUUCUUCAUCUGUCAU 200 2194-2214 ATGACAGAUGAAGAAAUUGAGGC 1802
 2192-2214 AD-1684724.1 UCAAUUCUUCAUCUGUCAU 194 2195-2215
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 2198-2220 AD-1684730.1 AAUUCUGCUUGGCUACAGAAU 185 2223-2243
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 2222-2244 AD-1684732.1 UUCUGCUUGGCUACAGAAUUU 206 2225-2245
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 2224-2246 AD-1684734.1 CUGCUUGGCUACAGAAUUAU 188 2227-2247
 AAUAAUTCUGUAGCCAAGCAGAA 1812 2225-2247 AD-1684735.1
 UGCUUGGCUACAGAAUUAUUU 189 2228-2248 AAUAATUCUGTAGCCAAGCAGA 1813
 2226-2248 AD-1684736.1 GCUUGGCUACAGAAUUAUUGU 190 2229-2249
 ACAATAAUUCUGUAGCCAAGCAG 1814 2227-2249 AD-1684737.1
 AUUAUUGUGAGGAUAAAUCU 204 2242-2262 AGAUTUTAUCCTCACAAUAAUUC 1815
 2240-2262

(645) TABLE-US-00021 TABLE 13 Modified Sense and Antisense Strand Sequences
 of KHK dsRNA SEQ SEQ ID Antisense ID Duplex Name Strand Sequence 5' to 3'
 NO. Strand Sequence 5' to 3' NO. AD-1290507.2 ususcuucAfuCfUfGfucaaauggauL96 1816
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 asAfsuuuGfacagaugAfaGfaaauusgsa 855 AD-1290514.2 csasuuuUfuCfAfUfcugucaaaauL96 630
 asUfsuugAfcagaugaAfgAfaauugsas 854 g AD-1290515.2 asgsgccuUfaUfAfAfuguaaagaguL96 606
 asCfsucuUfuacauuaUfaAfggcccususa 830 AD-1290516.2 gsgsauuaAfaUfUfCfugccauuuauL96 640
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 asAfsCagAfugaagaaAfuUfgaggcsasg 853 AD-1290524.2 asusuucuUfcAfUfCfugucaaauguL96 632
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(646) TABLE-US-00022 TABLE 14 KHK Single Dose Screen in Hep3b Cells 10 nM 1 nM 0.1 nM

Average %	Average %	Average %	message	message	message	Duplex ID	remaining	STDEV
remaining	STDEV	remaining	STDEV	AD-1290635.2	98.30	12.80	109.90	0.00
133.00	0.00	AD-1684641.1	25.30	0.00	51.50	14.40	18.60	10.60
AD-1684620.1	7.20	2.70	11.30	0.70	10.10	2.10	AD-1684619.1	6.60
1.50	13.70	3.60	17.90	2.90	AD-1684610.1	7.60	1.60	15.40
3.70	25.00	7.40	AD-1684617.1	7.60	1.40	13.30	5.40	12.50
4.00	AD-1613079.2	9.70	2.20	36.70	8.80	33.60	5.70	AD-1684618.1
5.60	0.70	13.00	4.60	9.50	1.60	AD-1684622.1	8.30	2.50
14.00	2.30	17.60	4.50	AD-1684607.1	8.30	0.90	14.40	2.70
12.10	2.10	AD-1684614.1	7.80	1.50	22.00	7.60	23.30	7.70
AD-1613059.2	8.50	0.50	11.70	1.40	11.80	1.50	AD-1613075.2	6.90
0.70	16.00	4.90	19.60	6.80	AD-1613061.1	5.90	2.20	8.00
2.50	10.50	6.10	AD-1684608.1	10.20	2.80	19.20	3.00	18.80
6.80	AD-1684624.1	16.50	3.50	31.80	5.70	50.70	12.80	AD-1423319.3
9.20	0.70	14.40	2.10	14.90	1.60	AD-1684606.1	8.10	2.00
12.40	2.20	14.10	4.40	AD-1684609.1	8.40	1.20	17.00	2.90
14.00	4.00	AD-1684616.1	15.20	2.90	29.70	8.40	31.50	3.00
AD-1613060.2	13.60	1.70	27.00	1.90	29.80	9.50	AD-1613087.1	16.80
2.90	35.50	6.30	39.80	10.40	AD-1290539.5	12.30	2.80	18.30
2.90	21.70	2.50	AD-1290885.5	17.50	5.90	41.70	7.50	63.60
1.40	AD-1613062.2	12.10	2.10	24.10	4.90	28.60	9.00	AD-1684611.1
14.70	3.70	22.30	2.40	26.50	6.10	AD-1684623.1	18.60	3.60
31.90	3.00	59.60	13.80	AD-1612969.2	15.40	4.00	30.40	10.90
52.70	6.80	AD-1613256.2	8.30	1.20	23.60	6.70	33.40	3.80
AD-1423312.3	5.70	2.30	20.50	5.50	16.00	5.80	AD-1548743.7	14.60
2.40	24.20	2.30	34.60	8.10	AD-1684596.1	21.40	8.00	20.40
0.70	24.30	6.00	AD-1423336.7	10.90	1.50	32.80	3.40	36.60
5.80	AD-1612963.2	12.30	4.10	26.50	5.90	28.70	7.10	AD-1290611.3
15.60	3.80	32.40	7.40	54.20	10.00	AD-1613254.2	11.60	2.20
20.70	4.50	24.00	2.80	AD-1684612.1	11.70	1.00	19.50	2.70
31.70	9.10	AD-1684645.1	14.90	2.80	28.80	6.70	35.70	4.70
AD-1684621.1	21.20	6.50	59.50	12.40	42.00	9.10	AD-1290837.5	17.70
6.20	49.70	7.60	69.90	15.10	AD-1684598.1	37.30	9.50	67.00
19.00	66.40	10.90	AD-1684638.1	23.00	6.60	47.50	13.90	40.40
7.20	AD-1290599.7	11.00	1.20	18.60	3.30	40.00	5.00	AD-1613400.2
7.80	1.70	19.10	7.80	23.00	2.20	AD-1684615.1	17.50	3.20
41.00	2.90	52.70	6.40	AD-1684630.1	14.20	1.30	45.40	10.30
51.20	10.00	AD-1613072.1	43.70	14.40	72.70	6.50	86.70	11.40
AD-1684632.1	17.50	5.60	36.80	7.70	53.40	3.40	AD-1613242.2	11.60
2.60	28.40	9.80	35.20	8.90	AD-1684605.1	53.80	6.00	72.00
8.60	102.80	17.40	AD-1684594.1	16.70	3.90	27.10	4.20	35.80
9.10	AD-1612957.2	14.90	5.10	29.60	7.40	25.90	6.30	AD-1290971.3
13.60	2.40	37.60	9.50	36.90	12.60	AD-1684595.1	29.60	9.90
49.80	12.20	52.40	9.80	AD-1613371.3	14.20	4.40	27.70	5.80
32.10	2.80	AD-1684593.1	20.10	5.10	32.70	10.70	39.60	13.90
AD-1684639.1	13.50	4.50	35.10	6.50	39.90	2.30	AD-1613094.2	16.80
5.20	38.70	4.90	58.50	17.50	AD-1684597.1	30.30	3.00	61.00
14.40	48.90	4.70	AD-1290563.2	50.40	7.90	92.60	14.00	95.40
26.70	AD-1684599.1	41.60	13.60	66.40	17.20	66.80	14.50	AD-1684640.1
17.80	3.60	50.90	10.60	62.70	11.10	AD-1290584.2	32.10	9.80
70.00	11.80	76.10	20.40	AD-1684627.1	20.10	10.50	54.00	12.10
71.60	11.20	AD-1290969.7	18.50	5.40	31.80	7.00	34.30	5.10
AD-1290651.2	51.80	11.50	100.80	20.90	117.60	24.10	AD-1684628.1	17.40
7.00	26.80	5.30	34.10	6.40	AD-1290884.5	20.80	7.20	37.60
7.60	39.00	5.00	AD-1684633.1	25.60	5.90	52.30	3.50	77.80
17.00	AD-1684637.1	31.90	10.90	55.00	9.20	68.60	17.60	AD-1684629.1
38.00	5.90	86.00	13.60	91.70	16.40	AD-1684635.1	32.60	4.30
87.20	14.10	45.90	4.10	AD-1684613.1	43.20	9.20	84.20	23.20
75.90	26.30	AD-1684604.1	47.30	8.80	78.10	15.50	83.00	11.40
AD-1290666.2	28.30	8.10	62.20	16.50	74.10	11.60	AD-1684634.1	39.90
11.00	76.80	9.60	86.20	17.40	AD-1684600.1	70.30	11.40	101.80
5.30	63.80	22.40	AD-1684644.1	68.00	19.10	159.30	17.60	131.10
15.50	AD-1684643.1	96.80	16.70	130.10	23.00	140.40	19.50	AD-1684601.1
92.10	22.20	136.00	33.00	115.30	33.70	AD-1684626.1	35.10	

7.20 65.30 10.50 66.90 5.70 AD-1684642.1 72.30 18.80 116.80 22.20 124.30 8.60 AD-1684592.1
68.50 13.90 120.10 23.20 85.70 6.80 AD-1684636.1 52.00 18.10 88.40 34.10 87.30 10.60 AD-
1684603.1 133.70 13.10 143.80 21.30 128.00 22.60 AD-1684631.1 114.50 35.20 163.20 10.70
162.80 24.90 AD-1684625.1 74.60 12.90 121.80 25.30 105.40 20.30 AD-1290523.2 38.30 4.50
44.10 3.20 49.20 15.10 AD-1684602.1 91.40 21.30 151.40 24.60 110.60 24.70 AD-1290570.2
113.50 16.00 137.90 22.90 94.90 11.30 AD-1684666.1 116.20 5.80 109.50 14.40 143.90 23.60 AD-
1290865.2 58.40 13.50 64.50 10.30 77.90 7.80 AD-1290589.2 92.60 17.80 108.80 5.90 113.00
7.50 AD-1684681.1 111.40 18.20 129.40 13.10 131.70 21.70 AD-1684689.1 148.40 36.10 91.30
3.00 122.40 27.70 AD-1684708.1 51.20 10.40 83.10 17.60 93.00 21.90 AD-1290557.2 118.30
12.80 76.90 5.70 125.20 33.40 AD-1290515.2 106.60 14.40 116.50 20.20 98.10 9.30 AD-
1290741.2 109.20 9.20 119.20 28.90 144.10 20.80 AD-1684713.1 91.70 14.60 84.70 7.90 116.30
17.60 AD-1290650.2 89.60 6.80 75.90 10.30 123.80 18.30 AD-1290897.2 54.80 10.60 81.70 22.90
75.20 24.00 AD-1290556.2 123.30 14.80 92.50 13.30 115.10 12.00 AD-1290750.2 93.70 13.30
95.10 14.20 100.10 12.50 AD-1684714.1 98.00 6.20 136.70 24.30 118.10 12.60 AD-1290654.2
102.80 15.70 119.50 14.80 134.00 32.20 AD-1290909.2 91.20 10.30 108.90 34.40 81.90 22.00 AD-
1684674.1 96.60 21.20 118.40 19.30 112.90 27.70 AD-1684712.1 46.90 8.60 88.20 13.40 75.30
13.50 AD-1290796.2 111.90 12.00 119.70 14.30 171.60 18.40 AD-1290612.2 65.20 11.20 94.10
16.20 87.40 11.00 AD-1290633.2 106.60 11.10 94.20 9.50 97.00 23.00 AD-1684691.1 142.10
25.10 112.20 13.20 129.20 9.80 AD-1290659.2 79.40 14.80 103.40 10.30 120.70 13.50 AD-
1684686.1 107.60 17.70 129.20 33.80 108.30 23.50 AD-1290604.2 60.90 12.20 64.30 15.40 76.30
16.10 AD-1290574.2 121.10 5.80 122.10 15.40 128.60 12.90 AD-1290609.2 87.60 13.30 111.90
17.40 91.70 17.20 AD-1290911.2 134.00 15.60 65.80 14.70 82.30 16.50 AD-1290615.2 110.30
11.00 127.90 11.60 158.20 13.00 AD-1684710.1 95.20 12.40 103.60 33.60 117.10 10.50 AD-
1684709.1 89.50 19.50 81.20 18.80 92.00 6.10 AD-1290533.2 91.20 19.70 104.90 13.90 99.00
13.30 AD-1684671.1 78.00 7.50 102.80 18.70 98.50 11.20 AD-1684650.1 108.50 3.10 127.80
12.40 118.90 14.50 AD-1684711.1 109.10 20.40 116.90 21.40 106.80 24.30 AD-1684670.1 73.90
14.70 104.70 17.20 127.80 8.90 AD-1684692.1 96.10 16.30 113.50 17.10 103.70 8.30 AD-
1290939.2 112.00 10.60 108.90 18.70 101.00 11.20 AD-1684662.1 113.70 21.40 115.40 20.80
117.00 19.60 AD-1684700.1 97.00 21.20 99.20 20.90 65.60 13.10 AD-1290742.2 120.00 18.60
140.80 26.00 97.10 12.40 AD-1684715.1 85.70 18.20 72.70 18.50 120.60 8.70 AD-1290894.2
102.60 11.70 125.50 28.90 149.60 44.30 AD-1684718.1 112.50 13.00 122.40 22.90 136.80 18.20
AD-1290702.2 112.80 29.70 73.50 8.90 80.50 18.40 AD-1684688.1 106.50 25.50 124.40 36.40
123.10 16.60 AD-1684690.1 110.50 5.50 94.50 13.40 106.60 11.50 AD-1684672.1 94.00 12.20
91.00 4.70 98.10 9.30 AD-1684684.1 124.60 21.80 134.40 17.30 121.20 11.40 AD-1684723.1
122.90 4.90 137.70 20.30 142.70 35.90 AD-1684651.1 121.50 44.40 94.60 19.10 107.00 15.50
AD-1684683.1 86.10 5.00 122.80 16.60 91.60 12.10 AD-1290973.2 135.00 19.60 121.30 17.90
129.00 27.10 AD-1290857.2 106.00 12.00 105.00 22.10 89.90 13.10 AD-1684649.1 61.80 9.50
90.40 9.30 67.00 31.50 AD-1290516.2 106.50 19.40 84.40 15.50 70.60 6.80 AD-1290554.2 71.60
21.20 72.80 12.90 71.10 20.50 AD-1290509.2 115.50 16.80 112.90 19.60 92.90 15.10 AD-
1290660.2 132.60 30.90 135.00 21.70 98.20 28.10 AD-1684698.1 116.30 8.30 111.20 6.20 90.70
25.00 AD-1290670.2 122.20 6.90 135.80 24.50 150.90 20.40 AD-1684673.1 101.80 11.20 103.80
5.00 105.40 30.20 AD-1684646.1 148.30 19.10 159.50 17.50 128.30 21.40 AD-1290597.2 109.30
23.60 97.80 15.50 109.10 34.10 AD-1290573.2 71.20 7.50 88.60 4.80 93.50 6.40 AD-1684707.1
88.20 7.50 94.50 13.40 91.10 19.00 AD-1684722.1 102.60 11.40 147.30 34.10 162.60 1.40 AD-
1290639.2 88.00 11.40 74.20 8.40 83.20 11.10 AD-1290551.2 100.60 27.30 74.50 19.50 105.80
31.00 AD-1684655.1 146.00 14.60 127.60 21.40 109.60 18.60 AD-1684678.1 148.20 38.60 144.00
24.60 160.70 11.20 AD-1290800.2 129.20 17.20 141.00 10.40 119.30 16.10 AD-1684726.1 83.90
6.80 95.10 21.20 153.80 21.30 AD-1290764.2 85.70 9.50 100.30 16.60 89.50 14.40 AD-1290672.2
90.20 14.50 108.20 6.50 122.60 29.80 AD-1684685.1 158.00 23.20 147.70 27.30 131.50 17.80
AD-1290528.2 17.30 2.30 26.80 4.40 35.90 7.40 AD-1684696.1 83.80 15.70 87.80 20.90 56.60

13.20 AD-1290836.2 120.80 24.10 138.00 20.80 153.80 22.50 AD-1684682.1 99.50 7.40 121.50
22.20 102.10 19.30 AD-1684659.1 61.20 13.60 89.50 3.40 102.00 18.30 AD-1684669.1 101.90
15.80 114.60 19.50 112.10 24.30 AD-1684704.1 115.60 9.50 143.60 36.20 111.10 34.60 AD-
1684716.1 83.10 11.60 87.30 7.90 106.30 22.40 AD-1684699.1 106.50 14.40 111.70 29.80 64.20
17.80 AD-1290510.2 112.30 18.30 137.60 27.30 155.40 21.70 AD-1290531.2 87.20 9.20 113.60
33.80 125.10 21.60 AD-1684703.1 101.00 14.00 104.40 15.90 81.80 17.40 AD-1290910.2 96.60
6.80 96.30 35.70 123.50 13.10 AD-1684705.1 98.30 15.30 108.60 32.30 96.40 21.30 AD-
1290618.2 109.80 9.20 117.20 13.60 72.30 17.00 AD-1684668.1 95.50 5.80 105.80 16.30 103.00
8.00 AD-1684701.1 61.40 12.80 59.40 11.60 81.70 15.50 AD-1290542.2 110.40 18.30 120.10
23.60 110.00 3.90 AD-1290626.2 92.50 9.90 95.20 10.10 105.00 8.60 AD-1684724.1 98.40 8.20
106.50 12.10 144.80 26.30 AD-1290535.2 131.60 18.00 137.80 13.30 127.70 33.60 AD-1290558.2
129.70 41.60 107.30 30.00 61.90 12.50 AD-1684647.1 118.10 43.70 124.50 25.80 94.60 17.50 AD-
1290763.2 91.60 6.90 121.20 18.40 105.70 16.10 AD-1684677.1 94.50 18.30 97.10 22.20 103.50
4.00 AD-1684658.1 115.10 25.50 108.90 16.00 126.20 12.60 AD-1684687.1 92.20 17.60 107.40
15.70 104.80 11.50 AD-1684719.1 97.00 5.10 121.20 5.20 106.00 10.70 AD-1290681.2 130.40
12.30 120.30 17.30 113.50 18.10 AD-1684653.1 133.20 34.50 112.90 7.20 94.80 25.20 AD-
1290841.2 166.20 35.70 140.10 1.50 147.10 15.70 AD-1290687.2 83.90 6.80 140.00 15.70 138.80
12.90 AD-1290592.2 75.50 4.00 63.30 22.80 53.40 4.00 AD-1290522.2 86.40 5.50 102.50 18.10
95.70 11.40 AD-1290880.2 125.50 24.70 94.30 11.10 126.00 12.60 AD-1684693.1 56.90 19.40
53.80 7.10 46.50 6.60 AD-1684679.1 96.50 11.30 111.50 11.40 115.70 21.40 AD-1290555.2
102.60 12.20 59.80 6.50 74.00 33.30 AD-1684648.1 64.70 25.60 86.20 8.10 69.90 28.10 AD-
1684725.1 97.40 10.30 109.20 10.50 143.70 21.00 AD-1684706.1 37.50 4.70 46.80 16.60 55.10
14.90 AD-1684663.1 111.70 15.50 118.30 30.30 86.10 35.10 AD-1684702.1 100.20 11.00 79.80
12.00 81.80 10.10 AD-1291003.2 110.40 24.90 108.30 14.50 122.30 20.40 AD-1684664.1 110.20
5.00 117.70 22.00 98.80 7.20 AD-1684697.1 58.70 13.20 66.60 14.80 62.10 15.80 AD-1290514.2
89.80 6.40 108.60 7.20 151.30 18.10 AD-1290989.2 93.40 7.70 101.00 18.70 110.30 13.70 AD-
1684657.1 90.30 11.70 98.80 16.00 99.70 29.30 AD-1684654.1 128.30 24.50 109.50 22.10 105.60
33.40 AD-1684695.1 106.60 19.70 91.90 14.50 58.70 7.60 AD-1290712.2 78.60 8.50 126.00 31.80
96.20 13.60 AD-1290931.2 144.80 16.70 127.50 23.50 120.60 18.50 AD-1684729.1 117.20 9.80
134.20 14.70 138.70 25.30 AD-1290805.2 87.90 18.00 94.50 33.50 102.90 20.50 AD-1290755.2
117.00 13.30 124.90 28.50 125.00 29.10 AD-1290527.2 124.10 2.80 76.20 21.60 92.30 31.20 AD-
1684652.1 102.00 14.00 96.20 28.30 72.30 18.50 AD-1290507.2 107.20 18.20 118.90 22.10
145.70 41.10 AD-1290747.2 119.20 36.80 103.60 5.30 102.40 14.10 AD-1290926.2 104.90 20.70
105.00 24.10 85.60 16.00 AD-1290983.2 93.70 26.10 136.80 26.50 117.40 15.60 AD-1684680.1
119.70 9.90 115.70 20.40 112.70 18.00 AD-1290605.2 82.10 14.20 96.30 22.40 90.80 9.80 AD-
1290842.2 147.60 23.00 145.10 24.30 151.50 27.20 AD-1290565.2 90.90 12.90 107.40 14.80
162.30 28.00 AD-1290661.2 81.70 18.70 82.70 7.80 73.00 24.20 AD-1684667.1 41.60 3.10 54.00
5.90 61.80 5.20 AD-1290684.2 85.00 8.00 92.90 11.80 101.60 7.90 AD-1290524.2 114.60 17.20
143.40 15.60 139.30 26.90 AD-1290655.2 142.20 24.00 128.40 16.50 136.10 21.10 AD-1684660.1
110.20 18.10 134.30 16.20 122.60 20.30 AD-1684656.1 87.70 37.40 75.00 26.10 76.20 25.20 AD-
1684676.1 37.60 2.80 58.60 12.60 72.40 5.00 AD-1684727.1 104.30 31.00 110.10 20.50 117.50
12.10 AD-1684665.1 115.00 28.50 115.70 16.40 105.90 4.90 AD-1290993.2 126.50 14.30 107.50
12.10 98.50 26.30 AD-1684721.1 84.80 4.40 115.10 9.00 144.20 10.10 AD-1290543.2 41.90 8.40
52.50 9.70 54.00 10.30 AD-1290719.2 102.80 11.70 107.60 10.40 136.50 13.40 AD-1290602.2
97.50 7.70 111.60 23.90 116.30 21.10 AD-1290564.2 51.40 1.80 55.70 6.00 75.50 15.00 AD-
1684737.1 96.80 16.70 111.80 9.20 105.40 18.20 AD-1684694.1 63.70 21.10 65.10 20.50 72.20
15.80 AD-1290875.2 116.90 11.10 123.80 27.30 72.10 27.80 AD-1684732.1 99.40 14.40 91.50
17.70 138.80 7.00 AD-1684717.1 49.00 11.30 101.40 15.10 104.80 3.30 AD-1290908.2 118.70
7.20 117.70 15.60 97.80 14.70 AD-1684720.1 86.00 2.70 90.20 15.90 112.10 6.60 AD-1290552.2
98.50 16.80 121.20 17.10 137.80 29.70 AD-1684728.1 101.60 25.20 113.60 20.10 136.80 29.60

AD-1684731.1 137.20 10.40 126.20 11.10 142.60 2.40 AD-1290722.2 137.00 19.70 166.40 13.80
170.50 18.70 AD-1684730.1 130.20 30.50 117.60 15.60 117.80 9.00 AD-1684736.1 128.90 12.90
137.40 13.70 135.60 30.10 AD-1684733.1 169.30 14.60 170.60 26.10 178.70 11.10 AD-1684661.1
31.60 6.40 41.70 8.40 48.80 15.50 AD-1684734.1 80.20 13.20 65.20 20.10 85.00 20.90 AD-
1684735.1 114.70 26.20 94.40 18.00 119.80 18.20 AD-1290600.2 120.90 18.80 101.00 13.30
118.60 9.70 AD-1290624.2 130.50 13.50 141.60 27.20 115.50 17.40 AD-1290643.2 111.40 14.00
62.10 21.50 114.00 21.40 AD-1684675.1 94.00 21.20 89.60 10.90 84.00 7.70
(647) TABLE-US-00023 TABLE 15 KHK Single Dose Screen in PCH Cells 10 nM 1 nM 0.1 nM
Average % Average % Average % message message message Duplex ID remaining STDEV
remaining STDEV remaining STDEV AD-1290635.2 23.50 10.50 0.00 0.00 21.60 5.20 AD-
1684641.1 0.00 0.00 0.00 0.00 6.00 0.20 AD-1684620.1 1.00 0.10 1.70 0.50 3.20 0.30 AD-
1684619.1 1.30 0.10 2.00 0.10 2.80 0.30 AD-1684610.1 2.00 0.60 2.10 0.20 4.20 0.50 AD-
1684617.1 1.70 0.10 2.10 0.40 3.00 0.50 AD-1613079.2 1.40 0.40 2.30 0.20 3.90 1.00 AD-
1684618.1 1.00 0.30 2.30 0.30 3.40 0.60 AD-1684622.1 1.80 0.20 2.40 0.20 4.20 0.80 AD-
1684607.1 1.60 0.10 2.50 0.40 5.60 0.80 AD-1684614.1 2.10 0.70 2.60 0.30 4.30 0.30 AD-
1613059.2 1.90 0.40 2.70 0.40 5.10 0.20 AD-1613075.2 1.10 0.30 2.70 0.30 3.50 0.20 AD-
1613061.1 1.90 0.70 2.80 0.30 5.00 0.80 AD-1684608.1 2.10 0.30 2.80 0.20 6.10 1.10 AD-
1684624.1 1.70 0.20 2.80 1.00 7.00 1.30 AD-1423319.3 1.90 0.40 2.90 0.80 4.90 0.70 AD-
1684606.1 2.00 0.30 3.10 0.20 5.80 0.80 AD-1684609.1 2.00 0.30 3.10 0.30 5.50 0.60 AD-
1684616.1 2.20 0.50 3.10 0.80 5.20 1.40 AD-1613060.2 2.00 0.70 3.30 0.20 6.30 1.40 AD-
1613087.1 1.50 0.20 3.30 0.10 5.30 0.70 AD-1290539.5 2.50 0.10 3.80 0.10 6.20 0.70 AD-
1290885.5 2.00 0.40 3.90 0.60 6.30 0.90 AD-1613062.2 2.80 0.40 3.90 0.10 6.90 0.80 AD-
1684611.1 2.00 0.20 4.00 0.80 7.70 0.80 AD-1684623.1 3.10 0.30 4.00 0.80 7.90 1.80 AD-
1612969.2 3.90 0.40 4.70 0.60 11.30 1.20 AD-1613256.2 3.60 0.40 4.70 1.10 7.20 1.00 AD-
1423312.3 3.60 1.70 4.80 0.90 7.60 2.20 AD-1548743.7 4.40 1.30 4.90 0.30 9.30 1.90 AD-
1684596.1 4.40 0.60 4.90 0.70 13.40 3.90 AD-1423336.7 2.40 0.70 5.00 1.30 8.00 2.30 AD-
1612963.2 4.40 0.70 5.00 0.30 12.50 1.60 AD-1290611.3 2.60 0.70 5.20 1.10 12.60 3.50 AD-
1613254.2 3.50 1.00 5.20 1.40 9.80 2.60 AD-1684612.1 3.40 1.00 5.30 0.90 9.60 0.40 AD-
1684645.1 3.00 0.60 5.50 1.20 6.20 0.40 AD-1684621.1 3.10 0.40 5.80 0.10 12.10 3.10 AD-
1290837.5 2.50 0.60 5.90 0.80 13.90 1.50 AD-1684598.1 4.60 1.10 6.00 0.70 14.80 2.90 AD-
1684638.1 4.00 1.30 6.30 1.10 8.60 1.80 AD-1290599.7 3.70 1.00 6.50 1.10 11.80 1.80 AD-
1613400.2 4.50 1.30 6.60 1.40 10.30 2.40 AD-1684615.1 2.70 0.80 5.60 1.10 13.00 1.90 AD-
1684630.1 4.00 1.70 6.70 1.30 9.60 1.70 AD-1613072.1 3.30 1.00 6.90 0.70 12.30 1.80 AD-
1684632.1 4.10 1.50 7.00 1.30 8.00 1.10 AD-1613242.2 5.10 2.20 7.10 1.20 9.20 2.30 AD-
1684605.1 7.30 1.30 7.60 0.90 17.10 1.90 AD-1684594.1 6.70 1.00 7.70 1.20 15.60 2.10 AD-
1612957.2 5.00 0.80 7.80 1.80 14.00 1.10 AD-1290971.3 6.20 1.90 8.10 1.50 12.80 2.00 AD-
1684595.1 5.90 0.70 8.20 2.20 18.90 1.30 AD-1613371.3 4.40 1.10 8.80 1.00 9.60 1.10 AD-
1684593.1 6.30 0.80 8.80 1.10 12.00 2.50 AD-1684639.1 5.70 1.90 8.80 1.70 19.10 5.30 AD-
1613094.2 6.90 1.20 9.70 0.90 19.90 4.60 AD-1684597.1 5.70 1.00 9.70 1.60 22.40 2.80 AD-
1290563.2 17.60 5.00 9.80 0.00 27.40 8.90 AD-1684599.1 4.90 1.80 9.90 1.80 19.80 2.70 AD-
1684640.1 6.40 2.70 10.10 1.50 18.20 4.00 AD-1290584.2 5.50 2.50 10.40 1.30 15.80 3.80 AD-
1684627.1 5.90 1.70 12.30 1.00 21.60 3.70 AD-1290969.7 6.70 2.40 12.60 4.00 16.80 5.60 AD-
1290651.2 3.90 1.10 13.30 1.50 24.70 4.10 AD-1684628.1 9.10 2.40 13.30 0.20 18.50 2.20 AD-
1290884.5 5.40 1.60 13.60 1.70 22.20 5.90 AD-1684633.1 8.20 3.80 14.20 0.90 27.10 5.90 AD-
1684637.1 7.80 3.50 15.10 2.00 23.00 2.60 AD-1684629.1 5.80 1.90 16.90 3.20 24.00 1.40 AD-
1684635.1 12.10 2.70 17.10 1.20 27.50 7.70 AD-1684613.1 7.10 2.10 18.20 3.80 35.50 5.20 AD-
1684604.1 11.80 0.40 19.90 0.60 41.60 2.50 AD-1290666.2 10.00 3.00 20.80 4.80 37.00 2.30 AD-
1684634.1 10.00 4.00 22.70 4.00 37.30 6.30 AD-1684600.1 11.60 1.90 27.90 0.80 53.10 5.40 AD-
1684644.1 15.60 4.80 28.80 3.10 55.40 19.80 AD-1684643.1 12.20 4.20 30.20 2.80 41.90 4.30
AD-1684601.1 16.00 2.10 31.20 3.80 60.10 6.00 AD-1684626.1 21.50 6.40 31.60 5.10 60.50 16.20

AD-1684642.1 13.50 2.60 35.50 5.20 56.50 8.40 AD-1684592.1 21.80 1.90 35.70 5.00 59.70 6.20
AD-1684636.1 11.30 4.50 38.10 6.60 59.60 14.00 AD-1684603.1 22.70 4.00 51.50 2.10 77.80
11.40 AD-1684631.1 21.40 7.60 53.80 9.50 68.00 7.30 AD-1684625.1 12.80 4.50 54.20 7.70 67.50
14.20 AD-1290523.2 68.70 17.50 67.80 25.20 105.40 19.60 AD-1684602.1 37.60 3.30 69.00 5.10
99.90 5.60 AD-1290570.2 113.90 17.40 74.40 15.30 118.80 10.50 AD-1684666.1 114.80 4.60
75.40 12.40 99.60 7.20 AD-1290865.2 90.60 6.20 76.50 35.10 107.90 14.00 AD-1290589.2 133.30
30.50 77.30 8.40 120.40 23.60 AD-1684681.1 118.30 35.00 80.70 18.10 115.30 14.10 AD-
1684689.1 145.00 30.40 82.20 13.40 100.00 21.80 AD-1684708.1 85.20 12.90 83.50 7.60 89.10
5.30 AD-1290557.2 143.10 27.50 85.40 13.60 142.40 27.30 AD-1290515.2 146.90 30.00 85.90
15.30 152.80 28.10 AD-1290741.2 77.20 17.70 86.30 19.50 88.50 16.50 AD-1684713.1 81.30
21.10 86.80 14.00 89.80 15.00 AD-1290650.2 70.10 15.20 87.00 13.50 93.80 15.50 AD-1290897.2
84.50 12.20 88.50 35.40 98.20 8.90 AD-1290556.2 130.10 24.90 88.60 21.40 149.90 29.90 AD-
1290750.2 41.60 15.10 89.20 24.30 131.10 34.80 AD-1684714.1 84.20 10.00 89.20 7.70 80.10
8.10 AD-1290654.2 117.30 18.70 89.50 16.40 107.40 13.50 AD-1290909.2 79.50 5.30 89.50 12.00
87.40 12.00 AD-1684674.1 109.80 18.20 89.60 26.20 94.20 6.20 AD-1684712.1 75.20 12.40 89.70
6.10 87.70 10.50 AD-1290796.2 94.60 12.30 90.70 8.80 104.30 13.60 AD-1290612.2 141.60 32.40
90.80 5.60 116.90 22.40 AD-1290633.2 92.10 16.50 90.80 4.20 91.90 11.20 AD-1684691.1 130.90
21.50 90.80 10.80 147.30 22.50 AD-1290659.2 77.40 11.60 91.50 10.90 83.40 8.80 AD-1684686.1
115.30 23.00 91.80 23.80 132.80 20.30 AD-1290604.2 102.20 21.70 92.60 11.90 91.50 11.60 AD-
1290574.2 118.10 16.60 93.80 4.40 138.30 11.10 AD-1290609.2 76.10 7.50 93.90 12.10 79.40 4.90
AD-1290911.2 83.20 10.90 94.10 22.60 85.00 20.70 AD-1290615.2 83.40 11.20 94.90 16.30 87.60
9.90 AD-1684710.1 83.80 5.80 95.20 11.00 89.50 7.60 AD-1684709.1 74.10 5.00 95.40 15.30
91.70 6.20 AD-1290533.2 60.80 2.90 96.00 21.40 98.10 5.80 AD-1684671.1 109.90 24.30 96.10
27.80 114.60 15.80 AD-1684650.1 88.00 6.10 96.50 24.20 100.70 11.50 AD-1684711.1 74.10 6.90
97.00 3.80 91.30 7.20 AD-1684670.1 87.20 10.00 97.30 28.90 90.90 5.20 AD-1684692.1 141.10
33.00 97.50 14.20 136.30 32.50 AD-1290939.2 122.30 10.60 98.30 11.70 128.50 6.70 AD-
1684662.1 99.20 17.30 98.60 8.40 100.00 20.40 AD-1684700.1 99.30 9.10 98.60 9.60 106.60
10.90 AD-1290742.2 129.20 16.80 98.90 12.20 128.20 17.60 AD-1684715.1 93.70 10.90 98.90
10.10 100.70 9.00 AD-1290894.2 68.90 12.30 99.60 29.10 80.80 16.30 AD-1684718.1 83.00 3.60
99.60 5.80 96.10 14.40 AD-1290702.2 82.00 17.20 100.60 20.70 90.80 15.10 AD-1684688.1
126.60 29.80 100.60 9.30 144.00 31.80 AD-1684690.1 156.10 29.90 101.30 11.40 156.50 25.20
AD-1684672.1 119.50 27.80 101.40 23.10 125.10 34.20 AD-1684684.1 109.90 17.40 102.00 9.20
116.50 8.10 AD-1684723.1 89.40 12.20 102.40 7.90 107.30 6.20 AD-1684651.1 77.60 7.90 102.50
16.10 86.80 14.20 AD-1684683.1 122.10 20.10 102.50 12.50 135.40 19.70 AD-1290973.2 66.80
15.60 103.80 13.80 93.70 11.40 AD-1290857.2 92.30 5.20 103.90 6.10 105.60 6.90 AD-1684649.1
78.90 6.80 103.90 27.90 103.80 11.00 AD-1290516.2 89.30 12.20 105.20 7.00 107.80 12.00 AD-
1290554.2 101.40 2.80 105.20 10.30 102.20 6.50 AD-1290509.2 93.00 10.80 105.40 4.20 113.40
8.80 AD-1290660.2 101.60 7.50 105.40 4.70 98.80 6.00 AD-1684698.1 105.40 4.50 105.60 11.70
110.20 7.20 AD-1290670.2 88.70 6.50 106.10 15.30 94.50 10.60 AD-1684673.1 88.30 5.20 106.10
2.50 94.50 11.50 AD-1684646.1 68.80 21.90 106.60 19.00 127.30 25.00 AD-1290597.2 105.80
10.00 107.40 3.40 110.00 12.20 AD-1290573.2 97.40 12.60 107.90 15.20 92.20 10.80 AD-
1684707.1 86.40 2.60 108.00 4.80 98.50 11.00 AD-1684722.1 96.60 11.70 108.00 12.10 96.90 8.80
AD-1290639.2 106.10 7.10 108.40 8.50 103.50 6.20 AD-1290551.2 111.70 7.50 109.50 7.80
104.80 7.70 AD-1684655.1 91.50 9.00 109.50 7.40 94.60 5.10 AD-1684678.1 91.10 9.20 109.50
5.60 88.50 3.60 AD-1290800.2 107.10 7.30 109.60 10.90 93.70 8.50 AD-1684726.1 104.90 16.30
109.60 14.60 114.50 8.20 AD-1290764.2 77.50 7.80 110.10 17.00 93.00 12.00 AD-1290672.2
99.10 10.80 110.70 9.90 104.30 13.40 AD-1684685.1 133.70 29.90 110.70 29.30 128.50 32.50 AD-
1290528.2 45.40 4.40 110.80 15.30 84.60 13.70 AD-1684696.1 114.00 11.70 111.00 5.70 120.50
6.60 AD-1290836.2 118.10 8.10 111.70 12.30 123.80 4.70 AD-1684682.1 111.90 18.40 111.80 9.50
124.20 10.20 AD-1684659.1 97.90 4.90 111.90 8.60 99.50 10.20 AD-1684669.1 64.50 9.70 111.90

15.10 81.40 7.10 AD-1684704.1 108.80 8.60 112.00 6.30 97.30 6.70 AD-1684716.1 102.80 16.80
112.00 22.40 98.40 12.60 AD-1684699.1 103.80 5.50 112.40 6.40 105.30 13.70 AD-1290510.2
99.80 10.70 112.70 7.50 115.90 10.10 AD-1290531.2 113.00 19.70 113.00 27.70 118.50 20.40 AD-
1684703.1 85.30 9.20 113.40 14.70 107.50 6.00 AD-1290910.2 109.50 15.50 113.50 20.80 133.70
28.50 AD-1684705.1 94.20 4.80 113.50 8.20 113.00 5.90 AD-1290618.2 113.70 12.60 114.20 7.30
103.40 5.70 AD-1684668.1 74.50 13.50 114.20 6.00 84.80 9.70 AD-1684701.1 92.50 14.90 114.30
10.30 105.10 6.80 AD-1290542.2 104.10 14.20 114.40 10.00 123.10 10.40 AD-1290626.2 119.70
13.70 114.40 15.00 144.90 14.70 AD-1684724.1 107.40 11.70 114.40 19.80 121.90 13.70 AD-
1290535.2 90.80 4.90 114.60 14.70 101.60 7.00 AD-1290558.2 109.40 6.40 114.90 15.30 115.40
12.00 AD-1684647.1 99.40 27.50 114.90 19.80 136.90 32.80 AD-1290763.2 75.90 7.30 115.50
16.90 111.20 14.60 AD-1684677.1 91.30 8.80 115.50 15.40 95.80 18.60 AD-1684658.1 92.50
14.90 116.50 15.90 109.20 12.60 AD-1684687.1 114.50 18.10 116.90 13.90 132.30 10.30 AD-
1684719.1 102.40 6.30 117.20 11.70 96.70 8.20 AD-1290681.2 107.00 9.00 117.60 11.00 117.40
4.40 AD-1684653.1 80.70 3.20 117.70 17.10 81.80 9.30 AD-1290841.2 75.10 4.00 118.00 11.00
85.70 4.70 AD-1290687.2 107.70 12.90 118.20 6.40 124.20 4.50 AD-1290592.2 119.00 9.20
118.30 4.30 120.60 11.50 AD-1290522.2 118.50 37.70 118.50 20.40 136.80 37.60 AD-1290880.2
96.70 19.90 118.70 21.80 110.10 20.90 AD-1684693.1 114.30 7.70 118.70 10.70 116.50 13.10 AD-
1684679.1 117.20 20.30 118.80 6.80 126.50 13.80 AD-1290555.2 125.00 8.30 118.90 4.70 113.50
16.10 AD-1684648.1 81.50 12.40 119.10 21.00 95.30 6.60 AD-1684725.1 114.50 12.10 119.50
17.20 127.60 7.20 AD-1684706.1 89.90 9.20 119.60 27.40 124.10 22.90 AD-1684663.1 82.10 7.70
120.20 13.70 84.40 10.90 AD-1684702.1 98.10 2.80 120.40 5.00 107.50 3.20 AD-1291003.2 85.70
8.50 120.50 8.30 95.90 9.20 AD-1684664.1 87.30 9.50 120.70 17.50 98.60 17.10 AD-1684697.1
86.40 5.20 121.00 8.60 109.80 7.60 AD-1290514.2 110.00 18.00 121.20 8.80 125.50 8.40 AD-
1290989.2 88.80 3.60 121.60 11.60 110.00 12.50 AD-1684657.1 82.50 2.60 121.60 14.20 104.60
5.40 AD-1684654.1 81.10 4.50 121.80 15.10 101.60 2.30 AD-1684695.1 103.90 12.30 122.10 6.90
105.10 10.90 AD-1290712.2 122.10 28.40 122.40 10.60 114.50 28.30 AD-1290931.2 82.00 3.10
122.40 14.10 89.10 11.20 AD-1684729.1 104.60 9.20 123.10 7.70 123.50 12.60 AD-1290805.2
101.60 3.30 123.20 9.40 100.50 5.90 AD-1290755.2 120.80 13.20 123.60 3.30 110.50 19.00 AD-
1290527.2 105.70 3.50 123.80 6.20 116.20 6.30 AD-1684652.1 81.80 9.60 123.80 14.40 91.50
12.50 AD-1290507.2 119.00 5.80 124.40 10.90 119.80 5.30 AD-1290747.2 91.90 6.10 124.50 3.30
102.50 17.60 AD-1290926.2 74.50 8.60 125.30 8.90 92.90 15.90 AD-1290983.2 88.30 10.80
125.70 4.10 96.00 6.30 AD-1684680.1 105.10 11.90 125.90 2.90 126.30 18.10 AD-1290605.2
99.20 7.80 126.50 19.90 107.40 14.50 AD-1290842.2 100.80 20.90 127.20 11.10 105.20 14.70 AD-
1290565.2 104.90 15.40 127.30 19.20 120.90 17.40 AD-1290661.2 115.40 2.60 127.70 5.70 116.60
9.50 AD-1684667.1 79.60 10.90 129.10 19.10 93.80 5.00 AD-1290684.2 100.20 19.10 129.20
10.90 111.00 9.40 AD-1290524.2 114.10 8.60 129.40 21.20 134.10 19.90 AD-1290655.2 84.20
4.60 129.90 7.20 91.70 2.40 AD-1684660.1 106.80 17.30 130.10 13.70 119.80 14.80 AD-
1684656.1 94.10 9.30 130.40 15.20 90.80 14.70 AD-1684676.1 125.80 31.60 130.70 10.10 141.50
42.50 AD-1684727.1 112.30 11.80 130.80 15.60 134.10 10.20 AD-1684665.1 82.50 19.00 131.30
9.90 103.40 6.40 AD-1290993.2 81.50 10.60 132.90 17.10 96.40 5.80 AD-1684721.1 118.30 29.70
133.40 22.80 129.60 16.70 AD-1290543.2 117.30 33.60 133.80 5.80 134.30 21.30 AD-1290719.2
125.50 4.70 134.50 18.60 135.30 13.50 AD-1290602.2 122.90 13.50 135.30 23.70 117.30 12.20
AD-1290564.2 105.80 15.20 136.20 13.00 109.20 11.40 AD-1684737.1 139.10 33.20 136.40 12.50
142.70 28.20 AD-1684694.1 113.60 7.10 136.60 11.80 118.50 7.50 AD-1290875.2 100.10 5.10
138.20 5.60 104.20 4.30 AD-1684732.1 115.40 13.50 139.80 6.40 131.10 6.80 AD-1684717.1
119.30 16.20 140.00 30.80 120.10 24.90 AD-1290908.2 96.80 28.00 140.10 27.80 147.20 37.70
AD-1684720.1 124.30 14.00 140.50 23.90 120.80 14.90 AD-1290552.2 123.10 4.70 142.50 18.90
137.50 12.20 AD-1684728.1 128.40 15.80 143.40 23.80 139.10 18.30 AD-1684731.1 131.10 12.90
147.20 5.60 148.60 29.20 AD-1290722.2 121.90 13.20 147.80 22.40 124.10 19.20 AD-1684730.1
144.30 31.10 150.10 41.90 121.70 5.50 AD-1684736.1 132.70 21.70 151.10 19.90 144.20 15.80

AD-1684733.1 130.50 19.30 151.20 22.00 141.30 22.00 AD-1684661.1 117.60 34.20 151.80 0.00
134.50 29.30 AD-1684734.1 140.80 29.20 153.70 37.90 150.30 28.10 AD-1684735.1 147.10 29.80
154.10 1.60 159.80 22.20 AD-1290600.2 130.70 35.00 160.60 27.90 150.40 19.70 AD-1290624.2
150.80 32.00 161.90 26.00 151.90 33.80 AD-1290643.2 139.30 26.20 163.80 27.20 160.60 26.00
AD-1684675.1 116.00 28.80 NA 0.00 130.20 26.80

(648) TABLE-US-00024 TABLE 16 KHK Single Dose Screen (Dual-Luciferase Assay)

RLuc/FLuc RLuc/FLuc RLuc/FLuc 10 nM 1 nM 0.1 nM Duplex Name Avg (%) SD (%) Avg (%)
SD (%) Avg (%) SD (%) AD-1290635.2 27.55 4.72 75.07 12.04 82.87 2.65 AD-1684641.1 110.35
9.02 108.94 4.27 110.57 4.47 AD-1684620.1 49.87 8.22 58.56 5.05 70.53 11.50 AD-1684619.1
52.62 8.75 75.71 10.40 80.49 2.91 AD-1684610.1 37.56 5.38 67.47 3.80 80.31 4.72 AD-1684617.1
55.49 13.45 70.80 1.77 75.55 1.72 AD-1613079.2 53.10 14.44 59.41 4.50 81.54 6.24 AD-
1684618.1 55.17 9.77 69.69 5.23 72.30 5.02 AD-1684622.1 63.23 13.55 78.66 4.27 89.31 3.28
AD-1684607.1 58.97 7.60 72.73 7.51 76.63 5.82 AD-1684614.1 64.70 10.39 80.88 8.02 79.18 5.68
AD-1613059.2 51.21 4.83 77.16 7.06 74.45 6.20 AD-1613075.2 60.08 8.05 77.04 7.40 83.17 6.93
AD-1613061.1 33.86 3.11 66.45 5.22 75.37 1.85 AD-1684608.1 57.25 9.23 78.62 6.26 78.71 3.21
AD-1684624.1 79.54 24.34 87.68 11.58 90.36 3.70 AD-1423319.3 36.18 4.92 71.09 4.29 81.57
7.77 AD-1684606.1 54.06 6.46 69.09 8.66 81.86 3.98 AD-1684609.1 50.63 11.26 69.42 5.35 75.07
4.46 AD-1684616.1 68.70 8.45 83.79 6.71 90.02 5.65 AD-1613060.2 66.82 13.96 88.93 7.79 85.14
3.13 AD-1613087.1 74.02 4.74 88.43 10.42 86.90 7.15 AD-1290539.5 54.13 8.10 76.99 5.02 77.45
2.84 AD-1290885.5 61.23 8.10 86.40 22.40 86.33 2.17 AD-1613062.2 41.02 9.84 61.06 10.45
78.05 7.96 AD-1684611.1 28.66 7.54 68.17 4.16 73.93 11.62 AD-1684623.1 70.49 21.68 78.20
8.68 93.68 4.29 AD-1612969.2 82.38 14.04 98.61 3.54 88.83 5.84 AD-1613256.2 63.57 23.77
80.73 3.90 85.86 7.48 AD-1423312.3 78.78 18.49 89.56 6.87 93.64 11.24 AD-1548743.7 60.42
7.72 80.40 4.82 87.16 5.21 AD-1684596.1 66.99 3.42 89.91 18.17 93.12 9.16 AD-1423336.7 55.41
10.18 75.06 10.07 81.43 5.09 AD-1612963.2 68.21 8.77 90.00 8.17 89.11 10.05 AD-1290611.3
72.79 17.88 87.65 5.54 89.02 6.88 AD-1613254.2 60.32 9.01 93.11 8.72 91.19 7.21 AD-1684612.1
57.05 5.54 68.77 6.05 77.64 4.46 AD-1684645.1 94.74 17.21 100.78 3.85 97.29 4.48 AD-
1684621.1 70.57 13.92 96.48 4.17 102.25 8.16 AD-1290837.5 75.64 11.34 87.27 7.07 89.95 5.79
AD-1684598.1 82.26 9.02 100.74 8.99 95.70 14.97 AD-1684638.1 90.83 19.35 99.80 1.21 88.22
5.61 AD-1290599.7 81.02 16.64 88.84 3.66 96.80 4.47 AD-1613400.2 78.98 6.10 101.81 4.90
97.90 11.65 AD-1684615.1 67.00 20.89 97.66 10.46 88.82 5.13 AD-1684630.1 92.58 21.02 102.41
5.76 95.70 6.29 AD-1613072.1 74.57 8.15 92.03 6.87 101.02 3.25 AD-1684632.1 123.67 22.63
120.79 8.89 111.58 5.69 AD-1613242.2 78.42 23.84 97.73 5.29 91.80 1.46 AD-1684605.1 100.37
8.25 97.95 10.50 93.62 9.84 AD-1684594.1 72.68 15.57 112.86 9.89 96.08 5.44 AD-1612957.2
84.90 22.12 94.33 10.29 95.29 8.77 AD-1290971.3 90.45 5.32 92.69 5.86 100.99 5.05 AD-
1684595.1 67.25 5.15 83.63 8.61 97.72 8.62 AD-1613371.3 84.73 8.63 93.73 5.20 92.88 3.87 AD-
1684593.1 95.59 10.06 105.09 8.84 99.26 9.30 AD-1684639.1 111.42 26.90 100.56 13.89 100.10
10.09 AD-1613094.2 109.88 20.02 106.26 7.18 91.52 4.75 AD-1684597.1 87.17 18.25 98.70 10.75
97.29 11.97 AD-1290563.2 57.60 6.91 70.25 6.46 93.86 10.89 AD-1684599.1 95.44 9.63 99.22
5.79 96.16 3.23 AD-1684640.1 68.26 5.61 95.79 2.54 102.60 4.75 AD-1290584.2 61.38 5.41 94.01
8.50 94.05 2.24 AD-1684627.1 90.44 15.14 92.95 9.50 93.69 9.49 AD-1290969.7 81.86 22.36
100.31 15.03 104.51 10.34 AD-1290651.2 97.29 8.97 93.94 5.58 90.64 2.47 AD-1684628.1 99.71
21.99 94.37 8.89 95.59 1.56 AD-1290884.5 89.60 2.17 98.52 11.66 105.58 5.25 AD-1684633.1
124.09 6.72 124.21 8.35 126.12 7.08 AD-1684637.1 87.99 25.10 99.86 3.94 98.08 2.65 AD-
1684629.1 96.24 14.82 108.74 6.76 98.94 1.93 AD-1684635.1 115.43 19.21 110.59 14.86 111.51
5.28 AD-1684613.1 77.44 17.90 86.13 7.31 96.31 11.48 AD-1684604.1 126.73 12.80 104.44 9.66
88.10 9.76 AD-1290666.2 77.77 7.66 85.66 5.31 91.09 7.78 AD-1684634.1 101.54 9.42 112.46
7.92 102.84 8.33 AD-1684600.1 92.36 15.27 102.10 14.83 98.62 7.39 AD-1684644.1 94.78 12.14
105.73 11.26 91.40 7.74 AD-1684643.1 134.90 25.23 106.06 3.63 114.45 5.37 AD-1684601.1
109.99 17.91 106.64 6.25 93.03 6.35 AD-1684626.1 105.90 5.90 97.95 4.84 97.84 3.22 AD-

1684642.1 121.11 34.83 118.91 13.95 106.61 2.62 AD-1684592.1 93.35 12.76 103.28 10.86 103.22
9.57 AD-1684636.1 101.15 23.41 102.25 10.70 108.30 10.50 AD-1684603.1 105.78 12.89 103.83
10.25 94.04 10.93 AD-1684631.1 88.92 2.51 98.87 14.74 103.10 3.82 AD-1684625.1 98.13 10.61
116.53 26.66 103.80 4.63 AD-1290523.2 36.07 3.49 83.69 18.88 78.13 21.59 AD-1684602.1
108.47 14.10 109.97 12.29 100.25 5.84 AD-1290570.2 69.96 7.85 86.80 14.21 101.61 7.01 AD-
1684666.1 77.63 4.44 85.12 7.88 89.66 10.91 AD-1290865.2 67.01 0.58 88.22 11.34 96.40 4.28
AD-1290589.2 67.56 4.89 85.24 5.97 97.93 8.15 AD-1684681.1 86.44 3.20 88.04 8.65 103.81
12.28 AD-1684689.1 49.57 6.76 86.48 5.18 90.62 11.35 AD-1684708.1 23.03 1.62 33.48 7.99
57.59 11.14 AD-1290557.2 94.58 3.42 98.87 10.22 103.18 8.22 AD-1290515.2 48.39 6.21 70.61
2.16 86.52 1.87 AD-1290741.2 62.89 8.45 108.61 21.31 83.31 16.19 AD-1684713.1 25.62 2.85
56.78 8.85 73.15 15.46 AD-1290650.2 45.96 4.20 75.32 14.73 88.04 15.00 AD-1290897.2 91.50
12.78 99.78 4.12 95.69 6.37 AD-1290556.2 46.60 6.23 72.83 12.88 95.77 2.43 AD-1290750.2
100.69 11.17 112.50 12.75 98.54 9.04 AD-1684714.1 28.09 1.61 63.87 15.58 58.74 14.71 AD-
1290654.2 59.38 5.24 82.45 5.68 92.90 11.63 AD-1290909.2 79.81 7.84 85.50 1.84 99.95 6.60 AD-
1684674.1 38.19 5.91 68.84 4.29 86.91 7.62 AD-1684712.1 27.74 1.92 51.61 7.17 81.25 10.84
AD-1290796.2 90.91 8.26 93.92 11.82 98.49 8.90 AD-1290612.2 42.68 2.55 66.55 14.42 68.94
7.63 AD-1290633.2 50.34 3.85 73.42 11.49 83.99 10.78 AD-1684691.1 66.75 8.42 91.12 4.34
97.18 11.63 AD-1290659.2 29.24 3.16 58.78 1.46 83.51 13.24 AD-1684686.1 55.60 5.55 81.33
10.32 90.49 12.00 AD-1290604.2 30.50 1.61 58.09 12.94 75.45 12.12 AD-1290574.2 62.06 4.41
79.92 5.27 89.83 8.75 AD-1290609.2 27.15 4.45 50.65 5.55 73.67 12.54 AD-1290911.2 96.37 7.95
93.75 11.00 96.22 5.42 AD-1290615.2 51.72 2.67 70.39 15.92 77.60 16.60 AD-1684710.1 42.53
5.32 73.27 11.74 68.13 11.94 AD-1684709.1 58.70 4.19 79.18 25.68 86.01 19.30 AD-1290533.2
47.59 4.90 68.99 11.94 77.30 6.48 AD-1684671.1 47.59 1.95 61.26 3.20 82.05 10.96 AD-
1684650.1 46.11 8.10 76.50 10.07 82.83 4.08 AD-1684711.1 25.37 2.16 39.63 9.66 66.61 6.31 AD-
1684670.1 61.98 4.96 79.82 3.32 86.14 2.50 AD-1684692.1 51.68 5.44 73.68 5.82 94.81 6.71 AD-
1290939.2 79.47 12.39 91.84 15.17 102.79 10.83 AD-1684662.1 92.36 5.74 89.87 6.57 98.46 7.31
AD-1684700.1 49.06 2.35 66.17 5.75 80.59 11.96 AD-1290742.2 97.05 12.34 108.91 9.16 98.27
8.80 AD-1684715.1 39.93 3.07 71.43 4.49 66.16 5.22 AD-1290894.2 69.53 5.96 86.15 6.51 89.41
8.71 AD-1684718.1 38.19 2.50 56.43 6.04 89.12 4.16 AD-1290702.2 63.45 4.81 94.75 15.47 93.98
25.50 AD-1684688.1 31.21 4.38 62.76 5.56 71.36 5.67 AD-1684690.1 37.78 7.44 65.65 6.32 81.53
5.19 AD-1684672.1 50.47 5.26 77.82 8.07 89.96 7.10 AD-1684684.1 34.84 3.57 66.09 6.84 81.50
13.30 AD-1684723.1 35.25 2.97 68.48 5.60 65.99 9.83 AD-1684651.1 70.03 6.39 85.13 4.79 94.44
5.75 AD-1684683.1 54.95 5.39 77.78 7.33 95.72 7.28 AD-1290973.2 98.50 6.54 92.81 14.03 98.60
4.58 AD-1290857.2 87.52 2.70 96.73 5.63 99.17 2.18 AD-1684649.1 72.47 0.92 84.56 8.23 89.82
2.21 AD-1290516.2 26.76 0.49 42.76 4.24 59.84 6.99 AD-1290554.2 61.58 4.05 88.87 14.21 74.37
3.21 AD-1290509.2 73.11 5.62 99.31 14.02 101.98 16.69 AD-1290660.2 54.56 9.39 83.38 0.31
86.16 6.94 AD-1684698.1 49.13 6.23 101.74 19.35 58.63 21.67 AD-1290670.2 70.45 5.20 85.30
12.39 83.73 9.68 AD-1684673.1 54.65 6.25 78.09 7.36 92.77 10.76 AD-1684646.1 88.86 5.66
109.84 2.99 100.07 6.89 AD-1290597.2 64.33 3.82 104.86 17.05 78.64 14.59 AD-1290573.2 21.93
2.25 51.90 11.20 61.08 10.20 AD-1684707.1 25.65 1.98 47.58 11.42 58.67 13.41 AD-1684722.1
42.02 2.86 72.20 13.41 91.03 17.74 AD-1290639.2 65.65 8.85 106.60 6.21 80.86 3.84 AD-
1290551.2 37.93 4.95 63.38 4.48 69.08 10.49 AD-1684655.1 82.73 4.01 88.85 9.58 98.35 13.91
AD-1684678.1 71.26 6.57 96.82 5.79 98.92 5.96 AD-1290800.2 81.68 8.76 95.82 8.76 99.82 11.38
AD-1684726.1 28.47 1.78 45.62 14.88 47.10 11.51 AD-1290764.2 67.56 8.24 67.75 7.72 79.61
6.78 AD-1290672.2 42.65 3.57 64.81 14.49 73.51 15.34 AD-1684685.1 52.36 4.45 80.46 8.33
90.09 6.80 AD-1290528.2 40.22 5.18 55.62 6.12 74.10 6.88 AD-1684696.1 36.27 4.80 43.87 3.17
104.26 16.60 AD-1290836.2 78.37 9.59 84.56 11.22 101.22 23.26 AD-1684682.1 80.66 18.04
114.19 18.74 92.36 3.20 AD-1684659.1 80.80 5.53 92.64 9.37 100.32 4.86 AD-1684669.1 61.93
5.52 72.55 4.18 83.30 4.78 AD-1684704.1 73.43 3.74 102.98 4.47 82.11 11.96 AD-1684716.1
49.39 6.90 78.93 10.67 88.70 8.71 AD-1684699.1 32.13 1.49 49.17 12.29 88.94 7.66 AD-

1290510.2 29.78 2.09 61.76 7.12 57.41 3.23 AD-1290531.2 30.64 2.31 61.41 11.62 88.80 10.45
AD-1684703.1 29.94 4.56 48.74 5.70 64.27 5.80 AD-1290910.2 78.87 4.68 83.35 7.19 102.58 9.62
AD-1684705.1 84.81 5.73 78.46 14.78 108.49 29.28 AD-1290618.2 38.80 2.21 91.85 24.03 70.60
14.74 AD-1684668.1 82.61 3.09 85.84 5.55 102.22 13.23 AD-1684701.1 33.37 3.20 59.65 10.73
79.48 7.48 AD-1290542.2 25.33 4.70 75.56 5.50 43.19 10.83 AD-1290626.2 42.99 7.92 66.24 9.90
78.99 6.15 AD-1684724.1 25.49 3.64 41.90 10.84 75.17 17.57 AD-1290535.2 39.61 4.59 86.53
3.83 75.49 9.54 AD-1290558.2 52.87 4.30 77.41 14.99 88.24 18.29 AD-1684647.1 63.31 20.44
88.14 11.46 99.17 9.42 AD-1290763.2 76.79 3.45 91.24 4.92 100.24 8.61 AD-1684677.1 49.89
7.84 62.12 3.31 80.47 5.14 AD-1684658.1 61.99 9.91 82.39 6.38 93.13 3.41 AD-1684687.1 56.84
10.02 73.94 6.08 90.30 8.39 AD-1684719.1 21.60 1.78 54.66 6.44 49.60 12.61 AD-1290681.2
70.01 13.50 102.05 13.79 97.11 8.26 AD-1684653.1 77.63 3.92 90.78 7.67 107.04 10.77 AD-
1290841.2 102.02 12.88 92.72 10.43 92.89 8.78 AD-1290687.2 24.79 5.58 47.49 9.56 54.96 7.96
AD-1290592.2 42.32 3.00 63.61 7.19 81.99 13.27 AD-1290522.2 63.08 8.25 80.77 2.61 89.07 3.72
AD-1290880.2 86.62 8.68 84.96 10.84 112.23 7.89 AD-1684693.1 33.57 4.76 72.88 8.53 56.56
8.52 AD-1684679.1 63.98 10.81 81.92 8.14 99.47 7.56 AD-1290555.2 29.21 2.95 57.32 7.73 76.70
14.67 AD-1684648.1 57.07 4.43 72.65 5.51 90.63 6.01 AD-1684725.1 37.50 5.23 60.12 13.34
75.61 7.15 AD-1684706.1 34.25 5.45 66.03 7.18 61.51 7.34 AD-1684663.1 100.81 7.18 100.10
13.78 98.87 4.95 AD-1684702.1 27.84 1.80 55.04 5.17 62.99 6.59 AD-1291003.2 99.82 7.41 90.75
6.78 96.81 11.67 AD-1684664.1 98.11 1.83 95.21 12.95 100.40 8.82 AD-1684697.1 63.74 8.80
90.40 16.93 79.86 15.55 AD-1290514.2 25.53 1.05 39.72 6.89 59.21 5.46 AD-1290989.2 80.25
6.00 95.06 4.87 94.19 9.09 AD-1684657.1 85.45 4.39 92.61 11.75 107.59 8.01 AD-1684654.1
93.92 6.75 95.83 4.68 99.57 13.80 AD-1684695.1 40.77 3.71 87.34 16.28 65.10 12.55 AD-
1290712.2 76.80 6.49 89.36 6.93 97.09 5.67 AD-1290931.2 74.20 8.48 91.78 7.74 93.55 7.40 AD-
1684729.1 32.72 5.04 71.92 14.70 63.69 5.82 AD-1290805.2 85.12 3.08 83.71 7.55 90.13 4.27
AD-1290755.2 82.43 5.17 101.69 9.25 96.50 4.82 AD-1290527.2 35.84 2.68 72.84 6.30 61.60
13.91 AD-1684652.1 85.57 2.57 95.34 9.86 91.15 4.65 AD-1290507.2 29.99 3.00 86.33 6.23 57.42
9.02 AD-1290747.2 93.73 11.80 93.12 7.86 93.91 7.92 AD-1290926.2 105.81 5.23 93.36 5.15
101.13 11.67 AD-1290983.2 83.97 2.62 84.46 5.64 102.69 8.80 AD-1684680.1 75.84 12.82 88.75
9.33 92.50 2.09 AD-1290605.2 71.92 10.68 111.52 16.71 79.13 8.92 AD-1290842.2 104.49 5.65
91.21 5.53 95.02 7.94 AD-1290565.2 34.69 2.42 75.14 21.84 83.67 22.69 AD-1290661.2 47.11
5.63 82.93 15.48 77.74 26.12 AD-1684667.1 50.51 6.46 74.42 2.87 89.86 7.71 AD-1290684.2
61.88 5.00 86.54 7.60 91.09 8.89 AD-1290524.2 55.34 5.89 100.08 30.59 97.04 35.30 AD-
1290655.2 55.60 8.94 79.31 8.29 86.56 4.09 AD-1684660.1 90.30 7.72 90.68 6.87 89.53 4.84 AD-
1684656.1 99.30 6.66 87.28 5.54 93.70 10.20 AD-1684676.1 42.60 3.34 64.98 4.82 81.15 8.48
AD-1684727.1 34.77 9.34 62.64 16.13 56.10 7.77 AD-1684665.1 88.07 7.37 85.96 2.62 96.42 4.35
AD-1290993.2 126.19 4.37 103.72 5.04 100.36 8.22 AD-1684721.1 30.93 4.69 73.02 5.08 74.51
8.53 AD-1290543.2 60.43 2.14 78.76 1.16 86.30 3.38 AD-1290719.2 72.55 5.47 106.96 11.38
89.56 13.94 AD-1290602.2 36.81 5.78 75.60 6.07 63.11 4.79 AD-1290564.2 46.18 1.47 88.84
15.84 89.88 12.30 AD-1684737.1 19.91 2.75 65.84 2.75 50.34 9.86 AD-1684694.1 37.30 6.64
65.54 8.09 68.75 11.54 AD-1290875.2 88.36 6.67 87.65 4.57 95.91 11.39 AD-1684732.1 21.92
1.86 70.21 17.18 48.70 2.36 AD-1684717.1 20.14 3.23 53.41 2.38 49.49 5.05 AD-1290908.2 69.14
8.90 91.09 8.96 105.33 6.07 AD-1684720.1 25.19 1.57 65.20 9.13 55.67 9.89 AD-1290552.2 60.07
5.86 110.60 5.95 68.03 4.85 AD-1684728.1 32.65 4.26 60.26 10.61 81.39 13.63 AD-1684731.1
71.00 4.39 97.76 10.63 103.47 10.13 AD-1290722.2 56.44 2.77 96.22 25.60 84.59 1.63 AD-
1684730.1 55.45 5.72 103.19 33.31 94.48 30.63 AD-1684736.1 22.63 1.19 55.13 6.21 63.03 19.64
AD-1684733.1 31.22 2.64 74.74 15.23 80.11 13.22 AD-1684661.1 88.94 5.85 86.32 8.94 96.70
2.89 AD-1684734.1 24.44 1.34 55.52 4.03 41.46 7.30 AD-1684735.1 27.36 1.31 62.50 9.06 50.38
6.57 AD-1290600.2 63.33 8.27 96.26 30.33 114.38 17.93 AD-1290624.2 34.72 1.33 92.38 34.13
71.36 13.25 AD-1290643.2 31.00 3.54 63.18 15.27 67.58 13.33 AD-1684675.1 43.08 8.00 63.69
6.05 73.37 3.64

Example 6: Effects of siRNA-GalNAC Conjugates in Non-Human Primates

(649) The effect of candidates identified from the in vitro studies described above, duplexes AD-1613062, AD-1613073, AD-1613242, AD-1613243, AD-1613246, AD-1613247, AD-1613400, AD-1634397, AD-1634424, and AD-1634425, were further investigated for their effectiveness in non-human primates. Specifically, a single dose of 3 mg/kg of AD-1613062, AD-1613073, AD-1613242, AD-1613243, AD-1613246, AD-1613247, AD-1613400, AD-1634397, AD-1634424, or AD-1634425 was subcutaneously administered to cynomolgous monkeys. Sera and tissue samples were collected at Days 29, 50, and 78 post-dose.

(650) mRNA was extracted from liver tissue by a magnetic bead-based extraction method using the NucleoMag RNA kit by Macherey-Nagel. cDNA was generated using Applied Biosystems SuperScript IV VILO master mix. Taqman probe-based qPCR was used to quantify KHK mRNA which was normalized to the geometric mean of two housekeeping genes, ARL6IP4 and PPIB.

(651) Targeted quantitation of KHK protein in cynomolgus monkey liver was performed via liquid chromatography coupled to mass spectrometer (LC-MS) using parallel reaction monitoring (PRM) in positive ion mode. The signature peptide sequence selected for quantitation was HLGFQSAGEALR (SEQ ID NO: 2044) (UniProtKB—A0A2K5V1R8-1, amino acid positions 198-209).

(652) FIG. 2 shows the effect of administration of a single 3 mg/kg dose of the selected duplexes on the level of KHK mRNA and protein at Day 29 post-dose.

(653) FIG. 3 shows the effect of administration of a single 3 mg/kg dose of the selected duplexes on the level of KHK mRNA at Day 50 post-dose.

(654) FIG. 4 shows the effect of administration of a single 3 mg/kg dose of the selected duplexes on the level of KHK mRNA at Day 78 post-dose.

(655) The data demonstrate that the indicated agents, e.g., AD-1613400 and AD-1613243, are effective in durably and potently inhibiting KHK mRNA and protein expression.

EQUIVALENTS

(656) Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments and methods described herein. Such equivalents are intended to be encompassed by the scope of the following claims.

Claims

1. A method of inhibiting expression of a ketohexokinase (KHK) gene in a cell, the method comprising contacting the cell with a double stranded ribonucleic acid (dsRNA) agent that inhibits expression of ketohexokinase (KHK), or a pharmaceutically acceptable salt thereof, wherein the dsRNA agent comprises a sense strand and an antisense strand forming a double stranded region, wherein the nucleotide sequence of the sense strand differs by no more than 4 bases from the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' of SEQ ID NO: 1420 and the nucleotide sequence of the antisense strand differs by no more than 4 bases from the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' of SEQ ID NO:1532, wherein a, g, c and u are 2'-O-methyl (2'-OMe) A, G, C, and U, respectively; Cf and Af are 2'-fluoro (2'-F) C and A, respectively; dC, dA, and dT are 2'-deoxy C, A, and T, respectively; and s is a phosphorothioate linkage, and wherein the sense strand of the dsRNA agent is conjugated to a ligand, thereby inhibiting expression of the KHK gene in the cell.

2. The method of claim 1, wherein the cell is within a subject.

3. The method of claim 2, wherein the subject is a human.

4. The method of claim 1, wherein the nucleotide sequence of the sense strand comprises the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' of SEQ ID NO: 1420 and the nucleotide sequence of the antisense strand comprises the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' of SEQ ID NO:1532.

5. The method of claim 1, wherein the nucleotide sequence of the sense strand consists of the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' of SEQ ID NO: 1420 and the nucleotide sequence of the antisense strand consists of the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' of SEQ ID NO:1532.
6. The method of claim 1, wherein the ligand is conjugated to the 3' end of the sense strand of the dsRNA agent.
7. The method of claim 1, wherein the ligand is an N-acetylgalactosamine (GalNAc) derivative.
8. The method of claim 7, wherein the ligand is one or more GalNAc derivatives attached through a monovalent, bivalent, or trivalent branched linker.
9. The method of claim 7, wherein the ligand is ##STR00040##
10. The method of claim 9, wherein the dsRNA agent is conjugated to the ligand as shown in the following schematic ##STR00041## wherein X is O or S.
11. The method of claim 10, wherein X is O.
12. A method of treating a subject having a ketohexokinase (KHK) associated disorder, comprising administering to the subject a therapeutically effective amount of a double stranded ribonucleic acid (dsRNA) agent that inhibits expression of ketohexokinase (KHK), or a pharmaceutically acceptable salt thereof, wherein the dsRNA agent comprises a sense strand and an antisense strand forming a double stranded region, wherein the nucleotide sequence of the sense strand differs by no more than 4 bases from the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' of SEQ ID NO: 1420 and the nucleotide sequence of the antisense strand differs by no more than 4 bases from the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' of SEQ ID NO:1532, wherein a, g, c and u are 2'-O-methyl (2'-OMe) A, G, C, and U, respectively; Cf and Af are 2'-fluoro (2'-F) C and A, respectively; dC, dA, and dT are 2'-deoxy C, A, and T, respectively; and s is a phosphorothioate linkage, and wherein the 3'-end of the sense strand is conjugated to a ligand as shown in the following schematic ##STR00042## wherein X is O or S, thereby treating the subject having a KHK associated disorder.
13. The method of claim 12, wherein the subject is a human.
14. The method of claim 12, wherein the KHK-associated disorder is a liver disease.
15. The method of claim 14, wherein the liver disease is selected from the group consisting of non-alcoholic fatty liver disease (NAFLD) and non-alcoholic steatohepatitis (NASH).
16. The method of claim 12, wherein the KHK-associated disorder is dyslipidemia selected from the group consisting of hyperlipidemia, high LDL cholesterol, low HDL cholesterol, hypertriglyceridemia, postprandial hypertriglyceridemia, adipocyte dysfunction, visceral adipose deposition, obesity, and metabolic syndrome.
17. The method of claim 12, wherein the KHK-associated disorder is a disorder of glycemic control selected from the group consisting of insulin resistance, type 2 diabetes, and glucose intolerance.
18. The method of claim 12, wherein the administration of the dsRNA agent to the subject causes a decrease in one or more serum lipid and/or a decrease in KHK protein accumulation.
19. The method of claim 12, wherein the administration of the dsRNA agent to the subject causes a decrease in fructose metabolism.
20. The method of claim 12, wherein the dsRNA agent, or a pharmaceutically acceptable salt thereof, is administered to the subject subcutaneously.
21. The method of claim 12, wherein the nucleotide sequence of the sense strand comprises the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' of SEQ ID NO: 1420 and the nucleotide sequence of the antisense strand comprises the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' of SEQ ID NO:1532.
22. The method of claim 12, wherein the nucleotide sequence of the sense strand consists of the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' of SEQ ID NO: 1420 and the nucleotide sequence of the antisense strand consists of the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' of SEQ ID NO:1532.

23. The method of claim 12, wherein X is O.

24. A method of treating a subject having type 2 diabetes, comprising administering to the subject a therapeutically effective amount of a double stranded ribonucleic acid (dsRNA) agent that inhibits expression of ketohexokinase (KHK), or a pharmaceutically acceptable salt thereof, wherein the dsRNA agent comprises a sense strand and an antisense strand forming a double stranded region, wherein the nucleotide sequence of the sense strand differs by no more than 4 bases from the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' of SEQ ID NO: 1420 and the nucleotide sequence of the antisense strand differs by no more than 4 bases from the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' of SEQ ID NO:1532, wherein a, g, c and u are 2'-O-methyl (2'-OMe) A, G, C, and U, respectively; Cf and Af are 2'-fluoro (2'-F) C and A, respectively; dC, dA, and dT are 2'-deoxy C, A, and T, respectively; and s is a phosphorothioate linkage, and wherein the 3'-end of the sense strand is conjugated to a ligand as shown in the following schematic ##STR00043## wherein X is O or S, thereby treating the subject having type 2 diabetes.

25. The method of claim 24, wherein the subject is a human.

26. The method of claim 24, wherein the dsRNA agent, or a pharmaceutically acceptable salt thereof, is administered to the subject subcutaneously.

27. The method of claim 24, wherein the nucleotide sequence of the sense strand comprises the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' of SEQ ID NO: 1420 and the nucleotide sequence of the antisense strand comprises the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' of SEQ ID NO:1532.

28. The method of claim 24, wherein the nucleotide sequence of the sense strand consists of the nucleotide sequence 5'-gscsaggaagCfAfCfugagauucgu-3' of SEQ ID NO: 1420 and the nucleotide sequence of the antisense strand consists of the nucleotide sequence 5'-asdCsgadAudCucagdTgCfuuccugcsasc-3' of SEQ ID NO:1532.

29. The method of claim 24, wherein X is O.
